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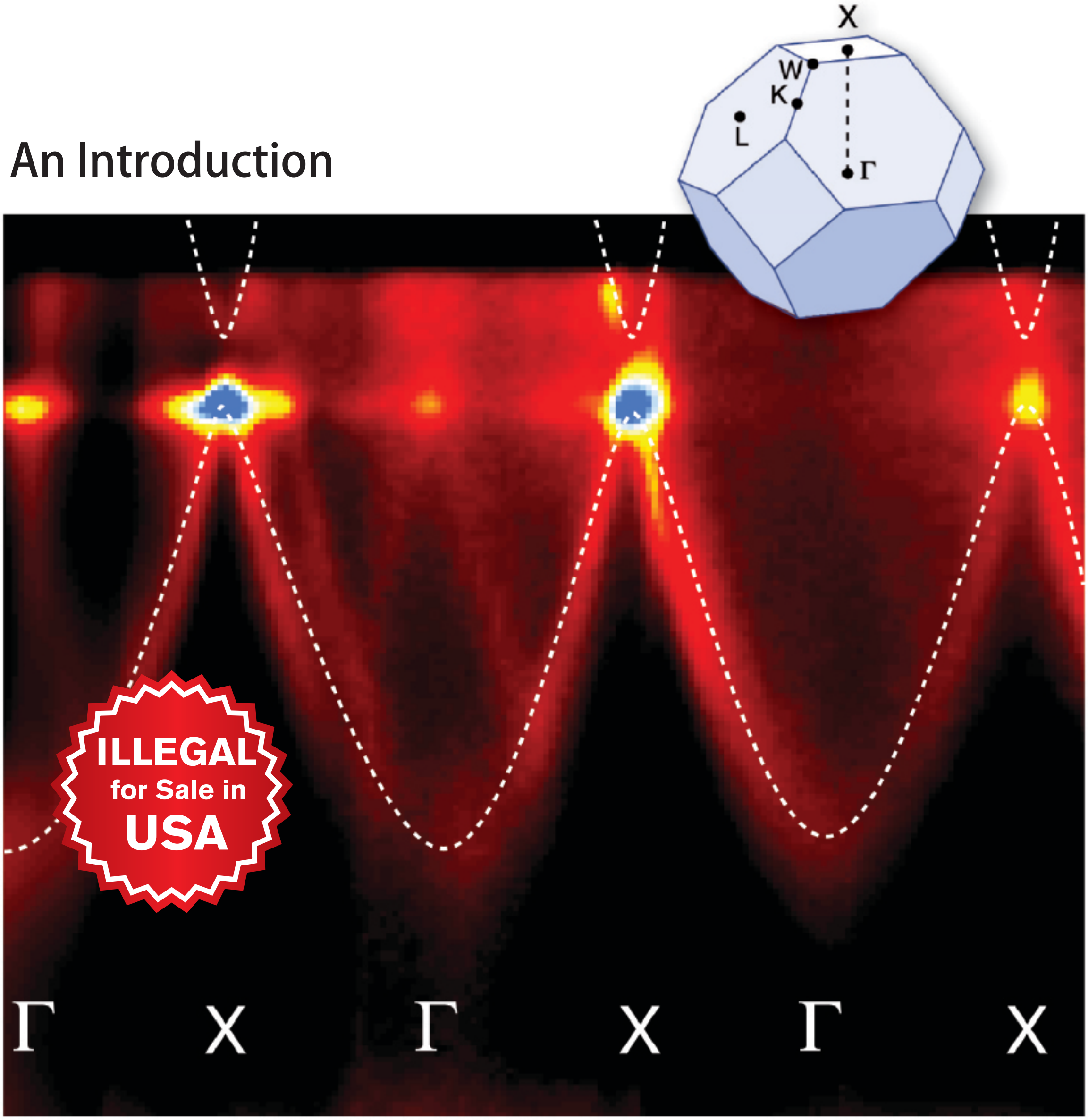
Philip Hofmann

Solid State Physics

An Introduction



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Solid State Physics

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Solid State Physics: An Introduction

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Cover
Band structure of aluminum determined by angle–resolved photoemission. The color plot shows the photoemission intensity along the Γ–X direction. The dashed lines qualitatively track the maximum of the intensity and correspond to the band structure. Data taken from *Physical Review* B **66**, 245422 (2002), see also Fig. 6.10 in this book.

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Solid State Physics: An Introduction. Philip Hofmann

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Preface

This book emerged from a course on solid-state physics for third-year students of physics and nanoscience, but it should also be useful for students of related fields such as chemistry and engineering. The aim is to provide a bachelor-level survey over the whole field without going into too much detail. With this in mind, a lot of emphasis is put on a didactic presentation and little on stringent mathematical derivations or completeness. For a more in-depth treatment, the reader is referred to the many excellent advanced solid-state physics books. A few are listed in the Appendix.

To follow this text, a basic university-level physics course as well as some working knowledge of chemistry, quantum mechanics, and statistical physics is required. A course in classical electrodynamics is of advantage but not strictly necessary.

Some remarks on *how to use this book*: Every chapter is accompanied by a set of "discussion" questions and problems. The intention of the questions is to give the students a tool for testing their understanding of the subject. Some of the questions can only be answered with knowledge of later chapters. These are marked by an asterisk. Some of the problems are more of a challenge in that they are more difficult mathematically or conceptually or both. These problems are also marked by an asterisk. Not all the information necessary for solving the problems is given here. For standard data, for example, the density of gold or the atomic weight of copper, the reader is referred to the excellent resources available on the World Wide Web. Several useful resources for lectures, including equations, suggestions for lectures and the solutions to the problems in this book, can be found on the publisher's web pages. More information, such as an updated list of relevant www pages, can be found on the author's web pages.

I would like to thank the people who have helped me with many discussions, encouragement and suggestions. In particular, I would like to mention my colleagues Arne Nylandsted Larsen, Ivan Steensgaard, Maria Fuglsang Jensen, Justin Wells, Jørgen Bøttiger and many other instructors and students involved in teaching the course in Aarhus. Finally I thank Jørn Lyngesen for a careful proof-reading of the entire manuscript.

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1
Chemical Bonding in Solids

In this chapter, we discuss different mechanisms that can lead to bonding between atoms so that they form solids. We will encounter different cases such as ionic, covalent, or metallic bonding. It has to be kept in mind that these are just idealized limiting cases. Often mixed bonding types are found, for example, a combination of metallic and covalent bonding in the transition metals.

As in conventional chemistry, only a restricted number of all the electrons participate in the bonding. These so-called valence electrons are the electrons in the outermost shell(s) of an atom. The electrons in the inner shells, or core electrons, are bound so tightly to the nucleus that they do not feel the presence of other atoms in their neighborhood.

1.1
Attractive and Repulsive Forces

Two different forces must be present to establish bonding in a solid or in a molecule. An attractive force is necessary for any bonding. Different types of attractive forces are discussed below. A repulsive force, on the other hand, is required in order to keep the atoms from getting too close to each other. An expression for an interatomic potential can be written as

$$\phi(r) = \frac{A}{r^n} - \frac{B}{r^m}, \tag{1.1}$$

where $n > m$, that is, the repulsive part has to prevail for short distances (sometimes this is achieved by assuming an exponential repulsion potential). Such a potential and the resulting force are shown in Figure 1.1. The reason for the strong repulsion at short distances is the Pauli exclusion principle. For a strong overlap of the electron clouds of two atoms, the wave functions have to change in order to become orthogonal to each other, because the Pauli principle forbids having more than two electrons in the same quantum state. The orthogonalization costs much energy, hence the strong repulsion.

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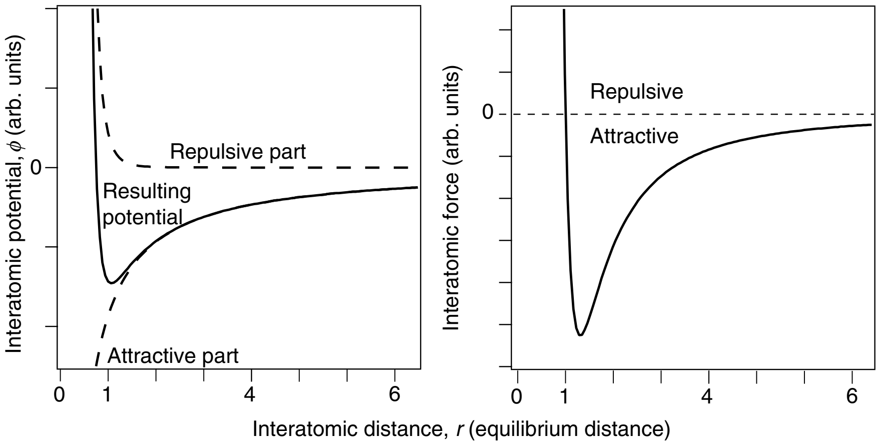

Figure 1.1 (a) Typical interatomic potential for bonding in solids according to (1.1) with $n = 6$ and $m = 1$. (b) Resulting force, that is, $-\mathrm{grad}\phi(\mathbf{r})$.

1.2 Ionic Bonding

Ionic bonding involves the transfer of electrons from an electropositive atom to an electronegative atom. The bonding force is the Coulomb attraction between the two resulting ions. Ionizing both atoms usually costs some energy. In the case of NaCl, the ionization energy of Na is 5.1 eV but the electron affinity of Cl is only 3.6 eV. The net energy cost for creating a pair of ions is thus $5.1 - 3.6 = 1.5$ eV. The energy gain is given by the Coulomb potential. For just one Na and one Cl ion separated by a distance $a = 0.28$ nm, this is $-e^2/4\pi\varepsilon_0 a$, which amounts to 5.1 eV. Potential energies for more complicated structures are discussed below. It is important to distinguish between the different energy contributions involved: The *ionization energy* is the energy needed to turn atoms into ions, the *lattice energy* is the electrostatic energy gain by assembling a lattice, and the *cohesive energy* is the lattice energy minus the ionization energy, that is, it represents the total energy balance to form the ionic solid.

The fact that the total potential energy curve in Figure 1.1 contains a repulsive part in addition to the Coulomb potential means that the actual potential minimum for a given interatomic distance a is a bit shallower than expected from the pure Coulomb potential (10% or so). Therefore, the lattice energy can be calculated rather accurately using classical physics if the interatomic distance a is known. Predicting this distance, however, requires the inclusion of quantum effects. In any event, ionic bonding is very strong. The cohesive energy per atom is in the order of several electron volts.

1.3 Covalent Bonding

Covalent bonding is based on the true sharing of electrons between different atoms. The simplest case is that of the hydrogen molecule that we will discuss quantitatively below. In solids, covalent bonding is often found for elements with a roughly half-filled outer shell. A prominent example is carbon that forms solids as diamond or graphite as well as complex molecules such as Buckminster Fullerene C_{60} or carbon nanotubes. The covalent bonds in diamond are constructed from a linear combination of the 2s orbital and three 2p orbitals. This results in four so-called sp^3 orbitals that stick out in a tetrahedral configuration from the carbon atoms. In graphite, the 2s orbital is combined with only two 2p orbitals, giving three sp^2 orbitals, all in one plane and separated by an angle of 120°, and one p orbital oriented perpendicular to this plane. This linear combination of orbitals already reveals an important characteristic for the covalent bonding: it is highly directional. In addition to this, it is also very stable and the bonding energies are several electron volts.

A very instructive example for covalent bonding is the hydrogen molecule H_2 for which we will sketch a solution here. We go into some detail, as much of this will be useful in later chapters. As a starting point, take two hydrogen atoms with their nuclei at $\mathbf{R}_A$ and $\mathbf{R}_B$ and we call $|\mathbf{R}_B - \mathbf{R}_A| = R$. We do, of course, know the solution of the Schrödinger equation for each of the atoms. Let these ground-state wave functions be Ψ_A and Ψ_B, respectively. The Hamilton operator for the hydrogen molecule can be written as

$$H = -\frac{\hbar^2\nabla_1^2}{2m_e} - \frac{\hbar^2\nabla_2^2}{2m_e} + \frac{e^2}{4\pi\varepsilon_0}\left\{\frac{1}{R} + \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} - \frac{1}{|\mathbf{R}_A - \mathbf{r}_1|} - \frac{1}{|\mathbf{R}_B - \mathbf{r}_2|} - \frac{1}{|\mathbf{R}_A - \mathbf{r}_2|} - \frac{1}{|\mathbf{R}_B - \mathbf{r}_1|}\right\}, \tag{1.2}$$

where $\mathbf{r}_1$ and $\mathbf{r}_2$ are the coordinates of the electrons belonging to the A and the B nucleus, respectively. The first two terms refer to the kinetic energy of the two electrons. The operators ∇_1^2 and ∇_2^2 act only on the coordinates $\mathbf{r}_1$ and $\mathbf{r}_2$, respectively. The electrostatic term contains the repulsion between the two nuclei and the repulsion between the two electrons as well as the attraction of each electron to each nucleus.

The solution of this problem is not simple. It would be greatly simplified by removing the electrostatic interaction between the two electrons because then the Hamiltonian could be written as the sum of two parts, one for each electron. This could then be solved by a product of the two wave functions that are solutions to the two individual Hamiltonians, that is, the two-particle wave function would look like $\Psi(\mathbf{r}_1, \mathbf{r}_2) = \Psi_A(\mathbf{r}_1)\Psi_B(\mathbf{r}_2)$. Actually, this is not quite right because such a wave function is not in accordance with the Pauli principle. Since the electrons are fermions, the total wave function must be antisymmetric with respect to particle exchange and the simple product wave function does not fulfill this requirement.

The total wave function consists of a spatial part and a spin part and therefore there are two possibilities for forming an antisymmetric wave function. We can either choose a symmetric spatial part and an antisymmetric spin part or vice versa. This is achieved by constructing the spacial wave function of the form

$$\Psi_{\uparrow\downarrow}(\mathbf{r}_1,\mathbf{r}_2) \propto \Psi_A(\mathbf{r}_1)\Psi_B(\mathbf{r}_2) + \Psi_A(\mathbf{r}_2)\Psi_B(\mathbf{r}_1), \tag{1.3}$$

$$\Psi_{\uparrow\uparrow}(\mathbf{r}_1,\mathbf{r}_2) \propto \Psi_A(\mathbf{r}_1)\Psi_B(\mathbf{r}_2) - \Psi_A(\mathbf{r}_2)\Psi_B(\mathbf{r}_1). \tag{1.4}$$

The plus sign in (1.3) returns a symmetric spatial wave function that we can take for an antisymmetric spin wave function with the total spin equal to zero (the so-called singlet state); the minus in (1.4) results in an antisymmetric spatial wave function for a symmetric spin wave function with the total spin equal to 1 (the so-called triplet state).

The antisymmetric wave function (1.4) vanishes if $\mathbf{r}_1 = \mathbf{r}_2$, that is, the two electrons cannot be at the same place simultaneously. This leads to a depletion of the electron density between the nuclei and hence to an antibonding state. For the symmetric case, on the other hand, the electrons have opposite spins and can be at the same place, which leads to a charge accumulation between the nuclei and hence to a bonding state (see Figure 1.2).

An approximate way to calculate the eigenvalues of (1.2) was suggested by W. Heitler and F. London in 1927. The idea is to use the known single-particle 1s wave functions for atomic hydrogen for Ψ_A and Ψ_B to form a two-electron wave function

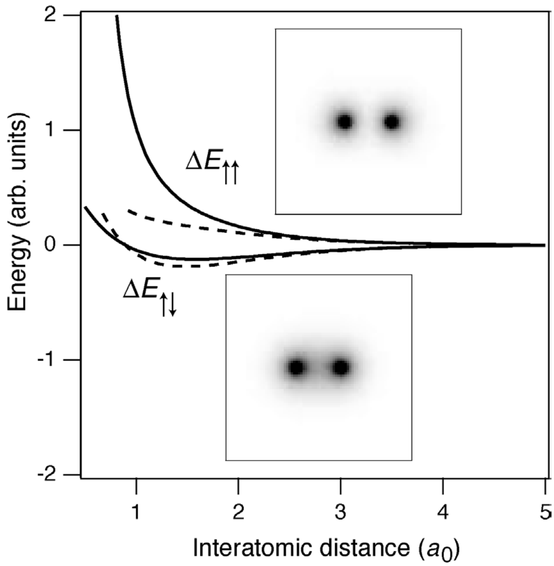

Figure 1.2 The energy changes $\Delta E_{\uparrow\uparrow}$ and $\Delta E_{\uparrow\downarrow}$ for the formation of the hydrogen molecule. The dashed lines represent the approximation for long distances. The two insets show gray scale images of the corresponding electron probability density.

Ψ ($\mathbf{r}_1$, $\mathbf{r}_2$), which is given by either (1.3) or (1.4). An upper limit for the ground-state energy eigenvalues can then be calculated by

$$E = \frac{\int \Psi(\mathbf{r}_1, \mathbf{r}_2) H \Psi(\mathbf{r}_1, \mathbf{r}_2) \mathrm{d}\mathbf{r}_1 \mathrm{d}\mathbf{r}_2}{\int \Psi(\mathbf{r}_1, \mathbf{r}_2) \Psi(\mathbf{r}_1, \mathbf{r}_2) \mathrm{d}\mathbf{r}_1 \mathrm{d}\mathbf{r}_2}. \tag{1.5}$$

The calculation is quite lengthy and shall not be given here. The resulting ground-state energies for the singlet and triplet states can be written as

$$E_{\mathrm{singlet}} = 2E_0 + \Delta E_{\uparrow\downarrow}, \tag{1.6}$$

$$E_{\mathrm{triplet}} = 2E_0 + \Delta E_{\uparrow\uparrow}. \tag{1.7}$$

E_0 is the ground-state energy for one hydrogen atom that appears here twice because we start with two atoms. The energies $\Delta E_{\uparrow\uparrow}$ and $\Delta E_{\uparrow\downarrow}$ are also shown in Figure 1.2. $\Delta E_{\uparrow\uparrow}$ is always larger than zero and does not lead to any chemical bonding. $\Delta E_{\uparrow\downarrow}$, however, shows a minimum below zero at approximately 1.5 times the Bohr radius. This is the binding state.

For long distances between the nuclei, (1.6) and (1.7) can further be simplified to give

$$E = 2E_0 + C \pm X, \tag{1.8}$$

that is, the energy change upon bonding has two parts, one that does depend on the relative spin orientations of the electrons ($\pm X$) and one that does not (C). The energy difference between the two states is then given by $2X$, where X is called the exchange energy. In the case of the hydrogen molecule, the exchange energy is always negative.

We will encounter similar concepts in the chapter about magnetism where the underlying principle for magnetic ordering is very similar to what we have here. The total energy of a system of electrons depends on their relative spin directions through the exchange energy, and therefore a particular ordered spin configuration is favored. For two electrons, the "magnetic" character is purely given by the sign of X. For a negative X, the coupling with two opposite spins is favorable (the antiferromagnetic case) whereas a positive X would lead to a situation where two parallel spins give the lowest energy (the ferromagnetic case).

1.4
Metallic Bonding

In metals, the outer valence electrons are removed from the ion cores, but in contrast to ionic solids, there are no electronegative ions to bind them. Therefore, they are free to migrate between the remaining ion cores. These delocalized valence electrons are involved in the conduction of electricity and are therefore often called conduction electrons. One can expect metals to form from elements for which the energy cost of removing outer electrons is not too big. Nevertheless, this removal always costs some energy that has to be more than compensated by the bonding. Explaining the energy gain from the bonding in an intuitive picture is difficult but we can at least try to make it plausible. The ultimate reason must be some sort of energy lowering.

One energy contribution that is lowered is the kinetic energy of the conduction electrons. Consider the kinetic energy contribution in a Hamiltonian, $T = -\hbar^2\nabla^2/2m$. A matrix element $\langle\Psi|T|\Psi\rangle$ measures the kinetic energy of a particle. $T\Psi$ is proportional to the second spatial derivative of the wave function, that is, the curvature. For an electron that is localized to an atom, the curvature of its wave function is much higher than that for a nearly free electron in a metal and this is where the energy gain comes from.

The other contribution to the electron energy is the potential energy. One should think that the average electrostatic potential of any single electron in a solid is almost zero because there are (almost) as many other electrons as there are ions with the same amount of charge. But this turns out to be wrong. In fact, the electrons see an attractive potential. The reason is again partly due to the Pauli principle that, loosely speaking, does not allow two electrons with the same spin direction to be at the same place and therefore the electrons go "out of each others way." In addition to this, there is also a direct Coulomb interaction between the electrons, which makes them avoid each other. We will discuss this in more detail when dealing with magnetism.

Typically, metallic bonding is not as strong as covalent or ionic bonding but it amounts to a few electron volts per atom. Stronger bonding is found in transition metals, that is, metals with both s and p conduction electrons and a partially filled d shell. The explanation for this is that we have a mixed bonding. The s and p electrons turn into delocalized metallic conduction electrons, whereas the d electrons create much more localized, covalent-type bonds.

1.5
Hydrogen Bonding

Hydrogen atoms have only one electron and can form one covalent bond. If the bond is to a very electronegative atom (like F or O), the electron is mostly located close to that atom and the hydrogen nucleus represents an isolated positive (partial) charge. This can lead to a considerable charge density because of the small size and it can therefore attract negative (partial) charges in other molecules to form an electrostatic bond. This type of bonding is called hydrogen bonding. It is usually quite weak but in some cases the cohesive energy can be up to several hundred meV per atom. It is responsible for the intermolecular attraction in water ice and for the bonding of the double helix in DNA.

1.6
van der Waals Bonding

The term van der Waals bonding refers to a weak, purely quantum mechanical effect. The electron cloud around an atom or a molecule has no static charge distribution but one governed by quantum mechanical fluctuations. For a simple atom with a closed shell, this can be viewed as a fluctuating dipole moment. This dipole moment can polarize other atoms nearby and the interaction of the two neighboring dipoles reduces the total energy, that is, it can lead to bonding. This type of interaction is present in any solid but it is much weaker than ionic, covalent, or metallic bonding. Typical binding energies are in the meV range and therefore van der Waals bonding is only observable for solids that do not show other bonding behavior, for example, noble gases. Pure van der Waals crystals can only exist at very low temperatures.

1.7 Discussion and Problems

1.7.1 Discussion

1. Why is a typical interatomic potential, such as in Figure 1.1, so asymmetric?
2. Which elements are likely to form crystals through ionic bonding?
3. What kind of forces are important for ionic bonding?
4. How does the lattice energy in an ionic crystal depend on the interatomic distance?
5. Explain the difference between cohesive energy and lattice energy.
6. Which elements are likely to form metals?
7. Where does the energy gain in metallic bonding come from?
8. What is the difference between a simple metal and a transition metal (definition and typical physical properties)?
9. Why is van der Waals bonding much weaker than most other bonding types?

1.7.2 Problems

1. *Metallic bonding*: One can use some simple-minded arguments to explain why metallic bonding is stable. One argument is that allowing the wave functions to be smeared out in space reduces the momentum uncertainty Δp and thereby the highest momentum (and energy) through $\Delta p \Delta x = \hbar/2$. Estimate how much Δp would be reduced by allowing the wave function to spread from typical atomic dimensions (Bohr radius) to the typical dimensions of the lattice unit cell (a few Angstrom or 10 times the Bohr radius). If you assume that the highest p would correspond to a highest kinetic energy $E = p^2/2m_e$, how much would this reduce the kinetic energy?

2. *van der Waals force*: Show that the bonding energy due to the van der Waals force between two atoms depends on their distance r as r^{-6}. *Hint*: The van der Waals interaction is due to the mutual interaction of fluctuating dipoles. Suppose that one atom forms a dipole moment, at some time. This can be modeled as two point

charges, separated by a distance d. What is the electric field due to this dipole at some distance r that is much greater than d? Assume that a second atom at the distance r is polarized in the field such that its induced dipole moment is proportional to the field (see (9.4) in Chapter 9) and calculate how the potential energy of the system depends on r.

2
Crystal Structures

In most of this book, we are concerned with perfectly crystalline solids, with the atoms arranged on a perfect lattice without any defects, impurities, or boundaries. It turns out that the perfect periodicity and translational symmetry allow us to develop very good models for the solid state despite the very many ($\approx N_A$) particles involved. It may seem that such perfect crystals are not particularly relevant for real materials but this is not the case. Many solids are actually composed of small crystalline grains. The fact that these grains are three dimensional means that the number of atoms on the grain boundary is quite small compared to the number of atoms in the grain. For instance, for a grain size in the order of 1000^3 atomic distances, only about 0.1% of the atoms are at the grain boundaries, the rest find themselves in a perfectly crystalline environment. There are, however, some solids that are not crystalline. These are called amorphous solids. The amorphous state is characterized by the absence of any long-range order. There may, however, be some short-range order between the atoms.

This chapter is divided into three parts. In the first part, we define some basic mathematical concepts needed to describe crystals. We keep things simple and mostly use two-dimensional examples to illustrate the ideas. In the second part, we discuss common crystal structures. Finally, we go into a somewhat more detailed discussion of X-ray diffraction. The reason is that this technique is widely used, also in the fields of microbiology and nanotechnology, and you will probably encounter it again many times.

2.1
General Description of Crystal Structures

Our description of crystals starts with the mathematical definition of the lattice. A lattice is a set of regularly spaced points with positions defined by multiples of generating vectors. In two dimensions, a lattice of points $\mathbf{R}_{mn}$ would be defined in terms of two vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ like

$$\mathbf{R} = m\mathbf{a}_1 + n\mathbf{a}_2, \tag{2.1}$$

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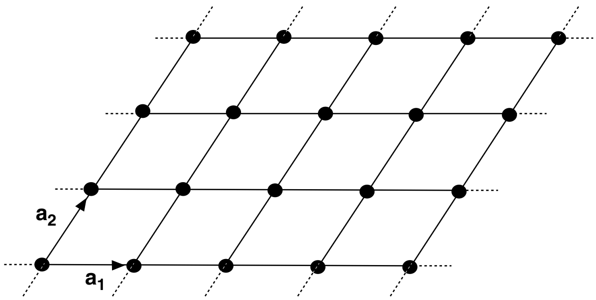

Figure 2.1 Example for a two-dimensional Bravais lattice.

where n and m are integers. In three dimensions, the definition would be

$$\mathbf{R} = m\mathbf{a}_1 + n\mathbf{a}_2 + o\mathbf{a}_3. \tag{2.2}$$

Such a lattice of points is also called a Bravais lattice. The number of possible Bravais lattices that differ by symmetry is limited to 5 in two dimensions and to 14 in three dimensions. An example of a two-dimensional lattice is given in Figure 2.1. The lengths of the vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are often called the lattice parameters.

Having defined the lattice, we move on to the definition of the primitive unit cell. It is any volume of space that, when translated through all the vectors of the Bravais lattice, fills space without overlap and without leaving voids. The primitive unit cell of a lattice contains only one lattice point. It is also possible to define non-primitive unit cells that contain several lattice points. These fill space without leaving voids when translated through a subset of the Bravais lattice vectors. Possible choices of a unit cell for a two-dimensional rectangular Bravais lattice are given in Figure 2.2. From the figure it is evident that a non-primitive unit cell has to be translated by a multiple of one (or two) lattice vectors to fill space without voids and overlap. A special choice of the primitive unit cell is the Wigner–Seitz cell that is also shown in Figure 2.2. It is the region of space that is closer to one given lattice point than to any other.

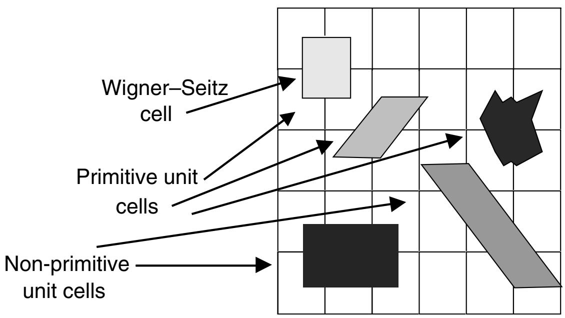

Figure 2.2 Illustration of unit cells (primitive and non-primitive) and of the Wigner–Seitz cell for a rectangular two-dimensional lattice.

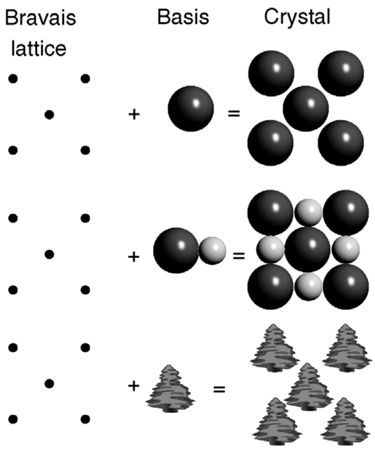

Figure 2.3 A two-dimensional Bravais lattice with different choices for the basis.

The last definition we need in order to describe an actual crystal is that of a basis. The basis is what we "put" on the lattice points, that is, the building block for the real crystal. The basis can consist of one or several atoms. It can even consist of complex molecules as in the case of protein crystals. Different cases are illustrated in Figure 2.3.

Finally, we add a remark about symmetry. So far, we have discussed the description of translational symmetry. But for a real crystal, there is also point symmetry. Compare the structures in the middle and the bottom of Figure 2.3. The former structure possesses a couple of symmetry elements that the latter does not have, for example, mirror lines, a rotational axis, and inversion symmetry. The knowledge of such symmetries can be very useful for the description of crystal properties.

2.2 Some Important Crystal Structures

After this rather formal treatment, we look at a number of common crystal structures. To some extent, it will also be possible to establish a link to the previous chapter. We will try to understand why a certain bonding type prefers a certain structure.

2.2.1 Cubic Structures

The simple cubic structure shown in Figure 2.4a is not very common, but it is an important starting point for many other structures. The reason why it is not common is its openness, that is, there are many voids if we think of the ions as spheres. In metals, directional bonding is not important and the atoms therefore prefer more close-packed structures. For covalent solids, directional bonding *is* important but six mutually perpendicular bonds on one atom are not common.

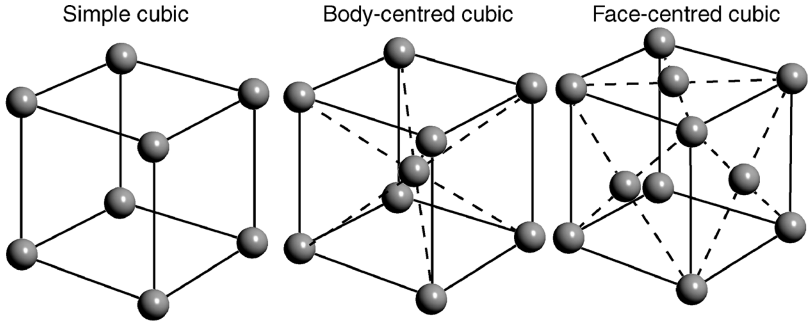

Figure 2.4 (a) Simple cubic structure; (b) body-centered cubic structure; (c) face-centered cubic structure. Note that the spheres are depicted much smaller than in the situation of most dense packing.

The packing density of the cubic structure is improved in the body-centered cubic (bcc) and face-centered cubic (fcc) structures that are also shown in Figure 2.4. In fact, the fcc structure has the highest possible packing density for spheres as we shall see later. These two structures are very common. Seventeen elements crystallize in the bcc structure and 24 elements in the fcc structure. Note that only for the simple cubic structure the cube is identical with the Bravais lattice. For the bcc and fcc lattices the cube is also a unit cell, but not the primitive one.

Cubic structures with a more complex basis than a single atom are also important. Figure 2.5 shows the structures of CsCl and NaCl that are both cubic with a basis containing two atoms.

Knowing the crystal structure for NaCl, we can now calculate the electrostatic contribution to the crystal binding, that is, the lattice energy discussed in Chapter 1. We had already calculated that the electrostatic energy gain for the bonding of a pair of ions is 5.1 eV. What is it for the crystal shown in Figure 2.5? Consider one Na ion at the center of the cube. It has six Cl ions at a distance of $a = 0.28$ nm. They decrease the energy by $-6e^2/4\pi\varepsilon_0 a$. At a distance of $a\sqrt{2}$, there are 12 other Na ions that give rise to

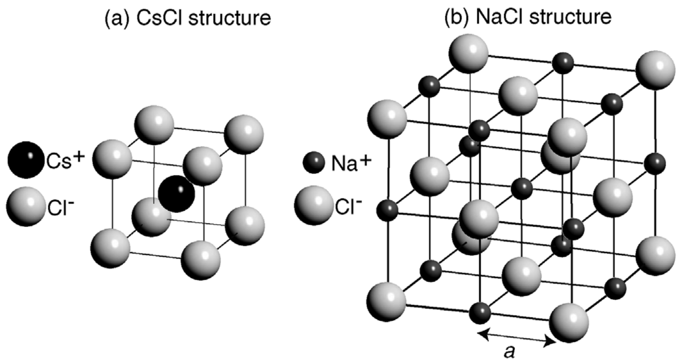

Figure 2.5 Structures of CsCl and NaCl. The spheres are depicted much smaller than in the situation of most dense packing, but the relative size of the different ions in each structure is correct.

an energy increase of $+12e^2/4\pi\varepsilon_0 a\sqrt{2}$. Then one finds eight Cl ions that again decrease the energy. Eventually, this series converges and the total energy gain is

$$E_{\mathrm{Na}} = -1.748\frac{e^2}{4\pi\varepsilon_0 a} = -M_{\mathrm{d}}\frac{e^2}{4\pi\varepsilon_0 a}. \tag{2.3}$$

M_{d} is called the Madelung constant. It is specific for a given structure. For calculating the lattice energy per mole, we have to multiply this energy gain by Avogadro's number N_{A}. We also have to multiply it by a factor of 2 to account for the fact that we have both Na and Cl ions in the solid. But at the same time, we have to divide it by a factor of 2 in order to avoid a double counting of bonds when we evaluate the lattice energy. So in the end, the lattice energy for 1 mol of NaCl is simply $-N_{\mathrm{A}}1.748e^2/4\pi\varepsilon_0 a$. The true lattice energy is somewhat smaller due to the repulsive part of the potential, as discussed in the previous chapter. Note that M_{d} is larger than 1 so that the lattice energy for the solid is higher than that for an isolated pair of ions. This is of course obvious since your salt shaker contains little crystals, not a molecular powder (for the calculation of M_{d}, see Problem 2.5).

2.2.2 Close-Packed Structures

Many metals prefer structural arrangements where the atoms are packed as closely as possible. Why is this? First of all, the metallic bonding does not have any directional preferences. Second, close-packed structures secure the highest possible overlap between the valence orbitals of the atoms, maximizing the delocalization of the electrons and thereby the kinetic energy gain. The structures also maximize the number of nearest neighbors for any given atom, again giving rise to strongly delocalized states.

In two dimensions, the closest possible packing of ions (i.e., spheres) is the hexagonal structure shown on the left side of Figure 2.6. For building a three-dimensional structure, one adds a second close-packed layer as in the middle of Figure 2.6. For adding a third layer, there are now two possibilities. One can either put

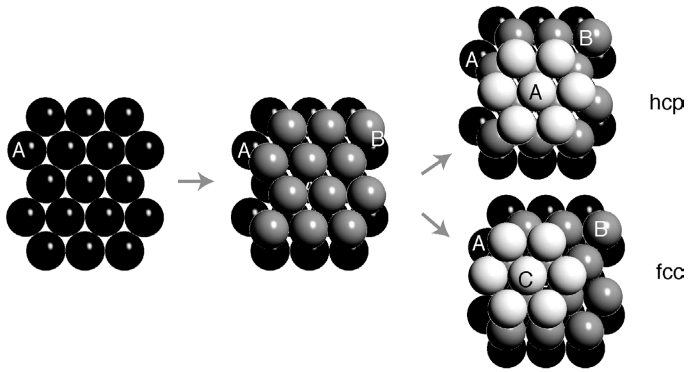

Figure 2.6 Close packing of spheres leading to the hcp and fcc structures.

the atoms in the "holes" just on top of the first layer atoms or one can put them into the other type of "holes." In this way, two different crystal structures can be built. The first has an ABABAB... stacking sequence, and the second has an ABCABCABC... stacking sequence. Both have exactly the same packing density, the spheres fill 74% of the total volume. The former structure is called the hexagonal close-packed structure (hcp) and the latter turns out to be the fcc structure we already know. An alternative sketch of the hcp structure is shown in Figure 2.15. The fcc and hcp structures are very common for elemental metals. Thirty-six elements crystallize as hcp and 24 elements as fcc.

As mentioned above, these structures also maximize the number of nearest neighbors for a given atom, the so-called coordination number. For both the fcc and the hcp lattice, the coordination number is 12.

An open question is why, if coordination is so important, not all metals crystallize as fcc or hcp structure. A prediction of the actual structure for a given element is not possible with simple arguments. However, we can collect some factors that play a role. Not optimally packed structures, such as the bcc structure, have a lower coordination number, but they bring the second-nearest neighbors much closer to a given ion than the close-packed structures. Another important consideration is that the bonding is not quite so simple, especially for transition metals. In these, bonding is not only achieved through the delocalized s and p valence electrons as in simple metals, but also by the more localized d electrons. Bonding through the latter has a much more directional character, so that the total bonding in a transition metal can be viewed as the sum of metallic bonding through the s and p electrons and covalent bonding through the d electrons.

The structures of many ionic solids can also be viewed as "close packed" in some sense. One can arrive at these structures by treating the ions as hard spheres that have to be packed as closely to each other as possible. The type of structure one arrives at then depends on the size difference of the ions; for example, one can explain the difference of the CsCl structure and the NaCl structure by the size difference of the Cs and Na ions.

2.2.3 Covalent Structures

In covalent structures, the bond length and direction are far more important than the packing density. Prominent examples are graphite and diamond as displayed in Figure 2.7. Graphite has three nearest neighbors in the graphene sheets forming the structure. The bonds to these atoms enclose an angle of 120°, and we have a clear example of sp^2 hybridization. In diamond, the carbon atoms have four nearest neighbors in a tetrahedral configuration, as expected for sp^3-type bonds. Interestingly, the diamond structure can also be described as an fcc lattice with a basis of two atoms.

The diamond structure is also found for Si and Ge. Many other isoelectronic materials (with the same total number of valence electrons), such as SiC, GaAs, and InP, also crystallize in a diamond-like structure but with each element on a different fcc sublattice.

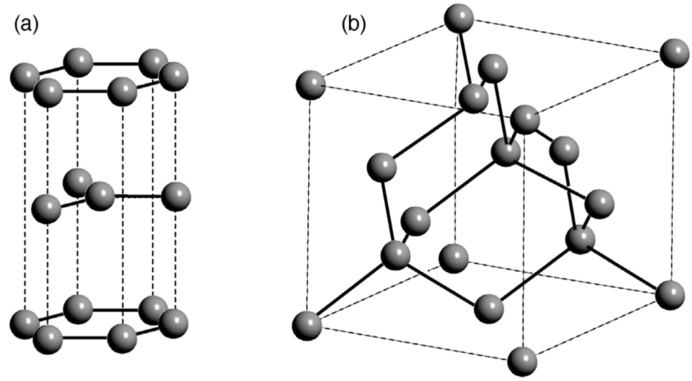

Figure 2.7 Structures for (a) graphite and (b) diamond. sp^2 and sp^3 bonds are displayed as solid lines.

2.3
Crystal Structure Determination

After having described different crystal structures, the question is of course how to determine these structures in the first place. By far the most important technique is X-ray diffraction. In fact, the importance of this technique goes far beyond solid-state physics, as it has become an important tool for fields such as structural biology as well. There the idea is that if you want to know the structure of a given protein, you can try to crystallize it and use the powerful methodology for structural determination by X-ray diffraction. We will also use X-ray diffraction as a motivation to extend our formal description of structures a bit.

2.3.1
X-Ray Diffraction

X-rays interact rather weakly with matter. A description of X-ray diffraction can therefore be restricted to single scattering, that is, incoming X-rays get scattered not more than once. This is called the kinematic approximation; it greatly simplifies matters and is used throughout the treatment here. In addition to this, we will assume that the X-ray source and detector are very far away from the sample so that the incoming and outgoing waves can be treated as plane waves. X-ray diffraction of crystals was discovered and described by M. von Laue in 1912. In 1913, W. L. Bragg and W. H. Bragg came up with an alternative description that is considerably simpler and serves as a starting point here.

2.3.1.1 **Bragg Theory**
The Braggs treated the problem simply as the reflection of the incoming X-rays at flat crystal planes. These planes could, for example, be the close-packed planes making up the fcc and hcp crystals or they could be alternating Cs and Cl planes

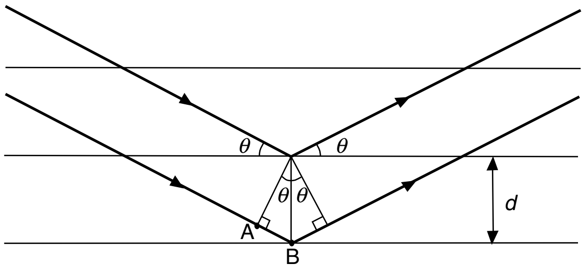

Figure 2.8 Construction for the derivation of the Bragg condition. The horizontal lines represent the lattice planes that are separated by a distance d. The heavy lines represent the X-rays.

making up the CsCl structure. At first glance, this has a very little physical justification because the crystal planes are certainly not "flat" for X-rays that have a wavelength similar to the atomic spacing. Nevertheless, the description is highly successful and we shall later see that it is actually a special case of the more complex Laue description of X-ray diffraction.

Figure 2.8 shows the geometrical considerations behind the Bragg description. A collimated beam of monochromatic X-rays hits the crystal. The intensity of diffracted X-rays is measured *in the specular direction*. The angle of incidence and emission is $90^\circ - \Theta$. The condition for constructive interference is that the path length difference between the X-rays reflected from one layer and the next layer is an integer multiple of the wavelength λ. In the figure,this means that $2\mathrm{AB} = n\lambda$, where AB is the distance between points A and B and n is a natural number. On the other hand, we have $\sin\theta = \mathrm{AB}/d$ such that we arrive at the Bragg condition

$$n\lambda = 2d \sin\theta. \tag{2.4}$$

It is obvious that if this condition is fulfilled for one layer and the layer below, it will also be fulfilled for any number of layers with identical spacing. In reality, the X-rays penetrate very deeply into the crystal so that thousands of layers contribute to the diffracted intensity. This results into very sharp maxima in the diffracted intensity, similar to the situation for an optical grating with many lines. The Bragg condition can obviously only be fulfilled for $\lambda < 2d$, that is, the X-rays have to have a wavelength that is somewhat shorter than the lattice spacing.

2.3.1.2 Lattice Planes and Miller Indices

The Bragg condition will work not only for a special kind of lattice plane in a crystal, such as the hexagonal planes in an hcp crystal, but for all possible parallel planes in a structure. We therefore come up with a more stringent definition of the term "lattice plane." It can be defined as a plane containing at least three non-colinear points of a given Bravais lattice. If it contains three, it will actually contain infinitely many because of translational symmetry. Examples for lattice planes in a simple cubic structure are shown in Figure 2.9.

The lattice planes can be characterized by a set of three integers, the so-called Miller indices. One arrives at these in three steps:

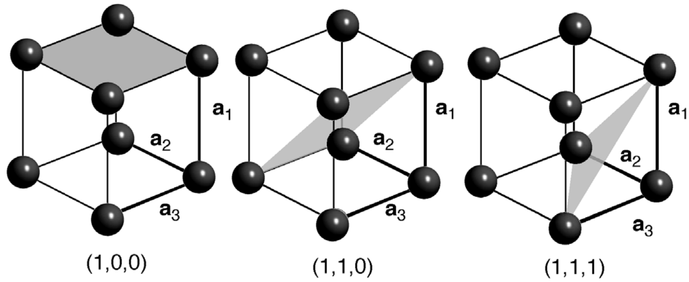

Figure 2.9 Three different lattice planes in the simple cubic structure characterized by their Miller indices.

1. One finds the intercepts of the plane with the crystallographic axes in units of the lattice vectors, for example, (1, ∞, ∞) for the leftmost plane in Figure 2.9.
2. One takes the "reciprocal value" of these three numbers. For our example, this gives (1, 0, 0).
3. By multiplying with some factor, one reduces the numbers to the smallest set of integers having the same ratio. This is not necessary in the example.

Such a set of three integers can then be used to denote any given lattice plane. Later, we will encounter a different and more elegant definition of the Miller indices.

In practice, the X-ray diffraction peaks are so sharp that it is difficult to align and move the sample such that the incoming and reflected X-rays lie in one plane with the surface normal. An elegant way to circumvent this problem is to use a powder sample of small crystallites instead of a single crystal. This will not only ensure that some crystallites are orientated correctly to get constructive interference from a certain set of crystal planes, it will automatically give the interference pattern for all possible crystal planes.

2.3.1.3 General Diffraction Theory

The Bragg theory for X-ray diffraction is useful for extracting the distances between lattice planes in a crystal, but it has its limitations. Most importantly, it does not give any information on what the lattice actually consists of, that is, the basis. Also, the fact that the X-rays should be reflected by planes is physically somewhat obscure. We now discuss a more general description of X-ray diffraction that goes back to M. von Laue.

The physical process leading to X-ray scattering is that the electromagnetic field of the X-rays forces the electrons in the material to oscillate with the same frequency as that of the field. The oscillating electrons then emit new X-rays that give rise to an interference pattern. For the following discussion, however, it is merely important that something scatters the X-rays, not what it is.

It is highly beneficial to use the complex notation for describing the electromagnetic X-ray waves. For the electric field, a general plane wave can be written as

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 e^{i\mathbf{k}\cdot\mathbf{r} - i\omega t}. \tag{2.5}$$

The wave vector $\mathbf{k}$ points in the direction of the wave propagation with a length of $2\pi/\lambda$, where λ is the wavelength. The convention is that the physical electric field is

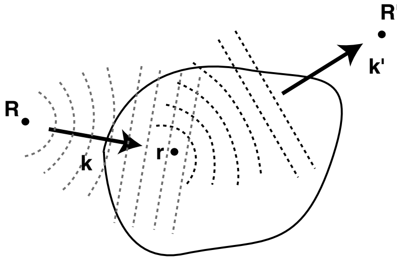

Figure 2.10 Illustration of X-ray scattering from a sample. The source and detector for the X-rays are placed at **R** and **R**$'$, respectively. Both are very far away from the sample.

obtained as the real part of the complex field and the intensity of the wave is obtained as

$$I(\mathbf{r}, t) = |\mathbf{E}_0 e^{i\mathbf{k}\cdot\mathbf{r}-i\omega t}|^2 = |\mathbf{E}_0|^2. \tag{2.6}$$

Consider now the situation depicted in Figure 2.10. The source of the X-rays is far away from the sample at the position **R**, so that the X-ray wave at the sample can be described as a plane wave. The electric field at a point **r** in space at time t can be written as

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{R})-i\omega t}. \tag{2.7}$$

Before we proceed, we can drop the absolute amplitude $\mathbf{E}_0$ from this expression because we are only concerned with relative phase changes. The field at point **r** is then

$$\mathbf{E}(\mathbf{r}, t) \propto e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{R})} e^{-i\omega t}. \tag{2.8}$$

A small volume element dV located at **r** will give rise to scattered waves in all directions. The direction of interest is the direction toward the detector that shall be placed at the position $\mathbf{R}'$, in the direction of a second wave vector $\mathbf{k}'$. We assume that the amplitude of the wave scattered in this direction will be proportional to the incoming field (2.8) and to a factor describing the scattering probability. We take this factor to be the local electron charge density $\rho(\mathbf{r})$.

$$\mathbf{E}(\mathbf{R}', t) \propto \mathbf{E}(\mathbf{r}, t)\rho(\mathbf{r}) e^{i\mathbf{k}'\cdot(\mathbf{R}'-\mathbf{r})}. \tag{2.9}$$

Again, we have assumed that the detector is very far away from the sample such that the scattered wave at the detector can be written as a plane wave. Inserting (2.8) gives the field at the detector as

$$\mathbf{E}(\mathbf{R}', t) \propto e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{R})}\rho(\mathbf{r}) e^{i\mathbf{k}'\cdot(\mathbf{R}'-\mathbf{r})} e^{-i\omega t} = e^{i(\mathbf{k}'\cdot\mathbf{R}'-\mathbf{k}\cdot\mathbf{R})}\rho(\mathbf{r}) e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} e^{-i\omega t}. \tag{2.10}$$

The total wave field at the detector can finally be calculated by integrating over the entire volume of the crystal V. As the detector is far away from the sample, the wave vector $\mathbf{k}'$ is essentially the same for all points in the sample. The result is

$$\mathbf{E}(\mathbf{R}', t) \propto e^{-i\omega t} \int_V \rho(\mathbf{r}) e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}\, dV. \tag{2.11}$$

In most cases, it will only be possible to measure the intensity of the X-rays, not the field, and this intensity is

$$I(\mathbf{K}) \propto |\int_V \rho(\mathbf{r}) e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \, dV|^2 = |\int_V \rho(\mathbf{r}) e^{-i\mathbf{K}\cdot\mathbf{r}} \, dV|^2. \tag{2.12}$$

In the last step, we have introduced the so-called scattering vector $\mathbf{K} = \mathbf{k}' - \mathbf{k}$, which is just the difference of outgoing and incoming wave vectors. Note that although the direction of the wave vector for the scattered waves $\mathbf{k}'$ is different from that of the incoming wave $\mathbf{k}$, the length is the same because we only consider elastic scattering.

Equation 2.12 is the final result. It relates the measured intensity to the electron density in the sample. This in turn, corresponds to the atomic structure. Except for very light elements, most of the electrons are located close to the ion cores and the charge density that scatters the X-rays is essentially identical to the geometrical arrangement of the ion cores. One could try to measure the intensity as a function of scattering vector $\mathbf{K}$ and to infer the structure from the result. This is a formidable task. It is greatly simplified by assuming that the specimen under investigation is a crystal with a periodic lattice. In the following, we introduce the mathematical tools that are needed to do this. The most important one is the so-called reciprocal lattice.

2.3.1.4 **The Reciprocal Lattice**

The concept of the reciprocal lattice is fundamental to solid-state physics because it greatly facilitates the description of periodic structures. We will use it to describe X-ray diffraction from periodic structures and we will meet it again and again in the next chapters. Unfortunately, the meaning of the reciprocal lattice turns out to be hard to grasp. Here, we choose to start out with a formal definition and description of some mathematical properties. We then discuss the meaning of the reciprocal lattice before we come back to X-ray diffraction. The full importance of the concept will become apparent throughout this book.

For a given Bravais lattice

$$\mathbf{R} = m\mathbf{a}_1 + n\mathbf{a}_2 + o\mathbf{a}_3. \tag{2.13}$$

We define the reciprocal lattice as the set of vectors $\mathbf{G}$ for which

$$\mathbf{R} \cdot \mathbf{G} = 2\pi l, \tag{2.14}$$

where l is an integer. Equivalently, we could require that

$$e^{i\mathbf{G}\cdot\mathbf{R}} = 1. \tag{2.15}$$

Note that this equation must hold for *any* choice of the lattice vector $\mathbf{R}$ and reciprocal lattice vector $\mathbf{G}$. We can write any $\mathbf{G}$ as the sum of three vectors

$$\mathbf{G} = m'\mathbf{b}_1 + n'\mathbf{b}_2 + o'\mathbf{b}_3, \tag{2.16}$$

where m', n', and o' are integers. It can be shown that this reciprocal lattice is again a Bravais lattice. The vectors $\mathbf{b}_1$, $\mathbf{b}_2$, and $\mathbf{b}_3$ spanning the reciprocal lattice can be constructed explicitly from the lattice vectors

$$\mathbf{b}_1 = 2\pi\frac{\mathbf{a}_2 \times \mathbf{a}_3}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}, \quad \mathbf{b}_2 = 2\pi\frac{\mathbf{a}_3 \times \mathbf{a}_1}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}, \quad \mathbf{b}_3 = 2\pi\frac{\mathbf{a}_1 \times \mathbf{a}_2}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}. \tag{2.17}$$

From this, one can derive the simple but useful property, [1)]

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}, \tag{2.18}$$

which can easily be verified by inserting the vectors spanning the reciprocal lattice as defined in (2.17). Equation 2.18 can then be used to verify that the reciprocal lattice vectors defined by (2.16) and (2.17) do indeed fulfill the fundamental property (2.14) defining the reciprocal lattice.

Another way to view the vectors of the reciprocal lattice is as wave vectors that yield plane waves with the periodicity of the Bravais lattice because

$$e^{i\mathbf{G}\cdot\mathbf{r}} = e^{i\mathbf{G}\cdot\mathbf{r}}e^{i\mathbf{G}\cdot\mathbf{R}} = e^{i\mathbf{G}\cdot(\mathbf{r}+\mathbf{R})}. \tag{2.19}$$

Finally, one can define the Miller indices in a much simpler way using the reciprocal lattice: the Miller indices (i, j, k) define a plane that is perpendicular to the reciprocal lattice vector $i\mathbf{b}_1 + j\mathbf{b}_2 + k\mathbf{b}_3$ (see Problem 2.8).

2.3.1.5 The Meaning of the Reciprocal Lattice

We have now defined the reciprocal lattice in a proper way and we will give some simple examples of its usefulness. The most important point of the reciprocal lattice is that it facilitates the description of lattice periodic properties.

We start with a one-dimensional lattice, a chain of points with a lattice spacing of a. Consider a function with the periodicity of the lattice, like the charge density along the chain $\rho(x) = \rho(x + a)$. We can write this as a Fourier series of the form

$$\rho(x) = C + \sum_{n=1}^{\infty}\{C_n \cos(x2\pi n/a) + S_n \sin(x2\pi n/a)\} \tag{2.20}$$

with real coefficients C_n and S_n. The sum starts at $n = 1$, that is, the constant part C has to be taken out of the sum. We can also write this in a more compact form

$$\rho(x) = \sum_{n=-\infty}^{\infty} \rho_n e^{ixn2\pi/a}, \tag{2.21}$$

using complex coefficients ρ_n. To ensure that $\rho(x)$ is still a real function, we have to require that

$$\rho^*_{-n} = \rho_n, \tag{2.22}$$

that is, the coefficient ρ_n must be the conjugate complex of the coefficient ρ^*_{-n}. This description is more elegant than the one with the sine and cosine functions. How is

1) δ_{ij} is Kronecker's delta that is 1 for $i = j$ and zero otherwise.

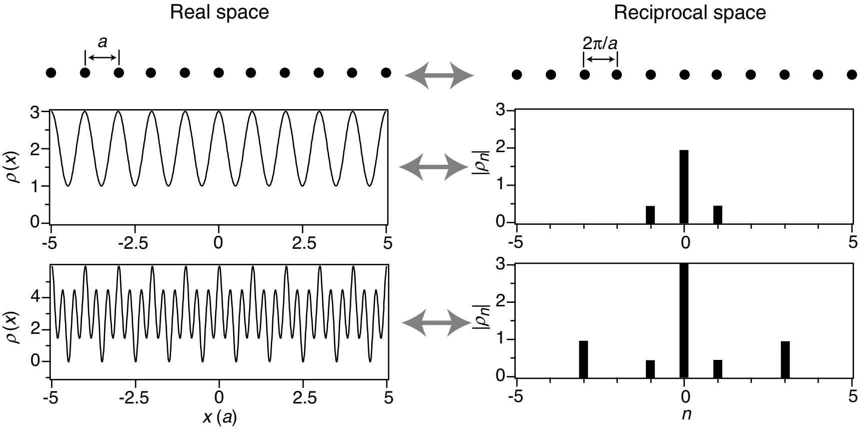

Figure 2.11 *Top*: Lattice for an infinite chain with a spacing of a as well as its reciprocal lattice, an infinite chain with a spacing of $2\pi/a$. *Middle and bottom*: Two lattice periodic functions $\rho(x)$ in real space as well as their Fourier transformation. The Fourier transformation is given such that the magnitude of the Fourier coefficients $|\rho_n|$ is plotted on the reciprocal lattice points they belong to.

this related to the reciprocal lattice? In one dimension, the reciprocal lattice of a chain of points with spacing a is also a chain of points with spacing $2\pi/a$ (see (2.18)). This means that we can write a general reciprocal lattice "vector" as

$$g = n\frac{2\pi}{a}, \tag{2.23}$$

where n is an integer. Exactly these reciprocal lattice "vectors" appear in (2.21). In fact, (2.21) is a sum of functions with the periodicity corresponding to the reciprocal lattice vectors, weighted by the coefficients ρ_n. Figure 2.11 illustrates these ideas by showing the lattice and reciprocal lattice of an infinite chain as well as two lattice periodic functions, as real space functions and as a number of Fourier coefficients on the reciprocal lattice points. The advantage of describing the functions by the coefficients ρ_n is immediately obvious: instead of giving $\rho(x)$ for *every* point in a range of $0 < x < a$, the Fourier description consists only of three numbers for the upper function and five numbers for the lower function. Actually, it is only two and three numbers because of (2.22).

The same ideas work also in three dimensions. In fact, one can use a Fourier sum for lattice periodic properties, which exactly corresponds to (2.21). For the charge density we get

$$\rho(\mathbf{r}) = \sum_{\mathbf{G}} \rho_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}}, \tag{2.24}$$

where $\mathbf{G}$ are the reciprocal lattice vectors.

With this we have seen that the reciprocal lattice is very useful for describing lattice periodic functions. But this is not all: it can also simplify the treatment of waves in crystals in a very general sense. Such waves can be X-rays, elastic waves, or even electronic wave functions. We will come back to this point at a later stage.

2.3.1.6 X-Ray Diffraction from Periodic Structures

Turning back to the specific problem of X-ray diffraction, we can now exploit the fact that the charge density is lattice periodic by inserting (2.24) in our expression (2.12) for the diffracted intensity. This gives

$$I(\mathbf{K}) \propto |\sum_{\mathbf{G}} \rho_{\mathbf{G}} \int_V e^{i(\mathbf{G}-\mathbf{K})\cdot\mathbf{r}}\, dV|^2. \tag{2.25}$$

Let us inspect the integrand. The exponential represents a plane wave function with a wave vector $\mathbf{G} - \mathbf{K}$. If the crystal is very big, the integration will average over the crests and troughs of the waves and the result of the integral will be very small (or zero for an infinitely large crystal). The only exception to this is the case where

$$\mathbf{K} = \mathbf{k}' - \mathbf{k} = \mathbf{G}, \tag{2.26}$$

that is, when the difference between incoming and scattered wave vector is equal to a reciprocal lattice vector. In this case, the exponential in the integral is 1, and the value of the integral is equal to the volume of the crystal. Equation (2.26) is often called the Laue condition. It is central to the description of X-ray diffraction from crystals in that it describes the condition for the observation of constructive interference.

Looking back at (2.25), the Laue condition can clearly only be fulfilled for one particular value of $\mathbf{G}$, corresponding to one particular value of the scattering vector $\mathbf{K}$. The intensity measured at the detector is proportional to the square of the Fourier coefficient of the charge density $|\rho_{\mathbf{G}}|^2$. One could therefore think of measuring the intensity of the diffraction spots appearing for all possible reciprocal lattice vectors, obtaining the Fourier coefficients of the charge density and reconstructing this charge density. This would give all the information needed and conclude the process of the structural determination. Unfortunately, this straightforward approach does not work because the Fourier coefficients are complex, not real numbers. Taking the square root of the intensity at the diffraction spot therefore gives the magnitude but not the phase of $\rho_{\mathbf{G}}$. The phase is lost in the measurement. This is known as the phase problem in X-ray diffraction. One has to work around it to solve the structure. One simple approach would be to calculate the charge density for a structural model, obtain the magnitude of the $\rho_{\mathbf{G}}$ values, and compare them to the experimental result. Based on the outcome, the model could be refined until the agreement is satisfactory.

2.3.1.7 The Ewald Construction

In 1913, P. Ewald published an intuitive geometrical construction for visualizing the Laue condition (2.26). The construction is shown in Figure 2.12, which represents a cut through the reciprocal lattice; the black points are the reciprocal lattice points. The construction works as follows:

1. Draw the wave vector $\mathbf{k}$ of the incoming X-rays such that it ends in the origin of the reciprocal lattice (you may of course choose the point of origin freely).

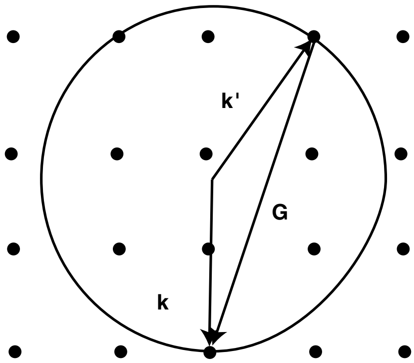

Figure 2.12 Ewald construction for finding the directions in which constructive interference can be observed. The dots represent the reciprocal lattice. The arrows labeled **k** and **k**′ are the wave vectors of the incoming and scattered X-rays.

2. Construct a circle of radius $|\mathbf{k}|$ around the starting point of $\mathbf{k}$.
3. Wherever a reciprocal lattice touches the circle, the Laue condition $\mathbf{k}' - \mathbf{k} = \mathbf{G}$ is fulfilled and a diffracted beam can be found in the direction $\mathbf{k}'$.

The figure clearly shows that (2.26) is a very stringent condition. For a given wavelength of the X-rays, not many reflections will occur. For a three-dimensional crystal, the circle must be replaced by a sphere, of course.

Practical X-ray diffraction experiments are often carried out in such a way that many constructive interference maxima are observed despite the strong restrictions imposed by the Laue condition (2.26). This can, for example, be achieved by using a wide range of X-ray wavelengths, that is, non-monochromatic radiation or by doing diffraction experiments not on one single crystal but on a powder of randomly oriented small crystals.

2.3.1.8 **Relation Between Bragg and Laue Theory**

We conclude our treatment of X-ray diffraction by showing that the Bragg description of X-ray diffraction is just a special case of the Laue description. We start by noting that the Laue condition (2.26) consists, in fact, of three separate conditions for the three components of the vectors. In the Bragg experiment, two of the conditions are automatically fulfilled because of the specular geometry: the wave vector change parallel to the lattice planes is zero. So, the vector equation (2.26) reduces to the scalar equation

$$k'_{\perp} - k_{\perp} = 2k_{\perp} = 2\frac{2\pi}{\lambda}\sin\Theta = G_{\perp}, \tag{2.27}$$

where $\mathbf{G}_{\perp}$ is the component of a reciprocal lattice vector perpendicular to the planes and we have used that the length of $\mathbf{k}$ is $2\pi/\lambda$. If we define a lattice vector $\mathbf{R}$ of length d, which is perpendicular to the lattice planes and connects these planes, a possible choice for a reciprocal lattice vector is also perpendicular to the planes and of length $m2\pi/d$ (see (2.14)). Inserting this for $\mathbf{G}_{\perp}$ into (2.27) gives the usual form of the Bragg condition (2.4).

2.3.2
Other Methods

While X-ray diffraction is arguably the most widespread and powerful method for structural determination, other techniques are used as well. Similar diffraction experiments can be carried out by making use of the wave character of neutrons or electrons. The former interact very weakly with matter because they are charge neutral. They are also difficult to produce. However, the use of neutrons has two distinct advantages over X-rays: first, that their relative interaction strength with light atoms is stronger and second, that they carry a magnetic moment. They can therefore interact with the magnetic moments in the solid, that is, one can determine the magnetic order. Electrons, on the other hand, have the advantage that one can use electron-optical imaging techniques, whereas making optical elements for X-rays is very difficult. On the other hand, their very strong interaction with matter causes a breakdown of the kinematic approximation, that is, multiple scattering events have to be taken into account. Low-energy electrons do not penetrate deeply into crystals. Therefore, they are more appropriate for surface structure determination.

2.3.3
Inelastic Scattering

Our discussion has been confined to the case of elastic scattering. In real experiments, however, the X-rays or particles can also lose energy during the scattering events. This can be described formally by considering the diffraction from a structure that does not consist of ions at fixed positions but is time dependent, that is, which fluctuates with the frequencies of the atomic vibrations. We cannot go into the details of such scattering processes here, but it is important to emphasize that the inelastic scattering, especially of neutrons, can be used to measure the vibrational properties of a lattice.

2.4
Discussion and Problems

2.4.1
Discussion

1. What mathematical concepts do you need to describe the structure of any crystal?
2. What are typical crystal structures for metals and why?
3. Why do covalent and molecular crystals typically have a much lower packing density than metal crystals?
4. How can you determine the structure of crystals?
5. What is the difference between the Bragg and von Laue description of X-ray diffraction?
6. How can you use the reciprocal lattice of a crystal to predict the pattern of diffracted X-ray beams?

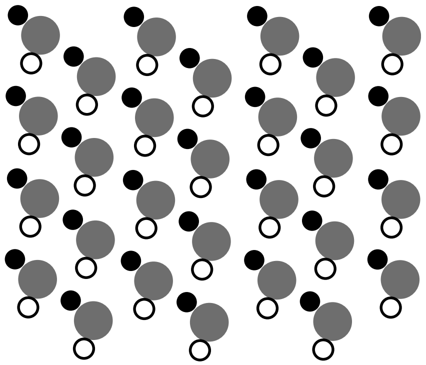

Figure 2.13 A two-dimensional crystal.

7. How can the reciprocal lattice conveniently be used to describe lattice periodic functions?

2.4.2
Problems

1. *Fundamental concepts*: In the two-dimensional crystal in Figure 2.13, find (a) a primitive unit cell, (b) a non-primitive, rectangular unit cell, and (c) the basis.

2. *Real crystal structures*: Figure 2.14 shows the structure for a hcp crystal and for graphite. (a) For the hcp crystal, draw a choice of vectors spanning the Bravais lattice. Show that the basis for the hcp crystal contains two atoms or, equivalently,

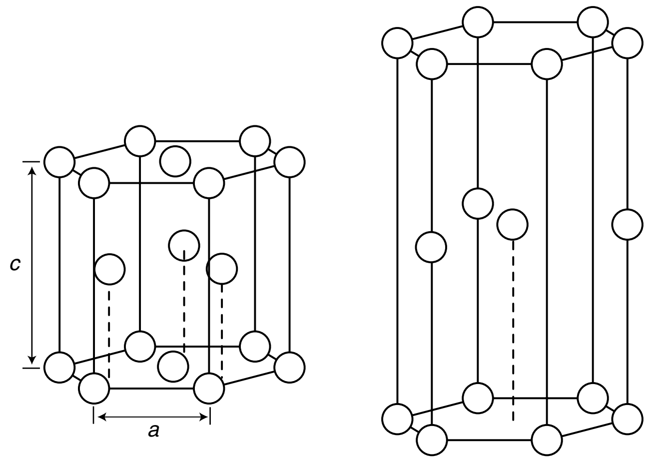

Figure 2.14 *Left*: Crystal structure of a hcp crystal. *Right*: Crystal structure of graphite. The lines are a mere guide to the eye, not indicating the size of the unit cell.

that there are two atoms in one primitive unit cell. (b) (*) Choose the vectors for the Bravais lattice for graphite and show that the basis contains four atoms.

3. *Real crystal structures*: Consider the hcp lattice shown in Figure 2.14. The Bravais lattice underlying the hcp structure is given by two vectors of length a in one plane, with an angle of 60° between them and a third vector of length c perpendicular to that plane. There are two atoms per unit cell. (a) Show that for the ideal packing of spheres, the ratio $c/a = (8/3)^{1/2}$. (b) (*) Construct the reciprocal lattice. Does the fact that there are two atoms per unit cell in the hcp crystal has any relevance? *Hint*: Use the result of Problem 2.7.

4. *Ionic bonding*: Calculate the potential energy for an ion in a sodium chloride crystal (the lattice parameter a is 2.81 Å) in units of electron volts and joules. Neglect the influence of the repulsive potential. From this, calculate the lattice energy of sodium chloride in units of joules per mole. Calculate also the cohesive energy in the same units.

5. *Ionic bonding*: The Madelung constant for a three-dimensional crystal of NaCl was presented in Section 2.2.1. (a) Derive the Madelung constant analytically for a one-dimensional chain of NaCl, as shown in Figure 2.15. (b) (*) Calculate the Madelung constant numerically for a one-dimensional, two-dimensional, and three-dimensional NaCl lattice and plot the result as a function of the number of iterations needed for the computation. Compare the result for the one-dimensional case with the analytical value from (a).

6. *X-ray diffraction*: (a) Determine the maximum wavelength for which Bragg reflection can be observed on a simple cubic crystal with a lattice constant of 3.6 Å. (b) What is the energy of the X-rays in electron volts? (c) If you were to do neutron diffraction, what would the energy of the neutrons have to be in order to obtain the same de Broglie wavelength? (d) You could argue that if you take X-rays with twice the wavelength, you would still get a Bragg peak because the X-rays would be diffracted from every other plane. Why is this argument not valid?

7. *The reciprocal lattice*: For a two-dimensional Bravais lattice

$$\mathbf{R} = m\mathbf{a}_1 + n\mathbf{a}_2, \tag{2.28}$$

the reciprocal lattice is also two dimensional:

$$\mathbf{G} = m'\mathbf{b}_1 + n'\mathbf{b}_2. \tag{2.29}$$

Often the most practical way to construct the reciprocal lattice is the relation

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}, \tag{2.30}$$

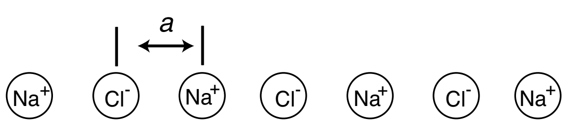

Figure 2.15 One-dimensional chain of ions.

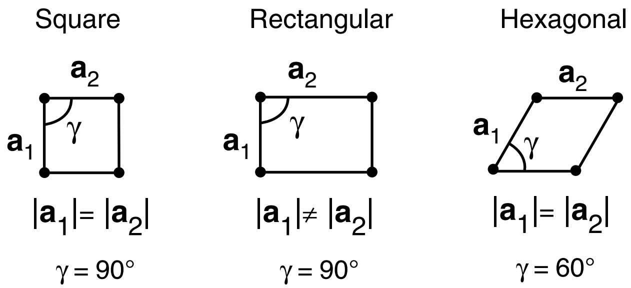

Figure 2.16 Two-dimensional Bravais lattices.

which remains valid in the two-dimensional case. Find the reciprocal lattice for the three cases given in Figure 2.16.

8. *Miller indices*: We have stated that the reciprocal lattice vector $m\mathbf{b_1} + n\mathbf{b_2} + o\mathbf{b}_3$ is perpendicular to the lattice plane given by the Miller indices (m, n, o). (a) Verify that this is correct for the lattice planes drawn in Figure 2.9. (b) (*) Show that this is true in general.

3
Mechanical Properties

In this chapter, we discuss the macroscopic behavior of a solid that is subject to mechanical stress and we will try to explain these properties in a microscopic picture. Our discussion is confined to isotropic solids, that is, solids for which the direction of the applied mechanical stress with respect to the crystal lattice is not important.

We start out with some fundamental definitions that are all illustrated in Figure 3.1. The applied stress on a solid σ is defined as the force F per area A perpendicular to the direction of the applied force (Figure 3.1a). Depending on the force direction, one can distinguish between tensile and compressive stress. The stress has the same dimension as pressure, that is, N m^{-2} or Pa. The solid responds to the stress by a deformation called strain ε. In the case of tensile stress, this would be a relative length extension $\Delta l/l$. The strain is therefore dimensionless, but in technical texts sometimes the unit m/m is found.

A shear stress τ is defined in a similar way as the applied force F per area A, but now the force is applied tangentially to the area (Figure 3.1b). Again, the material deforms according to the shear stress. The deformation is described by the angle α shown in the figure.

There are other mechanical deformations such as exposing the solid to hydrostatic pressure or torsion and these lead to similar definitions, but we do not discuss them here.

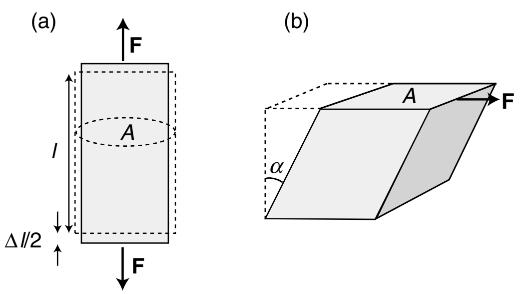

Figure 3.1 Definition of (a) stress $\sigma = F/A$ and strain $= \Delta l/l$ and (b) shear stress $\tau = F/A$ and shear angle α.

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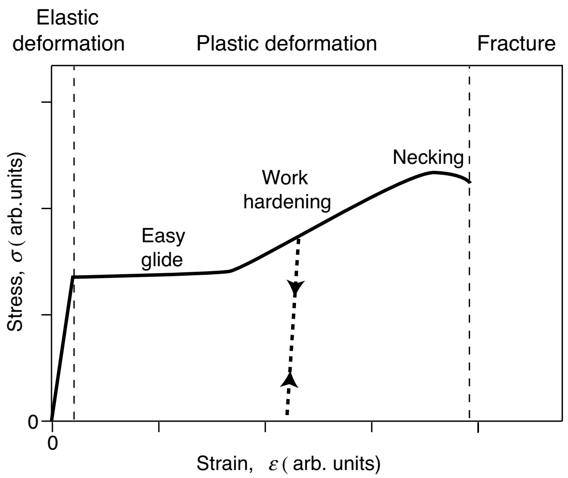

Figure 3.2 Typical stress of a solid as a function of applied strain. The dashed line illustrates the behavior when the stress is released and re-applied after increasing the strain into the work hardening region.

If we consider only the relation between stress and strain, the typical mechanical response of a solid is illustrated in Figure 3.2. It shows the resulting stress as a function of applied strain. This type of plot can seem rather odd at first. If you think of the strain as a consequence of the applied stress, you might be tempted to draw the curve with swapped axes. For the interpretation of the curve as it is displayed, one should adopt another point of view: The solid's length is increased and the strain that is "pulling back" is measured for every extension, very much like the force upon the extension of a mechanical spring.

Different regions in the curve can be distinguished. For a very small strain, typically much smaller than 1%, the deformation is elastic, that is, the solid goes back to its initial shape as the stress is released. In this region, the stress is also a linear function of the strain and this will allow the definition of various elastic constants in the next section. Beyond a certain yield strain or yield stress, plastic deformation sets in. This means that the deformation is permanent; once the stress is released, the solid does not return to its original shape. It only contracts slightly. The curve's shape in the region of plastic deformation will be discussed in a later section. Eventually, the strain becomes so high that the material fractures. This, naturally, defines the end of the stress/strain curve.

Although the region of elastic deformation is usually quite small, the amount of possible plastic deformation can vary widely from solid to solid. Some materials, such as glass or cast iron, will fracture immediately at the end point of the elastic limit. Such materials are called brittle. Materials such as most metals that do show plastic deformation before they fracture are called ductile.

3.1 Elastic Deformation

The elastic regime of deformation is small, but it is of high technical importance. For most applications, it is desirable that the deformation of the solids involved is elastic. Here, we define and discuss the concept of macroscopic elastic constants. We will also see that these constants can be connected to the picture of interatomic bonding that we have encountered earlier.

3.1.1 Macroscopic Picture

3.1.1.1 Elastic Constants

The linear behavior for the small deformations in the elastic regime leads to a few definitions of macroscopic elastic constants. The relation between stress and strain is given by Young's modulus Y:

$$Y = \frac{\sigma}{\varepsilon} = \frac{F}{A}\frac{l}{\Delta l}. \tag{3.1}$$

Young's modulus has therefore the same unit as the stress, that is, pascal. It is the slope of the initial stress/strain curve in Figure 3.2. The values of Young's modulus are usually very high, typically in the GPa region.

The possibility to define Young's modulus is equivalent to the validity of Hooke's law that is commonly used to describe a "springlike" force response. Suppose you extend a spring by some small amount. Then it responds by a force that is proportional to the extension. This is equivalent to

$$\sigma = Y\varepsilon \tag{3.2}$$

multiplied by A

$$F = \frac{YA}{l}\Delta l, \tag{3.3}$$

so that the usual spring constant is YA/l.

A linear material constant can also describe the shearing of a solid. The modulus of rigidity G is defined by

$$G = \frac{\tau}{\alpha}, \tag{3.4}$$

and similar definitions are made for the exposure of the solid to hydrostatic pressure or torsion.

3.1.1.2 Poisson's Ratio

When mechanical stress is applied to a solid, the strain in the direction of the stress is not the only consequence. In addition to this, the solid's dimensions may change in

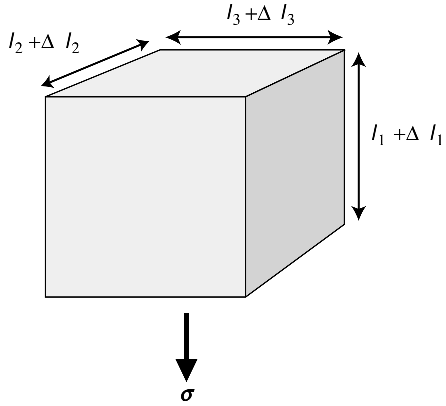

Figure 3.3 Sketch illustrating the definition of Poisson's ratio.

the directions perpendicular to the stress, as illustrated in Figure 3.3. Poisson's ratio ν is defined as the relation of perpendicular change to parallel change:

$$\frac{\Delta l_2}{l_2} = \frac{\Delta l_3}{l_3} = -\nu \frac{\Delta l_1}{l_1} = -\nu\varepsilon. \tag{3.5}$$

As we discuss only isotropic solids, the fractional change in the 2 and 3 directions is the same. The minus sign in the definition assures that ν is positive in the "normal" situation where the solid contracts sideways upon tensile stress. However, there are also exotic materials with a negative Poisson's ratio, for example, special types of molecular foam that expand sideways upon being exposed to tensile stress and contract when subject to compressive stress.

Poisson's ratio cannot take all possible values. It is limited to a range between -1 and $+0.5$. The lower limit is not so relevant as materials with a negative Poisson's ratio are rather rare. The upper limit is caused by the fact that a solid cannot decrease its volume when you pull on one side, or that it cannot increase its volume when you press it from one side. The volume of the solid after the application of a stress is

$$(l_1 + \Delta l_1)(l_2 + \Delta l_2)(l_3 + \Delta l_3). \tag{3.6}$$

For small changes, we can neglect higher order terms in the Δl's and this becomes

$$l_1 l_2 l_3 + \Delta l_1 l_2 l_3 + l_1 \Delta l_2 l_3 + l_1 l_2 \Delta l_3. \tag{3.7}$$

So the change in volume is

$$\begin{aligned} \Delta l_1 l_2 l_3 + l_1 \Delta l_2 l_3 + l_1 l_2 \Delta l_3 &= \Delta l_1 l_2 l_3 + l_1 \left(-\nu \frac{\Delta l_1}{l_1} l_2 \right) l_3 + l_1 l_2 \left(-\nu \frac{\Delta l_1}{l_1} l_3 \right) \\ &= (1-2\nu) \Delta l_1 l_2 l_3. \end{aligned} \tag{3.8}$$

For a positive Δl_1, this is required to be positive, which can only be achieved for values of ν smaller than or equal to 0.5.

Typical values of Poisson's ratio range between 0.2 and 0.4 for most materials. Rubber has a ν very close to 0.5, that is, it is an almost ideal non-compressible solid. Cork has $\nu \approx 0$, which is advantageous when you try to put a wine cork back into the bottle.

3.1.1.3 Relation Between Elastic Constants

In a broader mathematical context, all the macroscopic constants are different aspects of a few fundamental elastic properties. Not surprisingly, they are related to each other. For example,

$$G = \frac{Y}{2(1+\nu)}. \tag{3.9}$$

The intimate relation between the elastic properties has an advantage for our task to explain the macroscopic behavior in a microscopic picture. We will, in most cases, restrict ourselves to explaining one type of mechanical property and we are allowed to do so without great loss of generality.

3.1.2 Microscopic Picture

The elastic deformation of a solid can be explained in terms of changing interatomic distances. According to Figure 1.1, the potential ϕ between two atoms has a minimum at the equilibrium distance a and the interatomic force at this distance, which is just the negative spatial derivative of the potential, is zero. Upon the application of a compressive stress, the distance between the atoms is decreased. This results in a force that presses the atoms away from each other. For tensile stress, it is the other way round. Once the stress is released, the atoms return to their equilibrium distance.

This explains why the behavior is elastic but why is it linear? A linear force change close to the minimum can readily be seen in Figure 1.1b. More formally, we can expand the potential for distances x close to the minimum as a Taylor series:

$$\phi(x) = \phi(a) + \frac{\mathrm{d}\phi(a)}{\mathrm{d}x}(x-a) + \frac{1}{2}\frac{\mathrm{d}^2\phi(a)}{\mathrm{d}x^2}(x-a)^2 + \frac{1}{6}\frac{\mathrm{d}^3\phi(a)}{\mathrm{d}x^3}(x-a)^3 + \cdots. \tag{3.10}$$

The first term is simply an offset of the absolute energy scale and therefore irrelevant here. The second term is zero because the slope of ϕ, and hence the restoring force, vanishes at the equilibrium distance. The third term is responsible for the elastic behavior. It states that the potential close to the equilibrium depends on the square of the distance change, that is, the force depends linearly on the distance change. The fourth and higher terms are usually neglected.

For distance changes a that are sufficiently big, terminating the Taylor series after the third term may become imprecise and one would expect to see elastic deformation that is not linear. In such a deformation, the elastic constants would depend on the actual change of the atomic separation, for example, Young's modulus would depend on the applied stress. It turns out, however, that this is very rarely observed.

For most solids, the plastic deformation sets in for a strain of less than 1% and this is before higher order terms in (3.10) become important.

Equation 3.10 is strongly reminiscent of a Taylor series that is used to calculate the force constant and vibrational frequency in an oscillator. Indeed, we will later see that it is possible to relate the vibrational properties of a solid to its elastic properties.

We conclude our treatment of the elastic regime by looking at typical values of Young's modulus. As stated above, Y is very high for most materials, in the order of GPa. This means that the strain of typical materials in daily life is invisibly small. An overview is given in Figure 3.4 It is quite apparent how different bonding types relate to different values of Young's modulus. Metals and alloys are all in the range between 15 and 300 GPa. As a tendency, transition metals have a higher Y than simple metals, consistent with a stronger bonding due to localized d electrons. W and Mo have particularly high values, consistent with the fact that they also have high melting points, as we shall see later. Solids with covalent bonding span a much wider range. The sp^2 and sp^3 bonds in graphite and diamond lead to particular strength but note that graphite also appears at the lower end of the range. This is because we have neglected the possibility of anisotropy in solids. Graphite is strongly bonded parallel to the sp^2-linked planes but very weakly perpendicular to these planes. Polymers show small values of Y. The reason is that reversible length change in a polymer does not have to be achieved via the change of an interatomic bond length. It is sufficient to

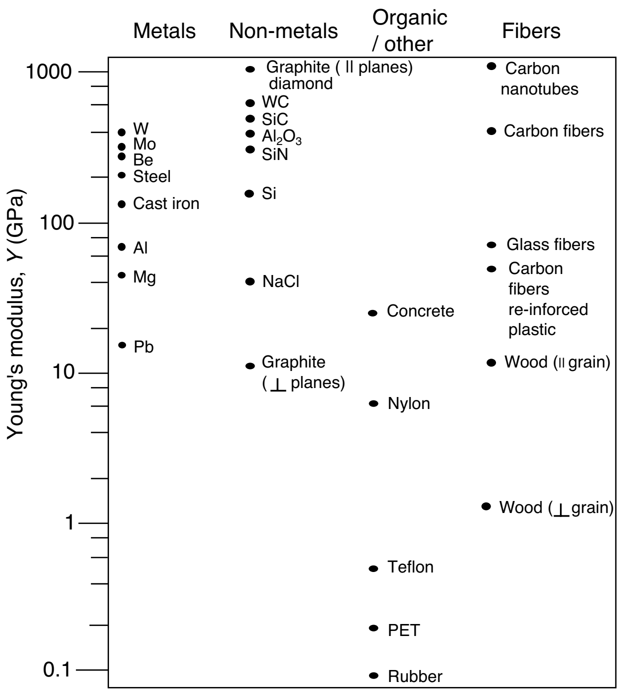

Figure 3.4 Young's modulus for different materials. The values are merely a guide, as strong variations are possible.

change the angles of the many bonds in a polymer, that is, to "unfold" it. Composite materials and fibers cover a wide range of Y from carbon nanotubes that have the highest Y of any material to wood perpendicular to the grains.

Note that the Young's modulus does not reflect the same physical quantity as the cohesive energy discussed in Chapter 1. The cohesive energy measures how deep the potential minimum in Figure 1.1 is. Young's modulus,on the other hand, corresponds to the curvature of the potential around the minimum. Obviously these two are related as the properties in Figure 3.4 correspond well to what we have discussed in connection with the cohesive energies. The covalent bonds in diamond, for example, give rise to a high cohesive energy and at the same time they strongly resist small changes in the bonding distance.

3.2 Plastic Deformation

Now we address the plastic deformation part of the stress–strain curve. We will be able to establish a link to microscopic models, but a detailed understanding of all phenomena cannot be accomplished on the basis of the perfect crystal. It will be necessary to introduce different types of imperfections in the crystal, such as point defects and dislocations. Such imperfections will also be important in our later treatment of electrical resistance.

3.2.1 Estimate of the Yield Stress

The most important practical issue associated with plastic deformation is arguably the end point of the elastic regime, that is, can we estimate the yield stress or yield strain? On the atomic scale, this is particularly simple for a shear deformation of a crystal.

Figure 3.5a shows two atom rows in a hexagonal crystal plane that in (b) is subject to shear stress. This leads to a deformation with a shear angle α in which the atomic rows are pulled away from their lowest energy position. For small shear angles, we can relate α to the interlayer distance a and the displacement x:

$$\alpha = \tan^{-1}\left(\frac{x}{a}\right) \approx \frac{x}{a}. \tag{3.11}$$

This expression of α is inserted in the definition of the modulus of rigidity (3.4):

$$\tau = G\alpha \approx \frac{Gx}{a}, \tag{3.12}$$

which directly establishes a link between macroscopic quantities and the microscopic displacements.

For small values of α, the atoms will go back into their equilibrium positions as soon as the shear stress is released; the deformation is elastic. But if the shear stress is

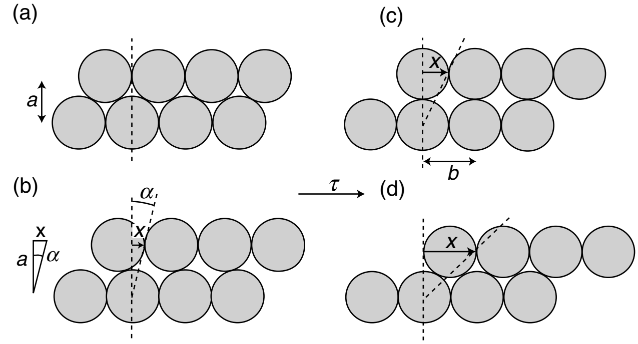

Figure 3.5 Estimate of the yield stress for shearing a solid. (a) Atoms in equilibrium position. (b) Distortion for a small shear stress. (c) Metastable equilibrium. (d) New stable equilibrium for the sheared solid .

increased more and more, the rows of atoms will eventually slip over each other. This leads to an unstable equilibrium when x is equal to half the interatomic distance b, as in Figure 3.5c. Stable equilibrium is reached as $x = b$ (Figure 3.5d). In fact, the microscopic positions of the atoms with respect to each other in the crystal do now correspond exactly to the starting point before applying any shear stress, but the crystal has undergone plastic deformation. Consequently, the shear stress must be a periodic function of x with a period of b. We write

$$\tau = C \sin\left(\frac{2\pi x}{b}\right), \tag{3.13}$$

where C is the highest value of the shear stress that can be reached (when $x = b/4$). In other words, when the applied $\tau > C$, the solid can be plastically deformed and thus C is the yield stress τ_Y. Equation 3.13 can be approximated for small x, giving

$$\tau = C\frac{2\pi x}{b} \tag{3.14}$$

and this is combined with (3.12) to give

$$\frac{Gx}{a} \approx C\frac{2\pi x}{b}, \tag{3.15}$$

that is,

$$\tau_Y = C = \frac{Gb}{2\pi a} = \frac{G}{\sqrt{3}\pi}, \tag{3.16}$$

where we have used that $a = b\sqrt{3}/2$. This equation allows us a direct estimate of the yield stress from the modulus of rigidity. It may have to be corrected by small geometrical factors depending on crystal structure and for three-dimensional crystals, but it should give the right order of magnitude for the yield stress.

But it does not. In fact, the estimated values of the yield stress are several orders of magnitude too high. Equation 3.16 gives values of the yield stress in the order of the modulus of rigidity or in the order of Young's modulus (see (3.9)), that is, 10^{10} Pa, but the measured yield stress for most materials is in the order of 10^6–10^9 Pa. What has gone wrong? It turns out that there is nothing wrong with this theory; the problem is that we have assumed a perfect defect-free crystal. Even a qualitative understanding of the stress/strain curve in the plastic regime requires the introduction of defects.

3.2.2
Point Defects and Dislocations

Defects, or crystal imperfections, in solids are a wide research subject of their own and we have to take them into account to some degree, even though we are mostly concerned with the perfect crystal. Defects are more than a small annoying perturbation of the perfect crystal. In the present context, they are essential for explaining the mechanical properties and later we will see that they are also indispensable for phenomena like electrical resistance in all materials or for the electronic properties of semiconductors. We broadly distinguish between very localized point defects and extended defects such as grain boundaries or dislocations.

A crystal can have different types of point defects and we mention only a few here. There can be an atom missing in the otherwise perfect crystal structure. Such a defect is called a vacancy. Atoms of a different kind can occur, either on the original lattice sites instead of the "correct" atoms or in between lattice sites. These defects are called substitutional or interstitial, respectively. The former play an important role in the doping of semiconductors; the latter are often used to design alloys with improved mechanical properties (see below).

Dislocations are much more extended because they are line-type defects. These lines can extend through the whole crystal or they can have the shape of a loop. For the mechanical properties of a solid, the edge dislocation shown in Figure 3.6 is of particular importance. This type of dislocation is caused by one extra sheet of atoms in

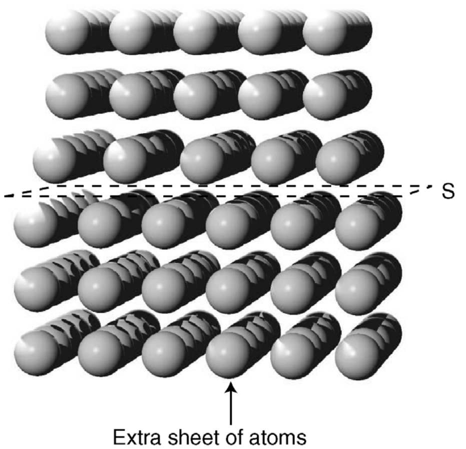

Figure 3.6 An edge dislocation. The plane denoted by S is called the slip plane.

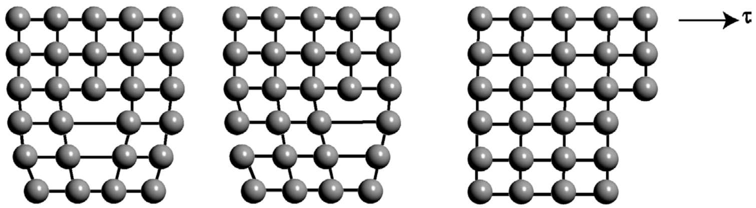

Figure 3.7 Shearing of a solid in the presence of an edge dislocation. The dislocation moves through the solid by breaking only one row of bonds at a time.

the crystal. The plane that is denoted S in the figure is called the slip plane for reasons that will become apparent in a moment.

3.2.3
The Role of Defects in Plastic Deformation

The presence of edge dislocations can explain why the yield stress of a real crystal is much smaller than predicted by (3.16). This is illustrated in Figure 3.7. As the solid with a dislocation is exposed to shear stress, a plastic yielding can be achieved by moving the dislocation through the crystal. It is immediately evident why this is much easier than the process discussed in connection with (3.16): When a dislocation is present in the solid, the plastic deformation can proceed by breaking one row of bonds at a time instead of all bonds between two planes simultaneously.

The problem now is that an estimation of the yield stress with dislocations present gives a value that is too low, in the order of 10^5 Pa. This is due to the presence of point defects that can "pin" a dislocation as shown in Figure 3.8. As the dislocation encounters a point defect, it needs a higher activation energy to move past it. In real materials, point defects that can pin dislocations are intentionally introduced. Examples are carbon that makes steel tougher or beryllium that has the same role in copper.

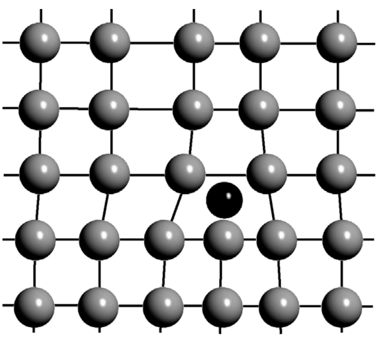

Figure 3.8 A point defect can pin a dislocation such that it cannot move.

The presence of dislocations and defects now allows us to understand the details of the plastic deformation in the stress/strain curve of Figure 3.2 up to the point of fracture. Once the yield stress is overcome, dislocation-assisted glide sets in. This is the so-called easy-glide region. The stress increases only very little for a big strain increase, that is, the curve is rather flat.

The next region is called the work hardening region. Here, the stress/strain curve is considerably steeper. The meaning of the term work hardening becomes obvious when we consider what happens as the stress is released (see Figure 3.2): the material will contract very little as the stress goes to zero. Upon a new application of stress, the material will deform *elastically* until the original stress/strain curve is reached again. But this point is reached at a higher stress than for the original material as the curve is steep in the work hardening zone. This means that the yield stress is higher and hence the term work hardening. The microscopic picture behind the work hardening is that the number of dislocations increases for higher strain values, for reasons not discussed here. At some point, the dislocations hinder each other's movement, and the slope of the stress/strain curve increases. Work hardening can be a useful technique to increase the strength of materials by pre-stressing them.

It is clear from both intuition and experience that the temperature is very important for the mechanical properties of materials. At elevated temperature the role of the entropy in a crystal becomes more important and defects will be generated in order to minimize the Gibbs free energy. In addition to this, activation barriers, such as the one needed to move a dislocation across a point defect, can be overcome more easily at higher temperatures.

3.2.4 Fracture

At the end of the stress/strain curve, the material will fracture. Immediately before fracturing, the stress might even decrease. This is due to a phenomenon called necking, in which the material narrows somewhere between the points at which the stress is applied. The narrower cross section means that the local stress is even higher. This causes a self-amplification that leads to fracture, even though the total applied stress on the sample might decrease.

So far we have only discussed ductile materials that have a stress/strain curve similar to the one shown in Figure 3.2. What happens in brittle materials that do not show any plastic deformation at all, but fracture at the end of the elastic part of the curve? This phenomenon is called brittle fracture. It is always associated with the presence of fine cracks in the material. At the end of the cracks, the local stress is much higher than the average stress in the material. More specifically, it is increased by a factor $2\sqrt{l/r}$, where l is the length of the crack and r the radius at the end. For a ductile material, the stress can be relieved by a local plastic deformation that work hardens the material around the end of the crack. Then the crack is stopped and cannot proceed further. If such a plastic deformation cannot happen, the crack will proceed through the whole material. Indeed, as the stress is increasing for deeper cracks, this is a self-amplifying process that can lead to the spontaneous breaking of very big structures (e.g., entire ocean-going ships).

Again, temperature is important for the behavior of materials. At elevated temperatures, the propagation of dislocations is easier and materials that behave brittle at low temperature can be ductile as the temperature is raised. A prominent example is glass, which is usually brittle, but at high temperature it is so ductile that its shape can be changed by blowing.

3.3 Discussion and Problems

3.3.1 Discussion

1. What happens typically when a crystal is exposed to a small stress?
2. How can an elastic deformation of a crystal be described microscopically and why would you expect Hooke's law to hold for a small strain?
3. How do the stress/strain curves look for a typical ductile and brittle material?
4. Young's modulus can be estimated from the microscopic force constants between atoms (or the other way round). Why and how? (*) How can you determine a typical interatomic force constant?
5. The yield stress of a solid estimated from a simple calculation is often much higher than the observed yield stress. Explain why.
6. Explain the phenomenon of work hardening.

3.3.2 Problems

1. *Elastic constants*: Another elastic constant that was not defined in this chapter is the bulk modulus K. It describes the response of a solid to applied hydrostatic pressure and is defined in a similar way as Young's modulus (3.1) as

$$K = -p\frac{V}{\Delta V}, \tag{3.17}$$

where p is the applied pressure and V and ΔV the volume and its change, respectively. The minus sign is needed for K to be positive because an applied pressure leads to a decrease of volume. K has again the dimension of pressure. (a) Consider a cube of isotropic material and show that the bulk modulus is related to Young's modulus and Poisson's ratio by

$$K = \frac{Y}{3(1-2\nu)}. \tag{3.18}$$

(b) What happens when ν reaches or exceeds its upper limit of 0.5?

2. *Elastic constants*: We have stated that the numerical value of the Poisson ratio is always between $+0.5$ and -1, but we have proven only the upper limit. Use Equation 3.9 to argue why -1 is the lower boundary for the Poisson ratio.

3. *Elastic constants*: In the text, we have treated only isotropic materials for which the elastic properties do not depend on the direction of applied stress. Graphite is an example of a solid that is highly anisotropic. The Young's modulus in two different directions differs by more than a factor of 100 (see Figure 3.4). Why is this so and which are the two directions?

4. *Elastic constants*: The metals with the highest values of Young's modulus in Figure 3.4 are also those with the highest cohesive energies and melting temperatures (see Figure 4.19a). Are these two aspects of the same thing?

4
Thermal Properties of the Lattice

Here, we discuss some thermal properties of solids such as their heat capacity, thermal conduction, thermal expansion, and melting. We restrict our treatment to the properties of the lattice. We neglect electronic contributions for now, in particular, from free electrons in metals. We shall come back to this in the next two chapters.

4.1
Lattice Vibrations

In our treatment of lattice vibrations, we restrict ourselves to the harmonic solid, that is, we consider only the first non-vanishing term in the series (3.10).

4.1.1
A Simple Harmonic Oscillator

When inspecting the interatomic potential in Figure 1.1 and the Taylor series (3.10) for the potential, one might be tempted to describe the vibrations of a solid with N atoms simply as $3N$ simple and independent harmonic oscillators. The factor of 3 comes from three different directions the atoms can oscillate in. It is surprising how far one gets with this very simple picture.

If the force constant γ is equal to $\mathrm{d}^2\phi(a)/\mathrm{d}x^2$ and x is the displacement from the equilibrium position, the equation of motion is

$$M\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\gamma x, \tag{4.1}$$

where M is the mass of the vibrating atom. This leads to a harmonic motion with the frequency[1)]

$$\omega = \sqrt{\frac{\gamma}{M}}. \tag{4.2}$$

1) For simplicity, we will often speak merely of "frequency" when "angular frequency" would be the correct term.

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The total energy for a one-dimensional harmonic oscillator is the sum of kinetic and potential energies:

$$E = \frac{1}{2}Mv^2 + \frac{1}{2}\gamma x^2. \tag{4.3}$$

Assuming classical motion, we can estimate the amplitude of the vibration by using the equipartition theorem of statistical mechanics. This theorem states that every generalized momentum or position coordinate, which appears squared in the Hamilton function (or in the total energy in our case), should contribute with a mean energy $k_BT/2$ to the system. Here, we have two squared coordinates and therefore the mean energy of the oscillator in contact with a heat bath is k_BT. At the highest displacement, the oscillator has only potential energy and so

$$\frac{1}{2}\gamma x_{\max}^2 = k_B T, \tag{4.4}$$

and thus

$$x_{\max} = \left(\frac{2k_B T}{\gamma}\right)^{1/2}. \tag{4.5}$$

This is usually a small percentage of the lattice spacing.

4.1.2
An Infinite Chain of Atoms

A more realistic model for crystal vibrations is a chain of atoms. If we restrict ourselves to longitudinal vibrations in this chain, the model is only one dimensional. However, we can learn a lot about vibrations in three-dimensional solids already from this. We start with infinite chains of one and two different atoms per unit cell before passing on to chains of finite length.

4.1.2.1 One Atom Per Unit Cell

An atomic chain with one atom per unit cell is shown in Figure 4.1. From a formal point of view, this is a one-dimensional atomic lattice with one atom per unit cell and a lattice vector of length a. This means that the reciprocal lattice is also one dimensional and the reciprocal lattice vector has a length of $2\pi/a$. The atoms at the lattice sites shall be connected with springs of a force constant γ. If we take only nearest neighbor interactions into account, the equation of motion for atom n is

$$M\frac{d^2u_n}{dt^2} = -\gamma(u_n - u_{n-1}) + \gamma(u_{n+1} - u_n), \tag{4.6}$$

or

$$M\frac{d^2u_n}{dt^2} = -\gamma[2u_n - u_{n-1} - u_{n+1}], \tag{4.7}$$

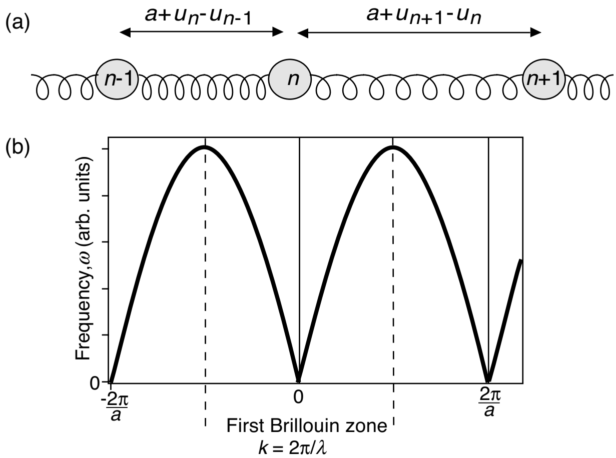

Figure 4.1 (a) One-dimensional chain with one atom per unit cell. (b) Allowed vibrational frequencies ω as a function of the wave vector k.

where u_n is the displacement of the nth atom in the chain. This can be solved by a kind of wave that is only defined on the lattice sites:

$$u_n(t) = u e^{i(kan-\omega t)}, \tag{4.8}$$

where $k = 2\pi/\lambda$ is the one-dimensional wave vector of the wave with the wavelength λ and u is the amplitude of the oscillation. Substituting this into the equation of motion gives

$$-M\omega^2 e^{i(kan-\omega t)} = -\gamma[2-e^{-ika}-e^{ika}]e^{i(kan-\omega t)} = -2\gamma(1-\cos ka)e^{i(kan-\omega t)}, \tag{4.9}$$

and this has a solution if we choose the $\omega = \omega(k)$ such that

$$\omega(k) = \sqrt{\frac{2\gamma(1-\cos ka)}{M}} = 2\sqrt{\frac{\gamma}{M}}\left|\sin\frac{ka}{2}\right|. \tag{4.10}$$

The resulting $\omega(k)$ is also plotted in Figure 4.1. The solutions describe waves propagating along the chain. What is special about the waves is that they are only defined on the actual lattice sites. The phase velocity ν of the wave is ω/k and the group velocity is $\partial\omega/\partial k$. Relations of the type (4.10), which connect a frequency or energy to a wave vector, are called dispersion relations. We will encounter them many more times, for example, in connection with electronic states. For a long wavelength, that is, a small k, the sine in (4.10) can be substituted by its argument and one gets the linear relation

$$\omega = \sqrt{\frac{\gamma}{M}}ak = \nu k. \tag{4.11}$$

Here, the group velocity and the phase velocity are equal. This limit corresponds to the propagation of sound waves with a velocity ν. The wavelength is very long, the atomistic structure becomes unimportant, and the waves propagate just as they do in a uniform elastic medium. The limit of short wavelengths is also very instructive. The shortest possible wavelength must be two lattice spacings, such that $\lambda = 2a$ and $k = \pi/a$. For $k = \pm\pi/a$, the group velocity of the wave becomes zero, meaning that the solutions of (4.7) are standing waves.

Equation 4.10 suggests that $\omega(k)$ is periodic in k with a period of $2\pi/a$. This is precisely the reciprocal lattice vector. The reciprocal lattice is not only useful to describe lattice periodic quantities like the charge density in a Fourier series. It is also fundamental for describing waves in a lattice. In the present case, as in general, the wave is unchanged when we add a reciprocal lattice vector to its wave vector k. This is illustrated in Figure 4.2, which shows the instantaneous displacement of atoms for two waves that differ by a reciprocal lattice vector. The displacement is the same. Another way of viewing this is to note again that the shortest possible wavelength in a lattice of spacing a is $\lambda = 2a$ or $k = \pi/a$, which corresponds to the situation where neighboring atoms move exactly out of phase. Any wave that has an even shorter wavelength can be equivalently described by a longer wavelength.

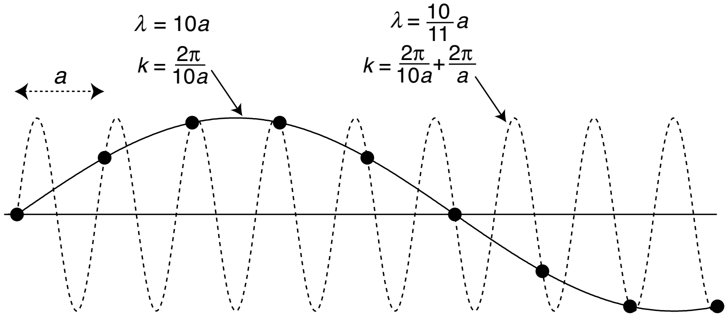

Figure 4.2 Instantaneous position of atoms in a chain for two different wavelengths with $\lambda = 10a$ and $\lambda = (10/11)a$. Note that the wave is transverse for illustrative purposes. Otherwise, we have only considered longitudinal waves in one-dimensional chains.

4.1.2.2 **The First Brillouin Zone**

The most remarkable result of the last section is probably that we have easily managed to describe the motion of all atoms in an *infinite* chain, just by making use of the chain's periodicity. In fact, it is sufficient to know the solution ω for k only in an interval of length $2\pi/a$ in order to know it everywhere. One could even argue that an interval of π/a is sufficient. This is due to the left/right symmetry of the chain. It does not matter if the wave travels to the left or to the right, that is, if k is positive or negative.

This motivates another definition that appears rather formal right now but turns out to be very useful. We call the region between $k = -\pi/a$ and $k = \pi/a$ the first Brillouin zone of the lattice. In a more general reciprocal lattice, the first Brillouin zone would be defined as all points that are closer to a given reciprocal lattice point (the origin) than to any other. We recognize that this definition corresponds exactly to that of the Wigner–Seitz cell in real space: the first Brillouin zone is the Wigner–Seitz cell of the reciprocal lattice. The first Brillouin zone and similar constructions are often said to be placed in reciprocal space or k-space, which means the same thing. The first Brillouin zone is indicated in Figure 4.1. Note again that $\omega(k)$ is flat at the boundaries of the first Brillouin zone, that is, for $k = -\pi/a$ and $k = \pi/a$.

4.1.2.3 **Two Atoms Per Unit Cell**

We also briefly discuss the situation with two atoms per unit cell as shown in Figure 4.3. The calculation of the vibrational frequencies is very similar to the case of one atom per unit cell. The lattice constant is now b and the length of the reciprocal lattice vector is $2\pi/b$. Of the two atoms per unit cell, one is at the origin and one at $b/2$. We write down the forces on each atom in a similar manner as above (see also Figure 4.3) and obtain two equations of motion.

$$M_1 \frac{d^2 u_n}{dt^2} = -\gamma[2u_n - v_{n-1} - v_n], \quad M_2 \frac{d^2 v_n}{dt^2} = -\gamma[2v_n - u_n - u_{n+1}], \tag{4.12}$$

where u_n and v_n are the displacements of the first and second atom in the nth unit cell, respectively. This can again be solved by wave-type functions of the form

$$u_n(t) = ue^{i(kbn - \omega t)}, \quad v_n(t) = ve^{i(kbn - \omega t)}. \tag{4.13}$$

When this is inserted into the equations of motion, we obtain a homogeneous linear system of equations for the amplitudes u and v:

$$-\omega^2 M_1 u = \gamma v(1 + e^{-ikb}) - 2\gamma u, \quad -\omega^2 M_2 v = \gamma u(e^{ikb} + 1) - 2\gamma v. \tag{4.14}$$

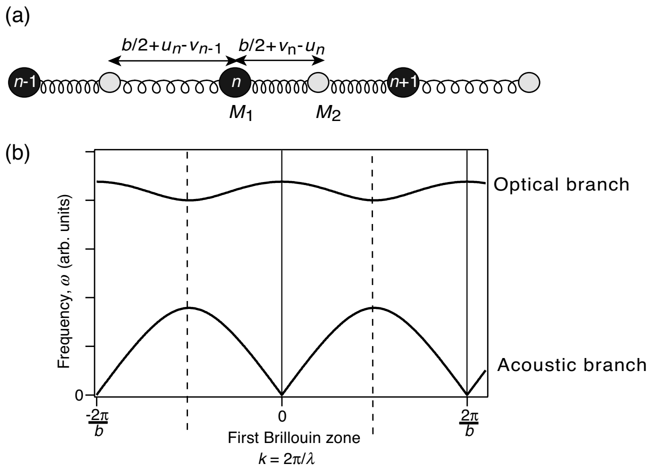

Figure 4.3 (a) One-dimensional chain with two atoms per unit cell. (b) Allowed vibrational frequencies ω as a function of the wave vector k.

A solution exists if the determinant of the coefficient matrix vanishes, that is,

$$\begin{vmatrix} 2\gamma-\omega^2 M_1 & -\gamma(e^{ikb}+1) \\ -\gamma(1+e^{-ikb}) & 2\gamma-\omega^2 M_2 \end{vmatrix} = 0. \tag{4.15}$$

This happens when

$$\omega^2 = \gamma\left(\frac{1}{M_1}+\frac{1}{M_2}\right) \pm \gamma\left[\left(\frac{1}{M_1}+\frac{1}{M_2}\right)^2 - \frac{4}{M_1 M_2}\sin^2\frac{kb}{2}\right]^{1/2}, \tag{4.16}$$

and these solutions are also shown in Figure 4.3. Again, the solutions have the periodicity of the reciprocal lattice vector $2\pi/b$, that is, it is sufficient to know them within the first Brillouin zone. Now we have two branches of solutions. The solution that goes to zero for small k is called the acoustic branch. As before, it corresponds to the propagation of acoustic waves through the crystal. The solution that has a finite ω at $k=0$ is called the optical branch. It is called like this because of the possibility to couple these vibrations to an electromagnetic field. At $k=0$, the optical branch gives rise to a vibration in which the two atoms per unit cell move exactly out of phase. If the two atoms per unit cell carry a different charge, this corresponds to a changing dipole moment that can couple to an external light field (see Problem 4.3).

4.1.3
A Finite Chain of Atoms

For describing the properties of real solids, the models discussed so far have a fundamental problem because of the fact that the chains are infinite. This will, for example, lead to infinite heat capacities, which is quite useless. What we really want is a finite but long chain of atoms. This can be done by limiting the length and introducing boundary conditions. For example, one can hold the atoms at the ends fixed. This leads to standing waves in the chain because we have fixed the nodes. Although this approach would not be wrong, it would be more convenient to start with traveling wave solutions if we are to describe phenomena such as heat transport later.

The most convenient boundary conditions solving the problem of what to do at the ends of the chain are those suggested by M. Born and T. von Kármán in 1912. For a chain with N atoms, these conditions state that

$$u_{N+n}(t) = u_n(t). \tag{4.17}$$

This can be visualized as a finite chain of atoms in which the end is tied to the beginning. Therefore, the conditions are also called cyclic boundary conditions or periodic boundary conditions. In three dimensions, this simple visualization does not work but we can think of a crystal of finite size that has identical crystals with identical motions attached to its sides. In this way, we end up with an infinite crystal lattice again but since it is made from small finite crystals, we can use it to describe the properties of one of these finite crystals.

The Born–von Kármán boundary conditions restrict the possible k values for the waves in the crystal. First of all, it becomes clear that the longest possible wavelength for a chain of N atoms with a spacing of a is restricted to Na. More precisely, we have to require that

$$e^{ikan} = e^{ika(N+n)} \tag{4.18}$$

so that

$$e^{ikNa} = 1 \tag{4.19}$$

and the possible values of k become

$$k = \frac{2\pi}{aN} m, \tag{4.20}$$

where m is an integer. We have seen that the vibrations are unaffected by adding multiples of $2\pi/a$ (i.e., reciprocal lattice vectors) to k and therefore we only get N possible different values for k and hence also N different vibrational frequencies ω. The possible values of n or k can be chosen to lie in the first Brillouin zone, that is, $-N/2 < m < N/2$, or between 0 and N if only positive values of k are desired. Physically, it does not make any difference. The k values for a finite chain are illustrated in Figure 4.4. Note that for a real finite solid, the number of atoms in one direction is very large. Therefore, the distances between the allowed k points is very small and the finite vibrational frequencies resemble closely the continuum of states for the infinite chain.

If we take N free atoms that can only move in one dimension, each atom has one degree of freedom. The total number of degrees of freedom is conserved when we put the atoms into a chain linked with springs, since we also get N different vibrations. These are the so-called normal modes. We can make the same argument for the chain

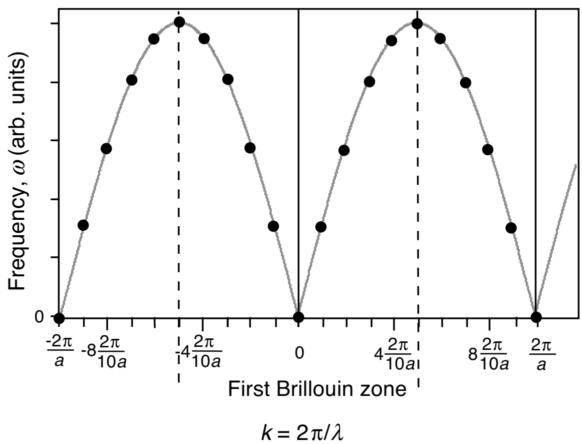

Figure 4.4 Vibrational spectrum for a finite chain of atoms with a length of 10 unit cells and a unit cell length of a. The light gray line represents the vibrational spectrum for an infinite chain of lattice constant a. The black markers represent the actual vibrational frequencies that are allowed.

with two atoms per unit cell. If we have N unit cells, we get N different k vectors. But now there are two vibrations corresponding to each k: the acoustic and the optical mode. Again the number of degrees of freedom is conserved.

An analysis in terms of normal modes can be done much more formally but this is beyond the scope of these notes. It turns out that the situation of a chain with N atoms and one atom per unit cell can be formally transformed to N equations for harmonic oscillators with different frequencies ω. These frequencies are those given by the dispersion relation $\omega(k)$.

4.1.4
Quantized Vibrations, Phonons

So far, we have neglected the quantized character of the lattice vibrations but this turns out to be essential for the correct description of many properties, for example, the heat capacity.

For one harmonic oscillator, as described by (4.1), the situation is very simple. The energy levels are quantized with

$$E_l = \left(l + \frac{1}{2}\right)\hbar\omega, \tag{4.21}$$

again with $\omega = (\gamma/M)^{1/2}$ and $l = 0, 1, 2, \ldots$. These energy levels are displayed in Figure 4.5a.

We have seen that a chain of N atoms can also be viewed as N harmonic oscillators, each with a different $k = 2\pi m/aN$ and vibrational frequencies $\omega(k)$. For such a system, the quantized energy levels have to be

$$E_l(k) = \left(l + \frac{1}{2}\right)\hbar\omega(k). \tag{4.22}$$

The notations here lends itself to an alternative interpretation of k. So far, we have viewed k as the one-dimensional wave vector. But here it becomes apparent

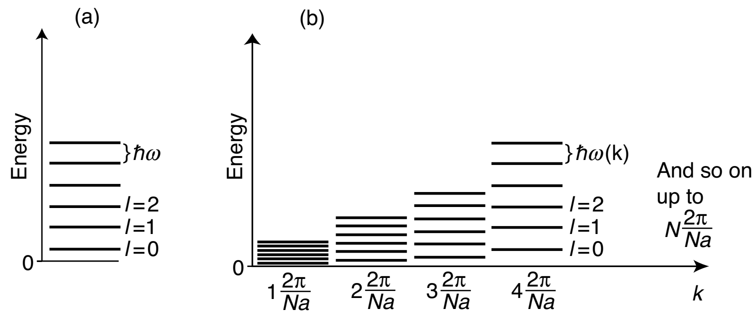

Figure 4.5 (a) Energy level diagram for one harmonic oscillator. (b) Energy level diagram for a chain of atoms with one atom per unit cell and a length of N unit cells.

that k also takes a role of a quantum number, just as l. k takes only discrete values and can be used to "label" different oscillators. The combination of k and l describes one vibrational state of the solid. It is the oscillator k that is excited to the level l. The interpretation of k as a quantum number has a lot to do with the symmetry of the system. In atoms, we have spherical symmetry that gives rise to the quantum numbers l and m. In solids, we have translational symmetry and the appropriate quantum number is k.

We can also come up with an alternative view of the whole vibrational spectrum. We have started to look at the vibrational motion of a single atom but for a system of many atoms, we have found vibrational modes in which all atoms participate, the normal modes of the system. Still (4.22) formally looks like the energy levels of many harmonic oscillators but these oscillators are not localized on any particular atom. All atoms participate. Often a single excitation of this oscillator spectrum is called a phonon.

A phonon is very similar, in concept[2] and name, to a photon. Depending on the type of experiments, photons can have wave character as well as particle character and the same is true for phonons. So far, our description mostly emphasized the wave character but if we want to describe properties like thermal conductivity, we need to think of phonons as "particles" that are generated at the hot end of some sample and conducted to the cold end. As in the case of photons, the wave and particle character can be reconciled if we describe the "particle" as a superposition of waves, which is localized in a certain volume of space. An additional similarity of phonons and photons is that both are bosons and therefore not subject to the Pauli exclusion principle. The quantum statistics for phonons will become important when we evaluate the heat capacity of solids.

4.1.5 Three-Dimensional Solids

Our simple discussion of atomic chains already contains most of the important physics for the vibrational properties of the lattice but we briefly discuss what happens in three dimensions, not least because we wish to establish a link to measured quantities of real solids.

4.1.5.1 Generalization to Three Dimensions

The ideas of our one-dimensional chains are quite easily generalized to three dimensions. We do not go into much detail because little additional physical insight is gained and the equations get quite messy because one has to keep track of many indices. The wave-type ansatz (4.13) also solves the equation of motion for three-dimensional solids. However, the notation has to be more complex in order to

2) The similarity is quite far-reaching. Even mathematically (in quantum field theory), photons can be viewed as excitations of quantum mechanical harmonic oscillators.

accommodate not only more atoms per unit cell but also more directions of motion. For instance, for a three-dimensional solid with two atoms per unit cell, the matrix corresponding to (4.15) will be a 6×6 matrix, giving six vibrational frequencies ω for every value of k. There will be three acoustic modes, one with longitudinal polarization as in one dimension and two with transverse polarization. Similarly, there will also be three optical modes.

Actually, the one-dimensional k turns into a real wave vector $\mathbf{k}$ with three components in three dimensions but it does of course retain its additional interpretation as a quantum number. For a simple cubic crystal with a lattice spacing a in all three directions and N atoms in every direction, one gets

$$\mathbf{k} = (k_x, k_y, k_z) = \frac{2\pi}{aN}(n_x, n_y, n_z) = \left(\frac{n_x 2\pi}{L}, \frac{n_y 2\pi}{L}, \frac{n_z 2\pi}{L}\right), \tag{4.23}$$

with n_x, n_y, n_z being integers and L the macroscopic side length of the crystal (the restriction to a macroscopic cube makes life easier without any loss of generality).

As in one dimension, it is sufficient to describe the vibrational states only within the first Brillouin zone. The definition of the first Brillouin zone is the same as in one dimension; it is the region of reciprocal space that is closer to a given reciprocal lattice point than to any other. However, the geometrical shape of the first Brillouin zone can be quite complicated in three dimensions. For an fcc crystal, it is the truncated octahedron that is shown in the inset of Figure 4.6. In the Brillouin zone and on the Brillouin zone boundary, points of high symmetry are abbreviated by certain letters. The letter Γ always stands for the center of the Brillouin zone, that is, for $\mathbf{k} = (0, 0, 0)$, and the zone center is often referred to as the Γ point.

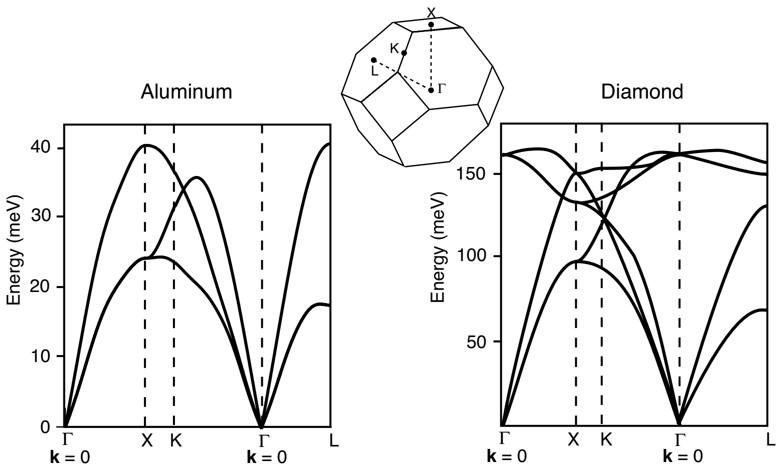

Figure 4.6 Phonon dispersion in Al and diamond along several directions in reciprocal space. The inset shows the Brillouin zone, which has the same shape for both materials. Reproduced from Refs [1, 2].

Figure 4.6 shows the phonon dispersion for both aluminum and diamond. In both cases, we can clearly identify the acoustic phonons with a linear dispersion near the Γ point. For diamond, there are also optical phonons, that is, phonons with a finite energy at Γ but these are not found for Al. The reason for this is simple: both materials have an fcc Bravais lattice but Al can be described with only one atom as basis, whereas two atoms are needed for diamond. This is the reason why optical phonons are found for the latter but not for the first.

4.1.5.2 **Estimation of the Vibrational Frequencies from the Elastic Constants**

The force constant between two atoms appearing in the vibrational properties is very closely related to our discussion of the microscopic origin of Young's modulus and it is easy to establish a link. Think about a rod of solid with a side length of only one lattice constant a, as shown in Figure 4.7. When we pull on this solid, the stress is

$$\sigma = \frac{F}{a^2}. \tag{4.24}$$

On the other hand, if every unit cell expands by x then we can write

$$F = \gamma x, \tag{4.25}$$

so that

$$\sigma = \frac{\gamma x}{a^2}. \tag{4.26}$$

The strain is simply $\varepsilon = x/a$ and

$$Y = \frac{\sigma}{\varepsilon} = \frac{\gamma x}{a^2}\frac{a}{x} = \frac{\gamma}{a} \tag{4.27}$$

(see also (3.3)).

From Young's modulus, we can therefore calculate the force constant and the vibrational frequency corresponding to this force constant. When comparing this to the experimental frequencies, we are faced with a dilemma: There is a whole range of frequencies, from zero to the highest frequency $2\sqrt{\gamma/M}$ (see (4.10)) for the chain of atoms, which corresponds to the situation in Figure 4.7. It can be argued that the right frequency to compare to is the one where we hold all atoms fixed and move only one. In a one-dimensional chain, the force constant for such a motion would be 2γ (see (4.7)) and give a vibrational frequency of $\omega = \sqrt{2\gamma/M}$, that is, a factor of $\sqrt{2}$ lower than the

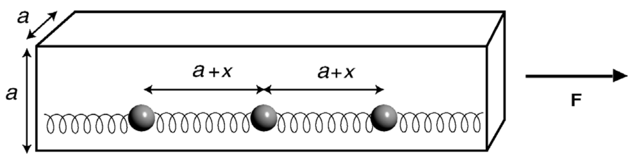

Figure 4.7 Illustration of the derivation of the interatomic force constant from Young's modulus.

Table 4.1 Measured vibrational frequencies of a solid and values estimated from Young's modulus.

	Diamond	**Pb**
mass	12 u	207 u
Young's modulus	1000 GPa	15 GPa
ω_{calc}	9×10^{13} Hz	4×10^{12} Hz
ω_{measr}	1.8×10^{14} Hz	7×10^{12} Hz

highest vibrational frequency in the solid.[3] In Table 4.1, we therefore compare the frequencies estimated from Young's modulus and the interatomic distance with the highest phonon frequency in the materials divided by $\sqrt{2}$. In view of the rough estimate, the agreement is rather good. Again we see that the very simplistic model of just an harmonic oscillator with $\omega = \sqrt{\gamma/M}$ gives a good idea about the vibrational properties of a solid. Diamond, which has light atoms and strong bonds, has much higher vibrational frequencies than lead, which has heavy atoms and weak bonds.

4.2 Heat Capacity of the Lattice

Historically, understanding the heat capacity of solids was one of the biggest early successes of quantum theory. In the beginning of the last century, the situation was extremely puzzling. Classical statistical mechanics explained the heat capacity of insulators at room temperature fairly well but it failed for lower temperatures and it totally failed for metals. Metals were expected to have a much higher heat capacity than insulators because of the many free electrons but it turns out that a metal's heat capacity at room temperature is similar to that of an insulator, as if the electrons were not there. We shall see later why this is so and we focus on the lattice now.

We ignore the difference between heat capacities at constant volume and at constant pressure. For solids, this difference is usually quite small but not entirely negligible. Experimentalists usually like to measure the heat capacity at constant pressure, for example, at ambient pressure. Theorists, on the other hand, prefer calculations at constant volume because otherwise all the quantum mechanical eigenvalues have to be recalculated for every temperature.

4.2.1 Classical Theory and Experimental Results

Classically, we can use two different approaches to calculate the heat capacity of a solid. The worrying result is that they do not give the same answer. Classical

3) This ignores the possibility of optical branches and the calculation should merely be taken as an estimate of the order of magnitude.

Table 4.2 Molar heat capacity (JK^{-1}) of different solids at liquid nitrogen temperature and room temperature compared to the Dulong–Petit value of $24.9\,JK^{-1}$.

Material	77 K	273 K
Cu	12.5	24.3
Al	9.1	23.8
Au	19.1	25.2
Pb	23.6	26.7
Fe	8.1	24.8
Diamond	0.1	5.2

thermodynamics does not tell us anything about the value of the heat capacity at finite temperature but it can be shown that it should vanish for zero temperature. The argument leading to this is based on very few general assumptions, notably on the requirement of a finite entropy at zero temperature (see section Further Reading in the Appendix).

Classical statistical mechanics, on the other hand, gives us the possibility to calculate a numerical value for the heat capacity of a solid: for a one-dimensional harmonic oscillator in contact to a heat bath, the mean energy is k_BT. For a three-dimensional oscillator, it must be $3k_BT$. This means that the heat capacity for one oscillator is $3k_B$ and for 1 mol of oscillators it is $3R = 24.9\,JK^{-1}$ *independent of the temperature and the material.* This result is called the rule of Dulong–Petit. Table 4.2 shows the heat capacities for a number of solids. The agreement with the Dulong–-Petit value is rather good at room temperature but less good at liquid nitrogen temperature where the heat capacity is generally smaller.

So there is already a conflict between different types of classical theories. The Dulong–Petit law, which is derived from the equipartition theorem of classical statistical mechanics, predicts a temperature-independent heat capacity whereas the heat capacity has to vanish at zero temperature according to classical thermodynamics. A vanishing heat capacity at zero temperature is also supported by experimental results. Already Table 4.2 points in this direction. Another example is shown in Figure 4.8, which shows the heat capacity of diamond as a function of temperature. At high temperatures, the heat capacity approaches the Dulong–Petit value but at lower temperatures it drops to zero. The figure is plotted using a double logarithmic scale. The low-temperature dependence is a line in this plot, suggesting a power law behavior. From the slope one directly reads that $C \propto T^3$. A microscopic theory of the heat capacity should be able to reproduce this behavior.

4.2.2 Einstein Model

The breakthrough to understanding the temperature-dependent heat capacity of solids was made by A. Einstein. His key idea was to solve the problem using quantum

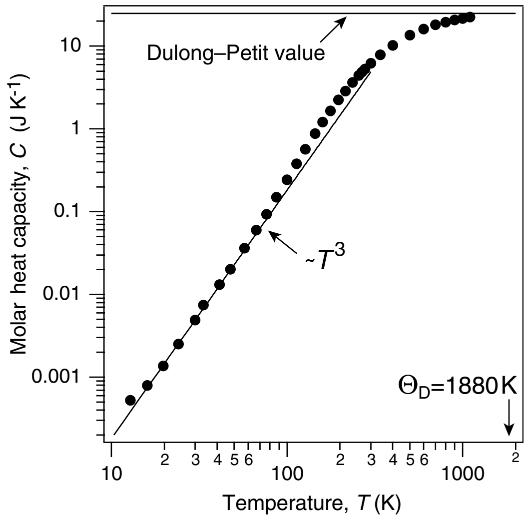

Figure 4.8 Temperature-dependent heat capacity of diamond. Data from Refs [3, 4].

theory to describe the oscillators in the solid. It starts out by assuming that the solid's vibrations are represented by independent harmonic oscillators that all have the same frequency, the Einstein frequency ω_E, so that their energy levels are

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega_E. \tag{4.28}$$

We are interested in the mean energy for $3N_A$ of these oscillators per mole of atoms, which are in contact to a heat bath. It is given by $3N_A$ times the mean energy for one oscillator:

$$\langle E\rangle = 3N_A\left(\langle n\rangle + \frac{1}{2}\right)\hbar\omega_E. \tag{4.29}$$

The mean quantum (or phonon) number $\langle n\rangle$ can be found using the Bose–Einstein distribution since lattice vibrations are of bosonic character, that is, there is no limit on the number of quanta n per state.

$$\langle n\rangle = \frac{1}{e^{\hbar\omega_E/k_B T} - 1}. \tag{4.30}$$

The resulting mean energy for $3N_A$ oscillators is therefore

$$\langle E\rangle = 3N_A\left(\frac{1}{e^{\hbar\omega_E/k_B T} - 1} + \frac{1}{2}\right)\hbar\omega_E. \tag{4.31}$$

The heat capacity is found by differentiation:

$$C = 3N_A\left(\frac{\partial\langle E\rangle}{\partial T}\right)_V = 3R\left(\frac{\hbar\omega_E}{k_B T}\right)^2 \frac{e^{\hbar\omega_E/k_B T}}{(e^{\hbar\omega_E/k_B T} - 1)^2}. \tag{4.32}$$

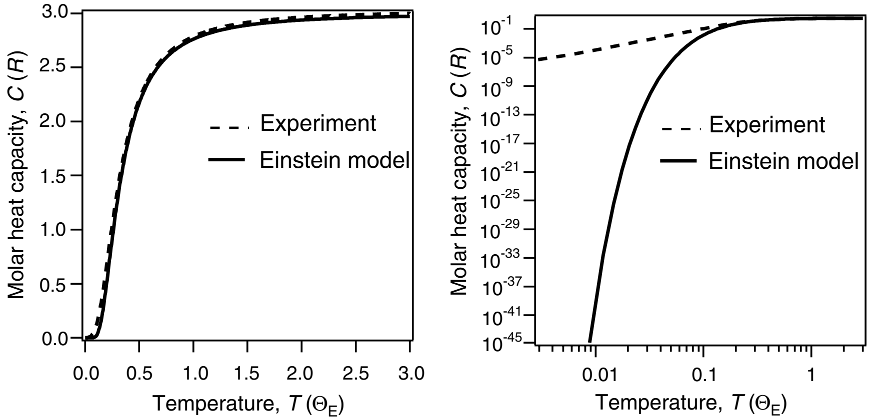

Figure 4.9 Temperature-dependent heat capacity in the Einstein model compared to a typical experimental result for an insulator: (a) linear scale; (b) log–log scale.

The result of this calculation is shown in Figure 4.9 together with a curve representing typical experimental results similar to those in Figure 4.8. In the high-temperature limit, the Einstein model correctly reproduces the Dulong–Petit value. "High temperature" means that the temperature must be at least as high as the Einstein temperature, which is the temperature corresponding to the vibrational frequency of the oscillators, that is, $\Theta_E = \hbar\omega_E/k_B$. The heat capacity also drops to zero at lower temperatures, in agreement with the experimental data. The only problem is that it drops too quickly to zero. At low temperatures it shows an exponential behavior whereas the experiment shows a power law behavior $C \propto T^3$. On the linear temperature scale, this does not show up too clearly but on the double log scale the problem is evident.

Can we understand this behavior in simple terms? In the high-temperature limit, the thermal energy is much bigger than the spacing between the energy levels $\hbar\omega_E$ and the quantized nature of the problem becomes insignificant. This is why we obtain the Dulong–Petit value for the specific heat. Note that this is independent from the nature of the solid and the details of the vibrational spectrum, as it should be. The only condition is that the temperature is higher than the Einstein temperature. At sufficiently low temperatures, almost all the oscillators are in their ground state. If the temperature is raised just a little, by much less than $\hbar\omega_E$, nothing will change and the heat capacity is zero. In fact, the probability of occupation of the first excited level of the oscillators follows an exponential behavior:

$$p_1 \propto e^{-\hbar\omega_E/k_B T}, \tag{4.33}$$

which is the low-temperature limit of (4.30). Therefore, the exponential decrease of the heat capacity when cooling an Einstein solid originates from the fact that the oscillators are "frozen" into their ground state.

4.2.3
Debye Model

We have seen that the key problem in the Einstein model is the low-temperature heat capacity. It falls off too quickly because all the Einstein oscillators get "frozen in" below Θ_E. In other words, all oscillators, which look like the one in Figure 4.5a, are in their ground state and the probability for exciting the first state is vanishingly low.

P. Debye noticed that this problem can be cured by using a more realistic model for the lattice vibrations at low energies. After all, these vibrations are the only ones that are really relevant at low temperatures. Consider again the finite chain model with one atom per unit cell, as shown in Figure 4.4. For a finite but macroscopic chain with a length L, the distance between k points is $2\pi/L$. The length of a reciprocal lattice vector is $2\pi/a$, where a is the lattice constant. Therefore, the spacing of allowed k points is very small compared to the size of the Brillouin zone. The oscillators with the smallest separation of the energy levels are found near $k = 0$. Figure 4.5b shows the energy levels for these oscillators and it is clear that the situation is different from the Einstein model (as in Figure 4.5a) because the level separation $\hbar\omega(k)$ is very small. Already here we can expect a low-temperature result that differs from the Einstein model. In the Einstein model, we got an exponential behavior once the level separation was getting large compared to the thermal energy. For a macroscopic crystal, this is not going to happen at any achievable temperature because the level separation is very small near $k = 0$.

The excitations with the lowest energies near $k = 0$ are those with the longest wavelength for which (4.11) holds, that is, sound waves with

$$\omega(k) = \nu k. \tag{4.34}$$

The basic assumption in the Debye model is now that this dispersion holds for all values of k. This is clearly inaccurate for the excitations at higher k, as we have seen in Figure 4.1 or Figure 4.4. It is totally incorrect for a unit cell containing more than one atom because it ignores the existence of the optical branches (see Figure 4.3). However, these modes are not excited at low temperatures anyway. At high temperatures, they may be excited but this does not matter so much. From the Einstein model, we have seen that the high-temperature heat capacity approaches the classical value, independent of the actual oscillator frequencies. For low temperatures, the Debye assumption is appropriate and it leads to good results as we shall see now.

We need to calculate the mean thermal energy for a set of oscillators with frequencies given by (4.34). For one oscillator with frequency ω, we know that the result is

$$\langle E \rangle = \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1} \tag{4.35}$$

(see (4.31)). The zero point energy $\hbar\omega/2$ is neglected right away as it does not contribute to the heat capacity. For a three-dimensional solid, it is now tempting to write the mean energy for all oscillators as

$$\langle E\rangle = 3\int \frac{\hbar\omega'}{e^{\hbar\omega'/k_B T}-1}\,d\omega'. \tag{4.36}$$

The factor of 3 stems from the fact that there are three possibilities for the wave polarization for a given ω: two different transverse and one longitudinal polarization. Here, we assume that all three waves follow the dispersion $\omega = \nu|\mathbf{k}|$. From the total energy, we could calculate the heat capacity.

But there are two problems: The first is that there may be more oscillators in, say, the frequency interval $\omega_1 + d\omega$ than in $\omega_2 + d\omega$. This must be included by a weighting factor $g(\omega)$ in the integral. This factor is called the density of states, for obvious reasons. The second problem is that we have to establish the upper limit ω_D for the integration, in other words, we have

$$\langle E\rangle = 3\int_0^{\omega_D} \frac{g(\omega)\hbar\omega}{e^{\hbar\omega/k_B T}-1}\,d\omega \tag{4.37}$$

and we are looking for $g(\omega)$ and ω_D.

We start by calculating the density of states $g(\omega)$ for a three-dimensional solid. It is insufficient to do this in one dimension because $g(\omega)$ depends on the dimensionality of the problem and we want to compare to experimental data for real solids. Consider a cube of solid with a macroscopic side length L. The vibrational solutions for such a cube are equivalent to those in one dimension: traveling waves. The three-dimensional wave vector can only have certain values that are given by (4.23), completely analogous to one dimension (see (4.20)). We can think of each vibrational state as being defined by a triple of ks (k_x, k_y, k_z) or, equivalently, by a triple of ns (n_x, n_y, n_z).[4)] How many possible states N do we get for a given highest $|\mathbf{k}|$ or n? If n is large, this has a simple geometric interpretation: We want to know how many states lie within a sphere of radius n_{max} or, alternatively, a sphere with radius k_{max} (see Figure 4.10). The total number of states is given by the volume of the sphere:

$$N = \frac{4}{3}\pi n^3, \tag{4.38}$$

or expressed in terms of $|\mathbf{k}|$:

$$N = \frac{4}{3}\pi\left(\frac{L|\mathbf{k}|}{2\pi}\right)^3, \tag{4.39}$$

or if we use the dispersion $\omega(|\mathbf{k}|) = \nu|\mathbf{k}|$:

$$N(\omega) = \frac{4}{3}\pi\left(\frac{L\omega}{2\pi\nu}\right)^3 = \frac{V}{6\pi^2\nu^3}\omega^3, \tag{4.40}$$

4) Actually, every such triple would characterize three different vibrational states but we have already taken care of this separately.

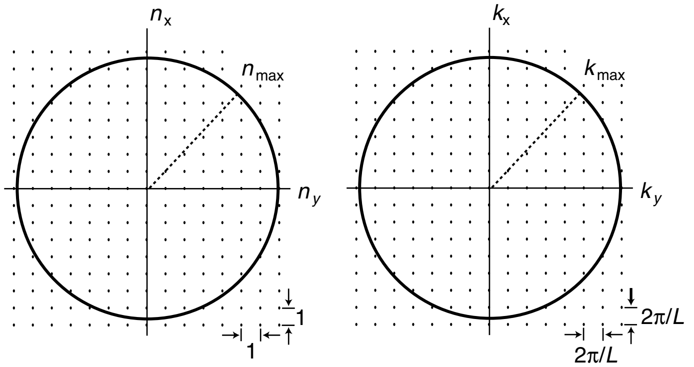

Figure 4.10 Two-dimensional cut, showing (a) the points of integers (n_x, n_y, n_z) that represent the allowed vibrational states and (b) the values of wave numbers (k_x, k_y, k_z) corresponding to (n_x, n_y, n_z). The circle represents a cut through a sphere for a certain highest value of n or k, such that a total of N states is enclosed.

where $V = L^3$ is the volume of the crystal. From this, we can get the density of states by differentiation:

$$g(\omega)\mathrm{d}\omega = \frac{\mathrm{d}N}{\mathrm{d}\omega}\mathrm{d}\omega = \frac{\omega^2 V}{2\pi^2 \nu^3}\mathrm{d}\omega. \tag{4.41}$$

Now we have to address the question of the upper integral limit ω_D for (4.37). Whatever, the nature of the excitations, the limit of the integral must be chosen such that we recover the correct number of normal modes, so for N atoms in the solid we must have

$$3N = 3\int_0^{\omega_D} g(\omega)\mathrm{d}\omega. \tag{4.42}$$

Using (4.41) and performing the integration results in

$$\omega_D^3 = 6\pi^2 \frac{N}{V}\nu^3. \tag{4.43}$$

ω_D is called the Debye frequency and the corresponding temperature $\Theta_D = \hbar\omega_D/k_B$ is called the Debye temperature.

With this (4.37) becomes

$$\langle E\rangle = 3\int_0^{\omega_D} \frac{\omega^2 V}{2\pi^2\nu^3}\frac{\hbar\omega}{e^{\hbar\omega/k_B T}-1}\mathrm{d}\omega = \frac{3V\hbar}{2\pi^2\nu^3}\int_0^{\omega_D}\frac{\omega^3}{e^{\hbar\omega/k_B T}-1}\mathrm{d}\omega, \tag{4.44}$$

and with the substitution $x = \hbar\omega/k_B T$ and $x_D = \hbar\omega_D/k_B T$, it becomes

$$\langle E\rangle = \frac{3Vk_B^4 T^4}{2\pi^2\nu^3\hbar^3}\int_0^{x_D}\frac{x^3}{e^x-1}\mathrm{d}x = 9Nk_B T\left(\frac{T}{\Theta_D}\right)^3\int_0^{x_D}\frac{x^3}{e^x-1}\mathrm{d}x. \tag{4.45}$$

From this, the heat capacity of the solid can be determined by differentiation with respect to temperature. Instead of writing down an expression for the heat capacity, we focus on the low- and high-temperature limits.

For high temperatures, x in (4.45) is small and the exponential function in the integral can be approximated by $1+x$. The integral is then merely over x^2 and the resulting energy is $\langle E\rangle = 3Nk_BT$. For 1 mol of atoms this is, equal to $3RT$, that is, it leads to the Dulong–Petit result. This was of course expected from the result in the Einstein model: The Dulong–Petit result is always reached at sufficiently high temperatures.

The less obvious limit is that for low temperatures. Here, x is large and we can make the approximation to carry out the integration to infinity instead of x_D. Then the integral has a value of $\pi^4/15$ and after differentiation we find the heat capacity to be

$$C = \frac{12\pi^4}{5} Nk_B \left(\frac{T}{\Theta_D}\right)^3. \tag{4.46}$$

This is the Debye T^3 law that fits the experimental data far better than the exponential behavior of the Einstein model (see Figure 4.8). The reason the Debye model works so well at low temperatures has been mentioned above: it provides only a good description of the vibrational modes at low energies and long wavelengths and is inaccurate for higher energies. But at low temperatures only the low-energy modes are excited anyway and therefore it works well.

The Debye temperature Θ_D plays a similar role as the Einstein temperature in the Einstein model: it sets a temperature scale. Only for temperatures sufficiently above the Debye temperature, the Dulong–Petit law is obeyed. On the other hand, the Debye temperature has been defined as the maximum vibrational frequency of the material, assuming a linear dispersion, and it is very often used as a measure of the maximum frequency or energies associated with vibrations in a given material. The Debye temperatures and frequencies for a number of selected materials are given in Table 4.3. The Debye temperatures follow a tendency that is consistent with intuition: heavy atoms and weak (i.e., metallic) bonds give rise to lower vibrational frequencies than light atoms with strong (i.e., covalent) bonds. The standard examples are again lead and diamond. It is interesting to compare ω_D to the measured highest vibrational frequency in Table 4.1 for lead and diamond. ω_D appears to be systematically too high. It is clear that the Debye model should overestimate the highest frequency because it simply assumes a linear dispersion for all wavelengths. For the linear chain, we have

Table 4.3 Debye temperatures and frequencies for selected materials at room temperature.

Material	**Θ_D(K)**	**ω_D (Hz)**
Pb	105	1.37×10^{13}
Cu	343	4.49×10^{13}
Si	645	8.45×10^{13}
Diamond	1860	2.44×10^{14}

seen that the real dispersion flattens out as k approaches $2\pi/a$ and the maximum value is therefore lower than ω_D (this argument is again ignoring the possibility of optical branches).

4.3 Thermal Conductivity

In this section, we address the transport of heat through a crystal by lattice vibrations (phonons), that is, the thermal conductivity κ_p. Daily experience tells us that metals are usually much better thermal conductors than insulators.Therefore, the contributions of the electrons to the thermal conductivity could be thought to be much more important than the lattice contribution. This, however, is not always the case. A classic example is the insulator diamond that has one of the highest thermal conductivities of all materials at room temperature. It may not be close to daily experience, but it would not be good to make teaspoons out of diamond. We will discuss the electronic contribution to the thermal conductivity in a metal later. Fortunately, the two contributions just add. The total thermal conductivity is the sum of the lattice and the electronic thermal conductivity:

$$\kappa = \kappa_p + \kappa_e. \tag{4.47}$$

First we have to state more precisely what we mean by thermal conductivity. Suppose you have a rod with a cross-sectional area A. One end of the rod should be in a heat bath with temperature T, while the other end is constantly heated with a power $\partial Q/\partial t$. Once dynamic equilibrium is established after some time, we measure the temperature difference ΔT between two points in the middle of the rod, which are separated by Δx. The thermal conductivity κ is then defined as

$$\kappa = \frac{1}{A}\frac{\partial Q}{\partial t}\frac{\Delta x}{\Delta T}. \tag{4.48}$$

Note the similarity of this expression with the usual definition of the electrical conductivity. Here, $\Delta T/\Delta x$ plays the role of the electric field and $(1/A)(\partial Q/\partial t)$ the role of the current density.

As discussed in our introduction of phonons, traveling wave solutions for systems with periodic boundary conditions cannot be used to describe thermal conduction. In fact, while the waves are traveling, their average amplitude is the same over the whole crystal such that a temperature gradient is not even possible. For thermal conduction, we want vibrational excitations that can be generated at one side of the solid and travel to the other side, that is, we need to describe the phonons as particles. This can be achieved by the superposition of traveling solutions to give wave packets that then travel through the crystal with a certain group velocity (Problem 6.10 is concerned with this subject for electronic waves). These wave packets are generated on the hot side of the solid and then propagate to the cold side.

In this sense, lattice vibrations can be viewed as a type of particles that travel through the solid. Often, such wave packets are also called 'phonons'. Indeed, it turns out that the thermal conductivity of the solid can be described using kinetic gas theory, as if the phonons were a gas of particles bouncing through the solid. For describing the thermal conductivity due to the gas of phonons, we take a result from kinetic gas theory:

$$\kappa_p = \frac{1}{3} c \lambda_p \nu_p, \tag{4.49}$$

where c is the heat capacity of the solid per unit volume, λ_p is the mean free path of the phonons, and ν_p is the phonon speed. In order to evaluate κ_p, we can take ν_p to be the velocity of sound ν and for the heat capacity we can take the results of the previous section. The only quantity that is unknown is the mean free path of the phonons. We discuss it in the following.

As the phonons propagate through the crystal, they can be scattered by imperfections in the lattice, such as point defects, dislocations, and the like. It is possible to grow crystals of such high perfection that these scattering effects become unimportant. Then the scattering of the phonons at the sample boundaries can be observed experimentally. Scattering from crystal imperfections is especially important at low temperatures.

At high temperatures another scattering process becomes dominant: the number of phonons increases and phonons can be scattered from other phonons. This causes the mean free path, and thereby also κ_p, to decrease. At low temperatures, on the other hand, the heat capacity in (4.49) decreases, causing κ_p to decrease as well. This means that there must be a temperature that produces a maximum in κ_p. This can typically be found at about 10% of the Debye temperature. An example is given in Figure 4.11.

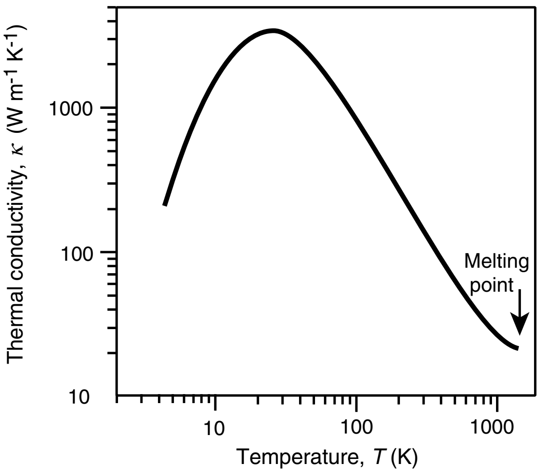

Figure 4.11 Temperature-dependent thermal conductivity of Si. Reproduced from Ref. [5].

Some numerical values for the thermal conductivity of solids are given in Table 4.4. Evidently, not only metals show high thermal conductivity but also some insulators,

Table 4.4 Thermal conductivity κ for metals and insulators at room temperature.

Material	κ(W m^{-1} K^{-1})
Copper	386
Aluminium	237
Steel	50
Diamond	2300
Quartz	10
Glass	0.8
Polystyrene	0.03

especially diamond. Diamond has a nearly perfect crystal structure such that defect scattering of phonons is unimportant. In this sense, it is similar to Si, also a nonmetal with the same structure, which shows a very high thermal conductivity at low temperatures (see Figure 4.11). The difference is that diamond also has a high Debye temperature such that phonon–phonon scattering becomes important at much higher temperatures than for Si. As a result, the $\kappa(T)$ curve for diamond looks qualitatively similar to that of Si but it is shifted to higher temperatures, explaining the good thermal conductivity at room temperature.

Phonon–phonon scattering deserves a comment: in a purely harmonic situation this cannot happen. Consider for example water waves or low-intensity light waves (i.e., anything but lasers). There the principle of superposition holds. The field amplitude at a given point of space is just the superposition of the field amplitudes from different waves propagating through space. The waves "propagate through each other." This principle also holds for phonons since they are lattice waves for the harmonic solid. Phonon–phonon scattering can only happen in the anharmonic case, that is, if the amplitude of the oscillations becomes so large that the fourth and higher terms in (3.10) become important. This may seem problematic because our whole treatment so far is based on the assumption that the vibrations are harmonic. In fact, the whole concept of a phonon only makes sense for a harmonic solid. If the anharmonic effects are not too strong, however, we can save the phonon picture to some extent. We can still think of phonons propagating through the crystal just that they now have a certain finite lifetime after which they decay into other phonons.

4.4
Thermal Expansion

We have seen that anharmonic effects are responsible for the finite thermal conductivity of a (perfect) crystal. In this section, we will encounter another effect of anharmonic vibrations, which is well known from daily experience: the thermal expansion of solids: The volume of a solid depends on the temperature. Here, we

Table 4.5 Coefficient of linear thermal expansion α at room temperature.

Material	$\alpha(10^{-5}\,\mathrm{K}^{-1})$
Pb	2.9
Al	2.4
Cu	1.7
Steel	1.1
Glass	0.9
Invar	0.09

restrict our attention to the linear expansion of isotropic solids. The coefficient of thermal expansion α canbe defined as

$$\frac{\Delta l}{l} = \alpha \Delta T, \tag{4.50}$$

where $\Delta l/l$ is the fractional length change of the solid and ΔT is the temperature change. α is always quite small (see Table 4.5). It is also temperature dependent and can be shown to vanish at zero temperature. At room temperature, most materials have an α in the order of $10^{-5}\,\mathrm{K}^{-1}$. A remarkable exception is Invar, a nickel–iron alloy that has one of the lowest coefficient of thermal expansion of all metallic compounds. Its discovery by C.-E. Guillaume around 1900 was a major technical breakthrough because it permitted the construction of highly stable measurement instruments, clocks and the like.

We can understand the thermal expansion of solids by an inspection of the interatomic potential (3.10). The first anharmonic term is the cubic term, which turns the potential asymmetric and therefore results in a change of the equilibrium distances for different temperatures. This is easily seen in a classical picture. Figure 4.12 shows a scaled-up version of the interatomic potential in Figure 1.1. We choose the energy scale such that the potential minimum has zero energy. The mean energy of the oscillator at a temperature T is $k_B T$. For a low temperature (T_1), the oscillation takes place between the positions r_1 and r_2. As the potential is roughly symmetric around its minimum, the average interatomic spacing is equal to the equilibrium distance. For a higher temperature (T_2), the oscillation takes place between r_3 and r_4. The potential is not symmetric anymore and the interatomic distances expands slightly on average. In effect, it follows the gray line for higher temperatures.

This picture is not changed qualitatively in a quantum mechanical treatment. The energy levels are then discrete. For a purely harmonic oscillator, they are equidistant and the mean interatomic distance is the same in all levels. For a non-harmonic oscillator, the energy level separation is not constant and the mean interatomic distance depends on the energy level.

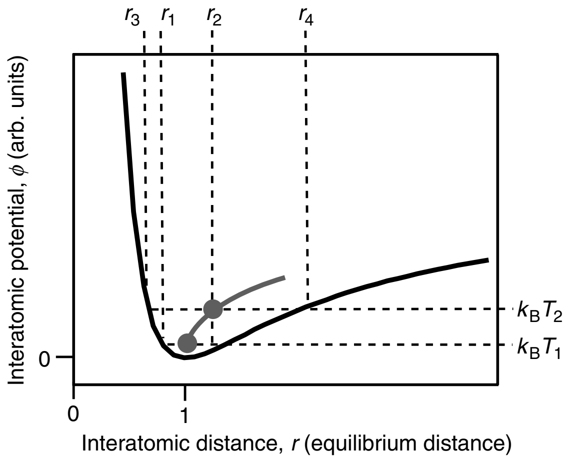

Figure 4.12 Classical picture for the thermal expansion of a solid. The interatomic potential Φ is shown as a function of interatomic distance. The gray line marks the temperature-dependent mean interatomic distance.

4.5
Allotropic Phase Transitions and Melting

In our description of crystal structures, we have merely argued that the optimal structure should be that with the strongest possible binding, that is, with the lowest total energy. This is only true at zero temperature. For higher temperatures, entropy effects have to be taken into account. Here, we discuss two types of structural changes at higher temperatures: allotropic phase transitions in which a crystal structure is transformed to another crystal structure and melting.

We start with a thermodynamic picture. In most cases, we want to study the structure at a given temperature, pressure, and particle number. This means that we have to minimize the Gibbs free energy:

$$G = U + PV - TS. \tag{4.51}$$

The situation for two competing phases is illustrated in Figure 4.13. For low temperatures, the A phase has the lower Gibbs free energy but at high-temperatures phase B has. There must be a phase transition between the two structures at a transition temperature T_C. This is the idea behind the so-called allotropic phase transitions from one crystal structure to another. As an example, iron crystallizes in a bcc structure at low temperatures but transforms into a fcc structure at 1185 K and again into a bcc structure at 1667 K.

Sometimes structures with a high Gibbs free energy exist even though alternative structures with a lower G are possible under the same conditions. An example is diamond that has a higher Gibbs free energy than graphite but there is an activation barrier for the transformation, which is quite high. Therefore, diamond is metastable.

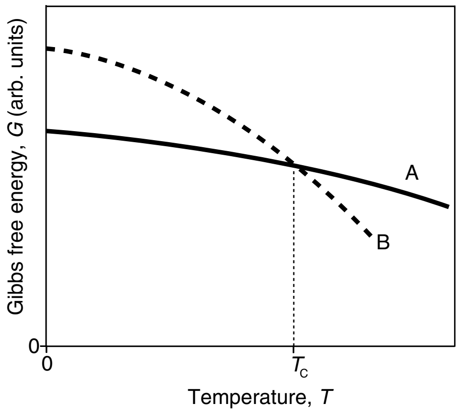

Figure 4.13 Gibbs free energy for two competing phases, A and B. At the temperature T_C a phase transition occurs.

The melting of a crystal can be described in the same picture, only that the B phase is taken to be the liquid. If we want to predict the melting temperature of a solid, we therefore have to consider the energy and entropy of both the liquid and the solid phase as a function of temperature, a formidable task.

There have also been attempts for a more simplistic prediction of the melting temperature T_m, which neglects entropy effects. First of all, one might suspect that T_m is related to the cohesive energy of a solid. This is indeed the case, as shown in Figure 4.14a. We can easily understand the trends in the figure: the noble gas crystals are merely bonded by the van der Waals interaction. This results in a low cohesive energy and melting temperature. On the other extreme, we have the refractory transition metals such W or Mo for which covalent bonding is important, as well as covalent materials such as Si. Simple metals like the alkali metals are found in the middle of the range.

For the prediction of the melting temperature of a solid, the relation to the cohesive energy is not very useful because the cohesive energy has to be known in the first place. An alternative idea was developed by F. Lindemann in 1910. He suggested that melting would occur when the amplitude of the interatomic vibration becomes too large, that is, when it reaches its certain fraction of the interatomic spacing. Using (4.5), we try this idea, stating that the solid melts once x_{max} reaches 5% of the interatomic distance a, that is,

$$T_m = \frac{(0.05a)^2\gamma}{2k_B} = \frac{(0.05a)^2\omega^2 M}{2k_B}. \tag{4.52}$$

For ω, we can use the Debye frequency ω_D. If we now write the above expression in terms of the Debye temperature, we get

$$T_m = \frac{(0.05a)^2\Theta_D^2 k_B M}{2\hbar^2}. \tag{4.53}$$

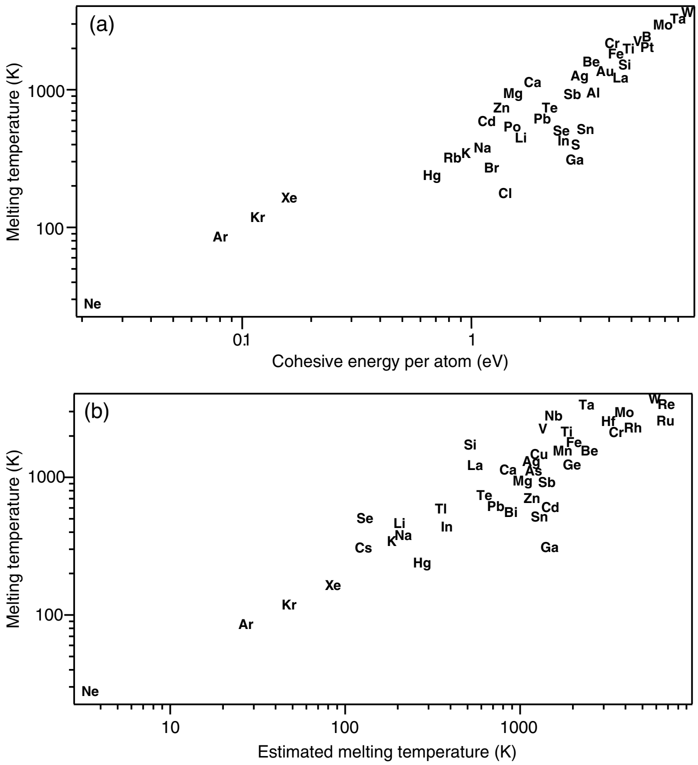

Figure 4.14 (a) Melting temperature as a function of cohesive energy for the elements. (b) Melting temperature as a function of the estimated melting temperature from the Lindemann criterion (4.53).

The result of this is shown in Figure 4.14b. With the given choice of 5% of the interatomic distance as a melting criterion, the simple model reproduces the trend correctly even though it neglects entropy effects as well as the influence of the detailed atomic structure.

4.6
Discussion and Problems

4.6.1
6.1. Discussion

1. What does the phonon dispersion, that is, the vibrational frequency as a function of wave number $\omega(k)$, look like for an infinite chain of atoms with one atom per unit cell?
2. What does it look like for two atoms per unit cell? Why does one speak of optical and acoustic branches?

3. The phonon dispersion for a one-dimensional chain with two atoms per unit cell is given by (4.10) and (4.16), respectively. What about the amplitude of the vibrations?
4. Why is the movement of the atoms in the chain the same when multiples of $2\pi/a$ are added to, or subtracted from, the wave number?
5. Explain the meaning and use of periodic boundary conditions to describe the properties of finite solids.
6. What predictions does the so-called Dulong–Petit law make about the heat capacity of a solid and its temperature dependence (and why)? How does it compare to the experiment?
7. At room temperature, do metals have a noticeably higher heat capacity than insulators because of the mobile electrons?
8. Explain the Einstein model for the heat capacity of a lattice. How do its predictions compare to the experiment?
9. Explain the Debye model for the heat capacity of a lattice. In which respect does it work better than the Einstein model and why?
10. What is the definition of the Debye temperature (or frequency) and what are the typical values?
11. Which one has a higher Debye temperature, lead or diamond, and why?
12. The thermal conductivity of an insulator has a maximum at about 10% of the Debye temperature. Why does it decrease for lower temperatures? Why does it decrease for higher temperatures?
13. Explain in a microscopic model why a solid undergoes thermal expansion.
14. Thermal expansion is a so-called anharmonic process. Why is it called like this?

4.6.2 Problems

1. *One-dimensional chain with one atom per unit cell:* We have determined the phonon dispersion relation for an infinite chain of atoms with lattice spacing a and one atom per unit cell (mass M). The result is (4.10). For longitudinal vibrations, sketch how the atoms would move for k close to 0 ($k \ll \pi/a$) and for k at the Brillouin zone boundary ($k = \pi/a$).

2. *One-dimensional chain with two atoms per unit cell:* For two atoms per unit cell of length b, we get two branches in the dispersion, the acoustic and the optical branch. The solutions are given by (4.16). (a) Plot these solutions inside the first Brillouin zone for $M_2 = 0.2M_1$, $M_2 = 0.9M_1$, and $M_2 = M_1$. (b) Consider the case where $M_1 \neq M_2$. For longitudinal vibrations, sketch how the atoms would move for k close to 0 and for k at the Brillouin zone boundary ($k = \pi/b$). (c) Which movement corresponds to which solution of (4.16)? (d) Explain what happens in the case of $M_2 = M_1$ discussed in (a).

3. *Optical phonons:* Optical phonons are called "optical" because they can in some cases couple to electromagnetic radiation. This can, for example, happen in a one-

dimensional chain with two ions of opposite charge per unit cell. A vibration of these two ions "against" each other would correspond to a changing dipole and could be excited by the changing electric field of an electromagnetic wave. Using energy and momentum conservation, show that such an excitation can only work for the optical phonon branch.

4. *Periodic boundary conditions:* Periodic boundary conditions lead to a restriction of the possible k values. Consider a linear chain of copper atoms. The length of the chain should be 1 cm, the lattice spacing should be 0.36 nm and the force constant 50 Nm^{-1}. Calculate the smallest possible finite wave number k. What would be the vibrational angular frequency for this wave number? What would be the corresponding energy in electron volts?

5. *The Debye model:* The phonon dispersion for a one-dimensional chain of atoms is given by (4.10) and shown in Figure 4.1. What would the dispersion look like in the Debye model?

6. *Atomic force constants and Debye temperature:* (a) Estimate the value of the force constant $\boldsymbol{\gamma}$ and the angular frequency $\boldsymbol{\omega}$ for the vibrations of the atoms in copper. Use that Young's modulus $Y = 130\,\mathrm{GPa}$ and that the cubic lattice constant is 0.36 nm. (b) Estimate the angular frequency for the vibrations that correspond to the Debye temperature of 343 K and compare to the result in (a). (c) Having obtained $\boldsymbol{\gamma}$, estimate the amplitude of the vibrations as a fraction of the lattice spacing at room temperature.

7. (*) *Heat capacity of layered solids:* Imagine a layered solid with a strong intralayer interaction and a very weak interlayer interaction, such as graphite. For graphite, the observed heat capacity between 15 and 50 K is not proportional to T^3 but to T^2. Show that you indeed get a T^2 temperature dependence if you regard graphite as a purely two-dimensional solid, neglecting any interaction between the sp^2 bonded sheets.

8. *Thermal expansion:* Explain why the coefficient of thermal expansion for a solid vanishes at $T = 0$.

9. *Thermal expansion:* In a Bragg reflection experiment using copper, a sharp peak is observed at an angle of 25.23° at 300 K. At 500 K, the same peak is observed at an angle of 25.14°. Use this information to calculate the coefficient of linear expansion for copper.

5
Electronic Properties of Metals: Classical Approach

Here and in Chapters 6, 7, and 9, we are concerned with the electrical properties of metals, semiconductors, and insulators. It would be natural to start out with a definition of the difference between these types of materials, but this is actually not so easy and we need to postpone it. Consider a couple of simple possibilities: One could argue that metals are good conductors of heat and electricity whereas semiconductors and insulators are not. In the case of heat conduction, we have already seen that diamond, which is an insulator, conducts even better than most metals. Electrical conductivity is also of little help: Some semiconductors such as silicon conduct electricity reasonably well. Yet another possibility is to define metals by the fact that they look "shiny" or "metallic." But this applies to some semiconductors, too, and again silicon can be taken as an example. It turns out that a proper definition has to wait until we treat electronic states in a quantum mechanical model in the next chapter.

5.1
Basic Assumptions of the Drude Model

In 1900, only 3 years after the discovery of the electron by J. J. Thomson, P. Drude suggested a simple model to explain many of the observed properties of metals. He did this by combining the existence of electrons as charge carriers with the highly successful kinetic gas theory. We will later see that the Drude model has many shortcomings, but it is still of fundamental importance for many of the concepts associated with electrical conductivity.

The model is based on the following assumptions:

- The electrons in a solid behave like a classical gas. They do not interact with each other at all: There is no Coulomb interaction and, as opposed to a classical gas model, they do not collide with each other either. This is known as the independent electron approximation. We will later see that this approximation is quite a good one: the electrons do indeed not interact much with each other.
- The positive charge is located on immobile ion cores. The electrons can collide with the ion cores. These collisions instantaneously change their velocity. However, in

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between collisions, the electrons do not interact with the ions either. This is known as the free electron approximation. We will see that this approximation is not very good. Indeed, the whole picture of the electrons colliding with the ions is problematic. In a perfect crystalline solid at low temperatures, the electrons do not collide with the ions at all, as we shall see later.

- The electrons reach thermal equilibrium with the lattice through the collisions. This means that their mean kinetic energy is

$$\frac{1}{2}m_e v_t^2 = \frac{3}{2}k_B T. \tag{5.1}$$

At room temperature, this results in an average speed of $v_t \approx 10^5\,\mathrm{ms^{-1}}$.

- One assumes that the probability per time for the electrons to undergo a scattering event with the ions is $1/\tau$. τ is called the relaxation time and plays a fundamental role in the theory. With the relaxation time τ, we can also define the mean free path between two scattering events as $\lambda = \tau v_t$. If the origin of the scattering is the collisions with the ions, one would assume that $\lambda \approx 1\,\mathrm{nm}$ so that $\tau \approx 1 \times 10^{-14}\,\mathrm{s}$.

For the description of almost all properties within the Drude model, it is necessary to determine how many electrons contribute to a process, for example, the electrical conductance. The relevant quantity is the conduction electron density n, that is, the number of conduction electrons per volume. n is calculated by assuming that every atom contributes Z_V conduction electrons, that is, electrons from its outermost shell, to the metallic bonding. The remaining electrons remain bound to the metal ions. For the alkali metals, Z_V would be 1, for the alkaline earth metals, it would be 2, and so on. There are ρ_m/A atoms per cubic meter, where ρ_m is the density of the solid in $\mathrm{kg\,m^{-3}}$ and A is the atomic mass in kilograms per atom. Consequently, the conduction electron density n is $Z_V\rho_m/A$. Values of n for selected metals are given in Table 5.1.

Table 5.1 Number of conduction electrons per atom, calculated conduction electron density n, and Hall coefficient of selected metals.

Metal	Z_V	n ($10^{28}\,\mathrm{m^{-3}}$)	Measured R_H in units of $-1/ne$
Li	1	4.7	0.8
Na	1	2.7	1.2
K	1	1.3	1.1
Rb	1	1.2	1.0
Cs	1	0.9	0.9
Cu	1	8.5	1.5
Ag	1	5.9	1.3
Be	2	24.7	−0.2
Mg	2	8.6	−0.4
Al	3	18.1	−0.3
Bi	5	14.1	≈40 000

5.2 Results from the Drude Model

Based on Drude's assumptions, we now explain several properties of metals. In the next section, we will discuss the limitations and problems arising in the Drude model.

5.2.1 DC Electrical Conductivity

Consider the behavior of the electrons when a static electric field E is applied. This will lead to a drift of the electrons in the electric field, which is superimposed on the thermal motion. The equation of motion for the drift is [1]

$$\frac{d\mathbf{v}}{dt} m_e = -e\mathbf{E}, \tag{5.2}$$

with the usual solution

$$\mathbf{v}(t) = \frac{-e\mathbf{E}t}{m_e}. \tag{5.3}$$

If we assume that the drift motion is destroyed in a collision with the ions and that on average the time since the last collision is τ, the average drift speed in the direction opposing the field is

$$\bar{\mathbf{v}} = \frac{-e\mathbf{E}\tau}{m_e}. \tag{5.4}$$

This is very slow compared to the thermal movement of the electrons.[2] For a typical field strength of $E \approx 10\,\mathrm{Vm}^{-1}$, we get a drift velocity of $|\bar{\mathbf{v}}| = 10^{-2}\,\mathrm{ms}^{-1}$.

Having the drift velocity, we can calculate the conductivity. Consider an area A perpendicular to the electric field. The number of electrons passing through the area per unit time is

$$n|\bar{\mathbf{v}}|A. \tag{5.5}$$

The amount of charge passing through the area is therefore

$$-en|\bar{\mathbf{v}}|A. \tag{5.6}$$

With this we can calculate the current density

$$\mathbf{j} = n\bar{\mathbf{v}}(-e), \tag{5.7}$$

1) Note that the charge of the electron is $-e$ throughout this book.

2) It is not obvious that (5.4) is the correct result for the average drift speed. One could be tempted to think that it is too high by a factor of 2. See Ashcroft and Mermin for a more detailed discussion of this.

and with the use of (5.4) we get

$$\mathbf{j} = \frac{ne^2\tau}{m_e}\mathbf{E} = \sigma\mathbf{E} = \frac{\mathbf{E}}{\rho}, \tag{5.8}$$

that is, the current density is proportional to the electric field strength. This result is the familiar Ohm's law and the constant of proportionality σ is called the conductivity of the material and its inverse ρ is called the resistivity.

Consider now the explicit expressions for the conductivity and resistivity that we have obtained. The conductivity is

$$\sigma = \frac{ne^2\tau}{m_e}, \tag{5.9}$$

and the resistivity

$$\rho = \frac{m_e}{ne^2\tau}. \tag{5.10}$$

Note that the elementary charge appears squared in these equations. The reason for this is that it is needed both to couple to the electric field, dragging the electrons along, and in the definition of the current. The fact that it is squared means that we would get the same result for carriers with a charge e instead of $-e$. We will come back to this when discussing semiconductors where positively charged carriers do in fact appear.

Another useful definition is the mobility of the carriers μ. It is given by

$$\mu = \frac{e\tau}{m_e}, \tag{5.11}$$

and the conductivity and resistivity can, of course, also be defined using this mobility:

$$\sigma = n\mu e, \qquad \rho = \frac{1}{n\mu e}. \tag{5.12}$$

Why do we need this definition? The concept of a carrier mobility can be useful for solids in which the carrier concentration can be changed by some external parameter without changing the scattering mechanism inside the solid, that is, without changing the relaxation time.

We can now compare the results to experimental data. Figure 5.1 shows the measured and calculated electrical conductivities of some selected metals at two different temperatures. The calculations have been made assuming a mean free path of $\lambda = 1$ nm for all elements, a distance in the order of the typical ion separation. This appears to be a justified value for the mean free path, given the dense packing of the ions in metals. For 273 K, the calculation appears to reproduce the right order of magnitude and the solid line lies in the middle of the scattered experimental data. Some elements lie far away from the calculation, notably the noble metals and the group V semimetals Sb and Bi. One could be tempted to conclude that the Drude model does not reproduce the details, but the general trend is correct anyway. For lower temperatures, the situation becomes more problematic. At 77 K, the calculated conductivity increases because v_t gets smaller, but the measured conductivity increases much more. At even lower temperature, the comparison becomes increasingly unfavorable .

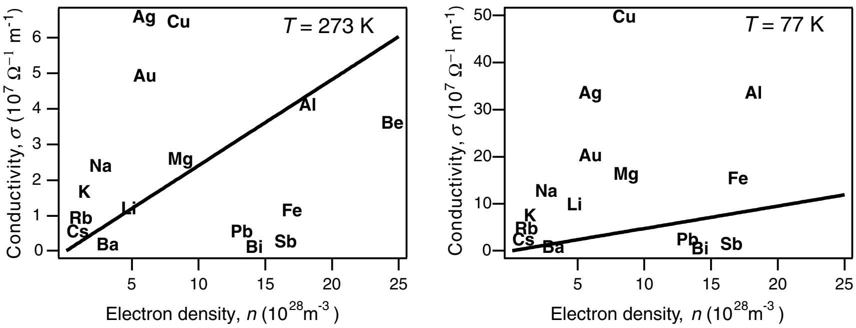

Figure 5.1 Measured and calculated electrical conductivities of metals as a function of conduction electron density for two different temperatures. The measured data are marked by the elements' names; the calculations are the solid lines.

5.2.2
Hall Effect

Another reasonably successful result of the Drude model is that it can explain the Hall effect. The effect was discovered by E. Hall in 1879 when he investigated the effect of a magnetic field on the current in a conductor. Hall found that an electric field E_{H} is built up, which is perpendicular to both the magnetic field and the current (see Figure 5.2a). The magnitude of the Hall field is proportional to both current density and magnetic field:

$$E_{\mathrm{H}} = R_{\mathrm{H}} j_x B_z, \tag{5.13}$$

where R_{H} is called the Hall coefficient. This is explained quite easily in the steady state (see Figure 5.2b). For the electrons to pass through the wire, the Hall field E_{H} must

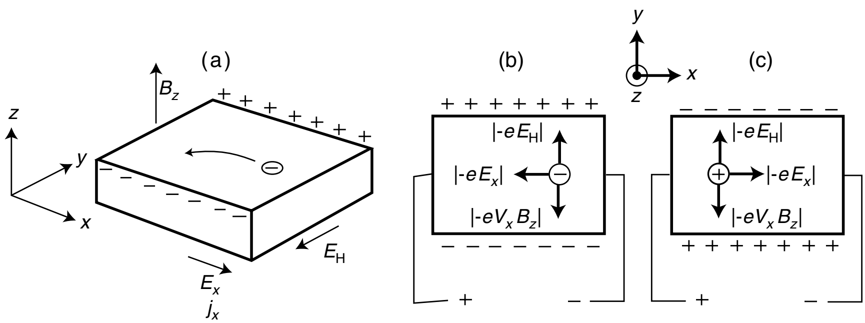

Figure 5.2 (a) Illustration of the Hall effect. (b) Equilibrium between Lorentz force and force caused by the Hall field for electrons passing through the sample. (c) The same for positively charged carriers (charge +e).

exactly compensate the Lorentz force in the opposite direction; thus,

$$-eE_{\mathrm{H}} = -eB_z v_x. \tag{5.14}$$

Now we get

$$R_{\mathrm{H}} = \frac{E_{\mathrm{H}}}{j_x B_z} = \frac{E_{\mathrm{H}}}{(-e)nv_x B_z}, \tag{5.15}$$

and finally

$$R_{\mathrm{H}} = \frac{v_x B_z}{-env_x B_z} = \frac{-1}{ne}. \tag{5.16}$$

Therefore, measuring the Hall coefficient provides direct experimental access to the electron density. In addition to the conduction electron density n, Table 5.1 gives the measured Hall coefficients, normalized by the expected value $-1/ne$.

The agreement between the measured R_{H} values and the ones calculated from the conduction electron density is rather good for the alkali metals and acceptable for the noble metals. It is terrible for Bi. The very high value means that for some reason the true conduction electron density must be much smaller than the calculated value. In a sense, the agreement is even worse for Be, Mg, and Al because not only the magnitude does not quite fit, the measured R_{H} is even positive, not negative. In this context, it is interesting to note again that the sign of the carriers is irrelevant for the conductivity of the sample. We would have obtained the same result as (5.9) if electrons were positively charged.

For the Hall coefficient, however, the sign does matter and it appears therefore that the current in Be, Mg, and Al is carried by positive charges: Imagine that we had positive carriers with a carrier density p (see Figure 5.2b). Then it is easy to show that

$$R_{\mathrm{H}} = \frac{1}{pe}, \tag{5.17}$$

that is, we get a positive R_{H}. The notion of positive carriers does not make any sense in the Drude model, but we will see that the quantum model of the electronic states is able to give an intuitive picture of positive carriers. This will be particularly useful for treating semiconductors and we come back to the positive Hall coefficient in metals in Chapter 7.

5.2.3 Optical Reflectivity of Metals

The Drude model can also explain why metals reflect light and therefore appear shiny. Before we explain this, we briefly state some fundamental relations from optics. We shall need these equations again when we discuss the optical properties of insulators in Chapter 9. Some of the concepts used here are explained in more detail in the beginning of Chapter 9. The Appendix gives you a reminder about the Maxwell equations in matter.

Light can be described as transverse, plane electromagnetic waves. We can write the electric field for a wave along the z direction as

$$\mathbf{E}(z,t) = \mathbf{E}_0 \mathbf{e}^{i(kz-\omega t)}, \tag{5.18}$$

with the wave number

$$k = \frac{2\pi N}{\lambda_0}, \tag{5.19}$$

where λ_0 is the wavelength in vacuum and

$$N = n + i\kappa \tag{5.20}$$

is the complex index of refraction. n, the real part of N, is needed to describe the refraction of the wave, and the imaginary part κ accounts for the damping inside the material.[3)]

An alternative way to describe the optical properties of materials is to use the complex dielectric function ε instead of the refractive index N. You may be familiar with the static dielectric constant ε that appears in the description of capacitors. The dielectric function is the same quantity, but it describes not only static properties but frequency dependence as well, that is, in general ε depends on ω and so, of course, does N. A familiar example of this frequency dependence is the phenomenon of dispersion that separates light of different colors when refracted by a glass prism. ε is related to N via

$$N = \sqrt{\varepsilon} = \sqrt{\varepsilon_r + i\varepsilon_i}. \tag{5.21}$$

With this (5.18) can be written as

$$\mathbf{E}(z,t) = \mathbf{E}_0 \mathrm{e}^{i((2\pi N/\lambda_0)z-\omega t)} = \mathbf{E}_0 \mathrm{e}^{i((\omega\sqrt{\varepsilon}/c)z-\omega t)}. \tag{5.22}$$

In the last step, we have used that $\lambda_0\omega/2\pi = c$, with c being the speed of light in vacuum.

Having these basic equations, we can proceed to explain the reflectivity of metals. Consider an electron in the electromagnetic AC field given by the optical light wave. If the angular frequency ω of the light is very small, we basically retain the DC behavior. If, on the other hand, ω is so high that $1/\omega$ is much shorter than the relaxation time τ, we can ignore the collisions with the ions altogether and treat the electrons as completely free. As τ is in the order of 10^{-14} s, this condition is fulfilled for optical frequencies.

3) This can be seen by inserting $N = n + i\kappa$ into (5.19) and (5.18), where it leads to a damping factor $\exp -2\pi\kappa z/\lambda_0$, that is, a damping for increasing z. One frequently finds an alternative definition of the refractive index as $N = n - i\kappa$. This corresponds to a wave that is damped in the direction of decreasing z.

We treat one single electron in the electric field of an electromagnetic wave. The polarization should be such that the **E** field lies in the x direction. The electron will move according to the equation of motion

$$m_{\mathrm{e}}\frac{\mathrm{d}^2x(t)}{\mathrm{d}t^2} = -eE(t). \tag{5.23}$$

The time dependence of the electric field is $E(t) = E\mathrm{e}^{-i\omega t}$, and as there are no other forces acting on the electron, a good ansatz for the solution of (5.23) appears to be $x(t) = x\mathrm{e}^{-i\omega t}$. With this we get

$$x = \frac{eE}{m_{\mathrm{e}}\omega^2}. \tag{5.24}$$

The dipole moment for one electron caused by this is $-ex$. For a solid with a conduction electron density of n, the polarization (the dipole density) is given by

$$P = -nex = -\frac{ne^2E}{m_{\mathrm{e}}\omega^2}. \tag{5.25}$$

On the other hand, we know the general relation between the E field and the dielectric displacement field D, which is

$$D = \varepsilon\varepsilon_0 E = \varepsilon_0 E + P, \tag{5.26}$$

such that

$$\varepsilon = 1 + \frac{P}{\varepsilon_0 E}. \tag{5.27}$$

Using this and our result for the polarization P (5.25), we obtain an expression for the dielectric function ε:

$$\varepsilon = 1 - \frac{ne^2}{\varepsilon_0 m_{\mathrm{e}}\omega^2} = 1 - \frac{\omega_{\mathrm{P}}^2}{\omega^2} \tag{5.28}$$

with the so-called plasma frequency ω_{P} given by

$$\omega_{\mathrm{P}}^2 = \frac{ne^2}{m_{\mathrm{e}}\varepsilon_0}. \tag{5.29}$$

This expression for the dielectric function is our final result. Why does this explain that metals reflect visible light? To see this consider the last term in (5.22). We have to distinguish between two cases: for $\omega < \omega_{\mathrm{p}}$, ε is a real and negative number. Therefore, $\sqrt{\varepsilon}$ is purely imaginary, the complex index of refraction (5.20) contains only the imaginary component $i\kappa$, and (5.22) represents an exponentially damped penetration of the wave into the solid. The damping cannot be due to inelastic processes because our (5.23) does not take such processes into account. Therefore, energy is conserved and since the lightis not transmitted into the solid, it must be reflected from it (for a

Table 5.2 Observed values of the plasma energy $\hbar\omega_P$ together with the values calculated from the Drude model.

Metal	Measured $\hbar\omega_P$(eV)	Calculated $\hbar\omega_P$(eV)
Li	6.2	8.3
K	3.7	4.3
Mg	10.6	10.9
Al	15.3	15.8

more formal treatment see Problem 9.4). For $\omega > \omega_p$, on the other hand, ε is real and positive and (5.22) represents a light wave that propagates into the metal. The bottom line is that metals are reflecting low-frequency light, but they become transparent for high-frequency light. The transition happens at the plasma frequency. The low-frequency behavior is not surprising because it should essentially be the same as in the electrostatic case for which it is assumed that metals are free of electric fields.

The plasma frequency can be calculated solely from the conduction electron density of the metal. Instead of the plasma frequency ω_P, one commonly uses the plasma energy $\hbar\omega_P$. Calculated and measured values are given in Table 5.2. We see that for most metals the plasma frequency lies in the far ultraviolet region, that is, most metals are reflecting visible light.

5.2.4 The Wiedemann–Franz Law

In Drude's time, one of the most convincing pieces of evidence for his theory appeared to be that it (apparently) yielded a quantitatively correct description of the Wiedemann–Franz law. In 1853, G. H. Wiedemann and R. Franz found that the ratio of thermal and electrical conductivity is constant for all metals at a given temperature. Later, it was found by L. Lorenz that this constant is proportional to the temperature; thus,

$$\frac{\kappa}{\sigma} = LT, \tag{5.30}$$

where L is the so-called Lorenz number.

In the Drude model, the ratio of thermal and electrical conductivity is readily calculated. The thermal conductivity is that of a classical gas and can be described by an equation similar to (4.49), only using corresponding properties (heat capacity, speed, and mean free path) for the electron gas. The electrical conductivity is given by (5.9). The result is

$$\frac{\kappa}{\sigma} = \frac{3}{2}\frac{k_B^2}{e^2}T = LT, \tag{5.31}$$

which is just the Wiedemann–Franz law with the only problem being that L is too small by about a factor of 2. Drude, however, had made a mistake of a factor of 2 in his calculation such that L came out almost correct. Therefore, his theory was in impressive quantitative agreement with the experimental data. It should be noticed that Drude's mistake was rather subtle (it had to do with the scattering probabilities of the electrons) and therefore not readily discovered by other researchers.

5.3
Shortcomings of the Drude Model

Despite its great success, the Drude model has a number of serious shortcomings. We discuss several of them here to motivate the quantum treatment of metals in the next chapter. Even before starting our work on the Drude model, several assumptions could have raised suspicion. Take for example the nature of the scattering. There is no justification for a missing electrostatic interaction between the electrons and the lattice, and it is also not clear why the electrons collide only with lattice ions and not among themselves. In addition to this, the de Broglie wavelength for thermal electrons is in the order of nanometers. The criterion for treating the electrons as classical particles, however, is that their de Broglie wavelength is much smaller than the typical dimensions of the structures they are moving in. This is clearly not the case.

As for a comparison to experimental data, we have already seen that the predicted conductivity at low temperatures is not high enough. In our model, we have calculated the relaxation time in the Drude model for a fixed mean free path and a temperature-dependent thermal velocity. In this way, the relaxation time increases for lower temperatures. The problem in the calculation is the assumption of a fixed mean free path, given by the atomic spacing. This turns out to be completely wrong. In fact, at low temperatures, the mean free path of electrons in very pure samples can become macroscopic, micrometers or even millimeters. Apparently, the electrons manage to sneak past all other electrons and all ions as well. This appears quite mysterious, but we will be able to explain it in the next chapter.

Another problem related to the conductivity is the conductivity of alloys. Alloying a small amount of impurities into an otherwise pure solid can drastically reduce the conductivity. In the framework of the Drude model, it is not at all clear why.

The historically most important issue associated with the classical treatment of electrons in a metal is that the electrons do not seem to contribute to the specific heat at room temperatures. We have seen in the previous chapter that the heat capacity for most solids, including metals, is given by the Dulong–Petit rule at room temperature (see Table 4.2). This is in sharp contrast to the classical expectation. For 1 mol of monovalent classical metal, the heat capacity of the lattice would be $3N_Ak_B = 3R$. Each electron has three translational degrees of freedom contributing with $k_B/2$ to the heat capacity. For 1 mol of electrons, the heat capacity would be $3R/2$, giving a total heat capacity of $9R/2$, which is not the observed Dulong–Petit value. For metals with more conduction electrons per atom, the situation would get even worse. The fact that the Dulong–Petit value is observed for many metals suggests that the electrons do not contribute to the heat capacity, a puzzle that can only be understood by a quantum mechanical treatment of the electrons.

5.4 Discussion and Problems

5.4.1 Discussion

1. Describe the basic assumption of Drude's model for metals. Explain the relaxation time and the mean free path of the electrons.
2. How fast do the electrons move in Drude's model and how does their speed depend on the temperature?
3. Describe the electrical conduction in Drude's model. Where does the electrical resistance come from?
4. How does the measured voltage drop along a metal wire as a function of current through the wire qualitatively compare to the prediction of Drude's model? How is the quantitative agreement?
5. (*) List cases in which Ohm's law is not valid.
6. When an electric field is applied, how does the additional speed of the electrons compare to their thermal speed (at room temperature)?
7. What is the Hall effect and what can be measured by it?
8. Explain qualitatively why the sign of the Hall coefficient depends on the charge of the carriers (positive or negative) and why it is inversely proportional to the carrier density.
9. Why do metals not transmit light? Is this so for all light frequencies?
10. What is the Wiedemann–Franz law?
11. Which properties of metals are not described adequately by Drude's model?
12. With some knowledge in quantum mechanics, why would you expect Drude's model not to work?

5.4.2 Problems

1. *Classical versus quantum description of metals*: Calculate the classical mean kinetic energy for the electrons in Na at room temperature. From this, determine their de Broglie wavelength λ. For a classical description to be valid, we have to require that λ is much smaller than the mean separation d of the particles. Show that this is not the case.

2. *Ohm's law*: Explain how expression (5.8) is related to the more familiar form of Ohm's law $I = U/R$.

3. *Optical reflectivity of metals*: (a) Suggest a way of measuring the plasma energy in a metal. (b) Suppose that you want to develop a metallic thin-film coating for

windows such that they would transmit visible light but not infrared radiation. What properties would you require the coating material to have?

4. *Optical reflectivity of metals*: Estimate the penetration depth of visible light into aluminum.

5. *Wiedemann–Franz law*: Show that the Wiedemann–Franz coefficient L in the Drude model is indeed given by (5.31), that is,

$$L = \frac{\kappa}{\sigma T} = \frac{3}{2}\frac{k_B^2}{e^2}. \tag{5.32}$$

6. *Resistivity*: We have seen that the Drude model gives the correct order of magnitude for the resistivity of many metals near room temperature but what about the temperature dependence of the resistivity? Experimentally, it is found that this temperature dependence is linear near room temperature, that is,

$$\rho(T) = \rho_0(1 + \alpha(T - T_0)), \tag{5.33}$$

where $\rho(T)$ is the temperature-dependent resistivity, ρ_0 the resistivity at room temperature T_0, and α the so-called thermal resistance coefficient. Show that this experimental finding is in qualitative disagreement with the Drude model, that is, that the Drude model does not give rise to a linear temperature dependence.

7. *Phonons in metals*: The Drude model can be used to estimate the force constants, vibrational frequencies, and related properties in metals. We use a crude model to describe the vibration of a single ion in a monovalent metal: The ion is assumed to be a positive point charge in a spherical unit cell that is filled with electrons of the appropriate density n. All the rest of the crystal is ignored. (a) Consider a small displacement of the ion from the center of the unit cell. Show that the restoring force is proportional to the magnitude of the displacement. *Hint*: Use Gauss' law to calculate the force. (b) What is the vibrational frequency for a single ion of sodium? (c) Perform a crude estimate of the velocity of sound in sodium and compare your result to the experimental value of $3200\,\mathrm{m\,s^{-1}}$.

6
Electronic Properties of Metals: Quantum Mechanical Approach

In the previous chapter, we have discussed the Drude model in which the electrons in a metal are treated as classical, free, and independent particles. We have seen the success and the limitations of this approach. Now we go one step further. We take the quantum mechanical nature of the problem into account and we will see how this fixes many of the shortcomings of the classical model. We will also see that, going beyond the assumption of free electrons, it is possible to explain the fundamental difference between metals and non-metals. Indeed, we will come up with a more formal definition of what a metal is in the first place. We will, however, retain the approximation that the electrons move independently from each other. This works surprisingly well for many solids, and we will try to understand why.

Finding the quantum mechanical eigenstates of all the electrons in the solid is a formidable problem: We would have to construct a wave function that depends on the coordinates of all the electrons and also of all the ions, which make up the positive part of the potential. This is clearly hopeless! The first approximation we make is to ignore the motion of the ions by “freezing” them into their equilibrium position. We know that there are thermal vibrations and that this approximation appears poorly justified. However, it turns out to work rather well. The reason is the mass difference between the electrons and the ions. Suppose that the ions are in some given position and the electrons are in their ground state. When the ions move out of position, their motion is so slow that the fast electrons will be able to readjust their distribution such that they stay in a modified ground state, but still in the ground state. When the ions move back, the electrons adjust themselves to the old ground state. Therefore, the electronic and ionic motions can be effectively separated. This is called the Born–Oppenheimer approximation and it is also often used for treating molecules.

The other fundamental simplification we make is that we do not consider the correlated motion of the electrons. We merely calculate the electronic states for one electron that is moving in an effective potential $U(\mathbf{r})$, given by all the ions and all the other electrons. The stationary Schrödinger equation for the one-electron states then becomes

$$-\frac{\hbar^2\nabla^2}{2m_e}\psi(\mathbf{r}) + U(\mathbf{r})\psi(\mathbf{r}) = E\psi(\mathbf{r}). \tag{6.1}$$

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Admittedly, the whole problem has now been shifted into finding the right $U(\mathbf{r})$! Amazingly, we will find that for many metals $U(\mathbf{r})$ is quite small, that is, the electrons behave as if they were nearly free.

Finally, when we have found the eigenstates in the one-electron picture, we fill them with all the electrons according to the Pauli principle. This gives the correct occupation of the states but only for zero temperature. At higher temperatures, the statistical occupation of the states is given by the Fermi–Dirac distribution.

One great help for finding the electronic energy levels is the symmetry of the lattice. No matter how complicated $U(\mathbf{r})$ is, at least we know that it must be lattice periodic, that is,

$$U(\mathbf{r}) = U(\mathbf{r}+\mathbf{R}), \tag{6.2}$$

where $\mathbf{R}$ can be any vector of the Bravais lattice.

Before we start with a detailed quantum mechanical description of metals along these lines, we consider very simple models for the electronic states in solids. We motivate the idea of electronic energy bands and we give an informal but intuitive picture of the difference between metals, semiconductors, and insulators.

6.1 The Idea of Energy Bands

Here, we consider the solid as a type of giant molecule and ask about the possible energy levels in such a molecule. This approach is intuitive and would give the correct results but it is not very practical for treating crystals, in particular because their high degree of symmetry is not exploited. For simplicity, we build the molecule from Na atoms that have only one valence electron. Figure 6.1 illustrates what happens as we build an ever bigger cluster of Na atoms. For two atoms, the situation is similar to that of the hydrogen molecule in Figure 1.2: As the atoms get close to each other, bonding and antibonding molecular orbitals are formed.[1] Each Na atom has one 3s electron. These two electrons are accommodated in the bonding orbital. They have opposite spins in order to fulfill the Pauli principle's requirements (Figure 6.1a). Figure 6.1b illustrates the position of the energy levels as a function of the interatomic separation. The energy zero is chosen such that it corresponds to a very large separation of the atoms. For the separation a, the bonding level reaches the lowest energy. Since only the bonding level is occupied by two electrons, the energy of the antibonding state is irrelevant and the energy gain is maximized for the separation a.

What happens if we take more than two atoms? The interaction of two atomic states leads to the formation of two levels that are delocalized over the entire molecule. The

1) There is a conceptual difference between our treatment of the hydrogen molecule and the treatment of a Na cluster here. In the hydrogen molecule, we have calculated the energy levels for a genuine two-electron wave function. In the present chapter, we stick to the one-electron approximation, that is, we always calculate the electronic states of one electron in an effective potential of all the others. Then we fill these one-electron states according to the Pauli principle.

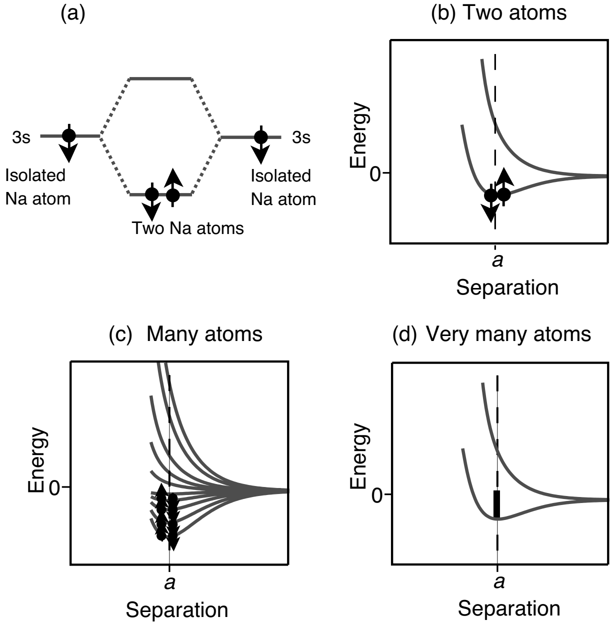

Figure 6.1 The formation of energy bands in solids. (a) Bonding and antibonding energy levels and their occupation for a molecule constructed from two Na atoms. The black dots and arrows symbolize the electrons with their spin. (b) The molecule's energy levels as a function of interatomic separation. (c) The energy levels for a cluster of many Na atoms as a function of their separation. (d) For very many energy levels there is a quasicontinuum between the lowest and highest energy levels. The band is half-filled with electrons as illustrated by the bar.

situation is similar for N atoms. The N atomic energy levels will split up into N non-degenerate molecular levels.[2] $N/2$ of these levels are then occupied by two electrons each. This is shown in Figure 6.1c. For very many atoms, the same principles apply. There is a quasicontinuum of states between the lowest and the highest level, which is half-filled (Figure 6.1d). This quasicontinuum is called a band.

Now we can qualitatively see why Na should show metallic behavior. The energy band of the valence electrons is exactly half-filled. When an electric field is applied to a sample of Na, the electrons experience a force opposite to the field direction. In order to move in that direction, they have to increase their kinetic energy by a bit, that is, they have to go into a state with slightly higher energy. For the electrons in the highest occupied states, this is easily possible because there are plenty of unoccupied states available at higher energies.

To see a quite different behavior, consider the formation of a band in Si, which is shown in Figure 6.2. The valence electrons involved in the bonding are the two 3s and

2) Some of these levels might be degenerate because of symmetry but this is not important for this qualitative discussion.

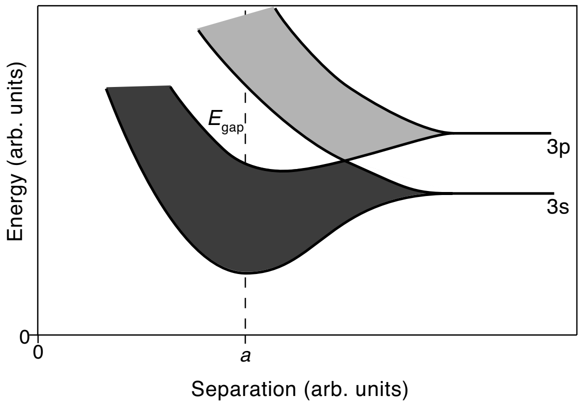

Figure 6.2 Band formation in Si. The lower band corresponds to the sp^3 states and is completely filled.

the two 3p electrons. As the Si atoms approach each other, the orbitals hybridize and form two bands of states at equilibrium distance a, each containing four states per Si atom, that is, a total of eight states per atom, derived from two and six atomic s and p states, respectively. The lower band consists of the sp^3 orbitals and these are fully occupied by the four valence electrons of each Si atom. The upper band is completely unoccupied and between the two bands there is an energy gap. This explains the insulating behavior of Si: When a voltage is applied, the conduction electrons cannot change their kinetic energy a little because there are no vacant states with energies slightly above the filled sp^3 band.

This picture allows us to group materials in two classes: metals and non-metals. Non-metals can be further divided into semiconductors and insulators (see Chapter 7). Unfortunately, the simple model lacks any predictive power. If we take carbon that has the same number of valence electrons as Si, we would come to the same picture for the bonding. This is correct for diamond that is sp^3 bonded and has a bandgap. But for graphite the situation is quite different. There are also bands but there is no bandgap.

6.2
Free Electron Model

6.2.1
The Quantum Mechanical Eigenstates

The free electron model is the quantum mechanical analogue to the Drude model. The electrons are free in the sense that they are not interacting with the ions or the other electrons. Treating free electrons in a quantum model comes down to the standard problem of a free particle in a box. The task is to solve (6.1) with $U(\mathbf{r}) = 0$, assuming certain boundary conditions. Note that the situation is very similar to that of a finite chain of atoms described in Section 4.1.3. The simplest boundary conditions are that the wave function has to vanish at the boundaries. This corresponds to holding the atoms at the end of a finite chain fixed. In both cases, it leads to standing waves. As in the case of lattice vibrations, it would be inconvenient to use such boundary conditions here because, ultimately, we will be interested in traveling solutions to account for electrical and thermal conductivity. Therefore, the Born–von Kármán boundary conditions (see (4.17)) are a better choice (for a more in-depth discussion of the different boundary conditions see Problem 11.1). We consider the three-dimensional case right away rather than discussing the problem in one dimension first, because a number of properties depend on the dimensionality. For simplicity, let us assume that we have a cubic box with a macroscopic side length L and volume $V = L^3$. Then the boundary conditions are

$$\psi(\mathbf{r}) = \psi(x, y, z) = \psi(x + L, y, z) = \psi(x, y + L, z) = \psi(x, y, z + L). \tag{6.3}$$

The solutions to the stationary Schrödinger equation are plane waves, normalized such that the integrated probability to find an electron in the box is 1:

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot\mathbf{r}}. \tag{6.4}$$

This is very similar to genuinely free electrons, but there are restrictions on the allowed values of $\mathbf{k}$, which are the same as for crystal vibrations with periodic boundary conditions (4.23), that is,

$$\mathbf{k} = (k_x, k_y, k_z) = \left(\frac{n_x 2\pi}{L}, \frac{n_y 2\pi}{L}, \frac{n_z 2\pi}{L}\right), \tag{6.5}$$

where n_x, n_y, and n_z are integers. The energy levels are

$$E(\mathbf{k}) = \frac{\hbar^2 k^2}{2m_e} = \frac{\hbar^2}{2m_e}(k_x^2 + k_y^2 + k_z^2). \tag{6.6}$$

These are shown as a function of n_x or k_x in Figure 6.3 for $n_y = n_z = 0$. In the figure, it appears that the level separation increases for higher energies or higher values of n_x but this is misleading and due to the fact that we have held n_x and n_y constant at zero. The separation between the energy levels is of the order

$$\frac{\hbar^2}{2m_e}\left(\frac{2\pi}{L}\right)^2, \tag{6.7}$$

which is very small because L is a macroscopic distance.

Since we neglect the electron–electron interaction, the calculated energy levels are one-electron levels. We can put in all the electrons according to the Pauli principle. We start by filling two electrons into the lowest energy state with $\mathbf{k} = (0,0,0)$ and $E(\mathbf{k}) = 0$. Then we proceed by occupying levels at higher $\mathbf{k}$, for example, $\mathbf{k} = (0,0,2\pi/L)$ until we have used up all the electrons. We want to know the highest occupied energy that we get when filling up the states. If the number of electrons to distribute is very large, we can use the same geometric interpretation as for the vibrational states (see Figure 4.12). Suppose that we have N electrons in our enclosed volume, such that the electron density is $N/V = N/L^3$. These can be accommodated on the $N/2$ states with

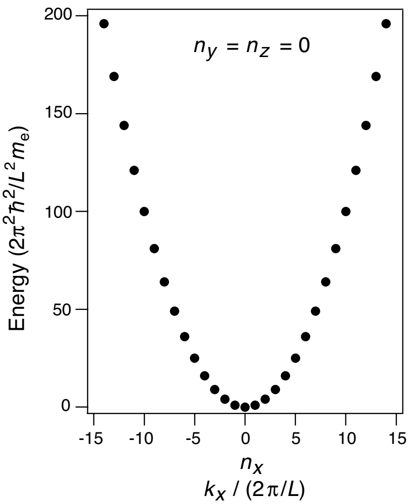

Figure 6.3 Electronic states in the free electron model. The increasing energy separations between the points at higher energies in an artifact caused by holding $n_y = n_z = 0$.

the lowest energy since we can have two electrons per state. We have to fill all states that lie within a sphere of radius $n_{\max}$ or, alternatively, a sphere with radius $k_{\max}$ such that

$$\frac{N}{2} = \frac{4}{3}\pi n_{\max}^3, \tag{6.8}$$

which gives

$$n_{\max} = \left(\frac{3N}{8\pi}\right)^{1/3}. \tag{6.9}$$

From this, we can calculate the energy of the highest occupied electron states to

$$E_{\max} = \frac{\hbar^2 k_{\max}^2}{2m_e} = \frac{\hbar^2}{2m_e}\left(\frac{2\pi}{L}\right)^2 n_{\max}^2. \tag{6.10}$$

This energy has a special name: it is called the Fermi energy E_F. For most metals, it is a few electron volts, that is, in the range of typical chemical binding energies. Similarly, $|\mathbf{k}|_{\max}$ is called the Fermi wave vector k_F. $n_{\max}$ is actually not used very much. A useful expression that follows from (6.10) is the relation between Fermi energy E_F and electron density n:

$$E_F = \frac{\hbar^2}{2m_e}\left(3\pi^2 n\right)^{2/3}. \tag{6.11}$$

The Fermi energy can be associated with the highest kinetic energy of the electrons in the solid. The highest speed v_F can then be calculated by $v_F^2 = 2E_F/m_e$. The result is in the order of $10^6\ \mathrm{m\,s^{-1}}$. This is a very high speed, especially if we keep in mind that everything has so far been calculated for a temperature of 0 K.

Finally, we can calculate the density of states $g(E)$ in the free electron model, that is, the number of available states per energy interval $\mathrm{d}E$. We will need this for the correct description of the situation at finite temperature and for many other things. From (6.9) and (6.10), the highest occupied energy for N electrons can be written as

$$E(N) = \frac{\hbar^2}{2m_e}\left(\frac{3\pi^2 N}{V}\right)^{2/3}, \tag{6.12}$$

from which we get the total number of states $N(E)$ for a given highest energy E and with this

$$g(E)\mathrm{d}E = \frac{\mathrm{d}N}{\mathrm{d}E}\mathrm{d}E = \frac{V}{2\pi^2}\left(\frac{2m_e}{\hbar^2}\right)^{3/2} E^{1/2}\mathrm{d}E. \tag{6.13}$$

This density of states for the free electron gas is shown in Figure 6.4a.

So far, we have only considered the situation at zero temperature. At any finite temperature, electrons will be thermally excited from their ground state. For fermions, the occupation probability of the states is given by the Fermi–Dirac distribution $f(E, T)$, which is given by

$$f(E,T) = \frac{1}{e^{(E-\mu)/k_B T}+1}, \tag{6.14}$$

where μ is the chemical potential. For metals, we can set $\mu = E_F$ and do not distinguish between the Fermi energy and the chemical potential at all. At zero temperature this is exactly true and at finite temperature it is a very good approximation. The Fermi–Dirac distribution is shown in Figure 6.4b. At zero temperature, it has a value of 1 for energies smaller than μ and 0 for energies higher than μ, meaning that all states below the chemical potential are occupied and all others are empty. This is consistent with our above discussion of how the states are filled. At finite temperature, the Fermi–Dirac distribution develops a "soft zone" around μ in which the occupation probability is no longer 1 or 0 but something in between. The soft zone is symmetric around μ and it has a width of about $4k_BT$. At room temperature, $k_BT \approx 25$ meV such that the soft zone is around 100 meV wide.

It is instructive to compare the electrons' mean kinetic energy in the quantum model to the result of the Drude model. In the quantum model, it must be some fraction of the Fermi energy E_F (see Problem 6.2) whereas it is given by (5.1) in the Drude model. The most important difference is not that the kinetic energy in the quantum model is fairly high, but that it is so (almost) *independent of the temperature*.

Finally, the density of occupied electron states at a given energy and temperature can be found by multiplying the density of states $g(E)$ with the Fermi–Dirac distribution $f(E, T)$, see Figure 6.4c. This definition gives us a way of calculating the chemical potential at any temperature because the total number of electrons N must be given by

$$N = \int_0^\infty g(E)f(E,T)\mathrm{d}E. \tag{6.15}$$

As pointed out earlier, however, the chemical potential in a metal depends only very little on the temperature (see Problem 7.3).

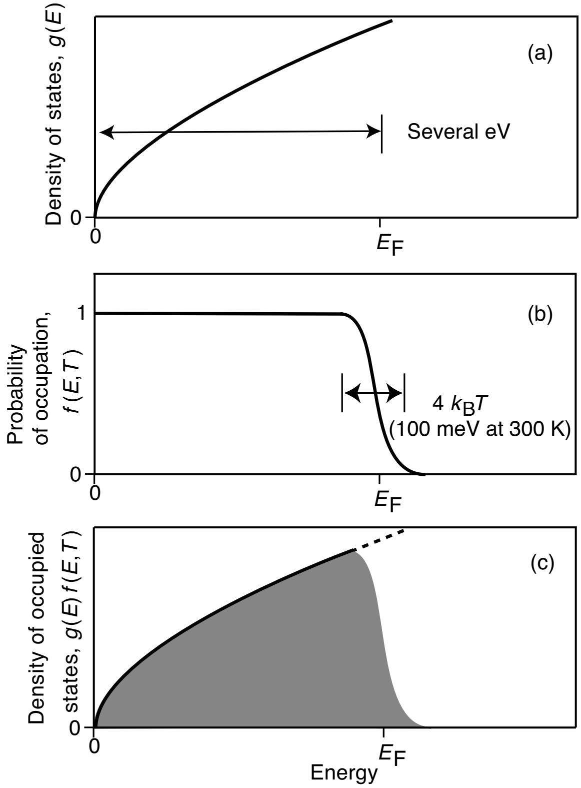

Figure 6.4 (a) Density of states for a free electron gas $g(E)$. (b) Fermi–Dirac distribution function at a finite temperature $f(E, T)$. (c) Density of occupied states $g(E)f(E, T)$. Note that the relative width of Fermi distribution's "soft zone" ($\approx 4k_BT$) is exaggerated in the sketch for temperatures around room temperature.

It is important to notice how different the energy scales are. The Fermi energy is several electron volts, while the soft zone of the Fermi–Dirac distribution is only 100 meV at room temperature. This means that the relative number of electrons in the soft zone is very small indeed. This turns out to be the key to understanding many properties of metals, for example, their heat capacity.

6.2.2 Electronic Heat Capacity

The fact that the Dulong–Petit rule is not only valid for insulators but also for many metals (see Table 4.2) suggests that the contribution of the free electrons to the heat capacity of a metal is very small. We can now understand why. When the temperature of the solid is raised, only a very small fraction of the electrons can be thermally excited. This is illustrated by Figure 6.5. Assume that the temperature of the solid is

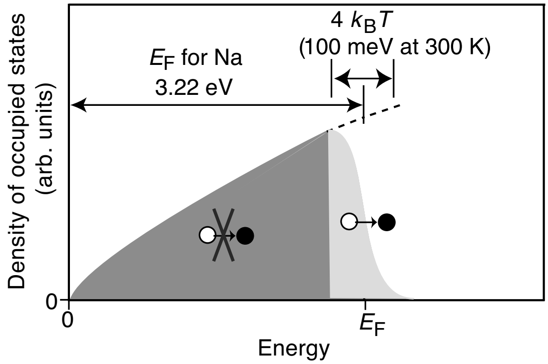

Figure 6.5 Illustration of the "Fermi trap" effect. Most of the electrons in a metal cannot change their energy by a small amount because the reachable states are already occupied by other electrons. As in Figure 6.4, the width of Fermi distribution's "soft zone" is not drawn to scale.

raised from zero to some finite temperature T. Classical particles would increase their kinetic energy by $3k_BT/2$. Here, this is impossible for most of the electrons because they are trapped: There are already other electrons occupying the states at slightly higher energies. A contribution to the heat capacity is, in fact, only possible for the electrons near the Fermi energy.

Let us try a "quick and dirty" estimate of the electronic heat capacity. The number of electrons in the soft zone is of the order $k_BTg(E_F)$. If we say that the mean thermal energy of these electrons is $3k_BT/2$, the total mean thermal energy is

$$\langle E \rangle = \frac{3}{2} k_B T g(E_F) k_B T \tag{6.16}$$

plus some offset which does not depend on the temperature. This gives

$$C = \left(\frac{\partial \langle E \rangle}{\partial T} \right)_V = 3k_B^2 T g(E_F). \tag{6.17}$$

This is quite close to the correct result, which is given by

$$C = \frac{\pi^2}{3} k_B^2 T g(E_F) \tag{6.18}$$

(see Problem 6.4). This expression has a number of interesting implications. The heat capacity is proportional to the density of states at the Fermi level $g(E_F)$. This is easy to understand because only the electrons close to the Fermi level can participate in thermal excitations. Since these electrons constitute only a small fraction of all electrons, the free electrons in a metal do not usually lead to a strong deviation from the Dulong–Petit behavior at high temperatures. On the other hand, (6.18) is linear in T whereas the (low temperature) heat capacity of the lattice (4.46) is proportional to T^3. This means that at low temperatures the lattice contribution vanishes faster than the electronic contribution and the latter can, in fact, be measured. This is illustrated in Figure 6.6.

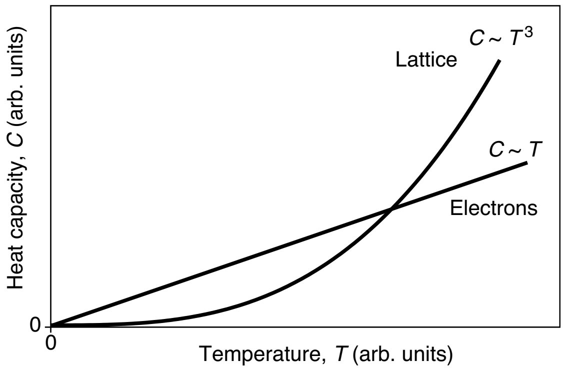

Figure 6.6 Qualitative sketch of the electronic and lattice contributions to the heat capacity at constant volume. At sufficiently low temperatures the electronic contribution dominates.

6.2.3 The Wiedemann–Franz Law

The free electron model correctly reproduces the Wiedemann–Franz law and also gives the correct Lorenz number L. This can be directly seen by inserting the appropriate expressions in (5.30). The thermal conductivity can be taken to have the same form as (4.49) with appropriate modifications for the velocity, which should be the Fermi velocity, and the heat capacity. The electrical conductivity is the same as in the Drude model. Both conductivities can be written such that they contain the relaxation time τ. We do not know anything about τ, but fortunately it cancels out in the final expression. Working out the details of this is left to the reader (see Problem 6.5). The final result is

$$\frac{\kappa}{\sigma} = \frac{\pi^2}{3}\frac{k_B^2}{e^2}T = LT. \tag{6.19}$$

This gives $L = 2.45 \times 10^8\ \mathrm{W\,\Omega\,K^{-2}}$, which agrees very well with the experimental data for many metals.

6.2.4 Screening

An important feature of metals is the ability to screen out electric fields. In fact, for the purpose of classical electrostatic field theory, it is commonly assumed that the metals are internally field free. On the atomic scale, this is not quite true but a metal is still very effective in screening out external fields. We do not want to derive the theory for this here, we merely state one result that quantifies it. Consider the Coulomb potential due to a positive point charge q in vacuum:

$$\phi_0(r) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}, \tag{6.20}$$

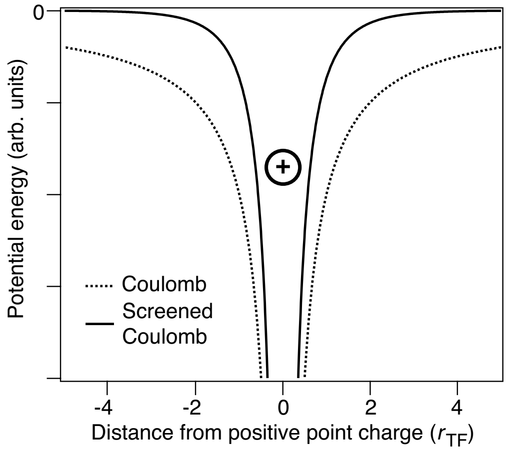

Figure 6.7 Potential energy of an electron due to a positive point charge in a metal compared to the Coulomb potential in free space.

where r is the distance from the charge. Inside the metal it is modified to give

$$\phi(r) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}e^{-r/r_{\mathrm{TF}}}, \tag{6.21}$$

where the so-called Thomas–Fermi screening length r_{TF} is given by

$$r_{\mathrm{TF}} = \sqrt{\frac{V\varepsilon_0}{e^2 g(E_{\mathrm{F}})}}. \tag{6.22}$$

r_{TF} is very small in metals, in the order of 1 Å, and the exponential part in (6.21) causes the screened potential to decay on the same length scale, much faster than the bare Coulomb potential (see Figure 6.7). This confirms the expected result that an electrostatic potential is screened out very rapidly in a metal. In fact, the effective screening can be used as an argument to explain why the electrons are free within a metal in the first place. With such an effective screening, it is not possible to localize them close to the potential of an ion because the screened potential is too weak.

6.3 The General Form of the Electronic States

The free electron model appears to describe certain phenomena in metals quite well but it has still some obvious shortcomings. In particular, it appears to be a fair description for metals but only if we want to describe a metal to start with. What about non-metallic compounds such as diamond or silicon? As we have seen in Figure 6.2, their characteristics is that the sp^3 band is completely filled and that there are no states immediately above the top of the band, in which electrons can be excited. This is not captured by the free electron model that gives a continuum of states from the lowest energy to infinity. One could argue that even for a metal this is incorrect: As we have seen in Figure 6.1d, the 3s band for Na is half-filled but it also has a finite width, that is, there are no states at all possible energies. This is hardly relevant for most physical phenomena such as conduction or heat capacity in which the excitation energies of the electrons are very small compared to the band width. But even for metals the free electron model poses some problems. Take for example Al that is a simple metal (not a transition metal) and should be described quite well by the free electron model. Yet, not even the sign of the Hall coefficient in Table 5.1 is correct and going from the classical to the quantum free electron description does not cure this problem.

Finally, the quantum mechanical free electron model is still not able to resolve some of the most basic questions of electron motion in solids. One example is that we still do not understand how the electrons at low temperature can have such long mean free paths in solids when there are lots of ions around. It is not surprising that the free electron model does not help here since it simply ignores the ions, but the question remains valid.

In order to make some progress, we have to describe the motion of the electrons in a non-vanishing lattice periodic potential (6.2) and solve (6.1). F. Bloch showed that the general wave function solving this problem has a very simple form

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r}), \tag{6.23}$$

where $u_{\mathbf{k}}(\mathbf{r})$ is a function with the periodicity of the Bravais lattice:

$$u_{\mathbf{k}}(\mathbf{r}) = u_{\mathbf{k}}(\mathbf{r}+\mathbf{R}). \tag{6.24}$$

The index $\mathbf{k}$ refers to the fact that the function $u_{\mathbf{k}}(\mathbf{r})$ and hence also $\psi_{\mathbf{k}}(\mathbf{r})$ can be changing, depending on the wave vector $\mathbf{k}$.

Before we prove Bloch's theorem, we mention one of its most important consequences. The solution (6.23) is very similar to the free electron solution; it is just a plane wave modulated by a lattice periodic function. This is an amazing fact because it means that the electronic states are spread out over the whole crystal, essentially as plane waves, even if we turn on the lattice periodic potential! In other words, (6.23) is a stationary solution for a perfectly periodic crystal. This means that the electrons travel through the crystal without bouncing into the lattice ions at all, immediately explaining why electrons travel through a metal with a very long mean free path, much longer than the distance between the ions. In fact, if the electrons are not scattered by the metal ions, we could expect the resistivity of a perfectly periodic metal crystal to be zero. We will later see that this would indeed be the case and we will discuss which mechanisms cause a finite resistivity.

We now prove Bloch's theorem (6.23). Like in the free electron model we use a cubic crystal of side length L and Born–von Kármán boundary conditions. The allowed values of $\mathbf{k}$ are then given by (6.5). Every solution of the Schrödinger equation (6.1) consistent with these boundary conditions can be written as a sum of plane waves:

$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} c_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}, \tag{6.25}$$

where $\mathbf{k}$ are the values consistent with the boundary conditions and the coefficients $c_{\mathbf{k}}$ are assumed to take care of the wave function's normalization. The lattice periodic potential can also be written as a Fourier series, using the reciprocal lattice vectors $\mathbf{G}$:

$$U(\mathbf{r}) = \sum_{\mathbf{G}} U_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}}. \tag{6.26}$$

Since we want the potential to be a real quantity, we must further require that

$$U_{-\mathbf{G}} = U_{\mathbf{G}}^{*}. \tag{6.27}$$

These two expansions can now be inserted into the Schrödinger equation (6.1). The kinetic energy term then becomes

$$-\frac{\hbar^2\nabla^2}{2m_e}\psi(\mathbf{r}) = \sum_{\mathbf{k}} \frac{\hbar^2 k^2}{2m_e} c_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}. \tag{6.28}$$

The potential energy terms becomes

$$\begin{aligned} U(\mathbf{r})\psi(\mathbf{r}) &= \left(\sum_{\mathbf{G}} U_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}}\right)\left(\sum_{\mathbf{k}} c_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}\right) = \sum_{\mathbf{kG}} U_{\mathbf{G}} c_{\mathbf{k}} e^{i(\mathbf{G}+\mathbf{k})\mathbf{r}} \\ &= \sum_{\mathbf{k}'\mathbf{G}} U_{\mathbf{G}} c_{\mathbf{k}'-\mathbf{G}} e^{i\mathbf{k}'\cdot\mathbf{r}}. \end{aligned} \tag{6.29}$$

In the last step, we have changed the summation index from $\mathbf{k}$ to $\mathbf{k}' = \mathbf{G} + \mathbf{k}$ in order to obtain the same plane wave form as in the expression for the kinetic energy. We are allowed to "shift" the indices by reciprocal lattice vectors as we wish since the sum in question extends over all vectors consistent with the boundary conditions and the reciprocal lattice vectors are clearly a subset of these. If we now rename the index $\mathbf{k}'$ in the potential energy expression back to $\mathbf{k}$, we can write the whole Schrödinger equation in the new form:

$$\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} \left\{ \left(\frac{\hbar^2 k^2}{2m_e} - E\right) c_{\mathbf{k}} + \sum_{\mathbf{G}} U_{\mathbf{G}} c_{\mathbf{k}-\mathbf{G}} \right\} = 0. \tag{6.30}$$

Since the plane waves with different $\mathbf{k}$ are orthogonal, every coefficient in the equation has to vanish in order for the sum to vanish. So the Schrödinger equation is reduced to a set of equations:

$$\left(\frac{\hbar^2 k^2}{2m_e} - E\right) c_{\mathbf{k}} + \sum_{\mathbf{G}} U_{\mathbf{G}} c_{\mathbf{k}-\mathbf{G}} = 0. \tag{6.31}$$

We can assume that every $\mathbf{k}$ in these equations lies in the first Brillouin zone. If it does not, we can shift the index $\mathbf{k}$ to $\mathbf{k} + \mathbf{G}'$ such that it does. Therefore, we can view (6.31) as a set of equations for every allowed $\mathbf{k}$ in the first Brillouin zone. Technically, the set consists of infinitely many equations but in practice, it is often sufficient to take only a few coefficients $c_{\mathbf{k}-\mathbf{G}}$ to be different from zero. We will see an example of this below. If we now take one such equation for a certain $\mathbf{k}$, we see that it only contains the coefficients $c_{\mathbf{k}}, c_{\mathbf{k}+\mathbf{G}}, c_{\mathbf{k}-\mathbf{G}}, \ldots$, that is, the coefficient for $\mathbf{k}$ itself and those for $\mathbf{k}$ plus all possible reciprocal lattice vectors. But this means that for a certain $\mathbf{k}$ the wave function (6.25) also only contains non-vanishing terms with these coefficients and therefore it can be written as

$$\psi_{\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{G}} c_{\mathbf{k}-\mathbf{G}} e^{i(\mathbf{k}-\mathbf{G})\cdot\mathbf{r}}. \tag{6.32}$$

This can also be written as

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{kr}} \left(\sum_{\mathbf{G}} c_{\mathbf{k}-\mathbf{G}} e^{-i\mathbf{G}\cdot\mathbf{r}} \right). \tag{6.33}$$

We now realize that the term in brackets is a Fourier series over the reciprocal lattice vectors and therefore a lattice periodic function, hence we have proven Bloch's theorem.

From this proof, we immediately get another important property of the Bloch functions. If we take (6.32) and shift the summation index by an arbitrary reciprocal lattice vector $\mathbf{G}'$, we get

$$\psi_{\mathbf{k}+\mathbf{G}'}(\mathbf{r}) = \sum_{\mathbf{G}+\mathbf{G}'} c_{\mathbf{k}-\mathbf{G}+\mathbf{G}'} e^{i(\mathbf{k}-\mathbf{G}+\mathbf{G}')\cdot\mathbf{r}}, \tag{6.34}$$

but since we sum over all reciprocal lattice vectors, this is exactly the same as the $\psi_{\mathbf{k}}(\mathbf{r})$ we started with. Therefore,

$$\psi_{\mathbf{k}+\mathbf{G}'} = \psi_{\mathbf{k}}(\mathbf{r}) \tag{6.35}$$

and when we insert this in the Schrödinger equation we get

$$E(\mathbf{k}+\mathbf{G}') = E(\mathbf{k}). \tag{6.36}$$

The periodicity of the potential in real space translates into a periodicity of the solutions in reciprocal space, as we have already seen for the lattice vibrations. This is quite important when we think about the set of equations (6.31). Since we get the same if we translate the whole problem by an arbitrary reciprocal lattice vector, we might just as well concentrate on only solving equations that have the $\mathbf{k}$ lying in the first Brillouin zone, greatly simplifying the problem.

6.4
Nearly Free Electron Model

Our proof of Bloch's theorem has also given us a rewritten form of the Schrödinger equation (6.31) that is actually a set of equations. Remarkably, all we need to do in order to determine the electronic states for any three-dimensional solid is to find the correct coefficients $c_{\mathbf{k}}$, assuming that we already know the potential. While formally simple, this is still difficult, especially since we do not know the potential for real solids.

However, as in the case of lattice vibrations, it is most instructive to solve the equations (6.31) for a one-dimensional solid with a lattice constant a. The reciprocal lattice for this solid is spanned by "vectors" of length $g = 2\pi/a$ and the potential can be written as a Fourier series

$$U(x) = \sum_n U_n e^{ingx}, \tag{6.37}$$

where the sum runs over all integers. We shall use a very simple potential: The Fourier coefficient for $n = 0$ can be set to zero because a constant potential offset does not change anything apart from a rigid shift of the energy eigenvalues. The only coefficients we are going to keep are $U_1 = U_{-1}$ and we call them simply U.

We start out by using a very small U. In practice, this means that we are treating free electrons but with the symmetry of the lattice. For a given k, we can write down many equations of the type (6.31). We start out by requiring that only c_k, c_{k-g}, and c_{k+g} are different from zero and we restrict ourselves to three equations. At present, there is no justification for this and we explore the consequences of allowing more coefficients and equations further down. We get

$$\left(\frac{\hbar^2(k-g)^2}{2m_e} - E\right)c_{k-g} + Uc_k = 0,$$

$$\left(\frac{\hbar^2 k^2}{2m_e} - E\right)c_k + Uc_{k-g} + Uc_{k+g} = 0,$$

$$\left(\frac{\hbar^2(k+g)^2}{2m_e} - E\right)c_{k+g} + Uc_k = 0. \tag{6.38}$$

This is a linear system of equations that has three solutions E_1, E_2, and E_3 for every value of k. The solutions are shown in Figure 6.8a. We find three parabolas that are identical to the free electron result in Figure 6.3. The parabolas are centered on the reciprocal lattice points 0, g, and $-g$. This is expected from (6.35) and (6.36). The only obvious problem is that there are no parabolas centered on higher order reciprocal lattice vectors, such as $2g$ and $-2g$, and this is actually caused by the fact that we have only used three coefficients and three equations in (6.38). We can extend (6.38) to five equations with five coefficients by also considering c_{k-2g} and c_{k+2g}. The result of this calculation is shown in Figure 6.8b. It is essentially the same as in Figure 6.8a, only that we do now also have parabolas centered on $2g$ and $-2g$ (but still none at the higher reciprocal lattice points). We see that neglecting higher coefficients has two consequences in the present case: We do not get the correct result in the higher Brillouin zones, and we do not get the correct result in the first Brillouin zone at high energies because we lack the parabolas from the higher Brillouin zones, which reach back into the first Brillouin zone. In any case, the result is very much like the free electron result but it also fulfils the symmetry requirement (6.36) imposed by the lattice, at least to some extent.

What happens when we now turn on the lattice potential U? This is shown in Figure 6.8c, again for the case of five equations of the type (6.31). The main consequence of the finite U is the opening of gaps between the parabolas at the

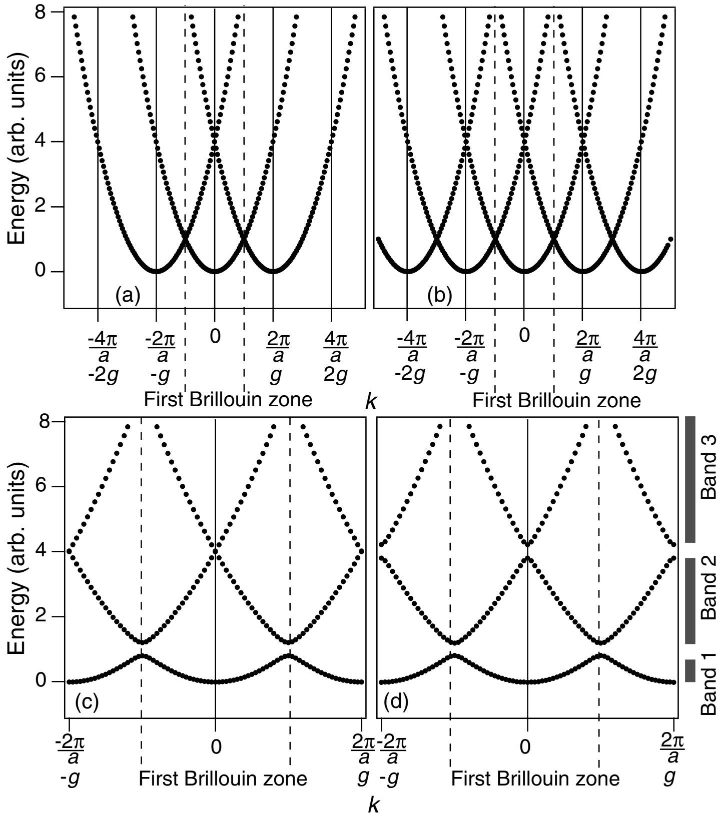

Figure 6.8 Electronic states in the nearly free electron model for a one-dimensional chain with unit cell length a. (a) Solutions for three equations (6.38), using a nearly vanishing $U = U_1 = U_{-1}$. (b) Solutions of five equations similar to (6.38), using a nearly vanishing $U = U_1 = U_{-1}$. (c) Same as (b) but for a larger value of U. (d) Same as (b) but for larger values for both $U_1 = U_{-1}$ and $U_2 = U_{-2}$. The gray bars symbolize the ranges where a quasicontinuum of energies is available (bands) and the gaps in between them.

Brillouin zone boundary; the other states are largely unaffected and appear still very much free electron like. However, this is a major change from the free electron model because it means that the solid no longer has a continuum of states from the lowest energy to infinity. In fact, now there is a range of energies for which there are no states at all.

The effect of turning on higher order contributions of the potential is shown in Figure 6.8d. Here, not only $U_1 = U_{-1}$ but also $U_2 = U_{-2}$ are chosen to have a finite value. The main effect of a finite $U_2 = U_{-2}$ is additional gap openings, now between the crossing points of the parabolas at higher energies. Note that now not only degeneracies at the Brillouin zone boundary are lifted but a finite U_2 also lifts the degeneracy between two states at the center of the Brillouin zone, at $k = 0$.

Quite general, the solution to a system of equations like (6.38) gives us n energy eigenvalues for every value of k or n relations of the type $E_n(k)$. n runs over all integers and k over the values consistent with the Born–von Kármán boundary conditions. The relations $E_n(k)$ are the dispersion relation for the electronic states and usually called the electronic band structure of the solid. In this context, n plays the role of a band index. In Figure 6.8d, for example, we see the two lowest bands and a part of the third band.

k can still be interpreted as the wave vector of the Bloch wave but we can also view it as a quantum number of the electronic states, in complete analogy with the role of k in the case of the lattice vibrations. Compare the situation to atomic physics. There n, the main quantum number, specifies the shell containing the electrons. The other quantum numbers l and m are the parameters of the spherical harmonics functions, which describe the angular part. In this sense, l and m are quantum numbers that are related to the spherical symmetry of the atom. In the solid, the symmetry is given by the periodic lattice structure and k can be viewed as the quantum number related to this symmetry.

We have now two different interpretations of k. It can be viewed as the wave vector of the Bloch wave or as a quantum number describing the state containing the electron. It is also very tempting to interpret $\hbar k$ as the momentum of the electron, as in the case of free electrons. But this is *wrong*. This can be seen quite easily. We apply the momentum operator $-i\hbar\nabla$ on the Bloch wave (6.23) to obtain

$$-i\hbar\nabla\psi_{\mathbf{k}}(\mathbf{r}) = \hbar\mathbf{k}\psi_{\mathbf{k}}(\mathbf{r}) - e^{i\mathbf{k}\cdot\mathbf{r}} i\hbar\nabla u_{\mathbf{k}}(\mathbf{r}), \tag{6.39}$$

which is only the same as in the free electron case when $u_{\mathbf{k}}(\mathbf{r})$ is constant, that is, when the Bloch wave is a free electron wave. We will understand the meaning of k somewhat better when we discuss the transport of electricity via Bloch states.

The occurrence of energy gaps at the zone boundary can also be made plausible by a very simple argument. Consider a free electron traveling perpendicular to a set of lattice planes separated by a distance a. If x is the direction perpendicular to the planes, the electron has the wave function $\psi(x) \propto e^{ikx}$ in this direction, that is, it behaves like a plane wave with a wavelength $\lambda = 2\pi/k$. Such a wave fulfills the Bragg condition (2.4) for a value of $k = n\pi/a$. This means that the lattice will reflect the wave back to some degree. As the solid is very big, the entire wave will eventually be reflected back, leading to two types of standing waves:

$$\psi(+) \propto e^{i(\pi/a)x} + e^{-i(\pi/a)x} = 2\cos\left(\frac{\pi}{a}x\right), \tag{6.40}$$

$$\psi(-) \propto e^{i(\pi/a)x} - e^{-i(\pi/a)x} = 2i\sin\left(\frac{\pi}{a}x\right). \tag{6.41}$$

This situation is shown in Figure 6.9. The two standing waves have probability densities that are shifted with respect to the positive ion potential. $\psi(+)$ shows an accumulation of probability near the ion cores, while $\psi(-)$ shows a depletion. Therefore, $\psi(+)$ has a lower energy than $\psi(-)$ even though both have the same wave vector $k = n\pi/a$. $\psi(+)$ and $\psi(-)$ thus correspond to the solution just below and above the energy gap at the Brillouin zone boundary, respectively.

When discussing lattice vibrations, we have stated that the group velocity of the lattice waves is given by $d\omega/dk$, where ω is the frequency of the wave and k the wave

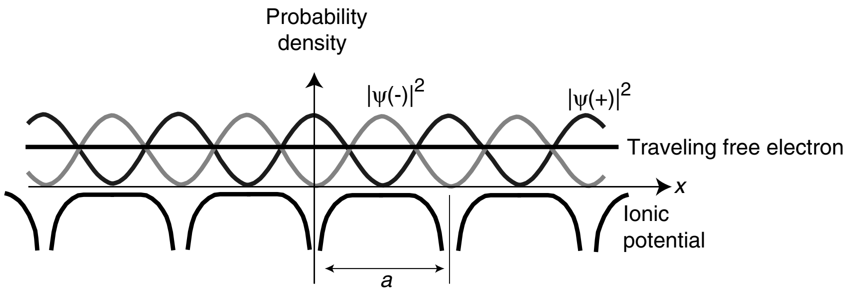

Figure 6.9 Qualitative explanation for the gap openings at the Brillouin zone boundary. Shown are two possible standing electron waves with k corresponding to the zone boundary π/a. Their probability density is either accumulated or depleted in the vicinity of the ion cores compared to a traveling free electron wave that has a constant probability density.

vector. This expression for the group velocity is of very general character in the theory of waves and it can be shown that it also holds for Bloch waves. There it is convenient to write it as

$$v_g = \frac{d\omega(k)}{dk} = \frac{1}{\hbar}\frac{dE(k)}{dk}. \tag{6.42}$$

In other words, the group velocity is given by the slope of the bands. If we now consider Figure 6.8c and d, we see that the group velocity of the bands with a finite periodic potential is zero at the Brillouin zone boundaries. This means that we have standing waves there, perfectly consistent with the argument we have just made. Note that a group velocity of zero has to be seen in a quantum mechanical sense. If we could measure the velocity v_g of an electron wave packet at the zone boundary, the expectation value would be zero, so we could not say if it moves to the right or the left side. However, this does not mean that the electron does not move. The expectation value for the kinetic energy ($\propto v_g^2$) would not be zero.

6.5 Energy Bands in Real Solids

So far, our discussion has been mostly confined to the situation in one dimension, except for the free electron model. Now we are in a position to understand the electronic energy bands in three dimensions and real materials, at least qualitatively. Introducing a lattice periodic potential had two main effects. The first was the symmetry in the bands (6.36), which allows us to consider the dispersion in the first Brillouin zone only, because it is identical around the other points of the reciprocal lattice. This symmetry also caused a backfolding of the bands starting at other reciprocal lattice points into the first Brillouin zone. The second effect was gap openings between bands that had been degenerate in the free electron model.

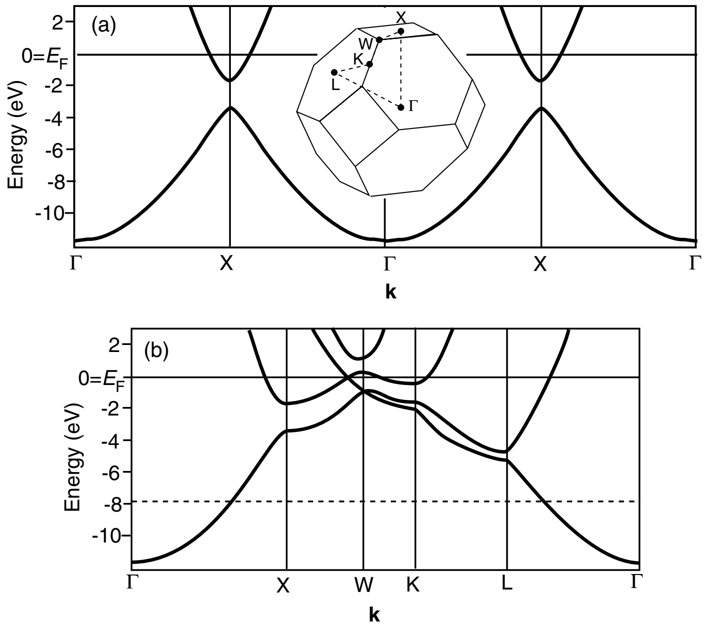

Figure 6.10 (a) Electronic energy bands in Al along the $\Gamma - X$ direction only. The inset shows the first Brillouin zone. (b) Energy bands in different directions given by the dashed path between high-symmetry points of the Brillouin zone. The horizontal dashed line represents the fictitious Fermi level for aluminum with the same structure but only one valence electron instead of three. Band structures taken from Ref. [6].

For three-dimensional materials these effects are very similar but the three-dimensional character of the problem makes it sometimes harder to keep the overview. The band energy now depends on a three-dimensional **k** and the Brillouin zone looks more complicated, too. As in the case of vibrational properties, we only discuss materials with an fcc Bravais lattice.

Figure 6.10 shows the energy bands of aluminum, a metal with only s and p electrons. The situation is still very similar to the free electron case. Consider first the dispersion in only one direction, as shown in Figure 6.10a. The dispersion is shown from the Γ to the X point at the zone boundary and beyond into the next zone, reaching Γ again and so on. At the X point, a gap is opened and above the gap another band is dispersing back toward the Γ point. This band can easily been recognized as the band stemming from the center of the next Brillouin zone, as in the one-dimensional model in Figure 6.8. The band structure of solids can be determined experimentally by angle-resolved photoemission. In this experiment, the sample is exposed to UV photons and electrons are emitted because of the photoelectric effect. The emitted electrons are sorted according to their **k**-vector and energy and from this it is possible to work back to **k** and the energy inside the sample, that is, to the band structure. The cover illustration of this book displays the experimental equivalent to Figure 6.10a.

Figure 6.10b shows the bands of Al in different high-symmetry directions in the first Brillouin zone. The continuation into the next zones, as in Figure 6.10a is usually not shown to avoid redundancy. The bands are filled up to the Fermi level that, for metals, is often used to define the energy zero. There are many bands crossing the Fermi level, which means that the electrons in the occupied states just below the Fermi level can be excited to states within the same band just above the Fermi level. In this way, they can contribute to the transport of electric and thermal current.

Aluminum has three electrons per unit cell. These electrons have been filled into the bands and this results in a separation of the Fermi level from the band bottom of about 12 eV. What would happen if we had only one electron per unit cell? The Fermi level would lie much lower, approximately at the dashed line in Figure 6.10b. In this case, the situation would be even more free electron like: The bands would cross the Fermi level at the same k distance from Γ in all directions and the electronic states at the Fermi level would therefore constitute a sphere in k-space. This is exactly the same as in the case of the free electron model.

We can now turn back to the initial question in this chapter: What characterizes a metal as opposed to a semiconductor? From what is said above, we would define a metal as a solid where bands cross the Fermi level, such that the energy of the electrons in these bands can be increased by a very small amount. Is this definition consistent with the band structure of typical semiconductors/insulators? To see this, we look at the band structures of Si and GaAs. Both have the same Brillouin zone shape as Al and the band structures are shown in Figure 6.11. The bands for these materials look considerably more complicated than those for the nearly free electrons or aluminum. However, we can still recognize several features. For the very lowest energies, the bands still look like parabolas. Bandgap openings at the Brillouin zone

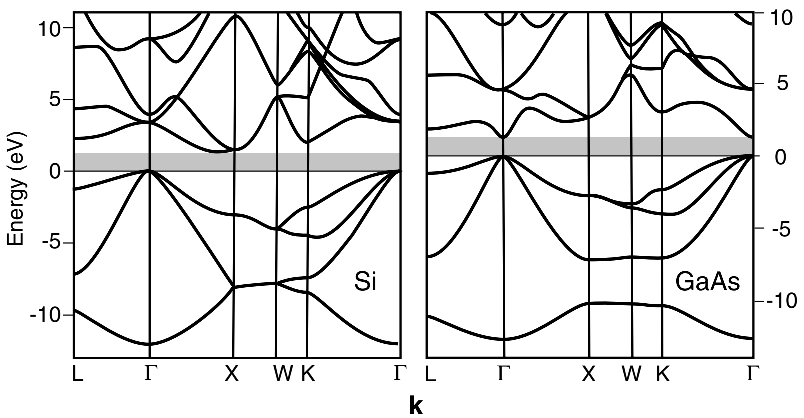

Figure 6.11 Electronic energy bands for Si and GaAs. These materials have the same Brillouin zone as Al (see Figure 6.10). The bands below the gray zone are completely filled and the bands above the gray zone are completely empty at zero temperature. The gray region represents an absolute bandgap in the electronic structure. Band structures taken from Ref. [7].

boundaries and the continuation of the (backfolded) bands can also be identified. For both materials, there is also an absolute bandgap emphasized by a gray area in the figure. "Absolute" means that this is not just a gap opening at some Brillouin zone boundary. It is a gap in the entire band structure; there are no states whatsoever, at any **k**, in the gray regions. When we fill the electrons of Si and GaAs into the bands according to the Pauli principle, the bands below the gray bandgap are exactly filled, there are no electrons left for the bands above the gap. Where this places the Fermi energy or, more precisely, the chemical potential is a subtle question that we address in the next chapter. Already now, we can say that it will be somewhere inside the gap region. In any event, we can certainly state that these materials are not metals in the sense of the definition made above: There are no bands crossing the Fermi level and no electrons that could increase their energy by a small amount in order to participate in the transport of electric current. In fact, if the energy of an electron is to be increased, it must at least be increased by the energy corresponding to the size of the gap.

An even simpler picture for seeing the difference between metals and insulators/semiconductors emerges when we look at their density of states, as in Figure 6.12. For free electrons, we have calculated the density of states to be proportional to the square root of the energy, as in 6.12a. The density of states increases as the energy increases. For the nearly free electron model, the density of states must be more complicated than this. Gaps appear in the band structure at the Brillouin zone boundaries and this

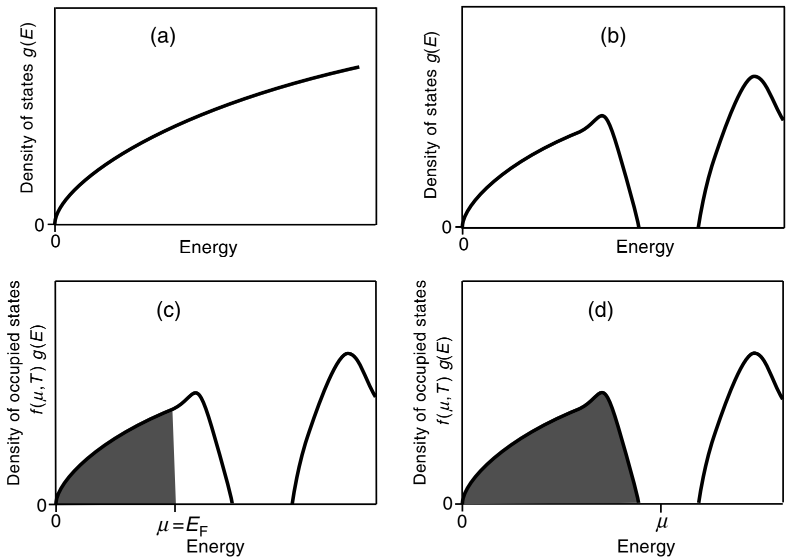

Figure 6.12 Illustration of the difference between metals and semiconductors/insulators. (a) Density of states in the free electron model. (b) Qualitative density of states in the nearly free electron model with the appearance of gaps. (c) Occupied density of states (gray area) for a metal at $T = 0$ K: the chemical potential (or the Fermi energy) cuts through a finite density of states. (d) The same for a semiconductor/insulator: the chemical potential is in a region of vanishing density of states.

can (but need not) lead to absolute gaps in the bands. Such a more complicated density of states is shown in Figure 6.12b. Now the difference between a metal and a semiconductor/insulator simply depends on how many electrons we have to fill into these states. If the highest energy we fill up to lies at a finite density of states, the solid is a metal (Figure 6.12c); if we just manage to fill the states up to a gap, the solid is a semiconductor/insulator (Figure 6.12d). Note that we have so far avoided to define the difference between a semiconductor and an insulator. We will address this in the next chapter.

Can we predict if a material is a metal or a semiconductor/insulator, starting from the known crystal structure? In the band picture discussed here, a material has to be a metal if a band is only partially filled with electrons. With a given number of electrons per unit cell, how many bands can we fill? It turns out that one band can accommodate exactly two electrons per unit cell (see Problem 6.8). Hence, we would expect a material with an uneven number of electrons to be a metal. Consider the example of Al that crystallizes in the fcc structure. The structure has one atom per unit cell and Al has three valence electrons per atom; hence, there are three valence electrons per unit cell. Two of these can completely fill one band, leaving another band half-filled. Thus, we would expect Al to be a metal. This is of course true and also consistent with Figure 6.10. The figure also shows the position of the Fermi level for the fictitious case of Al with only one valence electron. Following the same argument, this would also be a metal. Note, however, that the reverse argument is not valid: An even number of valence electrons per unit cell does not imply that the material is a semiconductor/insulator. The reason for this is that the electrons could be distributed into different bands in the three-dimensional band structure. Two electrons per unit cell, for example, could be placed in two different bands that overlap in energy. This would result in two partially filled bands and thus in a metal, despite the even number of electrons.

6.6 Transport Properties

We finally arrive at the description of transport in the quantum mechanical model and we will confine the discussion to the transport of electrical charges. The transport of heat via electrons proceeds along similar lines and we have some idea about the relation of electrical and thermal conductivity through the Wiedemann–Franz law. The transport properties of solids are a formidably complicated problem and we just give some very basic ideas about what is happening. These ideas can be presented by considering a one-dimensional solid.

When inspecting the mathematical solution of the Bloch wave (6.23), we see that it describes a modulated plane wave that is delocalized over the whole crystal, very much like a free electron. Such a solution cannot be used to describe the transport of electrons from one side of a crystal to the other. We will use the same approach as in the case of lattice vibrations: In order to describe transport, we think of the electrons traveling through the crystal as a wave packet, that is, as a superposition of Bloch waves within a certain Δk close to the k of interest (see Problem 6.10).

The most remarkable fact about the Bloch waves is that they represent electrons traveling through the crystal as a stationary solution of the Schrödinger equation. They are the solutions for the solid with the ions already present and these ions do not lead to any scattering. When discussing the shortcomings of the Drude model, we have asked how the electrons can manage to have a very long mean free path at low temperatures and to sneak past all the ions, completely incompatible with Drude's assumptions. Now we see why. A Bloch electron does not scatter off the lattice ions at all. Consequently, the electrical conductivity of perfectly crystalline metals should be infinite. This is obviously not the case and there must be some scattering mechanism for the Bloch electrons as well. We discuss possible candidates at the end of this section. For now we merely assume that there is some scattering present, which gives rise to a finite relaxation time τ. Note that the situation is again very similar to the transport of heat in a harmonic crystal. Phonons, which are packets of harmonic waves, can propagate undisturbed through the crystal and we had to invoke some effects (like defects) to obtain a finite thermal conductivity.

Viewing packets of Bloch electrons as particles with a certain wave vector k and (group) velocity v_g given by (6.42), we can arrive at a quasiclassical description of electrical conduction. Consider a particle with a velocity v_g and a charge $-e$ in an electric field that we denote here by $\mathcal{E}$, as not to confuse it with the energy. After a short time $\mathrm{d}t$, the particle has gained the energy $\mathrm{d}E$ given by

$$\mathrm{d}E = -e\,\mathcal{E} v_g \mathrm{d}t. \tag{6.43}$$

On the other hand, we have

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \frac{\mathrm{d}E}{\mathrm{d}k}\frac{\mathrm{d}k}{\mathrm{d}t}, \tag{6.44}$$

and combining this with (6.42) we get

$$\hbar\frac{\mathrm{d}k}{\mathrm{d}t} = -e\,\mathcal{E}. \tag{6.45}$$

This equation is quite plausible from the case of free electrons, where the momentum is $p = \hbar k$, but we have already shown that $\hbar k$ is not the momentum for Bloch electrons. Nevertheless, the equation of motion is correct. It means that an electric field causes the Bloch electrons to change their k and the rate of change is given by the field strength.

The situation for a partially filled band with no applied field is shown in Figure 6.13a. When the field is turned on, the electrons will have changed their k by δk after some short time δt, giving rise to the more asymmetric distribution in Figure 6.13b. Equation 6.45 suggests that the distribution would become increasingly asymmetric with time. In reality, however, there will be an inelastic scattering mechanism with a relaxation time τ, which prevents this from happening. Such a process, shown in Figure 6.13b, brings back the electrons close to k_F to unoccupied states at lower energy, close to $-k_F$. The combination of field acceleration and inelastic scattering leads to a stationary state in which all the electrons are

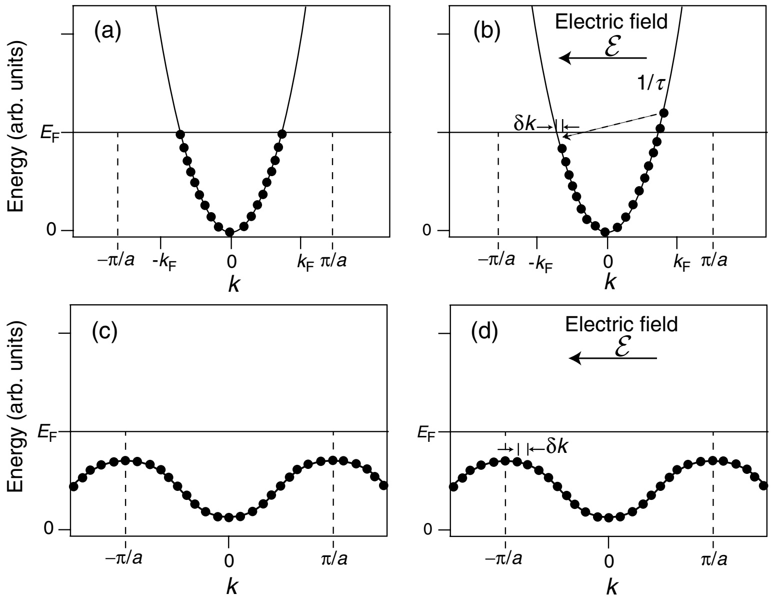

Figure 6.13 Simple picture of conduction in a metal. The circles symbolize filled electron states. (a) Situation for a partially filled band without an applied field. (b) Situation for a partially filled band with an applied field. After some time δt, all electrons are shifted by an amount δk due to the applied electric field. The asymmetric distribution in the electrons' group velocity gives rise to an electric current. The electrons at k_F can be scattered back to lower lying states at $-k_F$ with a probability proportional to the inverse relaxation time. (c and d) Corresponding situation for a completely filled band without and with electric field. All electrons are merely shifted into states that had been occupied already in the field-free case.

displaced by some δk. In most cases, δk will be rather small compared to the size of the Brillouin zone. The inelastic scattering also leads to a finite resistance.

According to (6.42), the asymmetric distribution in k corresponds to a asymmetric distribution of the group velocities of the electrons as well. While the whole distribution has been moved by δk, most electrons have ended up in states that had been occupied before by other electrons and nothing has changed. The only points where the change is important are around the Fermi level crossings $-k_F$ and k_F. In Figure 6.13b, the asymmetry of the distribution implies that there are more electrons with a group velocity to the right than to the left, that is, there is an electric current flowing. Obviously, the size of the current depends on the group velocity of the electrons at the Fermi level.

Figure 6.13c and d illustrates the situation for electrons in a full band. As the field is applied, these electrons are also moved by a certain δk. But this does not change the state at all because all electrons move into states that were also occupied before the field was turned on: the full band does not contribute to the conduction. Note that this is perfectly consistent with the fact that $\hbar k$ cannot be interpreted as the momentum of the Bloch electrons. Here, we increase $\hbar k$ for all electrons but the average momentum is obviously still zero. The picture is also consistent with our previous definition of metals and insulators. In an insulator, all the bands are completely full and hence no current can be passed through it.

It is quite instructive to combine (6.45) with (6.42) in the following way. Consider the acceleration of an electron initially traveling with the group velocity v_g:

$$a = \frac{dv_g}{dt} = \frac{1}{\hbar}\frac{d}{dt}\frac{dE(k)}{dk} = \frac{1}{\hbar}\frac{d^2E(k)}{dk^2}\frac{dk}{dt}. \tag{6.46}$$

If we substitute (6.45) for dk/dt in the last term, we get

$$a = -\frac{1}{\hbar^2}\frac{d^2E(k)}{dk^2}e\mathcal{E}. \tag{6.47}$$

This looks exactly like a classical equation of motion, if we define that the particles have a so-called effective mass

$$m^* = \hbar^2\left(\frac{d^2E(k)}{dk^2}\right)^{-1}. \tag{6.48}$$

The concept of the effective mass may appear rather artificial at first but it allows us to describe the motion of the electrons in a solid in the same semi-classical way as if they were free electrons. The only effect of the solid is that it can change the effective mass of the electrons. This effect can be dramatic: the effective mass can be much smaller or much bigger than the free electron mass. It can also be negative. For free electrons, the effective mass is of course equal to the electron mass m_e (see Problem 6.1).

The concept of the effective mass brings us back to the Drude formula for the electrical conductivity (5.9). If we treat conductivity in a quantum model, starting out with an expression like (6.47), the resulting conductivity will have the same form as (5.9). We have seen that only the electrons near the Fermi level in partially filled bands have to be considered to calculate the conductivity, so the mass in semiclassical version of (5.9) would have to be replaced by the effective mass at the Fermi level and the relaxation time would be that for the electrons at the Fermi level. The electron density n would still appear but not because all the electrons contribute to the current. The reason for having n in the equation is that the number of electrons at the Fermi level depends on the electron density. This is easily seen in the free electron model where the size of k_F and hence the size of the "Fermi sphere" depends on the electron concentration.

We have seen that Bloch waves travel through the perfect crystal without any scattering from the ions, in contrast to the electrons in the Drude model. Therefore, we still have to answer the question why metals have a finite resistivity. Ultimately, all the explanations come down to the fact that the lattice is not perfect. The most important contribution at higher temperature are lattice vibrations that destroy the perfect translational symmetry in the lattice and cause a scattering of the Bloch electrons. This also means that our initial assumption of the Born–Oppenheimer approximation is invalid since we have to consider the scattering of the electrons by the lattice atoms. Quite intuitively, the interaction between Bloch electrons and lattice vibrations is called the electron–phonon interaction. The scattering of electrons from lattice vibrations can be expected to be important if the temperature is not too low compared to the Debye temperature of the solid. Even at very low temperatures, the Bloch electrons are scattered because of remaining imperfections in the crystal. These can be all kinds of defects, point defects, dislocations, impurity atoms, and so on. Still, in a highly perfect crystal at low temperature, the conductivity can be several orders of magnitude higher than that at room temperature.

6.7 Brief Review of Some Key Ideas

Many of the problems we had with the Drude model have been caused by not taking into account that the electrons are fermions and thus underlie the Pauli principle. This has readily been cured by the quantum mechanical free electron model. Historically, the most important example for this is the heat capacity of the electrons, but the fact that only the electrons close to the Fermi energy can be excited by a small energy amount (and not all the electrons) is decisive in many properties of metals (conductivity, magnetism, screening, superconductivity, etc.). In formal terms, it is seen in the equations through the appearance of the density of states at the Fermi level $g(E_F)$.

The free electron model can also give us some hints as to why the electrons do not seem to interact much with each other: First, as the electrons move through the solid, many scattering processes between them are impossible because the states into which they could scatter are already occupied by other electrons. In addition to this, the Coulomb interaction in metals is strongly weakened by the very efficient screening. A real understanding of why the electrons do not interact much with each other, however, is quite difficult and far beyond the scope of this book.

Despite these successes, even the free electron model has some serious limitations, which are of course expected because of its simplicity. Many of these were qualitatively solved by the introduction of Bloch waves as a general solution to the Schrödinger equation for the periodic lattice and by the nearly free electron model as a particularly simple solution. This could account for different band structures that depended on the Fourier coefficients in the series describing the potential (6.26). It could explain the existence of band gaps and make it plausible that some materials are metals while others are not.

Our discussion of electrical conduction in a metal via Bloch states has also helped to understand several problems we had with the Drude model. The increase of the mean free path and the conductivity at low temperatures now follows naturally from the fact that the Bloch electrons do not scatter at all from the perfect lattice. At room temperature they are mainly scattered by lattice vibrations but at low temperatures these vibrations are frozen out. We can also understand why the resistivity of alloys can be much higher than for pure metals. If the alloys are built or mixed such that the two (or more) types of ions in the alloy do not form a periodic structure, this will lead to a strongly increased scattering of the Bloch electrons.

Finally, we have encountered the concept of the effective mass, leading to the perhaps most remarkable result of the whole chapter: The electrons in the periodic solid can be treated quite similarly to free electrons by a quasiclassical theory if we replace the free electron mass by an effective mass, which contains the information about the solid's band structure.

6.8 Discussion and Problems

6.8.1 Discussion

1. Describe qualitatively the physical origin of electronic energy bands in solids.
2. Describe the free electron model. How does the energy of an electron depend on the wave vector? How does the density of states depend on the energy?
3. How do the free electrons in the quantum model contribute to the heat capacity and how does this differ from the Drude model?
4. How does the heat capacity contribution of the electrons depend on the temperature?
5. Is it possible to measure the electronic contribution to the heat capacity despite the fact that it is usually quite small?
6. What is the form of the general solution of the Schrödinger equation for a lattice periodic potential?
7. The general solution of the Schrödinger equation for a lattice periodic potential suggests that a metal should not have any electrical resistance at all. Why is this so and where does the observed resistance come from?
8. What causes the existence of energy gaps in the electronic bands in a quantum model?
9. How does the electrical resistance of a metal depend on the temperature (qualitatively) and why?
10. (*) The density of states at the Fermi level of a metal, $g(E_F)$, appears in equations describing many different physical phenomena, such as electronic heat capacity, Pauli paramagnetism, screening, superconductivity, and others. Why is this so?
11. What is the speed of the electrons that contribute to the electrical current in the quantum model? How high is it compared to that in the Drude model and how does it depend on the temperature?
12. What is the physical interpretation (or several interpretations) of the vector $\mathbf{k}$ for an electronic state in a solid?

6.8.2 Problems

1. *Free electron model*: Show that the effective mass for free electrons is equal to the free electron mass m_e.

2. *Free electron model*: (a) Show that the mean kinetic energy of one electron in the quantum mechanical free electron model is $3/5\ E_F$ at $T = 0$ K. (b) Calculate the Fermi energy and the mean kinetic energy for potassium in electron volts. Use that K has a relative atomic mass of $A_r = 39.1$ and a density of $856\ \mathrm{kg\,m^{-3}}$. (c) Calculate the corresponding electron velocities. (d) Calculate the density of states at the Fermi level. How large is the number of electrons in the "soft zone" of about $4k_BT$ around the Fermi level at room temperature relative to the total number of electrons?

3. *Free electron model*: We have shown that the density of states for a free electron gas in three dimensions is given by (6.13). Show that the density of states for a free electron gas in two dimensions in independent of the energy.

4. *Free electron model*: (a) Calculate the electronic heat capacity for 1 mol of copper at 300 K. Use that the Fermi energy of Cu is 7 eV and the molar volume $7.11\ \mathrm{cm^3}$. (b) Compare the result of (a) to the Dulong–Petit value of the lattice and explain why it is so much smaller. (c) Below which temperature is the electronic contribution to the heat capacity higher than the contribution from the lattice? Use that the Debye temperature of Cu is 343 K. (d)(*) Derive the electronic heat capacity (6.18) in a proper way by calculating the total energy and differentiating. *Hint*: For this last part assume that $k_BT \ll E_F$. In this case, the density of states $g(E)$ in the vicinity of the Fermi energy can be taken to be $g(E_F)$, independent of the energy, and the chemical potential can be taken as independent of the temperature, as usual, that is, $\mu = E_F$.

5. *Free electron model*: Show that the Wiedemann–Franz law is indeed given by (6.19).

6. *Nearly free electron model*: We have seen that (6.38) gives an approximate solution of the Schrödinger for a weak potential with Fourier components $U = U_1 = U_{-1}$. The biggest deviation from the energies of free electrons was found at the Brillouin zone boundary ($k = \pi/a$). Using (6.38), show that c_k and c_{k-g} are much larger than c_{k+g} if $k = \pi/a$. Neglect then c_{k+g} and use (6.38) to show that the size of the gap opening at the Brillouin zone boundary is $2U$.

7. *Bloch electrons*: Show that the spacial probability density for one-dimensional free electrons is constant. (b) Show that it has the periodicity of the corresponding Bravais lattice for Bloch electrons.

8. *Band picture of solids*: Consider a one-dimensional chain of N atoms with one atom per unit cell. Each atom shall have Z valence electrons. (a) Show that you can fill exactly $Z/2$ bands with these electrons or, equivalently, that each band can accommodate $2N$ electrons. (b) Figure 6.11 shows that Si has four filled bands (for some values of $\mathbf{k}$ the energies of the bands are degenerate, but not for all). There are also four electrons per Si atom (not eight!). Explain why this is so. (c) Having an even number of electrons per unit is necessary but not sufficient for a solid to be a semiconductor/insulator. Give an example for an elemental solid that is a metal despite having an even number of electrons per unit cell.

9. *Transport properties*: Show, in one dimension, that the average group velocity for a filled band is zero. *Hint*: For a chain with lattice constant a and macroscopic length L, the distance between the allowed k values is $2\pi/L$, that is, it is very small. It is then useful to write the sum over the allowed k-points as an integral.

10. (*) *Transport properties*: An electron moving through the crystal can be described by a superposition of Bloch waves. Consider the time-dependent, one-dimensional Bloch wave functions

$$\psi_k(x,t) = e^{ikx - i\omega(k)t} u_k(x) \tag{6.49}$$

and the wave packet

$$\Psi(x,t) = \int_{\infty}^{\infty} e^{-(k-k_0)^2/b^2} e^{ikx - i\omega(k)t} u_k(x) dk. \tag{6.50}$$

If b is much smaller than the size of the Brillouin zone, the Gauss function $e^{-(k-k_0)^2/b^2}$ is strongly peaked around k_0 and we can assume that $u_k(x) = u_{k0}(x)$ does not depend on k and that the dispersion of the electronic states $\omega(k) = E(k)/\hbar$ is linear, that is,

$$\omega(k) = v_g(k - k_0) + \omega(k_0) = v_g k + \omega_0 \tag{6.51}$$

with $\omega_0 = \omega(k_0) - v_g k_0$ with $v_g = \partial\omega/\partial k$ at k_0. Show that the maximum of the probability distribution associated with this wave packet moves with a velocity v_g.

7
Semiconductors

In the previous chapter, we have defined the difference between metals and semiconductors/insulators based on the density of states at the chemical potential at 0 K. If the density of states is finite, the solid is a metal. Otherwise, it is an insulator or a semiconductor. We have not defined the character of the solid based on its conductivity. In fact, this would have been a very bad idea. The conductivity of solids spans over more than 27 orders of magnitude. As a general rule, metals are good conductors and semiconductors/insulators are not, but even among the metals, the variations are big. To make matters more complicated, the conductivity of solids is strongly temperature dependent, and this dependence is qualitatively different for different types of solids. A solid that is a poor conductor at low temperatures can be a good conductor at room temperature and vice versa.

The difference between semiconductors and insulators is not clearly defined as we shall see. However, the general idea behind a semiconductor is that the bandgap between the highest occupied states and the lowest unoccupied states is sufficiently small to get thermally excited carriers at reasonable temperatures. We know that the width of the soft zone in the Fermi–Dirac distribution is about 100 meV at room temperature. Therefore, a semiconductor gap cannot be much larger than a few electron volts because otherwise the number of excited carriers would be vanishingly small. Often one speaks of semiconductors in the case of materials with bandgaps below 3 eV and of insulators if the gap is larger. Gap sizes for some important semiconductors and insulators are given in Table 7.1.

The most common elemental semiconductors are the group IV elements Si and Ge with their characteristic tetrahedral sp^3 bonding. Many compound semiconductors are isoelectronic to Si and Ge and show the same bonding type. Examples are SiC that also contains only group IV elements or combinations of groups III and V such as GaAs, or II and VI like CdSe. The latter two groups of materials are commonly referred to as III–V and II–VI semiconductors.

The conductivity of semiconductors is strongly influenced by a very small amount of impurities, and a precise control of this impurity concentration turns out to be essential for the construction of semiconductor devices. In this chapter, we shall first consider the properties of pure or intrinsic semiconductors, and then those of

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Table 7.1 Gap sizes for common semiconductors (above the horizontal line) and insulators (below the horizontal line).

Material	Gap size (eV)
InSb	0.18
InAs	0.36
Ge	0.67
Si	1.11
GaAs	1.43
CdSe	1.74
SiC	2.3
Diamond	5.5
MgF_2	11

impure or doped semiconductors. Finally, we shall briefly discuss the most basic physics of some semiconductor devices.

7.1 Intrinsic Semiconductors

In this section, we study the basic properties of pure (intrinsic) semiconductors. Since a very small number of impurities have strong consequences on the behavior of semiconductors, this intrinsic state is hard to get and also of little technological relevance. However, many of the concepts we explore here can be transferred easily to the case of doped semiconductors that we treat in the next section.

A semiconductor or insulator is, by definition, a solid for which the chemical potential at zero temperature is placed at an energy where the density of states is zero. In other words, there is a certain energy band that is completely filled, and the next band at higher energy is completely empty. This situation is schematically shown in Figure 7.1a. The highest occupied band and the lowest unoccupied band are called the valence band (VB) and the conduction band (CB), respectively. At higher temperatures, the electrons have to be distributed according to the Fermi–Dirac distribution shown in Figure 7.1b. In order to apply this distribution, we have to know the position of the chemical potential.[1] Suppose that the chemical potential is just above the valence band maximum (VBM). This would result in a density of occupied states as shown in Figure 7.1c. Many states would be emptied in the valence band, but only very few states would be populated inthe conduction band. In other words, the

1) Some authors use the term “Fermi energy” instead of “chemical potential” for semiconductors. We speak of the Fermi energy only for metals.

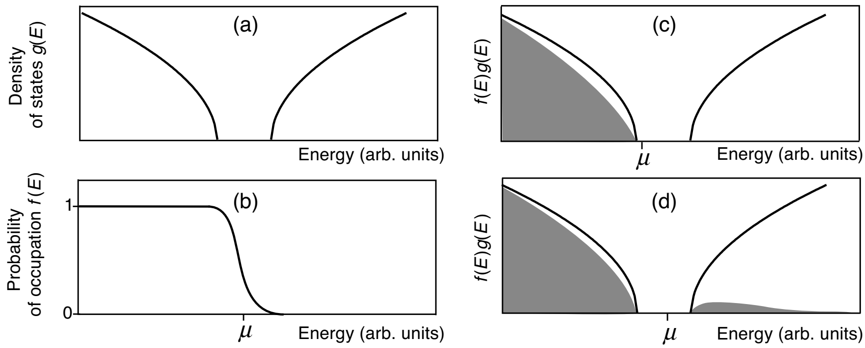

Figure 7.1 Charge neutrality and the position of the chemical potential in an intrinsic semiconductor. (a) Schematic density of states for a semiconductor. (b) Fermi–Dirac distribution at a finite temperature. (c) Occupied density of states (gray area) for the chemical potential just above the valence band maximum. (d) Occupied density of states for the chemical potential close to the middle of the gap. Note that the temperature in (c) and (d) is much higher than room temperature to make the presence of excited carriers visible.

total number of electrons would have decreased and the solid would be charged positively, just by warming it up! This is clearly unphysical as the total amount of charge has to be conserved. An equivalent argumentcan be made for a chemical potential just below the conduction band minimum (CBM), which would lead to a negatively charged solid. Consequently, we see that the chemical potential must lie close to the middle of the gap between the VBM and the CBM (Figure 7.1d). At a higher temperature, this leads to an equal number of missing electrons in the valence band and excited electrons in the conduction band.

At finite temperature the excited electrons in the conduction band cause this band to be partially filled. Therefore, we can apply the same concept for electronic transport as we have used for metals. The valence band is also partially filled at finite temperature and contributes to the conductivity of the solid through the remaining electrons.

There is another possibility to view the conduction by the valence band, which is illustrated in Figure 7.2. The semiconductor is schematically shown as atoms with valence 4 bound together. Each bond represents one electron in the valence band. At finite temperature, some of the electrons in the valence band are missing and these

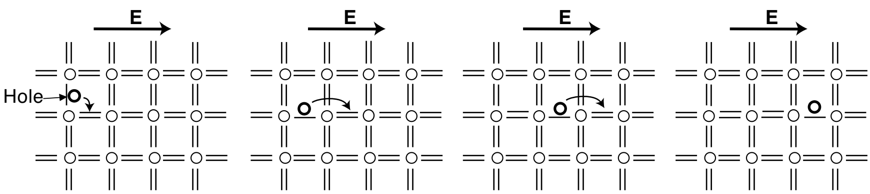

Figure 7.2 Transport of charge in an electric field for a partially filled valence band. The process can be interpreted as a motion of electrons in the direction opposing the field or as a motion of a hole in the field direction.

are represented by missing bonds and a circle. When an electric field is applied, electrons can use these missing states to travel to the positive potential as shown in the figure. But there is another, simpler way of viewing this. Instead of considering the complicated motion of the electrons, one can describe the conduction in terms of the moving missing electron or so-called hole.

In the following two sections, we examine these ideas more quantitatively. We calculate the position of the chemical potential in several circumstances. The central idea is always that a change of temperature cannot lead to the charging of the solid. We also consider the electrical transport through the valence band and the conduction band, and we shall see that the concept of holes also emerges from the mathematical side of the problem.

7.1.1
Temperature Dependence of the Carrier Density

Formally, it is very simple to write down the electron density n in the conduction band. It corresponds to the integrated occupied density of states in the conduction band:

$$n = \frac{1}{V}\int_{E_C}^{\infty} g_C(E) f(E,T)\,\mathrm{d}E, \tag{7.1}$$

where $g_C(E)$ is the density of states in the conduction band and E_C is the energy of the CBM. The density of missing electrons or holes in the valence band p can be calculated by an analogous formula:

$$p = \frac{1}{V}\int_{-\infty}^{E_V} g_V(E)[1-f(E,T)]\,\mathrm{d}E, \tag{7.2}$$

with E_V being the energy of the VBM. For the practical calculation, we have to make some approximations. The first is that we can use simple expressions for the two densities of states $g_C(E)$ and $g_V(E)$. We have noticed that if the chemical potential is roughly in the middle of the gap, most electrons in the conduction can be found very close to the CBM. At higher energies, the Fermi–Dirac distribution falls to zero very rapidly. An equivalent argument holds for the valence band: most holes will be found at the VBM. If we inspect the band dispersion close to these points, it is very nearly parabolic (see Figure 6.11). It is therefore sensible to simplify the relevant band structure of a semiconductor as in Figure 7.3a. Both the conduction and the valence band are represented by parabolas, but these are not necessarily free electron like. Instead, they can be described using the concept of the effective mass from (6.48). The effective mass is essentially the inverse curvature of a band, such that a parabolic band has the same effective mass everywhere. Hence, only two effective masses are needed, one for the CBM and one for the VBM. Furthermore, it is common practice to define the VBM as energy zero so that we have $E_V = 0$ and $E_C = E_g$ with the gap size E_g. The situation in Figure 7.3 resembles closely the dispersion for GaAs in Figure 6.11, but it is somewhat different from the dispersion of Si. In GaAs, both the VBM and the CBM are found at Γ, that is, at $\mathbf{k} = (0,0,0)$, and the gap is called a direct bandgap. In Si only, the VBM is found at Γ and the gap is called an indirect bandgap. But if we are

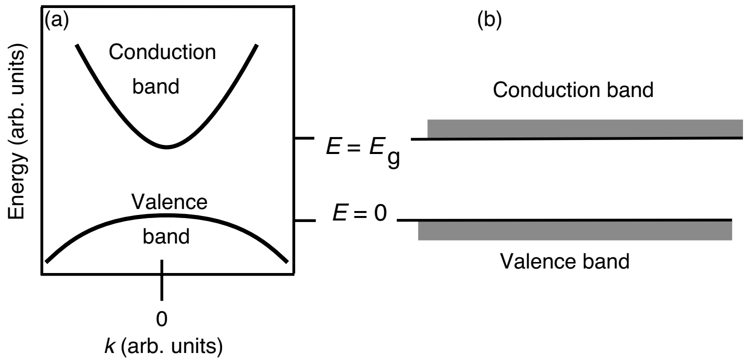

Figure 7.3 (a) Sketch of the valence band and conduction band in the vicinity of the gap. The bands are described as parabolas with different curvatures (effective masses). (b) Even simpler picture in which valence and conduction bands are represented by single energy levels.

only interested in the carrier densities, this difference does not play any role and the picture shown in Figure 7.3 will work for both.

Figure 7.3a can also give us a more mathematical derivation of the concept of valence band holes. The conduction band is almost empty apart from some electrons close to its minimum. When an electric field is applied, the situation is similar to that of a metal depicted in Figure 6.13b with a "Fermi level" just above the CBM. The electrons at the CBM are accelerated in an external electric field E according to $a = (-e)E/m_e^*$. Since m_e^* is positive, this is the "normal" motion of electrons in an electric field.

In the valence band, the situation is quite different. The band is almost filled, apart from some holes around the VBM. In this way, the situation is very similar to conduction in a metal where the "Fermi level" is just below the VBM. The electrons are also accelerated according to $a = (-e)E/m_h^*$, but now the effective mass is negative such that we can write $a = (-e)E/(-|m_h^*|)$. But this is the same as $a = eE/(|m_h^*|)$ and can therefore be interpreted as the motion of positive carriers with a positive effective mass. In the following, we assume that m_h^* is positive, and we consider holes rather than electrons as carriers in the VB.

The dispersion of the conduction band can now be written as

$$E = E_g + \frac{\hbar^2 k^2}{2m_e^*}, \tag{7.3}$$

where m_e^* is the effective mass for the electrons close to the CBM. The density of states in the free electron model is given by (6.13), and consequently the density of states in the conduction band is

$$g_C(E)\mathrm{d}E = \frac{V}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}(E-E_g)^{1/2}\mathrm{d}E. \tag{7.4}$$

For the valence band, we get analogous expressions:

$$E = -\frac{\hbar^2 k^2}{2m_h^*} \tag{7.5}$$

with the (positive) effective mass m_h^* of the holes (i.e., of the VBM) and

$$g_V(E)\mathrm{d}E = \frac{V}{2\pi^2}\left(\frac{2m_h^*}{\hbar^2}\right)^{3/2}(-E)^{1/2}\mathrm{d}E. \tag{7.6}$$

The density of states between valence band and conduction band is of course zero. Now we can formally write down the electron density (7.1) and the hole density (7.2), but unfortunately the integrals cannot be solved analytically.

It is therefore useful to introduce a further simplification. If the chemical potential is close to the middle of the gap, the gap size is at least a few hundred meV and we are interested in the properties of the material around room temperature, we can make the following approximations for the Fermi–Dirac distribution. For the calculation of the electron density, we write

$$f(E,T) = \frac{1}{e^{(E-\mu)/k_B T}+1} \approx e^{-(E-\mu)/k_B T}. \tag{7.7}$$

For the holes in the valence band, we get

$$1-f(E,T) = 1-\frac{1}{e^{(E-\mu)/k_B T}+1} \approx e^{(E-\mu)/k_B T} \tag{7.8}$$

(see Problem 7.1). Basically, these approximations mean that we have replaced the Fermi–Dirac distribution with a (shifted) classical Boltzmann distribution.

Now the integrals for the electron and hole density can be solved. For (7.1) we get

$$\begin{aligned} n &= \frac{1}{V}\int_{E_g}^{\infty}\frac{V}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}(E-E_g)^{1/2}e^{-(E-\mu)/k_B T}\mathrm{d}E \\ &= \frac{(2m_e^*)^{3/2}}{2\pi^2\hbar^3}e^{\mu/k_B T}\int_{E_g}^{\infty}(E-E_g)^{1/2}e^{-E/k_B T}\mathrm{d}E. \end{aligned} \tag{7.9}$$

The substitution $X_g = (E-E_g)/k_B T$ gives

$$n = \frac{(2m_e^*)^{3/2}}{2\pi^2\hbar^3}(k_B T)^{3/2}e^{-(E_g-\mu)/k_B T}\int_0^{\infty}X_g^{1/2}e^{-X_g}dX_g. \tag{7.10}$$

The last integral can be evaluated to give $\sqrt{\pi}/2$ so that the final result is

$$n = 2\left(\frac{2\pi m_e^* k_B T}{h^2}\right)^{3/2}e^{-(E_g-\mu)/k_B T} = N_{\mathrm{eff}}^C e^{-(E_g-\mu)/k_B T}, \tag{7.11}$$

where N_{eff}^C can be viewed as an effective number of states per volume for the conduction band. The same calculation for the hole density gives

$$p = 2\left(\frac{2\pi m_h^* k_B T}{h^2}\right)^{3/2}e^{-\mu/k_B T} = N_{\mathrm{eff}}^V e^{-\mu/k_B T}. \tag{7.12}$$

Equations 7.11 and 7.12 have a very intriguing and simple interpretation. Formally, they look like Boltzmann distributions for two levels with energies $E_g - \mu$ and μ. In this interpretation, the whole band character of the problem appears to be lost (it is still hidden in the effective masses), and it is quite sufficient to think of the valence band and the conduction band as two discrete energy levels (see Figure 7.3b). When considering the temperature dependence of the carrier density, this point of view is not entirely correct because the effective numbers of states are functions of the temperature themselves. Since their temperature dependence is, however, weak compared to the exponential term, it can often be neglected.

A useful relationship is derived by multiplying (7.11) and (7.12). We get

$$np = 4\left(\frac{k_B T}{2\pi\hbar^2}\right)^3 (m_e^* m_h^*)^{3/2} e^{-E_g/k_B T}, \tag{7.13}$$

meaning that the product of electron and hole concentrations is constant at any given temperature, independent of the chemical potential. This equation is often called the law of mass action and particularly useful when treating doped semiconductors.

From this, we can finally calculate the carrier concentration for an intrinsic semiconductor at a given temperature. Obviously, the intrinsic electron density n_i has to be equal to the intrinsic hole density p_i and from (7.13) we get

$$n_i = p_i = 2\left(\frac{k_B T}{2\pi\hbar^2}\right)^{3/2} (m_e^* m_h^*)^{3/4} e^{-E_g/2k_B T}. \tag{7.14}$$

Values for the important semiconductors Si and GaAs are given in Table 7.2. These densities are strongly temperature dependent and much smaller than those of typical metals (Table 5.1), and we can therefore expect intrinsic semiconductors to be rather poor electrical conductors.

Table 7.2 Intrinsic carrier densities for Si and GaAs at room temperature.

Material	Gap size (eV)	n_i at 150 K (m^{-3})	n_i at 300 K (m^{-3})
Si	1.11	4.1×10^6	1.5×10^{16}
GaAs	1.43	1.8×10^0	5×10^{13}

From the condition of charge neutrality $n = p$, we can get the position of the chemical potential

$$\mu = \frac{E_g}{2} + \frac{3}{4} k_B T \ln\left(\frac{m_h^*}{m_e^*}\right). \tag{7.15}$$

If the effective masses for electrons and holes are equal, the chemical potential lies in the middle of the gap, independent of temperature. If not, it must change as a function of temperature. If, for example, the holes are heavy and the electrons are light, that is, the valence band curvature is low and the conduction band curvature is high, many more holes than electrons would be generated at elevated temperature. In order to avoid this, the chemical potential has to move up as the temperature is raised.

To calculate all this, we needed essentially only three parameters: the gap size E_g and the effective electron and hole masses m_e^* and m_h^*. How can these quantities be measured? The gap size is relatively easy to measure, at least for direct gap materials such as GaAs.The basic idea is that the semiconductor cannot absorb light if the photon energy $h\nu$ is smaller than the gap size E_g. If $h\nu$ exceeds E_g, electrons from the conduction band can be excited into the valence band. The gap size can therefore be measured by studying the optical absorption of a semiconductor as a function of photon energy. Strong absorption sets in as $h\nu = E_g$.

The effective masses of electrons and holes can be determined by the technique of cyclotron resonance. When the semiconductor is placed into a static magnetic field B, the electrons move on circular (or spiral) orbits around the axis of the field with an angular frequency

$$\omega_c = \frac{Be}{m_e^*} \tag{7.16}$$

(see Figure 7.4). These angular frequencies and thus the effective masses can be measured by shining radio frequency waves from the side into the system and measuring the absorption on the other side. Strong absorption occurs when the radio frequency exactly matches ω_c. This is because the magnetic field causes a quantitation of the (nearly) free electron energy levels with a constant level spacing of $\hbar\omega_c$, similar to a harmonic oscillator, and the radio frequency wave can induce transitions between these levels. A similar argument applies for the holes.

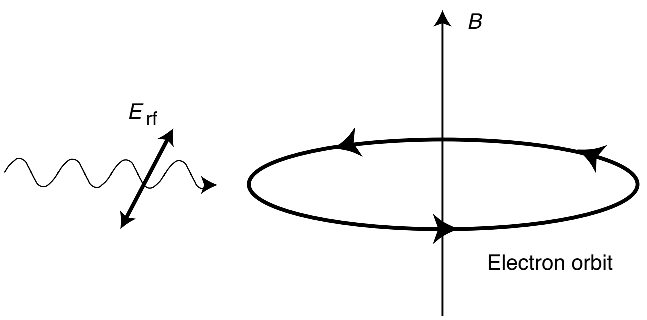

Figure 7.4 The measurement of cyclotron resonance. The electrons (or holes) are forced to move on circular orbits by a magnetic field B. Radio frequency radiation with the electric field vector E_{rf} can induce transitions between quantized circular orbits.

The effective masses in some semiconductor materials are given in Table 7.3. Note that the effective masses can be quite different from the free electron mass, in contrast to the case of many simple metals. In particular, the electron masses can be very small.

Table 7.3 Effective masses for semiconductors.

Material	m_e^*/m_e	m_h^*/m_e
Ge	0.60	0.28
Si	0.43	0.54
CdSe	0.13	0.45
InSb	0.015	0.39
InAs	0.026	0.41
GaAs	0.065	0.50

7.2 Doped Semiconductors

The carrier concentrations in most intrinsic semiconductors are too low to give any appreciable conductivity at room temperature. The situation can be changed by adding a very small amount of electrically active impurities, a process called doping. The dopant atoms change the conductivity by either donating (giving) electrons to the conduction band or accepting (taking) electrons from the valence band, that is, by generating holes. They are therefore called donors and acceptors, respectively. It is evident that a very small concentration of such impurity atoms changes the carrier density drastically. Consider the case of Si. There are about 5×10^{28} Si atoms per cubic meter, but at room temperature the intrinsic carrier concentration is only $1.5 \times 10^{16}\,\mathrm{m^{-3}}$. This means that a dopant concentration of more than $1.5 \times 10^{16}\,\mathrm{m^{-3}}$ would be sufficient to create more carriers than present in the intrinsic case, at least if every dopant atom gives rise to a free electron or hole. Thus, one dopant atom in 10^{12} silicon atoms would be sufficient to modify the carrier concentration significantly! It is, in fact, not possible to produce such pure samples so that there is always some amount of unintentional doping. The lowest impurity concentration that can currently be achieved is about $10^{18}\,\mathrm{m^{-3}}$.

7.2.1 n and p Doping

The two types of doping in a semiconductor are called n and p doping, for dopant atoms that give rise to free electrons in the conduction band (donors) and free holes in the valence band (acceptors), respectively.

n doping is achieved by putting pentavalent donor atoms such as P, As, or Sb into the Si lattice. These atoms have a valence configuration of s^2p^3, but only four of these five electrons are needed to form the sp^3 hybrid orbitals needed to place the dopant atom in the lattice. The remaining electron remains loosely bound to the dopant ion, attracted by one positive net charge. This is shown in Figure 7.5a.

The important point about the n doping is that the extra electron has a very small binding energy and can therefore easily be removed by thermal excitations. We can

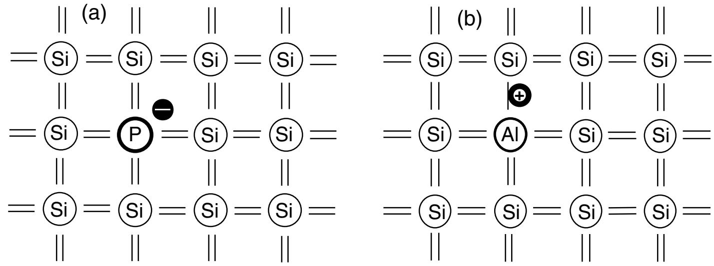

Figure 7.5 Non-ionized dopant atoms in a Si lattice: (a) donor and (b) acceptor.

estimate the binding energy by noticing the similarity of the problem to the hydrogen atom. The binding energies for hydrogen are

$$E_n = -\frac{m_e e^4}{8\varepsilon_0^2 h^2}\frac{1}{n^2}. \tag{7.17}$$

A binding energy of zero would correspond to a free electron, that is, an electron that has left the positively charged dopant atom and moves freely in the conduction band. The state corresponding to $n = 1$ is the state that is most tightly bound and its binding energy corresponds to the ionization energy of the impurity. This energy is well known to be 13.6 eV for a free hydrogen atom. For an impurity in Si, however, we have to change (7.17) in two ways. First, we have to replace the electron mass m_e by the effective mass of a conduction electron in Si, which is $0.43m_e$. More importantly, we have to take into account that the donor "atom" is not placed in vacuum but in solid Si. Therefore, the Si electrons screen the potential of the impurity atom and reduce the interaction. This is taken into account by replacing the dielectric constant of the vacuum ε_0 by $\varepsilon_0\varepsilon_{Si}$, where $\varepsilon_{Si} = 11.7$. The physical origin of ε is discussed in Chapter 9. The reduced effective mass and the screening both reduce the binding energy, to the order of $E_d = 40$ meV or so. Since the ionized state corresponds to the electron in the conduction band, this binding energy refers to the conduction band minimum. The energy levels are clarified in Figure 7.6a.

The reduced interaction due to the screening of the positive donor charge also affects the size of the donor "atom". We can estimate this by considering the Bohr radius, which is given by

$$a_0 = \varepsilon_0 \frac{h^2}{\pi m_e} e^2. \tag{7.18}$$

Applying the same substitutions as for the calculation of the energy, we see that this radius increases by a factor of about 30 with respect to the usual hydrogen Bohr radius.

Similar considerations apply to p doping with trivalent acceptor atoms such as B, Al, Ga, or In. This situation is shown in Figures 7.5b and 7.6b. The calculation of the binding energy is equivalent to the case of n doping, only that we now have a positively charged hole that is bound to a negatively charged ion.

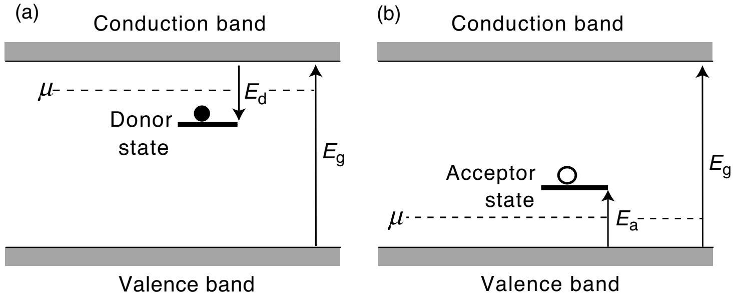

Figure 7.6 Energy levels for dopant atoms. (a) The donor levels are placed shortly below the conduction band minimum. Ionization of a donor atom corresponds to the transfer of the extra electron into the conduction band. (b) The acceptor levels are placed just above the valence band maximum. Ionization of an acceptor atom corresponds to accepting (taking) an electron from the valence band and thereby generating a mobile hole.

7.2.2 **Carrier Density**

The calculation of the carrier density in a doped semiconductor is rather complicated and only sketched here. The basic principle is the same as in the intrinsic case: We have to fulfill a charge neutrality condition to assure that a temperature change does not lead to a charging of the crystal. This can be formulated such that the number of electrons in the conduction band plus the number of (negatively) charged acceptor atoms must be equal to the number of the holes in the valence band plus the number of positively charged donor atoms. The problem is further complicated by the fact that replacement of the Fermi–Dirac statistics with the Boltzmann statistics does not necessarily apply because the chemical potential can be very close to the valence band or conduction band edges.

Here, we just give a qualitative discussion of an n-doped semiconductor. The situation is shown in Figure 7.7. At temperatures much lower than E_d/k_B, essentially none of the donor atoms is ionized, they are frozen out. When the temperature is raised, the donors are ionized and their electrons are moved to the conduction band. In some intermediate temperature range, essentially all donors are ionized and the carrier concentration in the conduction band corresponds to the concentration of the donors N_d. This region is called the extrinsic region. At much high temperatures, the excitation of intrinsic carriers across the bandgap will become important, leading again to a strong increase of carriers. This region is called the intrinsic region. From this, it is qualitatively clear how the chemical potential must behave. At very low temperatures, it must be situated between the CBM and the donor level E_d. At very high temperatures, it must lie in the middle of the gap, as in the intrinsic case. The calculated position of the chemical potential is also shown in Figure 7.7.

If we are only interested in a qualitative picture, as in the section about semiconductor devices, we may even use (7.11) and (7.12) to calculate the carrier density, but we must keep in mind that these equations are not correct quantitatively.

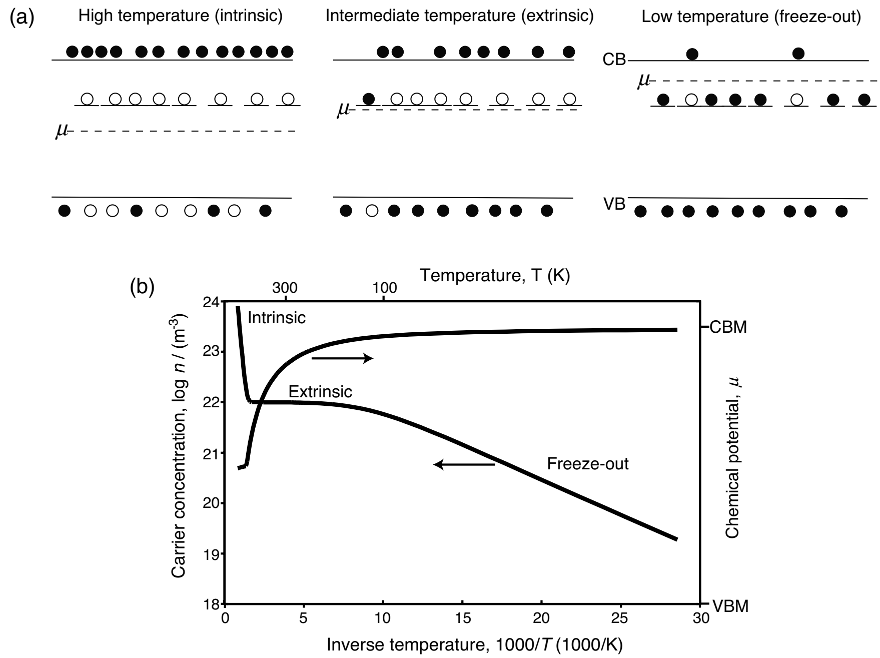

Figure 7.7 Electron density and position of the chemical potential for an n-doped semiconductor. (a) *Qualitative picture*: At very low temperatures, almost none of the donors is ionized (freeze-out region). At intermediate temperatures, almost all of the donors but very few intrinsic carriers are ionized (extrinsic region). At high temperatures, all the donors and some intrinsic carriers are ionized (intrinsic region). (b) *Quantitative picture*: The position of the chemical potential and the logarithmic carrier density are shown as a function of inverse temperature.

A more useful expression for calculating the hole and electron densities is the law of mass action (7.13), which remains valid because it is not based on any assumptions about the position of the chemical potential. The fact that np is constant at a given temperature has some interesting consequences. If we increase the number of electrons by n doping, the number of holes in the valence band has to be much lower than that in the intrinsic case because the product of n and p has to remain the same. The law of mass action can actually be used to calculate the hole concentration p.

In a doped semiconductor, the number of one type of carriers is greatly enhanced by the doping. These carriers are called the majority carriers. The other type of carriers is still there, but we have just seen that its concentration is even smaller than that in the intrinsic case at the same temperature. These carriers are called the minority carriers. One could be tempted to think that the minority carriers are utterly unimportant because there are so few. But this is not actually the case. The minority carriers are of essential importance for some semiconductor devices such as transistors.

The actual density of electrons and holes can be measured using the Hall effect described in Chapter 5. The Hall effect gives the sign and the density of the carriers. For a semiconductor, a positive R_H has an obvious interpretation as being caused by the holes in the valence band. If we have purely n- or p- doped samples and the intrinsic carriers do not play a role, the interpretation of R_H is simple. If both types of doping are present, however, both holes and electrons contribute to R_H and the situation becomes more complicated (see Problem 7.7).

Finally, the results from semiconductors have given us a plausible explanation of the fact that some metals show a positive Hall R_H, too. The sign of R_H is determined by the details of the band structure and the position of the Fermi energy in that band structure. If the Fermi energy of a metal is placed just below a band maximum similar to the VBM in a semiconductor, the electrons in that metal will behave like holes in a Hall measurement.

7.3 Conductivity of Semiconductors

In our discussion of the conductivity in the quantum model, we have seen that the simple Drude formula (5.9) can be viewed as approximately correct if we replace the free electron mass with the effective mass at the Fermi level. The conduction of electrical current in a semiconductor is somewhat different because it is carried by both electrons and holes. We therefore have to modify the expressions for the conductivity obtained in the Drude model and we do so using the concept of the mobility (5.11) and (5.12). The conductivity is then

$$\sigma = e(n\mu_e + p\mu_h), \tag{7.19}$$

with n and p representing the electron and hole concentrations and μ_e and μ_h their mobilities. Note that the mobility contains both the relaxation time and the effective mass of the carriers. Since for most semiconductors the electron effective mass is smaller than the hole effective mass, a higher conductivity can be achieved by n doping than by p doping (assuming that the relaxation time is the same for both conduction mechanisms).

One of the most important characteristics of a semiconductor is the temperature dependence of its conductivity as opposed to that of a metal. For metals, the conductivity decreases as the temperature is raised. This is because of the increasing probability for electrons to be scattered by phonons and the accompanying reduction of the relaxation time. Usually, the opposite behavior is observed for a semiconductor.[2] The conductivity increases as the temperature is raised. This is caused by the strongly temperature-dependent carrier density, which is illustrated in Figure 7.7: In the freeze-out region and in the intrinsic region, the carrier density changes so dramatically with temperature that the change of the mobility does not play a big role. At higher temperatures, the relaxation time also decreases, but the effect is more than compensated by the increased number of carriers. Indeed, the fact that carrier concentration has usually a much bigger effect than the mobility on the temperature dependence of the conductivity is one of the reasons to define the mobility in the first place.

2) Note that this is not true for all temperatures and doping concentrations. An example is given in Problem 7.8.

7.4 Semiconductor Devices

Semiconductor devices are arguably the most important application of solid-state physics to date. Most semiconductor devices are based on the physical phenomena that appear in inhomogeneous semiconductors, that is, semiconductors for which the doping concentration and type depend on the location. The simplest example is that of a pn junction (a diode). We treat diodes semi-quantitatively and the more complicated transistors just qualitatively.

7.4.1 The pn Junction

The simplest inhomogeneous semiconductor device is the junction between an n-and a p-doped semiconductor. In real devices, such junctions are not fabricated by joining bits of semiconductors but by starting with an undoped material and doping it inhomogeneously (e.g., through diffusion of impurities). In this way, no absolutely sharp boundary can be achieved, but we assume the existence of such a boundary here for simplicity.

Figure 7.8 shows what happens when we join p- (left) and n- (right) doped materials. In Figure 7.8a, the two semiconductors are still separated. At not too high temperatures, the chemical potential of the p-doped semiconductor will be close to the valence band, and in the n-doped semiconductor it will be close to the conduction band. In the p-doped sample, most of the acceptors will be negatively charged and the majority carriers are holes. There are also some minority electrons. In the n-type sample, most donors are ionized, the majority carriers are electrons, and there are also some minority holes.

Figure 7.8b shows what happens when the two semiconductors are joined. Electrons from the n side will diffuse into the p side and recombine with holes there and vice versa. This leads to a region of ionized donors and acceptors without the compensating charge of the generated carriers. This region is called the depletion layer. The ionized donors and acceptors form a region of localized charge with an electric dipole field between the two sides. The dipole field represents an obstacle for the p holes to move into the n part and for the n electrons to move into the p part. The process goes on until equilibrium is achieved.

The same phenomenon can be described more formally by stating that in thermal equilibrium the chemical potential has to be constant in the whole system. This implies a situation as in Figure 7.8c. The chemical potential is aligned in the whole system, but this can only be achieved by a macroscopic potential in the depletion region, which shifts the energy levels and "bends" the bands. Central questions in the treatment of the pn junction are now how big this potential change is and what size the depletion layer has.

The potential in the depletion region can be calculated by assuming a sharp transition between the depletion region and the non-depleted region as shown in Figure 7.9a. Then, we can state that charge densities in the p and n regions are $\rho_p = -eN_a$ and $\rho_n = +eN_d$, where N_a and N_d are the acceptor and donor concentrations, respectively. Charge neutrality furthermore requires that $N_a d_p = N_d d_n$, where d_p

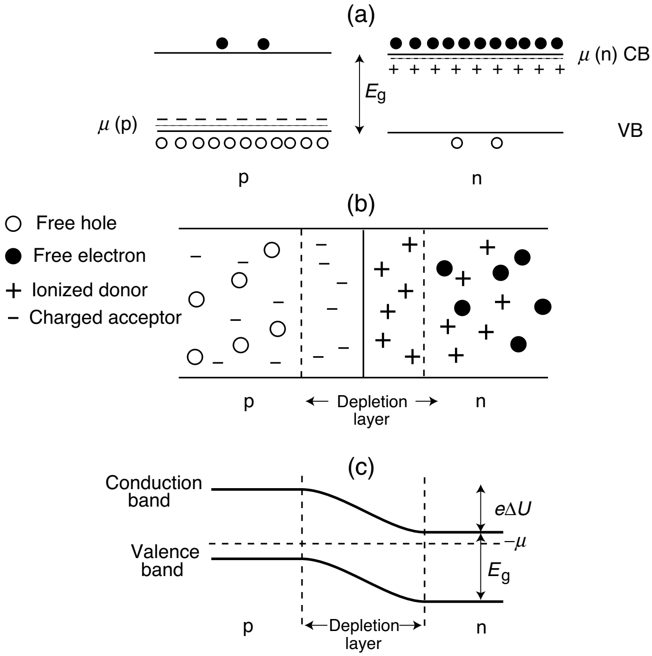

Figure 7.8 The pn junction. (a) Energy levels and carrier densities in separate p and n semiconductors. Ionized donors (acceptors) are symbolized by plus (minus) signs. The chemical potential is very close to the VB and the CB for p and n doping, respectively. (b) Formation of a depletion zone when the junction is established. (c) A position-independent chemical potential requires band bending over the depletion zone.

and d_n are the depths of the depletion layer in the p and n sides, respectively. If x is the direction perpendicular to the interface, it is possible to calculate the macroscopic potential $U(x)$ by solving the Poisson equation.

$$\frac{d^2 U}{dx^2} = -\frac{\rho}{\varepsilon\varepsilon_0} \tag{7.20}$$

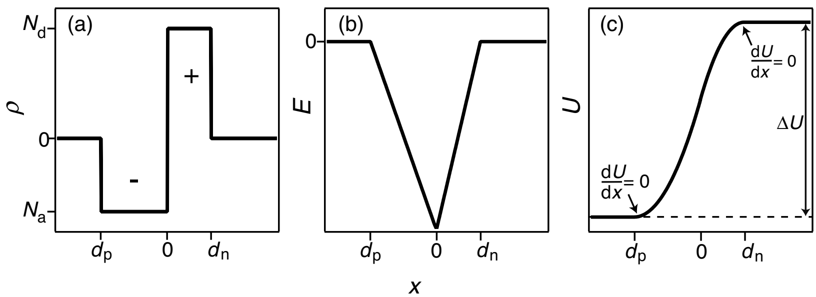

Figure 7.9 Idealized model of the depletion zone solved using the Poisson equation. (a) Charge density in the depletion zone. (b) Electric field. (c) Electrostatic potential.

with the boundary conditions that both $U(x)$ and $\mathrm{d}U(x)/\mathrm{d}x$ have to be continuous at $x=0$ and

$$\frac{\mathrm{d}U}{\mathrm{d}x}\Big|_{x=-d_\mathrm{p},d_\mathrm{n}}=0. \tag{7.21}$$

The solutions for the electric field and the potential are shown in Figure 7.9b and c. The calculation is left to the reader (Problem 7.10). The total potential difference across the space charge layer is

$$\Delta U=\frac{e}{2\varepsilon\varepsilon_0}\left(N_\mathrm{d}d_\mathrm{n}^2+N_\mathrm{a}d_\mathrm{p}^2\right). \tag{7.22}$$

From this and the charge neutrality conditions, the depth of the depletion zones can be calculated as

$$d_\mathrm{p}=\left(\frac{\Delta U 2\varepsilon\varepsilon_0}{eN_\mathrm{a}}\frac{N_\mathrm{d}}{N_\mathrm{a}+N_\mathrm{d}}\right)^{1/2},\quad d_\mathrm{n}=\left(\frac{\Delta U 2\varepsilon\varepsilon_0}{eN_\mathrm{d}}\frac{N_\mathrm{a}}{N_\mathrm{a}+N_\mathrm{d}}\right)^{1/2}. \tag{7.23}$$

The problem, however, is that we neither know ΔU nor d_p and d_n. What we do know is that the chemical potential at low temperatures must be very close to the VBM in the p-doped region and very close to the CBM in the n-doped region. Consequently, $\Delta U \approx E_\mathrm{g}$. From this, we can estimate the thickness of the depletion layer to be between 0.1 and 1 μm. This is very big on the atomic scale, justifying our macroscopic approach when using the Poisson equation.

We proceed with a strongly simplified semi-quantitative treatment of the pn junction, making some rather unjustified approximations, but illustrating the fundamental working principle. First we write down the approximate carrier densities in the conduction and valence bands on both sides of the junction. We use the notation that n_n and p_n are the electron and hole densities on the n side and the corresponding notation for the p side. Using the energy diagram in Figure 7.10 and the Boltzmann approximation to the Fermi function (which is evidently not a good approximation). From (7.11) and (7.12), we can readily write down the densities.

$$n_\mathrm{p}=N_\mathrm{eff}^C\mathrm{e}^{\left(\mu-E_g-e\Delta U\right)/k_\mathrm{B}T},\quad p_\mathrm{P}=N_\mathrm{eff}^V\mathrm{e}^{\left(e\Delta U-\mu\right)/k_\mathrm{B}T}, \tag{7.24}$$

$$n_\mathrm{n}=N_\mathrm{eff}^C\mathrm{e}^{\left(\mu-E_g\right)/k_\mathrm{B}T},\quad p_\mathrm{n}=N_\mathrm{eff}^V\mathrm{e}^{-\mu/k_\mathrm{B}T}. \tag{7.25}$$

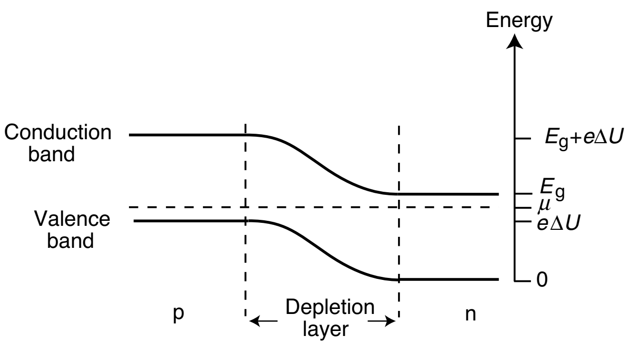

Figure 7.10 Definition of the energies in the pn junction. The VBM on the n side is taken as an energy zero.

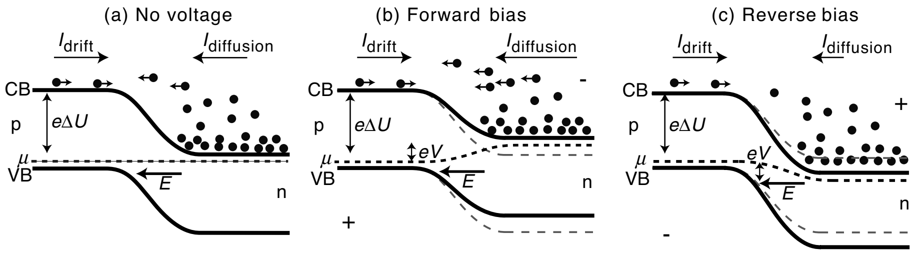

Figure 7.11 The pn junction as a diode (only considering the electrons, not the holes). (a) Without any bias voltage. (b) Forward bias. (c) Reverse bias. The gray dashed lines show the position of the n-side VB and CB without an applied voltage.

The equilibrium of the pn junction is not a static equilibrium in the sense that there are no currents across the junction. If minority electrons on the p side get close to or into the depletion layer, the electric field in this layer will transport them to the n side and the same applies for holes from the n side. This gives rise to so-called drift currents.

In the following, we treat only the electrons, and assume equivalent arguments can be made for the holes. The equilibrium situation is shown in Figure 7.11a. The drift of minority electrons from the p side is compensated by a so-called diffusion current of majority electrons from the n side into the p side. These majority electrons have to move against an electrostatic force in the depletion layer. Hence only electrons with a sufficiently high thermal energy can contribute to the diffusion current. If we say that the drift current and the diffusion current are proportional to the carrier densities, we can write.

$$|I_{\text{diffusion}}| = |I_{\text{drift}}| = |I_0| = C\mathrm{e}^{(\mu - E_g - e\Delta U)/k_B T}, \tag{7.26}$$

where C is the proportionality constant and the carrier density is taken as n_p from (7.24).

Now it is easy to extend these ideas to a pn junction with an applied external voltage, that is, a diode. If we apply a voltage V to the junction, we assume that the whole voltage drop occurs over the depletion zone, that is, we rigidly shift the bands on one side by an amount eV against the other side. This assumption is quite sensible because the resistance of the pn junction will be dominated by the depletion zone.

The situation for an external field opposing the field in the depletion zone (forward bias) is shown in Figure 7.11b. As an energy zero, we keep the position of the VBM on the n side without an applied voltage. The electron drift current is unchanged, but the diffusion current is modified because the chemical potential on the n side is shifted by eV:

$$|I_{\text{diffusion}}| = C\mathrm{e}^{((\mu + eV) - E_g - e\Delta U)/k_B T}. \tag{7.27}$$

The net current across the junction is the difference between drift and diffusion currents:

$$I = I_{\text{diffusion}} - I_{\text{drift}} = I_0\left(\mathrm{e}^{eV/k_B T} - 1\right). \tag{7.28}$$

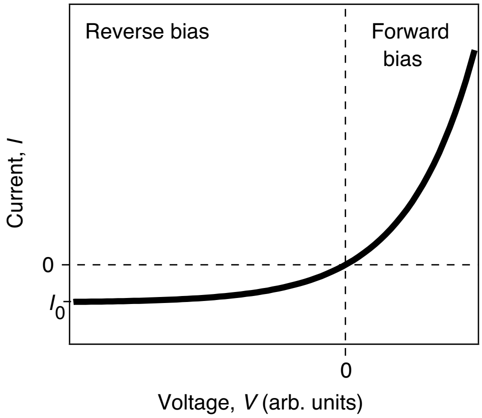

Figure 7.12 Characteristic $I(V)$ curve for a diode.

This current is zero without a bias voltage and increases exponentially as the bias voltage is raised. Qualitatively this is easy to understand because the external field lowers the barrier that has to be overcome by the majority electrons on the n-side to cross to the p-side.

Figure 7.11c shows the situation for the reverse-bias voltage. Now the chemical potential on the n side is lowered by eV, the diffusion current is

$$|I_{\text{diffusion}}| = C\mathrm{e}^{((\mu - eV) - E_\mathrm{g} - e\Delta U)/k_\mathrm{B}T} \tag{7.29}$$

and the resulting current is

$$I = I_{\text{diffusion}} - I_{\text{drift}} = I_0\left(\mathrm{e}^{-eV/k_\mathrm{B}T} - 1\right). \tag{7.30}$$

For this polarity of the external voltage, the current is always small because it can never exceed I_0, which is about 100 nA. The resulting current–voltage curve of the pn junction is plotted in Figure 7.12. It shows the expected "valve" or rectifying behavior for a diode.

7.4.2 Transistors

Transistors form essential building blocks for microelectronic devices because they can act as electrical amplifiers and switches. Here, there is not even space for a superficial discussion of the various transistor types and applications. We just pick out one specific type of transistor, the metal oxide field effect transistor (MOSFET), in order to give you the flavor of transistor physics.

The MOSFET is typically a part of an integrated circuit such as a computer memory or processor chip. A sketch of one transistor is shown in Figure 7.13a. The transistor is built onto a p-doped substrate. It consists of two n-doped regions called source and drain and an oxide region (usually SiO_2) between them on the top of the substrate. The top of the oxide, the so-called gate, is contacted with a metal electrode and so are source and drain.

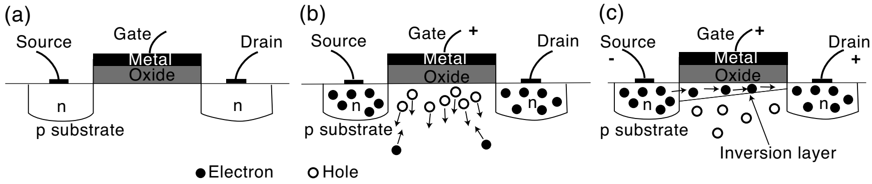

Figure 7.13 Design and working principle of a MOSFET transistor: (a) without applied voltage; (b) with a small positive gate voltage; (c) with a gate voltage large enough to generate an inversion layer and an applied voltage between source and drain.

The switching behavior of the MOSFET is shown in Figure 7.13b and c. The basic idea is to use the gate voltage in order to switch the current from the source to the drain. If no voltage is applied to the gate, no appreciable current of electrons can flow from the source to the drain, independent of the voltage between them. This is because one of the two pn junctions in the MOSFET is always biased in the reverse direction. If, however, a positive voltage is applied, two things happen: The majority holes in the p-type material are repelled from the oxide layer, and the minority electrons are attracted. If the voltage at the gate is larger than a threshold voltage, the minority electrons in a small channel below the oxide layer actually become the majority carriers, and an effective conduction is possible.

We can see how this switching is possible in Figure 7.14. For no applied voltage, the chemical potential inside the semiconductor is equal to the Fermi level of the gate metal. When a positive voltage is applied, the electric field caused by this bends the semiconductor bands close to the interface. For a sufficiently large gate voltage, the band bending is so strong that the conduction band gets very close to the chemical potential, even though the material is p doped. Therefore, many electrons in the conduction band become populated and one speaks of an inversion layer. The inversion layer in a MOSFET does not have a uniform thickness (see Figure 7.13) because, in addition to the field induced by the gate, the field between the drain and source electrodes has to be taken into account.

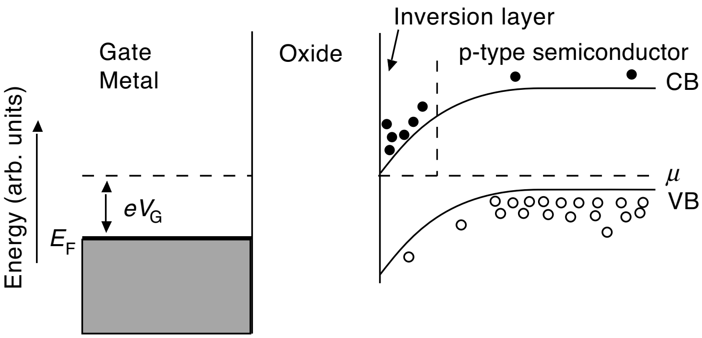

Figure 7.14 Generation of an inversion layer in the MOSFET. The positive gate voltage leads to a band bending that is strong enough to turn electrons from being minority carriers to being majority carriers.

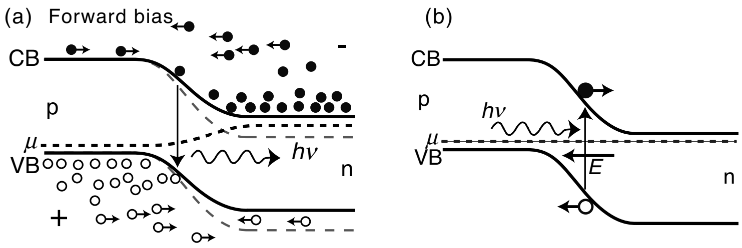

Figure 7.15 Optoelectronic devices. (a) A light emitting diode works because of carrier recombination. (b) A photodetector or solar cell is based on carrier separation in the depletion zone.

7.4.3 Optoelectronic Devices

As for transistors, we can only give the main ideas for some devices. There are two important classes: devices that turn electrical power into light and those that turn light into an electrical voltage. Their principles of operation are closely related and sketched in Figure 7.15.

The light emitting diode shown in Figure 7.15a is essentially a normal diode operated in forward bias. The current is transported by transferring majority electrons from the n side to the p side and majority holes from the p side to the nside. Once these carriers have reached the depletion zone or the other side, there is a high probability for them to recombine with the other type of carrier that is in the majority. When this happens, the energy of the recombination process can be emitted as light. What is essential in a light emitting diode is that the device material and design favor the recombination by light emission and that the gap energy, which is the same as the energy of the emitted photons, is chosen at the desired optical frequency.

Solar cells (Figure 7.15a) work in the opposite way. The setup is also a diode. Photons of the correct energy, which hit the device, can generate electron–hole pairs. If an electron–hole pair is created in the depletion region, the build-in electric field sweeps out the electron to the n side and the hole to the p side. This gives rise to a voltage difference that can be used to drive a current.

One of the most important technological problems associated with optoelectronic devices is that Si, the most important semiconductor by far, is not a good material to build them. For the generation of an electron–hole pair by a photon, both energy and momentum have to be conserved. The energy conservation can be taken care of by choosing photons with the energy of the gap, that is, $h\nu = E_g$. The momentum conservation means that the generated electron must have the same k as the hole state (plus or minus a reciprocal lattice vector) plus the momentum of the photon. The wave vector of a photon is $k_{h\nu} = \nu/2\pi c$, where c is the speed of light. For optical photons, $k_{h\nu}$ is negligibly small. Therefore, the electron and the hole have to have the same k, that is, the transition has to be vertical in k space. An inspection of the band structure of Si in Figure 6.11 shows that such a transition is not possible between the

VBM and CBM because they lie at different values of **k**. Si is said to have an indirect bandgap. But this is not the case for all semiconductor materials. The band structure of GaAs, also shown in Figure 6.11, reveals that this material has a direct bandgap with the CBM and the VBM at the same **k**, and it is therefore well suited for optoelectronic applications.

7.5 Discussion and Problems

7.5.1 Discussion

1. What is the difference between a metal, a semiconductor, and an insulator?
2. How does the resistance of a semiconductor (typically) change as a function of temperature and why?
3. In an intrinsic semiconductor, the chemical potential lies in the middle of the gap at low temperatures. Why?
4. Explain the difference between "electrons" and "holes."
5. How many mobile electrons and holes do you have as a function of temperature for an intrinsic semiconductor?
6. What is the meaning of the effective mass and how can it be related to the electronic band dispersion?
7. If you have a semiconductor with heavy holes and light electrons in the VBM and CBM, respectively, can you draw the bands schematically?
8. How can you measure the effective mass?
9. How can you measure the concentration of electrons or holes?
10. What effect have so-called acceptor or donor atoms in a semiconductor?
11. Why does a very small number of donor or acceptor atoms in a semiconductor have a big impact on the concentration of free carriers?
12. The interaction between a donor (acceptor) ion and its extra electron (hole) can be described in a way very similar to the Bohr model for the hydrogen atom but the binding energy is much smaller and the radius much bigger. Why?
13. Where does the chemical potential in an n type semiconductor lie at low and high temperatures?
14. Explain why a pn junction works as a rectifier.
15. Explain why GaAs is a more appropriate material for optoelectronic applications than Si.

7.5.2 Problems

1. *Non-degenerate semiconductors*: Derive the simplified expressions for the Fermi function (7.7) and (7.8), assuming that the chemical potential is situated approximately in the middle of the gap.

2. *Intrinsic semiconductors*: The chemical potential in an intrinsic semiconductor is given by (7.15). Explain qualitatively why it depends on the effective masses in this way.

3. *The chemical potential in metals*: In Chapter 6, we have simply identified the chemical potential μ in a metal with the Fermi energy E_F and stated that this is a good approximation for all temperatures, that is, that the chemical potential is independent of the temperature. Use the charge-neutrality arguments introduced in this chapter to show that this is indeed a justified approximation.

4. *Doped semiconductors*: Consider a non-ionized phosphorus donor atom in a Si crystal. (a) What is the "Bohr radius" of the resulting "atom"? (b) How many Si atoms are contained within this "Bohr radius"? (c) Estimate how high the concentration of impurities would have to be for the "Bohr radii" to overlap and what would you expect to happen in this case?

5. *Doped semiconductors*: When deriving the law of mass action (7.13), we did not make any assumption about the doping and the position of the chemical potential, and (7.13) is therefore also valid for doped semiconductors. In the case of a n-doped semiconductor, the electron concentration is much higher than that in the intrinsic case, and (7.13) therefore implies that the hole concentration is much smaller. Use a simple physical argument to explain why one should expect this.

6. *Doped semiconductors*: (a) Calculate the effective density of states $N_{\mathrm{eff}}^{\mathrm{C}}$ for the conduction band of silicon at $T = 150$ K and $T = 300$ K. (b) In the intermediate temperature (extrinsic) case of Figure 7.7a, it appears that almost all the electrons in the donor atoms have been excited into the conduction band. On the other hand, the chemical potential in this situation must be close to the middle of the gap, which means that the Fermi–Dirac distribution is small at the donor levels but even smaller in the conduction band. So, why are the electrons in the conduction band and not in the donor levels? (*Hint*: Keep in mind that the doping concentration is of the order $10^{20}\,\mathrm{m}^{-3}$).

7. *Doped semiconductors*: It is technically very difficult to produce semiconductor crystals showing truly intrinsic behavior. For the same reason, it is hard to fabricate a truly n- or p-doped semiconductor, and in general both types of dopant atoms will be present (although the concentration of one will generally be much higher). Show that in the case of both n and p doping, the Hall coefficient is given by

$$R_{\mathrm{H}} = \frac{E_{\mathrm{H}}}{Bj_x} = \frac{p\mu_{\mathrm{h}}^2 - n\mu_{\mathrm{e}}^2}{e(p\mu_{\mathrm{h}} + n\mu_{\mathrm{e}})^2}. \tag{7.31}$$

8. *Conductivity of semiconductors*: The electrical conductivity of a semiconductor is given by $\sigma = e(n\mu_{\mathrm{e}} + p\mu_{\mathrm{h}})$. The crucial difference between a metal and a semiconductor is that the carrier concentration for the former can increase very strongly as the temperature is raised whereas it is constant for the latter. In a very simple picture, this leads to the conclusion that semiconductors become

more conductive at higher temperature whereas metals become less conductive. The decrease in the conductivity of a metal is due to a stronger scattering of the electrons from phonons. This process is also present for semiconductors but unimportant compared to the increased number of carriers. But is this really always the case, in particular, for doped semiconductors? Consider for example the carrier concentration in Figure 7.7 around $T = 300$ K. Speculate how the conductivity would change around this temperature.

9. *Optical properties*: For an indirect gap semiconductor such as silicon, the difference in wave vector between the VBM and the CBM is in the order of $\delta k = \pi/a$, where a is the lattice constant of the material. (a) Explain why this is so. (b) Estimate δk for Si, where the lattice constant is 0.543 nm. (c) For an optical transition to take place, the photon needs to have at least the gap energy (1.11 eV for Si) and it has to have a wave vector correspondingto δk. What is the actual size of the wave vector of 1.11 eV photons? For which type of electromagnetic radiation does the modulus of the wave vector become comparable to the reciprocal lattice distances in a solid?

10. *The pn junction*: Solve the Poisson equation (7.20) to calculate the potential for the pn junction and the absolute potential difference (7.22) across the junction.

8
Magnetism

In this chapter, we are concerned with the magnetic behavior of solids. We can divide this into two categories. We shall first discuss how solids react to an external magnetic field. For most solids, these effects are quite weak and can largely be understood by the properties of the atoms making up the solids. The far more interesting case is that of spontaneous magnetic ordering in the absence of an applied magnetic field. Obviously, this is a genuine property of the solid and cannot be derived from atomic properties. Understanding magnetic ordering is quite complicated and can only be treated superficially here. One problem is that the electrons that give rise to magnetic ordering are often fairly localized on the ions, but they also show some itinerant behavior and they have to interact with the moments on the neighboring ions. It is therefore neither a good approximation to treat them as fully localized nor as completely delocalized. Another difficulty is that it is no longer a good approximation to consider one electron as it moves in a mean potential of all the other electrons because the magnetic moments of electrons on neighboring atoms are important and this cannot easily captured by a mean potential.

8.1
Macroscopic Description

Before we start to explore the magnetic properties of solids, it is quite useful to review the basics of magnetostatics in solids. In general, we assume that Gauss' law for magnetostatics

$$\oint \mathbf{B}\,d\mathbf{A} = 0, \qquad \operatorname{div} \mathbf{B} = 0 \tag{8.1}$$

is obeyed, that is, the sources of the magnetic induction **B** are magnetic dipoles, not monopoles. In vacuum, the magnetic induction is related to the magnetic field **H** by

$$\mathbf{B} = \mu_0 \mathbf{H}, \tag{8.2}$$

with the permeability of vacuum μ_0 ($\mu_0 = 4\pi \times 10^{-7}\,\mathrm{VsA}^{-1} = 4\pi \times 10^{-7}\,\mathrm{T}^2\mathrm{m}^3\mathrm{J}^{-1}$). Note that the SI unit of **B** is $\mathrm{T} = \mathrm{kg\,s}^{-2}\,\mathrm{A}^{-1}$ and that one tesla is a rather strong

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magnetic field. The Earth's magnetic field is typically in the order of 5×10^{-5} T, and strong magnetic fields in medical magnetic resonance scanners are only a few tesla.

In matter, we have

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mathbf{B}_0 + \mu_0\mathbf{M}, \tag{8.3}$$

where $\mathbf{M}$ is the macroscopic magnetization of the solid, that is, the density of magnetic dipole moments μ,

$$\mathbf{M} = \mu\frac{N}{V}. \tag{8.4}$$

The unit of $\mathbf{M}$ is J T^{-1} m^{-3}. For our purpose it is useful to regard $\mathbf{B}_0 = \mu_0 \mathbf{H}$ as the "external field." The unit of this field is T, like for $\mathbf{B}$. In many (but not all!) cases, there is a linear relation between the external field and the magnetization:

$$\mu_0\mathbf{M} = \chi_\mathrm{m}\mathbf{B}_0, \tag{8.5}$$

where χ_m is called the magnetic susceptibility.[1)] If the susceptibility is negative, the solid is called diamagnetic, and if it is positive, the solid is called paramagnetic. Instead of using the susceptibility, one can describe the magnetic properties of matter by the relative permeability $\mu = 1 + \chi_\mathrm{m}$. We do not use μ in this chapter but be aware of the possible confusion that can arise because the relative permeability and the magnetic moments are both denoted by μ.

Like for an electric dipole in an electric field, the potential energy U of a fixed magnetic dipole μ in a field $\mathbf{B}_0$ is $U = -\mu \cdot \mathbf{B}_0$. We want to know the potential energy of a solid of volume V in an external field, taking into account that the magnetization itself depends on the field. When the field is turned on from zero to B_0 and we assume the linear relation (8.5), we get

$$U = -V\int_0^{B_0} M\mathrm{d}B_0' = -V\int_0^{B_0} \frac{\chi_\mathrm{m}}{\mu_0} B_0'\mathrm{d}B_0' = -V\frac{\chi_\mathrm{m}}{2\mu_0} B_0^2. \tag{8.6}$$

For a paramagnetic solid, this corresponds to an energy decrease. Paramagnetic solids therefore experience a force toward locations of higher magnetic fields, that is, they are attracted to either pole of a permanent magnet. For diamagnetic solids, the opposite is true; they are expelled from regions of high magnetic fields. As we shall see later, χ_m is usually very small, so that these effects are not noticeable when you play with a permanent magnet and diamagnetic or paramagnetic solids.

1) Note that χ_m is dimensionless here. In the literature, other units for χ_m can be found, depending on the definition of M as magnetization per volume, mass, or mol of substance. Under some circumstances, χ_m is also defined as $\mu_0\partial\mathbf{M}/\partial\mathbf{B}_0$.

8.2
Magnetic Effects in Atoms

The magnetic effects in atoms do not really belong to a course of solid-state physics. However, since they are essential for the behavior of diamagnetic and paramagnetic solids, we review them very briefly anyway. For more detailed explanations, the reader is referred to the literature about quantum mechanics or atomic physics.

Paramagnetism in atoms is caused by the alignment of existing magnetic moments in an external field. What is the magnetic moment of an atom? In a hydrogen atom, the orbital angular momentum of a state is described by the quantum number l and its value is $\hbar\sqrt{l(l+1)}$. The possible values of l, in turn, depend on the principal quantum number n and take the values from 0 to $(n-1)$. The magnetic moment corresponding to l is μ, which has a modulus of

$$|\mu| = \frac{e\hbar\sqrt{l(l+1)}}{2m_e} = \sqrt{l(l+1)}\mu_B \tag{8.7}$$

with the Bohr magneton $\mu_B = 9.2741 \times 10^{-24}\,\mathrm{J\,T^{-1}} = 5.7884 \times 10^{-5}\,\mathrm{eV\,T^{-1}}$. This magnetic moment precesses around the direction of an applied field as shown in Figure 8.1a. The component of the magnetic moment μ_l in the direction of the field (or the angle between field and magnetic moment) is quantized and given by the magnetic quantum number m_l that takes $2l+1$ values $-l, \ldots, 0, \ldots l$:

$$\mu_l = -\frac{e\hbar m_l}{2m_e} = -m_l\mu_B. \tag{8.8}$$

This is illustrated for $l=2$ in Figure 8.1b. Finally, the electron has a spin quantum number $s = 1/2$, leading to magnetic moments in the field direction of approximately $-\mu_B$, $+\mu_B$. The hydrogen atom in the ground state has $n=1$ and $l=0$, and therefore only the spin magnetic moment matters.

What about the more complex atoms with many electrons? The problem is greatly simplified by the observation that for a filled shell, that is, for a set of n, l, which is

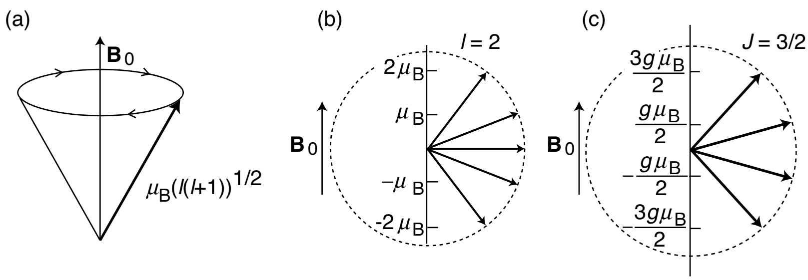

Figure 8.1 (a) Precession of an atomic magnetic moment in an external field. (b) Possible orientations for the magnetic moment in field direction for hydrogen with $l=2$. (c) Possible orientations for the magnetic moment in field direction for Cr^{3+}.

completely occupied, the total orbital magnetic moment is zero because the components in field directions and opposite to the field direction are equally strong. The same is true for the total spin magnetic moment because there are equally many electrons with spin $+1/2$ and spin $-1/2$. So, we only have to worry about non-filled shells. For these, we have to proceed in two steps. First, we have to find the total angular momentum that is described by the quantum number J and can have values of $\hbar\sqrt{J(J+1)}$. Then, we have to calculate the magnetic moments associated with J. Similarly as for the case of the orbital magnetic moment, there are $2J+1$ possibilities for the orientation of the angular momentum with respect to a magnetic field with magnetic moments of

$$\mu_J = -g m_J \mu_B, \tag{8.9}$$

where g is the so-called Landé splitting factor and m_J is the magnetic quantum number belonging to J. The total angular momentum of the electrons can be calculated from a vector sum of the spin and orbital moments. For light atoms (with weak spin–orbit coupling), these are independent and the so-called $L-S$ coupling scheme can be applied. In this, one obtains the quantum numbers for the total orbital and spin moments by

$$L = \sum m_l, \qquad S = \sum m_s. \tag{8.10}$$

Again, we can see that L and S are zero for filled shells. The actual calculation procedure is outlined by Hund's rules:

1. The spins of the electrons are arranged such that the maximum value of S consistent with the Pauli principle is achieved.
2. With the given S, the quantum numbers m_l have to be chosen such that the maximum value of L is achieved.
3. J in the ground state is now calculated as $J = L - S$ when the shell is less than half full, as $J = L + S$ if the shell is more than half full, and as $L = 0$, $J = S$ if the shell is half full.

The first rule arises from an effect similar to the exchange interaction, which we have encountered for the hydrogen molecule in the first chapter: The energy of a many-electron system is lower when the spins are parallel because this means that the spacial wave function is antisymmetric; the electrons keep out of each others way and this reduces the Coulomb interaction. The origin of the second rule is a bit less obvious, but it is qualitatively related to the fact that the energy is lowered when many electrons "rotate in the same direction." The third rule is related to spin–orbit coupling. The rules can be justified by both experiment and theory and shall not be discussed in further detail here.

As an example, consider the ion Cr^{3+} that has the electronic configuration [Ar]$3d^3$. The first of Hund's rules requires $S = 3/2$. The possible m_l values for the 3d shell are $-2, -1, 0, 1, 2$. Hund's second rule requires the largest possible value of L, that is, we have to choose $m_l = 0, 1, 2$ and therefore $L = 3$. Finally, Hund's third rule states that for less than half-filled bands, $J = L - S = 3 - 3/2 = 3/2$. The magnetic quantum number m_J therefore takes the values $-3/2$, $-1/2$, $1/2/$, $3/2$.[2)]

For the calculation of the possible magnetic moments, we only lack the Landé splitting factor that is given by

$$g = \frac{3J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}. \tag{8.11}$$

The resulting possible orientations of the magnetic moment for Cr^{3+} are shown in Figure 8.1c.

The bottom line is that the paramagnetic behavior of atoms depends on the presence of permanent magnetic moments. We have seen that such moments might be present if the atoms have open shells. The diamagnetic behavior is different. It is always there because it is caused by the electrons in the atoms and their reaction to the magnetic field. However, the diamagnetism of atoms and solids is usually very weak. We sketch its origin by first-order perturbation theory for an atom in a weak magnetic field.

Before we can do this we have to discuss how the Schrödinger equation of a particle changes in the presence of an electromagnetic field. To describe this in a convenient way, we need the concept of the so-called vector potential, which you may not be familiar with. The key idea is as follows. In electrostatics, the electric field $\mathbf{E}(\mathbf{r})$ can be generated by means of a potential $\phi(\mathbf{r})$ such that $\mathbf{E}(\mathbf{r}) = -\mathrm{grad}\,\phi(\mathbf{r})$. The introduction of this potential greatly simplifies many calculations as we only have to calculate the (scalar) potential instead of the (vectorial) field. Due to the non-existence of magnetic monopoles, it is impossible to define a similar scalar potential for the magnetic field, but one can define a so-called vector potential such that

$$\mathbf{B} = \mathrm{curl}\,\mathbf{A}. \tag{8.12}$$

As we shall see in a moment and later in Chapter 10, the introduction of $\mathbf{A}$ greatly simplifies the coupling of an external electromagnetic field into the Schrödinger equation. Using $\mathbf{A}$ also simplifies the notation in many other situations, for example, for obtaining the wave equation of the electromagnetic field from the Maxwell equations.

Having the vector potential $\mathbf{A}(\mathbf{r})$ and the scalar potential $\phi(\mathbf{r})$, the rules for coupling an external electromagnetic field to the Schrödinger equation are quite simple: The scalar potential obviously only acts as some addition to the potential that is already present, so we have to multiply it with the charge q of the particle described by the Schrödinger equation and add it to the Hamiltonian. The magnetic field is included by substituting the momentum operator $\mathbf{p} = -i\hbar\nabla$ by

$$\mathbf{p} \to \mathbf{p} - q\mathbf{A}. \tag{8.13}$$

2) Note that even for partially filled shells one can obtain $J = 0$ if $S = L$. Equation 8.9 would therefore suggest that this does not lead to any magnetic moment. This is not entirely true but we do not treat this case here.

We now come back to the original problem to find out how a weak external magnetic field changes the energy of an electron in an atom. Let us say that we have a magnetic field of strength B_0 only in the z direction, that is, $\mathbf{B}_0 = (0, 0, B_0)$. A vector potential generating this field is

$$\mathbf{A} = -\frac{1}{2}\mathbf{r} \times \mathbf{B}_0, \tag{8.14}$$

which is easily verified by an explicit calculation of curl $\mathbf{A}$. We know that the vector potential only affects the kinetic energy term of the electron, so we do not have to bother with the potential energy here. The original kinetic energy part of the Hamiltonian is now modified such that

$$H_{\text{kin}} \to H'_{\text{kin}},$$

$$\frac{\mathbf{p}^2}{2m_e} \to \frac{1}{2m_e}(\mathbf{p} + e\mathbf{A})^2 = \frac{1}{2m_e}\left(\mathbf{p} - e\frac{\mathbf{r} \times \mathbf{B}_0}{2}\right)^2. \tag{8.15}$$

Evaluating this expression gives

$$H'_{\text{kin}} = \frac{1}{2m_e}\left(\mathbf{p}^2 + e\mathbf{B}_0 \cdot (\mathbf{r} \times \mathbf{p}) + \frac{e^2}{4}(\mathbf{r} \times \mathbf{B}_0)^2\right), \tag{8.16}$$

where we have used that $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = -\mathbf{c} \cdot (\mathbf{b} \times \mathbf{a})$. Now we exploit that $\mathbf{B}_0$ has only a component in the z direction, that is, $\mathbf{B}_0 = (0, 0, B_0)$.

$$H'_{\text{kin}} = H_{\text{kin}} + H' = \frac{\mathbf{p}^2}{2m_e} + \frac{e}{2m_e}B_0(\mathbf{r} \times \mathbf{p})_z + \frac{e^2}{8m_e}B_0^2(x^2 + y^2). \tag{8.17}$$

The first term in this expression is the usual kinetic energy of the electron H_{kin}. The second and the third term represent the perturbation caused by the magnetic field. The energy change due to the perturbation is $E' = \langle\psi|H'|\psi\rangle$:

$$E' = \frac{e}{2m_e}B_0\langle\psi|(\mathbf{r} \times \mathbf{p})_z|\psi\rangle + \frac{e^2}{8m_e}B_0^2\langle\psi|(x^2 + y^2)|\psi\rangle. \tag{8.18}$$

The first term contains the angular momentum of the electron in the z direction. This is again the paramagnetism, caused by the angular momentum that is already present. It is just the orbital angular momentum, not the total angular momentum, because the spin does not appear in the non-relativistic Schrödinger equation. We disregard this term in the following and concentrate on the second part that represents the desired diamagnetism. According to (8.6), we can calculate the magnetic moment of the electron in the field as

$$\mu = -\frac{\partial E'}{\partial B_0} = -\frac{e^2}{4m_e}B_0\langle\psi|(x^2 + y^2)|\psi\rangle. \tag{8.19}$$

The minus sign is consistent with the expected diamagnetism. In order to estimate the magnitude of the magnetic moment, we now introduce some approximations: For a spherically symmetric electron distribution, the mean square distance for an electron from the nucleus is $r^2 = x^2 + y^2 + z^2$, hence $x^2 + y^2 = (2/3)r^2$ and we take r to be the atomic radius r_a. In addition to this, an atom contains not only one electron but Z electrons that we all take, somewhat crudely, to have a spherical probability distribution of radius r_a. With this, we arrive at an expression for the diamagnetic moment of an atom, which is

$$\mu = -\frac{Ze^2}{6m_e} r_a^2 B_0. \tag{8.20}$$

8.3 Weak Magnetism in Solids

As we shall see below, the magnetic effects in solids tend to be quite weak except for the cases of magnetic ordering and superconductivity (see Chapter 10). Here, we discuss the sources of this weak magnetism and find them directly related to the results for atoms in the previous section.

What do we expect for the magnetism in solids related to that in atoms? The diamagnetism discussed above arises from all electrons in the atom, the valence electrons and the core electrons. Therefore, it is not going to change much as we form the solid. In fact, we can view the solid simply as a dense cloud of atoms and calculate the diamagnetic susceptibility for this cloud.

The situation is more difficult in the case of paramagnetism. Even though many atoms with open outer shells should have a non-zero J and therefore a permanent magnetic moment, not so many solids actually show this behavior. It appears that the magnetic moment disappears upon the formation of the solid. The reason for this is particularly easy to understand for ionic solids. Although the contributing atoms generally have an atomic magnetic moment, the ionic solid does not, because it basically consists only of ions with closed shells. A similar situation is found for covalent bonds. This is intuitively clear from our discussion of the H_2 molecule in Chapter 1. Although both individual hydrogen atoms have a net spin of 1/2, the molecular ground state has zero spin and it does not have any magnetic moment either.

For having a "good" paramagnetic solid, we need atoms with open shells, which do not participate in the bonding and therefore do not change their properties much upon the formation of a solid. Possible candidates could be the 3d and 4d transition metals but in these the d electrons still participate in the bonding to a large extent. The best examples for quasiatomic paramagnetism in solids are therefore found in compounds of the 4f rare earth elements because the 4f electrons are very localized indeed.

8.3.1
Diamagnetism

8.3.1.1 Diamagnetism of the Ions

The diamagnetic susceptibility of a solid can be estimated directly from (8.20):

$$\chi_m = \mu_0 \frac{M}{B_0} = -\frac{\mu_0 Z N e^2}{6 V m_e} r_a^2 \tag{8.21}$$

and this is always very small, 10^{-5} or so, much smaller than 1, that is, the magnetization in the sample is much weaker than the external magnetic field. Since it is purely an atomic effect, it is also independent of the temperature.

8.3.1.2 Diamagnetism of Free Electrons

Apart from the atomic diamagnetism, it should be noted that free electrons also show a diamagnetic contribution to the susceptibility in a quantum mechanical picture. This contribution is given here without further derivation. It is

$$\chi_m = -\frac{1}{3V} \mu_B^2 \mu_0 g(E_F) \left(\frac{m_e}{m^*}\right)^2. \tag{8.22}$$

The main ingredients of this contribution are quite intuitive. First of all, there is the ubiquitous density of states at the Fermi level, which stems from the fact that only the electrons close to the Fermi level can respond to a magnetic field (or perform any other low-energy excitation, for that matter). In addition, the susceptibility depends on the ratio of electron mass and the effective mass. The lighter the effective mass, the stronger the diamagnetic contribution. In total, the diamagnetic contribution of the free electrons is very small, in the same order as the diamagnetic susceptibility of the ions.

8.3.2
Paramagnetism

We treat two contributions to paramagnetism. One is the alignment of magnetic moments of the atoms. The other stems from the free electrons. We will see that the first is, when present at all, usually much stronger than the second and the diamagnetic response, and it is therefore the dominant contribution to the magnetic properties. The second is of the same order as the diamagnetic contribution of free electrons, but it is easier to understand and the discussion is useful for our later description of spontaneous magnetic ordering.

8.3.2.1 Curie Paramagnetism

Calculating the susceptibility for a system of independent, distinguishable, localized magnetic moments is a standard example in the subject of statistical physics and details can be found in the literature. We only state the results here.

We know that the possible energy levels of an ion in an external magnetic field are given by $g\mu_B m_J B_0$ (see Figure 8.1c). Having these, it is straightforward to write down the partition function

$$Z = \sum_{m_J=-J}^{J} e^{-g\mu_B m_J B_0/k_B T}. \tag{8.23}$$

From this, one can calculate the mean energy of the magnetic moments and therefore also the mean moment in the field direction, which is

$$\bar{\mu}_C = \frac{1}{Z}\sum_{m_J=-J}^{J} g\mu_B m_J e^{g\mu_B m_J B_0/k_B T}. \tag{8.24}$$

From $\bar{\mu}_C$, one can calculate the susceptibility $\bar{\mu}_C \mu_0 N/V B_0$. Although the details of this are quite involved for a system with more than two states, there are two important limiting cases.

The first case, which is also the most important one, is $g\mu_B B_0 \ll k_B T$. Then, the susceptibility is found to be inversely proportional to the temperature with

$$\chi_C = \frac{C}{T}. \tag{8.25}$$

This is known as Curie's law. The Curie constant C is given by

$$C = \frac{\mu_0 N g^2 \mu_B^2 J(J+1)}{3 V k_B}. \tag{8.26}$$

The other limiting case is $g\mu_B B_0 \gg k_B T$, which can only be achieved at very low temperatures. In this case, all the magnetic moments are aligned as much as possible parallel to the field and the magnetization saturates. Figure 8.2 shows the complete dependence of the magnetization as a function of $g\mu_B B_0/k_B T$. The limit of Curie's law is indicated as a dashed line.

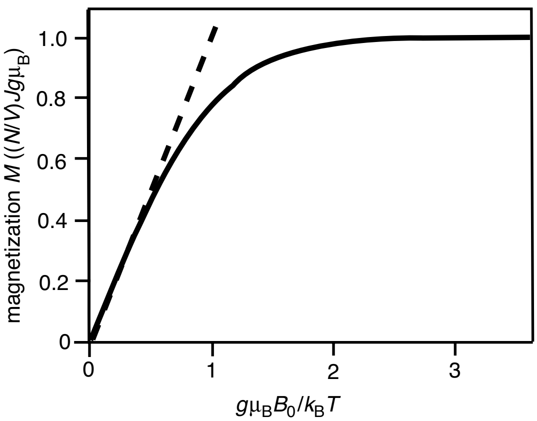

Figure 8.2 Paramagnetic susceptibility of a solid with localized magnetic moments. The limit of Curie's law is indicated as a dashed line.

The calculated values of the Curie constant C compare very well with the experimental data from solids containing rare earth ions, as expected. The comparison is much less favorable for the 3d transition metal compounds (see Problem 8.2). We have already addressed this issue above: The 3d electrons participate in the bonding and the states have a different character than the atomic orbitals assumed in the calculation of C. In fact, the potential in which these electrons move is far from the spherical potential that is assumed for atoms and much more dictated by the crystal symmetry. For the elements of the iron group, one observes values of C suggesting that the 3d electrons have $J = S$, that is, only spin and no orbital angular momentum at all. This effect is known as the quenching of the orbital angular momentum. If present, it allows us to think about the magnetization as the alignment of spin moments only.

The Curie paramagnetism is much stronger than the diamagnetism discussed previously, but it is still weak. Typical values of χ_m at room temperature are in the order of 10^{-3}–10^{-2}. Note that χ_m is temperature dependent, in contrast to the diamagnetic susceptibility discussed above. Actually, one might wonder why Curie paramagnetism is included in a section called “Weak Magnetism in Solids” at all. At very low temperatures and high magnetic fields, it can be a strong effect. The reason for describing it as “weak” is that it is so under “normal” condition, as opposed to spontaneous magnetic ordering that is discussed below.

8.3.2.2 **Pauli Paramagnetism**

Free electrons also exhibit paramagnetic behavior. If every free electron has a spin of 1/2 and a magnetic moment of μ_B, one could expect that they contribute to the saturation magnetization of the solid with μ_B times the density of the electrons. This saturation is achieved when all the magnetic moments align parallel to the field (or the spins align antiparallel to the field).[3] But this is not the case at all and the paramagnetic susceptibility of free electrons is actually very small indeed. This has been very puzzling for a long time.

The paramagnetic susceptibility of free electrons can be understood and even be calculated using the simple picture given in Figure 8.3. In Figure 8.3a, the density of states for the free electrons (6.13) is divided into two parts: one with the orientation of the magnetic moments parallel to an external field B_0 and one with the orientation antiparallel. This external field is assumed to be essentially zero. Figure 8.3b shows what happens when the field is increased to the value B_0. The electrons raise or reduce their energy by $\mu_B B_0$, depending on the orientation of their magnetic moments with respect to the field. Since μ_B is so small, this energy change is tiny for any achievable field, only 10^{-5} eV or so, much smaller than the distance from the bottom of the band to the Fermi level. Once this shift happens, the electrons that have moved above the Fermi level can lower their energy by flipping their spin and becoming electrons with a magnetic moment parallel

3) Here and in the following, we speak loosely of spins aligning parallel or antiparallel to a field or to each other but of course they do not. Only the z component of the spin can align with the field. The actual spin precesses around the field direction.

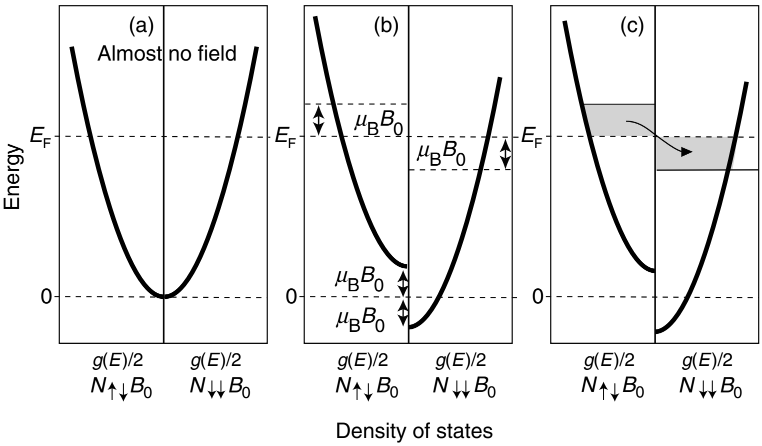

Figure 8.3 (a) Density of states for free electrons, split up into electrons having their magnetic moment parallel or antiparallel to an external field, but the field is almost zero. (b) When the field is increased to the value B_0, the energy of the electrons is increased or reduced by $\mu_B B_0$ depending on the orientation of their magnetic moments. (c) The electrons with a magnetic moment antiparallel to the field can go to a lower energy state by flipping their spin. In this way, a constant Fermi level is maintained. Note that the energy shift induced by B_0 is not drawn to scale.

to the field, as shown in Figure 8.3c. This gives rise to more electrons with a magnetic moment parallel than antiparallel to the field, that is, to paramagnetic magnetization.

In order to calculate χ_m, we must figure out the number of electrons that move over to the side with the magnetic moment parallel to the field. The moving electrons are represented by the gray areas on either side of the vertical axis in Figure 8.3c. The size of each area is $g(E_F)\mu_B B_0/2$. Therefore, we have a difference between electrons with their magnetic moment parallel and antiparallel to the field, which is

$$N_{\downarrow\downarrow B_0} - N_{\downarrow\uparrow B_0} = g(E_F)\mu_B B_0, \tag{8.27}$$

where the arrows denote whether the moments and the field are parallel or antiparallel. The net magnetization is

$$M = \frac{1}{V}(N_{\downarrow\downarrow B_0} - N_{\downarrow\uparrow B_0})\mu_B = \frac{1}{V} g(E_F)\mu_B^2 B_0, \tag{8.28}$$

and the susceptibility becomes

$$\chi_m = \frac{1}{V}\mu_0 \mu_B^2 g(E_F), \tag{8.29}$$

which is small, in the same order as the diamagnetic susceptibilities. We can of course see why the small size was hard to understand from a semiclassical point of view. As in the case of the electronic heat capacity, it has to do with the Fermi–Dirac statistics. The field required to align all the spins would not only have to be such that $\mu_B B_0 \gg kT$ but also such that $\mu_B B_0 > E_F$, which is huge.

8.4
Magnetic Ordering

So far, we have studied the diamagnetic and the paramagnetic behavior of solids. Neither of them leads to magnetic effects of appreciable size. In this section, we look at a much more spectacular phenomenon: long-range magnetic ordering without any applied field. The detailed understanding of this is very difficult and we will not attempt it. We just concentrate on some main ideas behind the mechanism of magnetic ordering.

Different types of magnetic ordering are shown in Figure 8.4. The ordering you are probably most familiar with is the ferromagnetic type that is observed for the iron group. It is also found for the rare earth elements gadolinium and dysprosium. Ferromagnetic ordering originates from the parallel alignment of the magnetic moments in the crystal. It gives rise to a macroscopically observable magnetization. A rather different case is antiferromagnetic ordering, which also entails a long-range ordering of magnetic moments, but the orientation of the moments on neighboring sites is opposite, such that no net magnetization is observed. Many transition metal oxide insulators show antiferromagnetic ordering. A mixture between the two cases is ferrimagnetic ordering where there is antiferromagnetic ordering between moments of different sizes in one unit cell, but ferromagnetic ordering between the unit cells, such that a net magnetization remains. An example for a material showing ferrimagnetic ordering is magnetite (Fe_3O_4).

How can we know that there is a phenomenon such as antiferromagnetism when it does not produce an observable macroscopic field? As discussed in Chapter 2, one can determine the microscopic magnetic ordering by neutron diffraction. Whereas X-rays "see" only the structure of the material, neutrons carry a magnetic moment and are therefore sensitive to the magnetic structure. The difference between geometric and magnetic structure is pronounced because the magnetic unit cell is bigger than the geometric unit cell (see Problem 8.8).

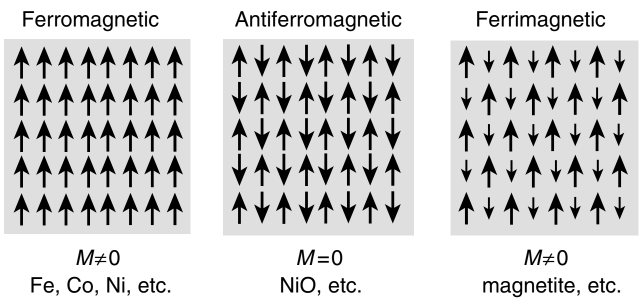

Figure 8.4 Types of magnetic ordering. The arrows denote direction and size of the localized magnetic moments.

8.4.1
Magnetic Ordering and the Exchange Interaction

Now we try to understand, at least qualitatively, where the magnetic ordering comes from. It is obviously related to some interaction between the magnetic moments. This is holding the moments aligned, despite the disordering effects of external magnetic fields and entropy (temperature). Knowing that ferromagnetic ordering exists even far above room temperature, we see that the energies needed to destroy the ordering are at least of the order of $k_B T$ at room temperature, that is, 25 meV or more.

Before we describe what causes the ordering, it is a good idea to mention a mechanism that is *not* causing it. This is the direct dipole–dipole interaction of the localized magnetic moments. It is a common misconception that the "little magnets" in the solid align themselves like an array of compass needles due to their magnetic interaction but in fact they do not. We can easily see this by estimating the size of the magnetic dipole–dipole interaction. It is very small and the energy difference corresponding to parallel and antiparallel alignment of two magnetic dipoles at typical atomic distances corresponds to temperatures in the order of 1 K (see Problem 8.4).

If the magnetic interaction is not the cause for alignment, where does the alignment come from? The responsible interaction is the exchange interaction, a funny type of force that stems from a combination of Coulomb interaction and the Pauli principle, that is, the need to have antisymmetric wave functions for fermions. We have encountered the exchange interaction already in our discussion of the hydrogen molecule in Chapter 1 where we have seen that the difference between the singlet and the triplet states is approximately twice the value of the exchange energy X or

$$E_{\uparrow\uparrow} - E_{\uparrow\downarrow} = -2X. \qquad (8.30)$$

For the hydrogen molecule, the exchange energy X is always negative (see Figure 1.2), which means that the triplet state has a higher energy than the singlet state and the ground-state ordering is therefore "antiferromagnetic." An inspection of Figure 1.2 shows that the exchange energy X (or the separation between the singlet and triplet state) is by no means small. Even for large distances, it amounts to a substantial fraction of an electron volt.

This already gives the basic ingredient for the coupling in solids as well. The magnetic dipole moments order because of the exchange interaction that favors either a parallel (positive X) or an antiparallel (negative X) alignment. The exchange energy is typically smaller than that in the hydrogen molecule but it is still of the order of 100 meV for the ferromagnetic elements. It is, however, very difficult to make qualitative predictions about the size or even the sign of X, as the following considerations illustrate.

In contrast to what we see in the hydrogen molecule, a positive exchange energy X could be expected for a multi-electron system on very general grounds. The reason is the Pauli principle: For two electrons, the triplet wave function (1.4) vanishes when the electrons have the same spacial coordinates, meaning that they will never be at the same time at same place. This reduces their mutual Coulomb repulsion, leading to an energy lowering compared to the singlet state and thus to a positive exchange energy. The most prominent example for this general idea is Hund's first rule that states that the electron states have to be occupied such that the highest possible value of S is realized. We have discussed the example of Cr^{3+}, which has a partially filled d shell with three electrons in it. The electrons are placed into different sub-shells to achieve the highest possible S, which also means that the electrons are kept apart from each other and the total potential energy is lowered. The same principle applies to free electrons in a metal (see Problem 8.5). Electrons with the same spin direction will not be at the same place. If there is a majority of spin-up electrons, each of these electrons feels the presence of the other electrons less and it is attracted more strongly to the ions in the crystal. This leads to an energy gain.

However, the situation is not that simple because other energy considerations apply as well, even if we neglect the disordering effect of finite temperature. For a free electron gas, a complete spin polarization would lower the Coulomb repulsion between the electrons, but it would increase the kinetic energy by a very large amount (in the order of the Fermi energy), as we have already seen in our discussion of Pauli paramagnetism. In fact, spontaneous magnetization is never observed for free electron-like metals. For the hydrogen molecule, not only the kinetic energy plays a role, but also the decrease of potential energy for the electrons that are attracted by two nuclei instead of one.

For a system of localized magnetic moments that interact via the exchange energy, the energy change of the solid due to magnetization was formulated by W. Heisenberg. For simplicity, we assume that the orbital magnetic momentum of the states under consideration is quenched and we only deal with spins. The energy of a spin S_i in the presence of all the other spins is then written as

$$E = -\mathbf{S}_i \cdot \sum_j 2X_{ij}\mathbf{S}_j, \tag{8.31}$$

where j runs over all the other atoms in the solid and X_{ij} is the exchange interaction between the spins $\mathbf{S}_i$ and $\mathbf{S}_j$. A quantitative treatment of this is very difficult even if further assumptions can be made, such as a restriction of the interactions to nearest neighbors. Nevertheless, (8.31) is a starting point for many modern theories of magnetism.

A condition for the Heisenberg model to be applicable is that we are in fact dealing with moments localized on the lattice sites, and this is by no means clear. For the 4f transition metals Gd and Dy, it is a good approximation because the 4f electrons are strongly localized to the metal ions. Indeed, if we divide the highest magnetization of a Gd or Dy solid by the number of its ions, the magnetic moment per ion is very close to the expected value from (8.9). This does not work for the elements of the 3d group, such as Fe, and the problem cannot even be cured by assuming that the orbital magnetic moment is completely quenched.

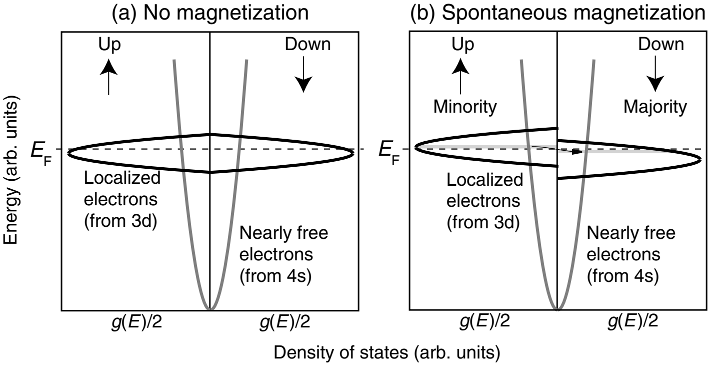

Figure 8.5 (a) Density of states in a transition metal, separated in two spin directions but without a net magnetization. (b) In the case of a spontaneous magnetization the electrons with one spin direction (here the "down" direction) will be in the majority. This is described by a relative shift of the d bands for different spin directions.

The 3d electron wave functions are much more spread out over the crystal than those of the 4f electrons. These states cannot therefore be treated as localized moments on the ions, but their itinerant character has to be taken into account. A description based on this idea goes back to E. C. Stoner and E. P. Wohlfarth. Figure 8.5a shows a sketch of the density of states in a transition metal. The s (and sometimes also p) electrons form the familiar free electron like band. The density of states looks quite different for the d electrons. The band only exists in a small energy range, its density of states is relatively high, and it is almost centered on the Fermi level. This can be understood by considering the character of the d electrons: The band is narrow in energy because the localized nature of the d states gives rise to a smaller overlap of the wave functions and a smaller splitting in energy (see Figure 6.1). The density of states is high because the narrow band has to accommodate 10 d states per atom in the crystal. Finally, it is almost centered on the Fermi level because it is just over half full. It is occupied by 6 out of 10 d electrons for Fe.

Figure 8.5b shows what happens when many of the electrons in the d band change their spin direction from "up" to "down." In this case, it will be more likely for every single electron that the electrons in the vicinity also have a "down" spin and this will lower its energy. The "down" spin direction is then called the majority spin and the "up" spin the minority spin. In the case of fully magnetized Fe, there are about twice as many electrons in the majority spin band than in the minority spin band. The energy of the shift between the two bands has to be in the order of the exchange energy. The tricky part, however, is that the energy gain when an electron changes its spin depends on how many other electrons are already in the minority and majority states.

A formal starting point for the description of magnetism in a band model is to assume that we have two bands of electronic (d-like) states with an identical dispersion $E(\mathbf{k})$ but split by a $\mathbf{k}$-independent magnetic interaction I.

$$E_{\uparrow}(\mathbf{k}) = E(\mathbf{k}) - I\frac{(n_{\uparrow}-n_{\downarrow})}{N},$$
$$E_{\downarrow}(\mathbf{k}) = E(\mathbf{k}) + I\frac{(n_{\uparrow}-n_{\downarrow})}{N}, \tag{8.32}$$

where $n_{\uparrow}$ and $n_{\downarrow}$ are the number of electrons with up and down spin, respectively, and N is the total number of electrons in the band. Unlike the bands we have encountered previously, the spin-resolved bands $E_{\uparrow}(\mathbf{k})$ and $E_{\downarrow}(\mathbf{k})$ can only accommodate one electron per $\mathbf{k}$ point, not two. Note that the magnetization of the sample is proportional to the relative occupation

$$R = \frac{n_{\uparrow}-n_{\downarrow}}{N}, \tag{8.33}$$

which appears in the above expression. In order to obtain R, we have to calculate the number of electrons in the $E_{\uparrow}(\mathbf{k})$ and $E_{\downarrow}(\mathbf{k})$ bands. These are given by

$$\begin{aligned} n_{\uparrow} &= Nf(E_{\uparrow}(\mathbf{k}), T), \\ n_{\downarrow} &= Nf(E_{\downarrow}(\mathbf{k}), T), \end{aligned} \tag{8.34}$$

where $f(E, T)$ is the Fermi–Dirac distribution. From these, we can calculate

$$\begin{aligned} R &= \frac{1}{N}(n_{\uparrow}-n_{\downarrow}) = \sum_{\mathbf{k}} (f(E_{\uparrow}(\mathbf{k}), T) - f(E_{\downarrow}(\mathbf{k}), T)) \\ &= \sum_{\mathbf{k}} \left(\frac{1}{e^{(E_{\uparrow}(\mathbf{k}) - IR - E_F)/k_B T} + 1} - \frac{1}{e^{(E_{\downarrow}(\mathbf{k}) + IR - E_F)/k_B T} + 1} \right). \end{aligned} \tag{8.35}$$

We must now seek the values of R for which these equations can be fulfilled. The trivial case is obviously $R = 0$, that is, a non-magnetized sample. However, there may also be combinations of I and T such that (8.35) is solved for non-zero values of R. An example will be considered in Problem 8.7.

At low temperatures, when the Fermi–Dirac distribution has a sharp edge, it is clear from Figure 8.5 why the ferromagnetic state is favored: Instead of having two sharp density of states that peak almost at the Fermi energy, one of the peaks has moved below E_F and the other above E_F. This means that many electrons that have been at E_F in the non-magnetic state have lowered their energy by an amount in the order of the exchange energy. This is a considerable energy gain and therefore the ordered state is favored. It appears obvious, and it is also confirmed by a more detailed treatment, that the total energy gain is proportional to both the exchange energy and the density of states at the Fermi level.

8.4.2
Temperature Dependence of the Ordering

If we raise the temperature sufficiently, $k_B T$ will eventually become so high that it is comparable with the exchange interaction energy that creates the magnetic ordering.

In the band picture of Figure 8.5b, the Fermi distribution would not have a sharp cutoff at E_F, but so broad that its width becomes comparable to the splitting between the majority and the minority band. In this case, there is no appreciable difference in occupation and the ordering gets lost. The temperature at which this happens is called the Curie temperature for a ferromagnet and the Néel temperature for an antiferromagnet.

Above the Curie temperature Θ_C, a ferromagnetic material behaves like a paramagnet and obeys the so-called Curie–Weiss law for the susceptibility:

$$\chi_m = \frac{C}{T-\Theta_C}. \tag{8.36}$$

Note that this is very similar to Curie's law (8.25), only the origin is shifted by the Curie temperature.

It is also interesting to consider the degree of magnetic ordering in a ferromagnet below the Curie temperature. A typical magnetization is shown in Figure 8.6. Due to the statistical nature of the alignment, that is, the competition between entropy effects and the exchange interaction strength, one could expect the transition to be rather gradual. But it is found that quite the opposite is true. The transition temperature is sharply defined and ferromagnetic ordering sets in swiftly as the material is cooled below Θ_C. This can be understood if we imagine what happens as we approach Θ_C from higher temperatures. At some point, very localized ordering sets in. This gives rise to an internal magnetic field that leads to further ordering and in this way a self-amplified process is triggered. A sharp transition is already predicted by the band model of ferromagnetism in its most rudimentary form (see Problem 8.7), and the formal reason for this is that a non-zero relative occupation of the bands R lowers the energy of the majority band and therefore favors an even higher value of R, such that a self-amplifying transition sets in. The detailed temperature dependence of the magnetization near the transition temperature is, however, not correctly reproduced by the simple form of the band model.

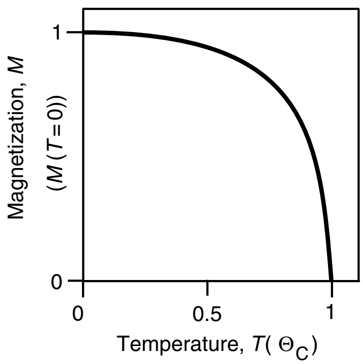

Figure 8.6 Magnetization of a ferromagnetic material below the Curie temperature Θ_C.

8.4.3 Ferromagnetic Domains

Not all ferromagnetic materials appear to show an external magnetic field below the Curie temperature. Sometimes it is necessary to "magnetize" them by an external magnetic field. The reason for this is the existence of magnetic domains, as suggested by P. Weiss in 1907 and shown in Figure 8.7a. The domains have different magnetization directions such that the total average magnetization of the sample is zero or small. The domains are separated by so-called Bloch walls in which the magnetization rotates from one direction to another. The Bloch walls are typically around 50–100 nm thick. Magnetic domains can be made visible by very fine iron powder on the magnet (the so-called Bitter method), or optically by the so-called Kerr effect or directly by spin-polarized scanning tunneling microscopy.

The existence of domains can be understood by considering what happens when the material is cooled below the Curie temperature. Ferromagnetic ordering sets in spontaneously at different places in the sample and the Bloch walls are found where the so-formed domains meet. There is, however, another way of viewing this. Consider the single-domain magnet in Figure 8.8a. It leads to a strong magnetic field outside the material with a certain energy density. By introducing a few domains as in Figure 8.8b, the external field is strongly reduced, leading to an energy gain, but the cost for this is the formation energy of the domain walls. If the latter is not too high, the state with a few domains is favorable. If now an external field is applied, the sample can be magnetized by moving the domain walls relative to each other, as in Figure 8.8c.

8.4.4 Hysteresis

By moving the domain walls in a piece of ferromagnetic material using an external field B_0, different magnetizations can be achieved. Figure 8.9 shows the situation for an initially unmagnetized sample in an oscillating field B_0. In the beginning, the magnetization M is zero and it increases as the field is turned on. For a certain field strength, the sample is completely magnetized in one direction, that is, the saturation

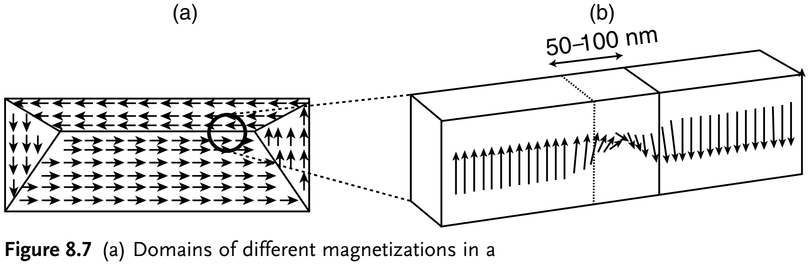

Figure 8.7 (a) Domains of different magnetizations in a ferromagnetic solid. (b) Detailed picture of the magnetization rotation in a Bloch wall between two domains.

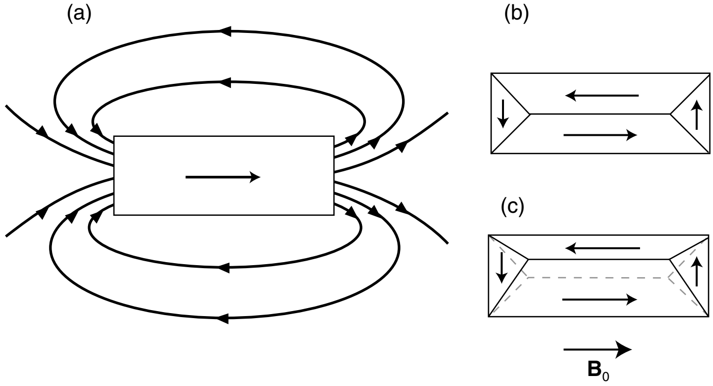

Figure 8.8 (a) **B** field inside and outside a piece of magnetic material (schematically). (a) Material with a single domain leading to a strong external field. (b) The introduction of a few domains greatly reduces the external field. (c) The material can be magnetized by the movement of domain walls caused by an external field $\mathbf{B}_0$. The dashed gray lines correspond to the situation in (b) before the exposure to $\mathbf{B}_0$.

magnetization M_S is reached. When the field is lowered again, the magnetization decreases, but for $B_0 = 0$ it has not reached zero but the so-called remanent magnetization M_R. A magnetization of $M = 0$ is first reached at the coercive field B_C in opposite direction. For an even stronger field in the opposite direction, saturation is reached again.

The cause for this hysteresis is partly that the movement of Bloch walls through the sample is not a simple reversible process. Sometimes the Bloch wall has to pass defects and this costs energy. The energy dissipation for one closed loop of the

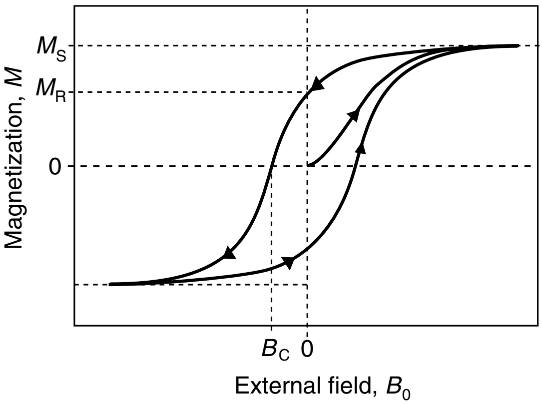

Figure 8.9 Magnetization of a ferromagnetic sample as a function of externally applied field B_0. The starting point of the curve is the origin.

hysteresis curve can be read directly from the curve if it is displayed in a slightly different way as $B(H)$ instead of $M(B_0)$. Then the dissipation is simply $\oint B\,\mathrm{d}H$, that is, the area enclosed by the hysteresis curve. The $B(H)$ curve looks quite similar to the $M(B_0)$ curve, but it does not show saturation in the same sense because B still increases as H increases even if M is saturated (see (8.3)).

Hysteresis can also be obtained in a simple model for a defect-free crystal. Consider a one-domain ferromagnet at a very low temperature. All the spins are aligned with an external magnetic field. The alignment will not be lost when the field is turned to zero. In fact, a field in the opposite direction with considerable strength is required to reverse the magnetization because all the spins have to be flipped over. Once this is achieved, the external field can again be turned to zero without further changes in the magnetization.

The exact shape of the hysteresis curve depends strongly on the type and structure of the material. It can be tailored to meet the needs of specific applications. If the goal is to have a good permanent magnet, both a large remanent magnetization and a high coercive field are desirable. These properties characterize so-called hard magnets. One can play certain tricks to achieve them. For example, one can make the grain size of the material smaller than the typical size of a magnetic domain. Then the grains cannot change their magnetization by moving domain walls. They are forced to flip over as a whole, an expensive process.

In the opposite extreme, one needs so-called soft magnets, for example, for applications in transformers. From what we have seen above, the energy dissipated in each magnetization circle is the area enclosed by the hysteresis loop. In a typical transformer, this energy is lost 50 (or 60) times a second, so the area should be small. This means that both a small coercive field and a small remanent magnetization are desirable. At the same time, one is looking for a high saturation field and a material with a high resistance in order to minimize eddy currents.

8.5 Discussion and Problems

8.5.1 Discussion

1. What is the difference between diamagnetism and paramagnetism? What are the basic physical reasons for both? How do diamagnetic/paramagnetic materials react when placed into an inhomogeneous magnetic field?
2. Can you give an example where the relation between the external field and magnetization is not linear, that is, where it deviates from (8.5)?
3. Describe the most important magnetic properties of atoms.
4. How do the nearly free electrons in metals contribute to the magnetic properties?
5. How would a gas of classical electrons contribute to the magnetic properties of a solid?
6. What solids are likely to express Curie-type paramagnetism?

7. In a Curie-type paramagnet, all the ions with magnetic moments contribute to the total susceptibility, but in a metal only very few nearly free electrons contribute to Pauli paramagnetism. Why?
8. The Curie formula for the paramagnetic susceptibility $\chi_C = C/T$ is only valid under certain conditions. What are these conditions and why is this so?
9. What interaction is responsible for the magnetic ordering in ferromagnets and antiferromagnets and how strong is it?
10. Most ferromagnetic chunks of material do not show any macroscopic magnetization? Why?
11. When a chunk of ferromagnetic material is magnetized in an external field and then the field is switched off, why does it keep a macroscopic magnetization?
12. What is the difference between a magnetically soft and a magnetically hard material?
13. How does a ferromagnet behave magnetically above its Curie temperature?

8.5.2
Problems

1. *Diamagnetic susceptibility of atoms:* (a) Give an upper estimate for the (dia) magnetic moment of an atom. Use an atom with many electrons such as Bi. From the magnetic moment, calculate the diamagnetic susceptibility and compare to the experimental value, which is $\chi_m = -1.7 \times 10^{-4}$. (b) Calculate the magnetic moment for a hydrogen atom in an external field exactly, starting from (8.18). You can use that the ground-state wave function of hydrogen is

$$\psi(r, \theta, \phi) = (\pi a_0^3)^{-1/2} e^{-r/a_0}. \tag{8.37}$$

You can also use that

$$\int_0^\infty x^n e^{-x} dx = n! \tag{8.38}$$

2. *Curie paramagnetism:* One often compares not the measured and calculated Curie constants but rather the so-called effective magneton number p where $p^2 = g^2 J(J+1)$. (a) Calculate p for the following ions and compare your result to the experimental value given in brackets: Nd^{3+} with three 4f electrons (3.5), Gd^{3+} with seven 4f electrons (8.0), Cr^{2+} with four 3d electrons (4.9), and Fe^{2+} with six 3d electrons (5.4). (b) Can the agreement for the 3d transition metals be improved by assuming that the orbital angular momentum is quenched, that is, that there is just the total spin?

3. *Ferromagnetic ordering:* For the ferromagnetic elements Gd^{3+} and Fe^{2+}, calculate the highest possible magnetization per atom and compare it to the experimental values taken at $\approx$0 K, which are $7.6\mu_B$ and $2.2\mu_B$, respectively. Discuss the results.

4. *Ferromagnetic ordering:* We have argued that ferromagnetic ordering is caused by the exchange interaction. It is a common misconception that it is caused by the

alignment of spins due to their mutual magnetic interaction. Can you disprove this? Estimate the magnetic interaction energy for two spins separated by a typical atomic distance. (*Hint:* If you want to estimate the magnetic field due to a microscopic moment, you can use an expression for the on-axis field of a circular wire loop and assume that the radius of the loop is very small). For which temperatures would you expect ordering due to this effect?

5. *Ferromagnetic ordering:* Consider two free electrons in a box of volume V. The Hamiltonian is

$$H = -\frac{\hbar^2\nabla_1^2}{2m_e} - \frac{\hbar^2\nabla_2^2}{2m_e} + \frac{e^2}{4\pi\varepsilon_0}\frac{1}{\mathbf{r}_1 - \mathbf{r}_2}. \tag{8.39}$$

It simplifies things greatly, and it also appears reasonable for a macroscopic box, to neglect the electron–electron interaction, such that

$$H = -\frac{\hbar^2\nabla_1^2}{2m_e} - \frac{\hbar^2\nabla_2^2}{2m_e}. \tag{8.40}$$

As in the case of the hydrogen molecule, we can take the two-electron wave function as a product of the single-electron wave functions (6.4). This, however, does not take into account the Pauli principle. In order to do this, we have to make the spacial part of the two-electron wave functions symmetric and antisymmetric for spin singlet and spin triplet states, respectively. We get

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}V}\left\{e^{i\mathbf{k}_1\cdot\mathbf{r}_1}e^{i\mathbf{k}_2\cdot\mathbf{r}_2} \pm e^{i\mathbf{k}_1\cdot r_2}e^{i\mathbf{k}_2\cdot r_1}\right\}. \tag{8.41}$$

(a) Show that these two-electron wave functions do indeed solve the Schrödinger equation with the Hamiltonian given by (8.52). What is the corresponding energy? (b) Concentrate on the triplet state and calculate the probability for the electron 1 to be found in the volume element $d\mathbf{r}_1$ and the electron 2 in the volume element $d\mathbf{r}_2$. Interpret the result.

6. *Ferromagnetic ordering:* What would be the qualitative difference between the density of states for a 3d transition metal shown in Figure 8.5a and a 4f transition metal?

7. *Ferromagnetic ordering:* Consider a further simplification of the band model for ferromagnetism shown in Figure 8.5. We assume that the bands given by (8.32) are so narrow that we can neglect their $\mathbf{k}$ dependence altogether and write

$$\begin{aligned} E_\uparrow &= E' - I\frac{n_\uparrow - n_\downarrow}{N} = E' - IR, \\ E_\downarrow &= E' + I\frac{n_\uparrow - n_\downarrow}{N} = E' + IR. \end{aligned} \tag{8.42}$$

Derive an expression for the relative occupation R and solve this expression graphically for different combinations of magnetic interaction I and temperature T. Illustrate that the magnetization of your system sets in rather abruptly as it is

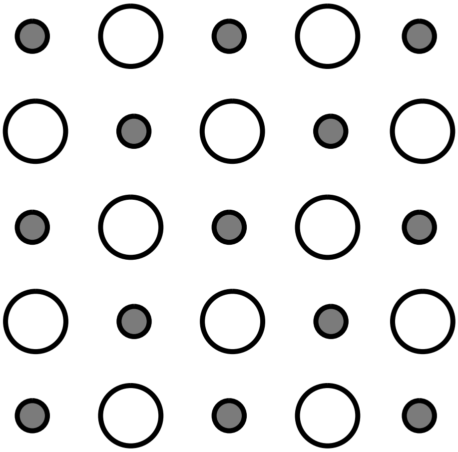

Figure 8.10 A two-dimensional version of NiO. The large circles represent Ni, and the small ones O.

cooled below the Curie temperature. For further simplicity, assume that the energy $E' = E_F$.

8. *Antiferromagnetic ordering:* NiO is antiferromagnetic with a Néel temperature of approximately 500 K. The magnetic moments are localized on the Ni ions only. A two-dimensional version of NiO is shown in Figure 8.10. (a) How would the magnetic moments be oriented in the antiferromagnetic state? What would the unit cell and the reciprocal lattice look like, when the magnetic ordering is taken into account and when it is not taken into account? (b) What experimental technique would reveal the magnetic ordering directly (and why) and thus allow you to study the transition between the ordered and non-ordered state.

9
Dielectrics

In the previous chapter, we have discussed the response of a solid to a magnetic field. In this one, we do the same thing for an electric field. We already know what an electric field does to metals: it causes a current to flow. Here, we are mainly concerned with insulators. In this way, the present chapter can also be seen as an extension of our discussion of materials (metals, semiconductors, and now insulators). At first glance, one could suspect that nothing interesting happens when an insulator is exposed to an electric field but this is not quite true. We shall discuss important phenomena such as dielectric polarization of materials in capacitors and the piezoelectric effect, but also phenomena at higher frequencies such as the interaction of electromagnetic waves with matter.

The formal description is in strong analogy with that of magnetism and many phenomena can be described using the same ideas. But there are also some important differences. The interaction of magnetic fields with solids is almost always very weak (except in ferromagnetism) but for the electric field this is not so. This has some advantages and disadvantages for the formal treatment. For example, the interaction of matter with electromagnetic waves is greatly simplified because we can almost always neglectthe magnetic part of the interaction. On the other hand, the calculation of the electric field inside an insulator becomes difficult because we have to take into account not only the external field but also the field created by the polarized solid itself.

9.1
Macroscopic Description

The macroscopic description of dielectric effects is similar to that for magnetism but it is not quite the same. Again we start by a few formal definitions. An electric field $\mathbf{E}$ leads to a dielectric polarization $\mathbf{P}$ of the solid of the form

$$\mathbf{P} = \chi_e \varepsilon_0 \mathbf{E}, \tag{9.1}$$

where χ_e is the electric susceptibility and $\varepsilon_0 = 8.854187817 \times 10^{-12}\,\mathrm{C^2\,J^{-1}\,m^{-1}}$ is the vacuum permittivity. In vacuum, there is nothing to be polarized so $\chi_e = 0$.

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The microscopic mechanism for the polarization will be discussed in the next section. It is somewhat similar to the case of magnetism. It mainly stems from the alignment of microscopic electrical dipoles that are either already there or induced by the field. We write microscopic electric dipole moments as $\mathbf{p} = q\boldsymbol{\delta}$, where q is the magnitude of the charges and $\boldsymbol{\delta}$ the separation vector. As usual for electric dipoles, this vector is defined as pointing from the negative to the positive charge. The polarization can then be written as

$$\mathbf{P} = \frac{N}{V}\mathbf{p} = \frac{N}{V}q\boldsymbol{\delta}. \tag{9.2}$$

From this, it is evident that $\mathbf{P}$ has the dimension of a surface charge density.

A material constant closely related to χ_e is the relative permittivity or dielectric constant ε, which is given by

$$\chi_e = \varepsilon - 1. \tag{9.3}$$

Defined in this way, both χ_e and ε are dimensionless. When dealing with the dielectric properties of solids, it is much more common to use ε rather than χ_e. This is different from the case of magnetic materials where mostly the susceptibility χ_m is used, rather than the relative permeability μ.

As in the case of magnetism, the linear relation (9.1) represents a limit and is only correct for small fields. The trouble is that the electric interaction with matter is not necessarily weak and non-linear effects are often encountered when using strong electric fields from laser light. This is a very interesting research field but we do not discuss it any further here.

A standard example for the electric polarization is the plane plate capacitor shown in Figure 9.1. For the empty capacitor in Figure 9.1a, Gauss' law can be used to figure out that the electric field between the plates of size A and distance d has the constant value $|\mathbf{E}| = \sigma/\varepsilon_0$, where σ is the surface charge density on the plates. With this it immediately follows that the capacitance is $C = A\varepsilon_0/d$. When a dielectric material is placed between the plates, it is locally polarized leading to macroscopic polarization $\mathbf{P}$. We can imagine this as arising from a very high density of microscopically small

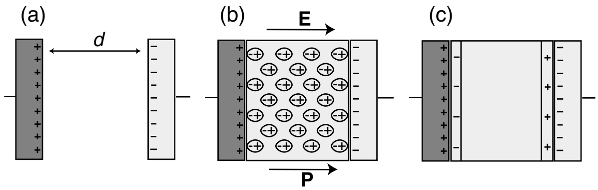

Figure 9.1 A plane plate capacitor. (a) Charges on the plates of the capacitor with no dielectric present between the plates. (b) Polarization of the dielectric material between the plates. (c) The net effect of the polarization is surface charge densities of the dielectric at the plate–dielectric interface.

electric dipoles (see Figure 9.1b). The most important effect of these dipoles is that they lead to a net surface charge density where the dielectric meets the planes. When this is taken into account, the resulting average electric field in the capacitor is $|\mathbf{E}| = \varepsilon_0^{-1}(\sigma - |\mathbf{P}|) = \sigma/\varepsilon\varepsilon_0$, that is, it is reduced by a factor of ε. The capacity, in contrast, is increased by the same factor. Indeed, this is often used to define a value of ε in the first place.

The seemingly simple problem of changing the capacitance of a plane plate capacitor is currently a big challenge in semiconductor device design. Consider the MOSFET in Figure 7.16. The oxide below the gate is in effect the dielectric material in a plane plate capacitor. This capacitor has to be able to store enough charge to make the MOSFET work (without needing too high a gate voltage), that is, it needs a sufficiently large capacitance. The big challenge is now to design ever smaller transistors and here the problem sets in. The capacitance is $C = A\varepsilon\varepsilon_0/d$, where ε is the dielectric constant of SiO_2. For a decrease in size A, C also decreases. This can be compensated by decreasing the thickness of the oxide layer d and this is what the semiconductor industry has been doing for the last 30 years. But now this does not work anymore because d is at its limit (a few nanometers). For an even thinner oxide, the film becomes "leaky" due to tunneling. Therefore, a current field of research is to find a material with a much higher ε than SiO_2 to replace SiO_2. Then one could have the same capacitance with a much thicker gate oxide.

9.2 Microscopic Polarization

There are several mechanisms causing the microscopic electric dipole moments that lead to a macroscopic polarization. They are shown in Figure 9.2. Figure 9.2a illustrates a mechanism that is really an atomic effect and has little to do with the fact that the atoms are placed in a solid. In an electric field, the spherical symmetry of

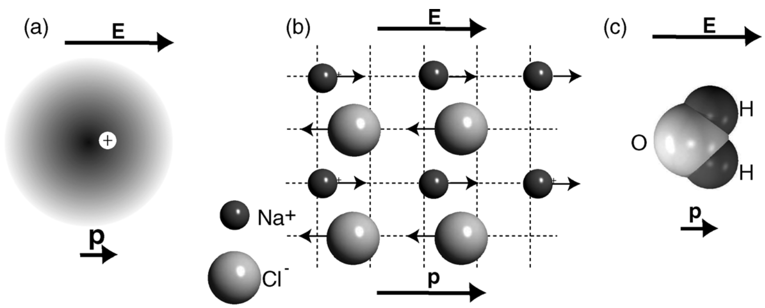

Figure 9.2 Mechanisms leading to microscopic electric polarization. (a) The electric field polarizes all atoms in the solid. (b) In ionic solids, like NaCl, the lattice can be polarized, giving rise to local electric dipoles. The dashed grid gives the position of the ions without an applied field. (c) If there are permanent dipoles in the solid and these are free to rotate, they orient themselves parallel to the field. A molecule with a permanent dipole is, for example, water.

the atom is lifted and the negative and positive charges are displaced, giving rise to an electric dipole

$$\mathbf{p} = \alpha \mathbf{E}, \tag{9.4}$$

where α is the atomic polarizability. This effect is present in all solids and is called electronic polarization. The trouble with α is that it is hard to calculate and therefore it is hard to predict the size of the atomic dipole moments based on the type of atoms in the solid (for a simple calculation see Problem 9.1).

The next polarization mechanism is relevant to ionic solids, where something very similar happens on a larger scale, as shown in Figure 9.2b. In the electric field, the lattice itself gets polarized since the positive ions are displaced in the direction of the external field and the negative ions opposite to the field. This effect is called ionic polarization.

Finally, there is the possibility that already existing dipoles are oriented in the field. Then the sum of these dipole moments does not average out but gives rise to a macroscopic polarization instead. Such permanent dipoles could be molecules such as water or HCl. This polarization mechanism is called orientational polarization. However, it is much more common in liquids or gases than in solids because the dipoles have to be free to move.

With (9.1) and (9.3) in mind, the presence of different polarization mechanisms should give rise to different electric susceptibilities χ_e and dielectric constants ε. Values of ε for a range of materials are listed in Table 9.1. Some trends are immediately clear. Under normal conditions, air has a dielectric constant close to 1 (an electric susceptibility close to 0), simply because the density of air is very small (see (9.2)). The dielectric constant is markedly higher for solids but there are no strong differences between materials with only electronic polarization (diamond) or additional lattice polarization (NaCl). Liquids with polar molecules have a much higher dielectric constant and some solids like barium titanate ($BaTiO_3$) have an even higher dielectric constant. We will discuss these special solids later on.

Table 9.1 Dielectric constants ε of selected materials at room temperature.

Material	Dielectric constant, ε
Vacuum	1
Air	1.000 573 (283 K, 1013 hPa)
Rubber	2.5–3.5
Glass	5–10
Diamond	5.7
Si	11.7
SiO_2	3.9
CdSe	10.2
NaCl	6.1
Ethanol (liquid)	25.8
Water (liquid)	81.1
$BaTiO_3$	100–1200

These microscopic mechanisms for the polarization of the solid are somewhat reminiscent of the mechanisms leading to a magnetization. There is, however, one important difference. All the mechanisms described above lead to a polarization parallel to the external field, that is, to "para-electric" behavior, not to "dia-electric" behavior.

9.3 The Local Field

Suppose that we know the dielectric constant ε of some solid and we want to calculate the microscopic polarizability α of the atoms making up the solid, knowing that there are no other mechanisms of polarization. We can use (9.1)–(9.4) to write

$$\mathbf{P} = (\varepsilon - 1)\varepsilon_0 \mathbf{E} = \frac{N}{V}\mathbf{p} = \frac{N}{V}\alpha\mathbf{E}, \tag{9.5}$$

where $\mathbf{E}$ is the average electric field in the dielectric material, that is, the sum of the external field and the average internal field due to the polarization. In the case of a dielectric material inside a plane plate capacitor, we know this average field. It is the external field reduced by a factor ε. This would give us the desired relation:

$$\alpha = \frac{(\varepsilon - 1)\varepsilon_0 V}{N}. \tag{9.6}$$

Unfortunately, this does not work because a microscopic dipole in the solid does not feel the average field $\mathbf{E}$ in the solid but the local field $\mathbf{E}_{\mathrm{loc}}$ at its site and this can be quite different. We can think of this local field by simply taking out the dipole under consideration and inspect the effect of all the other dipoles in its neighborhood. This is shown in Figure 9.3. It becomes clear that this local effect gives rise to an additional field $\mathbf{E}'$ that is parallel with the average field, that is, the local field felt by every dipole is stronger than the average field in the solid.

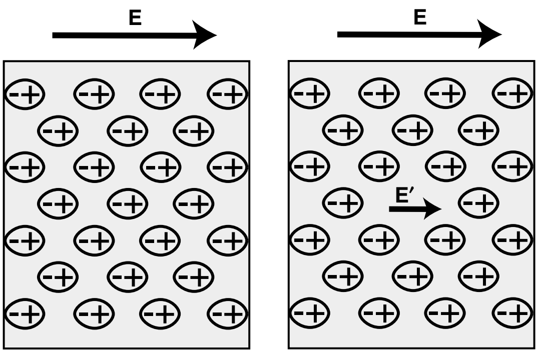

Figure 9.3 The local field on microscopic polarizable units. *Left*: Polarized microscopic dipoles in a dielectric placed in an external field. The electric field **E** is the average internal field in the dielectric. *Right*: The local field felt by every single dipole is not just **E** but $\mathbf{E} + \mathbf{E}'$ because the surrounding dipoles lead to a field increase.

We do not derive the strength of the local field in this approximation here, we merely give the result, which is

$$\mathbf{E}_{\text{loc}} = \frac{1}{3}(\varepsilon + 2)\mathbf{E}, \tag{9.7}$$

so we get

$$\mathbf{P} = \frac{N}{V}\alpha\mathbf{E}_{\text{loc}} = \frac{N\alpha}{3V}(\varepsilon + 2)\mathbf{E}. \tag{9.8}$$

On the other hand, we have (9.1) as an expression for **P** and by setting the two equal we obtain the so-called Clausius–Mossotti relation, which relates the atomic polarizability to the dielectric constant:

$$\alpha = \frac{\varepsilon - 1}{\varepsilon + 2}\frac{3\varepsilon_0 V}{N}. \tag{9.9}$$

Experimentally, the relation can be best tested for gases where the density can be varied.

9.4 Frequency Dependence of the Dielectric Constant

So far, we have only studied electrostatic behavior. Far more interesting is the dynamic behavior for fields with a time dependence, in particular, in the region of optical frequencies. We know from optics that ε is actually a complex number[1] and in the Drude model we have already encountered a considerable frequency dependence (see Chapter 5 and in particular (5.28)). We have seen that this frequency dependence can explain why metals become transparent for light above the plasma frequency ω_P. The frequency dependent $\varepsilon(\omega)$ is usually called the dielectric function.

For insulators, it is found that $\varepsilon(\omega)$ is also complex and frequency dependent and that energy can be resonantly transferred to the solid at some frequencies. The frequency dependence of $\varepsilon(\omega)$ implies a frequency dependence of the refractive index $N(\omega)$ through (5.21) and this effect is well known as dispersion in optical materials such as glass. Most of the frequency dependence can be explained by a simple idea combined with our knowledge about the polarization mechanisms.

In the electrostatic case, all types of microscopic polarization will be important: electronic polarization as well as ionic polarization and orientational polarization (the latter two obviously only if they are possible). If the frequency of the electric field is sufficiently high, neither the lattice nor the orientation of existing dipoles can follow it anymore. Only the electronic polarization remains and therefore the total

1) For nonisotropic materials, ε is not even a scalar quantity but a complex second rank tensor. As everywhere else in this book, we ignore the effect of anisotropy, unless it is strictly needed for a specific phenomenon.

Table 9.2 Dielectric constants ε of selected materials in the electrostatic case and at optical frequencies.

Material	Static ε	Optical ε
Diamond	5.68	5.66
NaCl	6.1	2.34
LiF	11.95	2.78
TiO_2	94	6.8

polarization of the solid decreases greatly. Therefore, $\varepsilon(\omega)$ should be lower at higher frequencies than in the electrostatic case.

In case of the ionic polarization, we have a good idea about the timescale involved. In Chapter 4, we have seen that lattice vibrations have a frequency in the order of 10^{13} Hz. So for optical frequencies (10^{14} Hz), the lattice atoms will not be able to follow the field anymore. This is confirmed by the data in Table 9.2 in which the electrostatic and the optical ε are given for different materials. The static ε is significantly larger than the optical ε, apart from the case of diamond, for which only electronic polarization can play a role. Similar arguments can be made for the orientational polarization but the orientation of dipoles is much slower and therefore turned effectively off at much lower frequencies. As orientational polarization is of little importance for most solids, we will not consider it in the following semi-quantitative treatment.

We can describe the frequency dependence of ε more quantitatively for a simple but instructive model. We have discussed that light can only couple to optical phonons very close to the center of the Brillouin zone at $\mathbf{k} = 0$ (see Problem 4.3). In a simple ionic crystal, these phonons correspond to an out-of-phase vibration of the positive and negative ions in the unit cell, whereas the motion in between unit cells is in phase (see Problem 4.2). We can therefore approximate the vibrations as vibrations of independent harmonic oscillators with one such oscillator per unit cell of the crystal. Each oscillator shall be driven by an electric field[2)] of the form $E \exp(-i\omega t)$ and have a resonance frequency $\omega_0 = (2\gamma/M)^{1/2}$, where γ is the force constant and M the reduced mass of the two ions (see Problem 4.2). We also include a damping term $\eta \mathrm{d}x/\mathrm{d}t$ that is proportional to the velocity, that is, the rate at which the interatomic distance changes. The physical meaning of this is that if this particular motion gets very strong, it is likely to excite other vibrations and thus be damped. The equation of motion is that of a damped and driven harmonic oscillator:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \eta\frac{\mathrm{d}x}{\mathrm{d}t} + \omega_0^2 x = \frac{eE}{M}\mathrm{e}^{-i\omega t}. \tag{9.10}$$

2) The electric field used here should actually be the local field, not the average internal field. This is not important for the conclusions from this simple model but it is important for the phenomenon of ferroelectricity, which is discussed later on.

Note the similarity to the problem of a free electron in an external field where we had the equation of motion (5.23), which has no restoring force and no damping. As in that case, a good ansatz for the solution is

$$x = A\mathrm{e}^{-i\omega t}, \tag{9.11}$$

resulting in an expression for the amplitude A:

$$A = \frac{eE}{M}\frac{1}{\omega_0^2 - \omega^2 - i\eta\omega}. \tag{9.12}$$

It will later be useful to split A into real and imaginary part by expanding the fraction with the complex conjugate of the denominator to give

$$A = \frac{eE}{M}\left(\frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + \eta^2\omega^2} + \frac{i\eta\omega}{(\omega_0^2 - \omega^2)^2 + \eta^2\omega^2}\right). \tag{9.13}$$

The oscillation of the ions will be accompanied by an ionic polarization of $eA\mathrm{e}^{-i\omega t}$ for every unit cell. From this, we can calculate the total polarization for a crystal with N unit cells and volume V. Apart from the ionic polarization we consider the electronic polarization of the ions. This gives

$$P = P_\mathrm{i} + P_\mathrm{e} = \frac{N}{V}eA\mathrm{e}^{-i\omega t} + \frac{N}{V}\alpha E\mathrm{e}^{-i\omega t}. \tag{9.14}$$

For simplicity, we have assumed that there is only one type of ions with a density of N/V and an effective atomic polarizability α. The two different ions in a NaCl crystal and their different polarizability can be taken care of by a suitable definition of α. Now we can calculate the dielectric function

$$\epsilon = \frac{P}{\epsilon_0 E\mathrm{e}^{-i\omega t}} + 1 = \frac{NeA}{V\epsilon_0 E} + \frac{N\alpha}{V\epsilon_0} + 1. \tag{9.15}$$

At sufficiently high frequencies (in the optical limit), we know that $P_\mathrm{i} = 0$ so that

$$\epsilon_\mathrm{opt} = \frac{N\alpha}{V\epsilon_0} + 1, \tag{9.16}$$

and therefore

$$\epsilon(\omega) = \frac{NeA}{V\epsilon_0 E} + \epsilon_\mathrm{opt}. \tag{9.17}$$

Combining this with (9.13), we get the final expression for the dielectric function $\epsilon(\omega) = \epsilon_\mathrm{r}(\omega) + i\epsilon_\mathrm{i}(\omega)$ with the real and imaginary parts:

$$\epsilon_\mathrm{r}(\omega) = \frac{Ne^2}{V\epsilon_0 M}\frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + \eta^2\omega^2} + \epsilon_\mathrm{opt} \tag{9.18}$$

and

$$\epsilon_\mathrm{i}(\omega) = \frac{Ne^2}{V\epsilon_0 M}\frac{\eta\omega}{(\omega_0^2 - \omega^2)^2 + \eta^2\omega^2}. \tag{9.19}$$

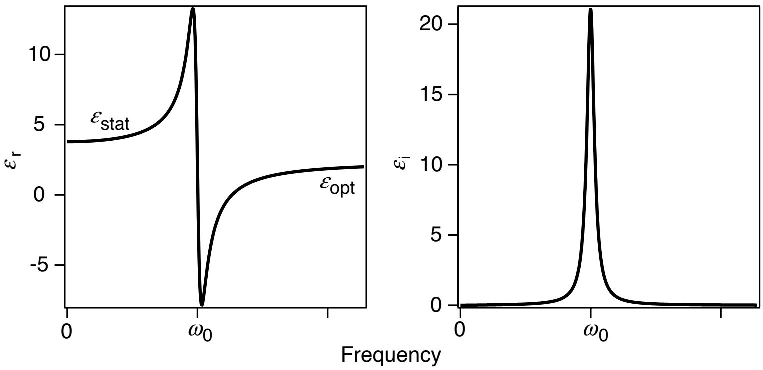

Figure 9.4 Dielectric function for the damped, driven harmonic oscillator close to the resonance frequency ω_0. The left side shows the real part and the right side the imaginary part of ε. ε_{stat} and ε_{opt} stand for the static and optical values of the real part of ε, respectively.

Both parts of $\varepsilon(\omega)$ are plotted in Figure 9.4 in the vicinity of the resonance frequency ω_0. The real part of $\varepsilon(\omega)$ is almost constant much below and much above ω_0 but its value is higher at lower frequencies, consistent with what we have said above. The imaginary part of $\varepsilon(\omega)$ is zero almost everywhere apart from the immediate vicinity of ω_0 where it shows a peak with a width given mainly by η.

What is the meaning of ε_i? To see this, consider the energy dissipation in the system. The instantaneous electrical power dissipated per unit volume is given by

$$P(t) = j(t)E(t), \tag{9.20}$$

where $j(t)$ is the (AC) current density and $E(t)$ the electric field.[3] As usual, we write $E(t) = E\exp(-i\omega t)$. In an insulator, there are no free currents. The only currents are polarization currents so that

$$j(t) = \frac{\partial D}{\partial t} = \frac{\partial}{\partial t}\varepsilon\varepsilon_0 E(t) = \varepsilon_0 E(t)(-i\omega\varepsilon_r + \omega\varepsilon_i). \tag{9.21}$$

The average dissipated power per cycle can now be calculated by

$$P = \frac{1}{T}\int_0^T E(t)j(t)\mathrm{d}t, \tag{9.22}$$

where $T = 2\pi/\omega$ is the period of one oscillation. We can easily see what happens in two limiting cases. If the dielectric function is purely imaginary, $j(t)$ is in phase with $E(t)$, the product of the two is always positive, and the integral will give a non-zero value that turns out to be

$$\frac{1}{2}\varepsilon_0\varepsilon_i E^2. \tag{9.23}$$

3) We work with scalar quantities here because we assume $j(t)$ and $E(t)$ to be in the same direction but not necessarily with the same phase.

If, however, ε is purely real, there will be a phase shift of $\pi/2$ between $E(t)$ and $j(t)$, the integrand oscillates around zero, and the integration gives $P=0$. We therefore see that ε_i measures the degree of power dissipation in the solid. It is obviously highest at the resonance.

So far, we have adopted the point of view that the low-frequency behavior of $\varepsilon(\omega)$ is given by the coupling of the electric field to the vibrations of the solid and that the high-frequency limit is purely due to electronic polarization (see (9.17)). This is, however, an oversimplification. In the visible and UV regime, $\varepsilon(\omega)$ is still not constant but has some pronounced structures for most materials. This is already evident from the different colors materials have. The frequency dependence of $\varepsilon(\omega)$ in the visible and UV region is due to the valence electrons and an understanding requires a proper quantum mechanical treatment of the solid's electronic structure. With what we have learned about the band structure of solids, we can gain a qualitative understanding. Consider the simplified semiconductor band structure in Figure 9.5a. We assume that the chemical potential lies somewhere between the valence band and the conduction band. Absorption of photons can occur when a photon has an energy of at least $h\nu = E_g$. As we have discussed previously, the photons carry very little momentum and therefore they can change the energy of an electron but not its **k** vector (plus or minus a reciprocal lattice vector). Possible photon-induced transitions are indicated by gray arrows in Figure 9.5. These transitions correspond to the absorption of electromagnetic energy by the solid and we have seen that this absorption is mainly given by ε_i. In the spirit of the simple model for the dielectric function we can write that

$$\varepsilon_i(h\nu) \propto \sum_{\mathbf{k}} M^2 \delta(E_C(\mathbf{k}) - E_V(\mathbf{k}) - h\nu), \tag{9.24}$$

where the sum runs over all the permitted **k** values in the first Brillouin zone, M is a matrix element determining the transition probability, and δ stands for the Dirac

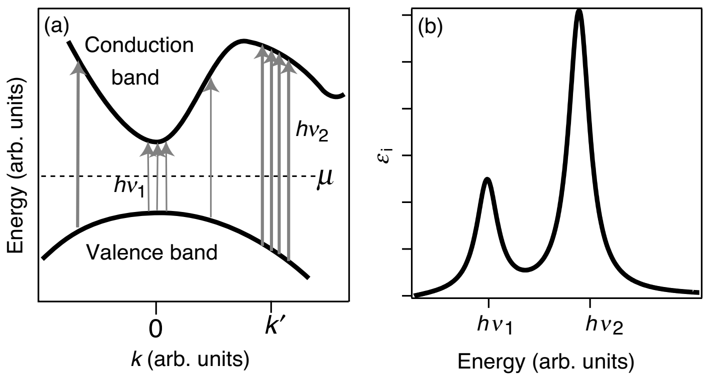

Figure 9.5 (a) Contributions to the imaginary part of ε by transitions between occupied and unoccupied states. The gray lines denote possible transitions. The density of such transitions is high at $\mathbf{k}=0$ and $\mathbf{k}=\mathbf{k}'$ because valence band and conduction band are parallel there. (b) ε_i resulting from this band structure.

delta function. This expression basically counts the possible transitions for a certain $h\nu$ and it will have maxima for energies $h\nu$ for which many different transitions between valence and conduction bands are possible. In Figure 9.5a, this is the case for the onset of the absorption at $h\nu_1 = E_g$ ($\mathbf{k} = 0$) and around $\mathbf{k} = \mathbf{k}'$, where the valence and conduction bands are parallel in some parts of the first Brillouin zone. Maxima in ε_i can thus be expected for $h\nu_1$ and $h\nu_2$, as illustrated in Figure 9.5b.

9.5 Other Effects

9.5.1 Impurities in Dielectrics

A small amount of impurities can have a pronounced effect on the properties of dielectrics, very much like in the case of semiconductors. We give two examples here.

The most "visible" case, as it were, is the change of optical properties that can be caused by impurities. Due to their large gap, most insulators/dielectrics tend to be transparent, for example, NaCl, diamond, and sapphire. If there are impurities in the material with electronic states inside the gap, this can lead to a strong change in the optical properties because now transitions from or into the impurity states are possible, and ε_i will show a resonance at the corresponding energy (see (9.24)). Good examples of such impurity states exist in sapphire. Depending on the type of impurity, sapphire can have many different colors (and names), for example, topaz (yellow), amethyst (purple), ruby (red), emerald (green), or sapphire (blue).

Impurities can also be used to make insulators conductive, exactly like doped semiconductors. Donor or acceptor levels have to lie close to the conduction band and valence band, respectively, in order to give rise to a finite density of electrons and holes and thereby conductivity. The important advantage over a usual semiconductor material is that high-temperature applications are possible. In a narrow-gap semiconductor, high temperatures are a problem because of the exponentially increasing number of intrinsic carriers. In a doped insulator, this is not an issue for practically relevant temperatures.

9.5.2 Ferroelectricity

Ferroelectric materials are solids that exhibit a spontaneous electric dipole moment, very much like ferromagnetic materials exhibit a spontaneous magnetic moment. Otherwise, the term "ferroelectric" is quite misleading because the typical ferroelectric does not contain any iron and the mechanism leading to the polarization can be quite different, too. A typical ferroelectric material is barium titanate ($BaTiO_3$) (Figure 9.6) in which the electric dipole moment stems from a shift of the (negatively charged) oxygen lattice against the positively charged Ba and Ti ions. In an external

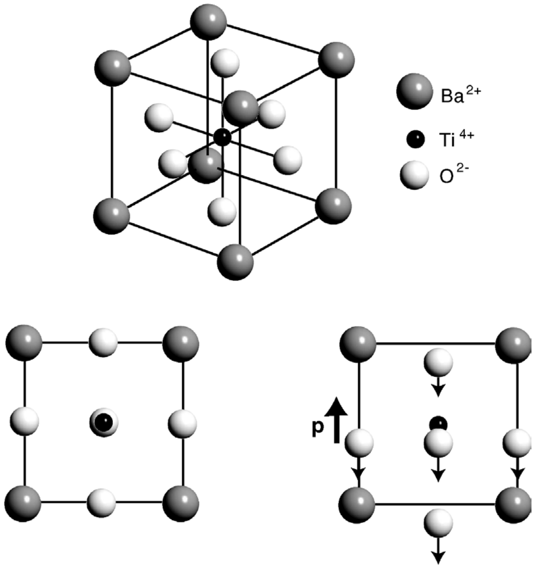

Figure 9.6 *Upper part*: The unit cell of barium titanate $BaTiO_3$. *Lower part*: In the ferroelectric state, the (negative) oxygen sublattice is displaced from the sublattice containing the (positive) Ba and Ti ions.

electric field, the orientation of a ferroelectric's polarization can be reversed and there is hysteresis, very much like in the case of ferromagnetism.

A detailed understanding of the ferroelectric effect can be quite complicated but the basic idea is given by the following argument. We have discussed the equation of motion for two ions in a unit cell, which leads to lattice polarization, by using a simple harmonic oscillator as a model (see (9.10)). In this model, we have considered the motion of the ions under the influence of the average electric field in the solid. We should have used the local field but we have argued that this does not change the course of the argument. In the case of ferroelectric materials, this distinction does matter because when we move an ion out of its equilibrium position, the local field force can pull the atom even further out if it is stronger than the harmonic restoring force. Eventually, force equilibrium is reached but in this way, a distortion and a permanent electric dipole are generated. At a certain temperature, the thermal fluctuations become strong enough to destroy the ferroelectric state. As in the case of ferromagnetism, this temperature is called the Curie temperature. For barium titanate, the Curie temperature is about 130 °C.

An interesting property of ferroelectric materials is that the dielectric constant can be very high below the Curie temperature (see barium titanate in Table 9.1). Given our explanation of ferroelectricity above, the undistorted crystal structure would be unstable and even a very small electric fluctuation would cause a distortion. In this sense, χ_e and thereby ε would be infinite. In reality, χ_e and ε are finite but high. The high dielectric constant of ferroelectric materials could have interesting applications for the gate insulator in MOSFETs.

9.5.3
Piezoelectricity

Piezoelectricity is an effect in which applying stress to a material leads to an electric polarization. This, in turn, gives rise to net surface polarization charges and these can be detected by measuring the voltage across the sample (see Figure 9.7a). The converse effect also exists. When a voltage is applied across the material, this leads to a macroscopic strain (Figure 9.7b). The figure also shows a possible microscopic structure that can give rise to this effect. The structure contains three dipoles that are arranged such that the resulting dipole moment is zero. A deformation of the unit gives rise to a net microscopic dipole moment. Equivalently, the unit will deform in an applied electric field. It is important to note that ferroelectric materials also show piezoelectricity but the opposite is not necessarily true. What is special about ferroelectricity is the spontaneous electric polarization of the solid.

Piezoelectric materials have many applications and both sides of the effect are used. Some examples are sensors (like in microphones), high-voltage sources (cigarette lighters), or actuators (loudspeakers). Piezocrystal-based actuators are especially important for nanotechnology because they permit positioning with unrivalled precision. They are used for positioning the tip in scanning tunneling microscopes, for example.

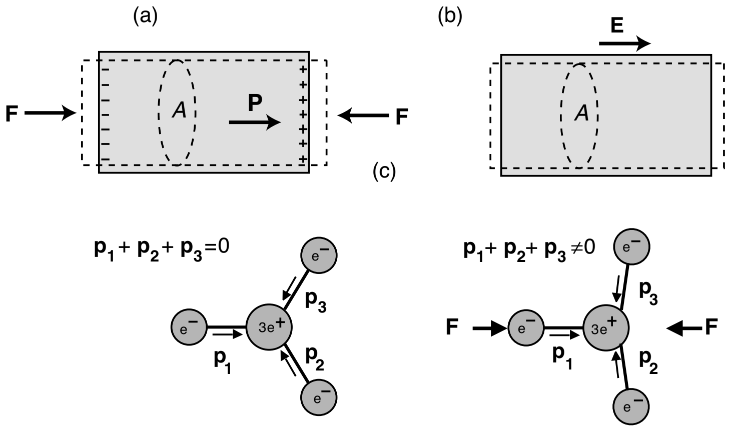

Figure 9.7 (a) Exposing a piezoelectric material to mechanical stress results in a macroscopic electric polarization. (b) Conversely, an electric field across the sample leads to a mechanical strain. (c) This is caused by the deformation of microscopic units in the crystal. Such a unit is shown without applied field or stress on the left. The unit consists of three dipole moments that sum up to a total moment of zero. On the right, stress is applied. This results in a distortion of the unit and a net dipole moment that is no longer zero.

9.5.4 Dielectric Breakdown

If the electric field across an insulator is too high, the insulator will start to conduct a current. This phenomenon is known as dielectric breakdown. The most important mechanism for the breakdown is that some free carriers (e.g., caused by impurities) are accelerated in the field, so much that they can ionize other atoms and generate more free carriers. Then the breakdown proceeds like an avalanche. The breakdown can be facilitated by operating the material close to a resonance frequency where much energy is dissipated, the material is heated, and the probability of having free carriers is increased.

9.6 Discussion and Problems

9.6.1 Discussion

1. What are the physical mechanisms that can lead to a dielectric polarization of solids?
2. If we want to calculate the polarization of a dielectric from the polarizability of the atoms, we have to use the local electric field, which is different from the average internal electric field and from the external electric field. Explain all three fields. Why could we ignore these complications in the case of Curie-type paramagnetism?
3. Qualitatively describe the frequency dependence of the dielectric function for a material with both electronic and ionic polarization.
4. What is the physical meaning of the imaginary part of the dielectric function?
5. Describe the role of impurities in dielectrics. Why are sapphire and diamond transparent and how can impurities change this?
6. What is the difference between ferroelectricity and piezoelectricity?

9.6.2 Problems

1. (a) *Electronic polarization*: Assume that an atom consists of a uniform sphere of negative charge with radius R surrounding a positive point charge. Show that the atomic polarizability is equal to $4\pi\varepsilon_0 R^3$. The negative charge in the sphere should be taken to remain uniform in an applied field. (b) Calculate the atomic polarizability for Ne and compare it to the experimental value of $4.3 \times 10^{-41}\ \mathrm{F\,m^2}$. Take the atomic radius of Ne to be 0.51 Å. (c) Estimate the index of refraction for Ne gas under normal conditions.

2. *Dielectric function*: Estimate the static dielectric constant of NaCl. Use that NaCl has a density of $\rho = 2170\,\mathrm{kg\,m^{-3}}$, a lattice constant of $a = 5.6\,\text{Å}$, and a Young's modulus of $Y = 40\,\mathrm{GPa}$. The optical ε can be taken from Table 9.2 and the result can be compared to the static ε from the same table.

3. *Dielectric function*: We have argued that the imaginary part of the dielectric function $\varepsilon_i(\omega)$ is responsible for the dissipation of energy as an electromagnetic wave travels through the solid. If this is so, one would also expect that such a wave is strongly damped in regions where $\varepsilon_i(\omega)$ is high. We know that the damping of the wave is closely linked to the imaginary part of the refraction index $N = n + i\kappa$. (a) Show that for a given complex ε, n, and κ can be calculated as

$$n = \sqrt{\frac{|\varepsilon| + \varepsilon_r}{2}}, \quad \kappa = \sqrt{\frac{|\varepsilon| - \varepsilon_r}{2}}. \tag{9.25}$$

(b) Plot $n(\omega)$ and $\kappa(\omega)$ for one harmonic oscillator described by (9.18) and (9.19) and discuss the result. (c) Is a strong damping of the waves (or a high κ) always associated with a high imaginary part of the dielectric function?

4. *Reflectivity of Metals*: After having studied the dielectric properties of insulators more closely, we can go back to the reflectivity of a metal, which we discussed in connection with the Drude model. In Chapter 5, we have merely argued that the a low-frequency wave cannot enter a metal (because of the high κ) but that it cannot be absorbed either (because of the lack of damping in (5.23)) and so energy conservation dictates that it should be reflected. Now we can calculate the reflectivity explicitly. The combination of the Maxwell equations with the appropriate boundary conditions leads to the so-called Fresnel equations, which describe the reflection and transmission through an interface. The normal-incidence reflectivity of a solid in vacuum, for instance, is given by

$$R = \frac{(n-1)^2 + \kappa^2}{(n+1)^2 + \kappa^2}. \tag{9.26}$$

Calculate and plot the reflectivity $R(\omega)$ for a free electron metal. To do this, derive a general expression for n and κ for a given ε (see Problem 9.3(a)) and use (9.51) in order to obtain the reflectivity.

10
Superconductivity

Superconductivity, that is, the fact that a current can flow in materials with zero resistance, was discovered in 1911 by H. Kamerlingh Onnes, shortly after it became possible to liquefy helium and thereby to reach the required low temperatures. The discovery was made when Kamerlingh Onnes wanted to study the low-temperature resistivity of mercury. At a temperature of 4.2 K, the resistivity dropped to an unmeasurably small value. The discovery is a good example of a spectacular, totally unexpected and practically important result from very basic research. Later, it should be followed by many other surprises in the behavior of solids at low temperatures.

Superconductivity implies several phenomena in addition to the fact that the solids have zero resistivity. These phenomena will be discussed in the present chapter. It will turn out that the change from the normal state to the superconducting state is a phase transition of the metal and that several properties change due to this transition.

Even though a lot of experimental and theoretical effort was made in order to arrive at a microscopic explanation of superconductivity, it took more than 40 years until the fundamental aspects of such a theory were laid out by J. Bardeen, L. N. Cooper, and J. R. Schrieffer. Their theory is now known as the BCS theory of superconductivity. The basic idea behind the theory is that charged carriers in the superconductor all condense into a single ground state, forming a coherent and macroscopic matter wave, that is, a quantum mechanical wave function that exists on a macroscopic scale. Today, macroscopic wave functions are known from other branches of physics. One example is laser light, where many photons are in the same quantum state and macroscopic coherence is achieved. Other examples are superfluidity, a low-temperature quantum state of liquid ^{4}He, which allows flow without any friction, or Bose–Einstein condensation, in which many atoms can condense into a single quantum state at very low temperatures. The obvious problem, however, is that the particles forming these macroscopic quantum states are bosons but the electrons in a metal are fermions. Bosons do not underlie the Pauli exclusion principle and can all condense in the same quantum state. The electrons in a metal should not be able to do this. It turns out that the way around this problem is to form two-electron pairs that have an integer spin and therefore behave as bosons, with the possibility to condense in a single ground state.

Solid State Physics: An Introduction. Philip Hofmann

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In this chapter, we will first look at some basic experimental observations from superconductors before we turn to the theory and physical principles behind the phenomenon. After this we will discuss some more advanced experimental observations and their explanation as well as at the so-called high temperature superconductors. We conclude the chapter with some comments about the richness of the field, which we can touch only very briefly here.

10.1 Basic Experimental Facts

10.1.1 Zero Resistivity

The most prominent phenomenon associated with superconductivity is of course the vanishing resistivity below a certain critical temperature T_C. The temperature-dependent resistivities of a superconductor and a "normal" metal that does not show superconductivity are shown in Figure 10.1. The resistivity of a normal metal decreases as the temperature is lowered and eventually levels off to a constant value. We have discussed the physical origin of this behavior already. Finite resistivity in a metal is caused by imperfections in the lattice such as impurity atoms, lattice defects, and thermal vibrations. For temperatures much lower than the Debye temperature, the vibrations are effectively "frozen in" but even at zero temperature impurities and defects are present and hence the resistivity is expected to remain finite.

For a superconductor, the picture is entirely different. Above the critical temperature T_C, the same behavior as for a normal metal is observed. At T_C, however, the resistivity drops to an unmeasurably small value. The temperature interval for the transition is very small, usually below 10^{-3} K. The width of the transition range depends somewhat on the quality of the sample but the transition temperature is a characteristic constant of the material.

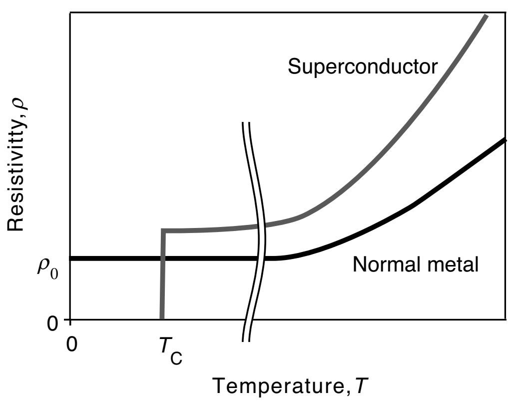

Figure 10.1 Typical temperature-dependent resistivity for a normal metal and for a superconductor.

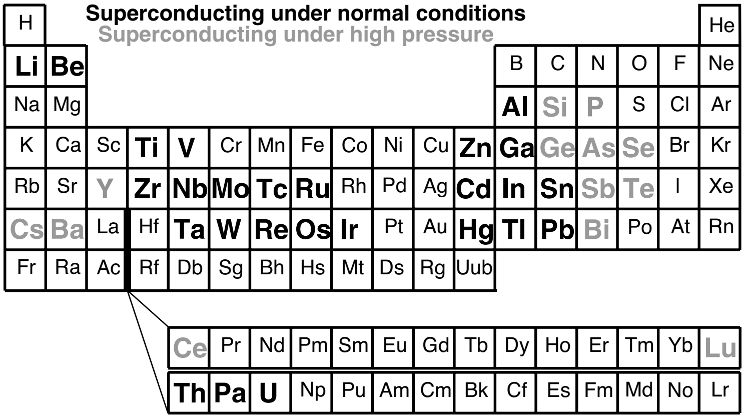

Figure 10.2 Periodic table of the elements with the superconducting elements in bold black letters. Gray letters indicate that the elements do only become superconducting in a high-pressure modification.

Superconductivity appears to be a common phenomenon as seen from Figure 10.2, which is a periodic table of the elements in which superconducting elements are highlighted. The table does of course not give any certainty that the other elements are not superconducting. It is possible that the necessary low temperature has just not been reached yet. In fact, superconductivity in Li at normal pressure was discovered as recent as 2007 at a temperature below 0.4 mK. In any case, a few trends emerge quite clearly from the Figure. The first is that metals that are good conductors are not necessarily also superconductors. In fact, the very best conductors such as Ag and Cu have not been found to be superconducting. Another is that the ferromagnetic elements are not superconducting either, suggesting that ferromagnetism and superconductivity are somehow mutually exclusive. In fact, it also turns out that contaminating a superconducting sample with a very small amount of magnetic impurities can destroy the superconductivity.

The critical temperature T_C is quite small for all elements. It is highest for Nb, which has $T_C = 9.2$ K. T_C can be higher for some metallic alloys, up to around 40 K, and it can be much higher for some transition metal oxides, the so-called high-temperature superconductors.

Some elements do not undergo a transition to a superconducting state in their crystalline structure at ambient conditions, but they can be made superconducting by exerting pressure. These elements are indicated in light gray in Figure 10.2. High pressure could possibly change the structure of a crystal in order to turn it into a superconductor but it has to be clear that the structure as such is not important in the superconducting phase transition. It is the electronic structure and the vibrations that count, as we will see later. However, if the elements do crystallize in different structures, for example, due to high pressure, these structures also have different electronic structures and, to a lesser degree, different vibrational properties.

Another possibility to achieve superconductivity in materials that are normally not superconductors is to grow them in ultrathin films or as small clusters. The films or clusters may still have the same structure as the bulk material under normal conditions but their electronic structure and the superconducting behavior can be different (see Chapter 11). Finally, superconductivity can also be achieved in solids that do not have any long-range crystalline order, so-called amorphous solids. In some cases, this may actually favor the transition. One example is the semimetal Bi, which is not superconducting in its normal crystalline bulk structure but which is superconducting as an amorphous film.

How does one know that the resistivity in the superconducting state is really zero and not just very small? In fact, one does not know this and it is not possible to answer this question from any experiment. One can, however, try to find an upper limit for the resistivity. The experimental approach to this is to induce a current in a superconducting loop using a magnetic field. One should then expect this current to go on forever without any decay. Observing the decay (or rather the lack of it) over a long time (years!) gives the possibility to give an upper limit for the resistivity. Currently, this upper limit is thought to be around $10^{-25}\,\Omega\,\mathrm{m}$.

Apart from the temperature, there are two other important parameters that can destroy the superconducting state. These are a magnetic field and a current through the sample. The combined effect of a magnetic field and the temperature is shown in Figure 10.3a. For low temperatures and small magnetic fields, in the region below the curve, the solid is in its superconducting state. For too high a magnetic field or too high a temperature, the superconductivity is lost, corresponding to the region above the curve. For a given temperature $T < T_C$, we can assign a critical magnetic field $B_C(T)$, which turns out to be

$$B_C(T) = B_C(0)\left[1 - \left(\frac{T}{T_C}\right)^2\right]. \tag{10.1}$$

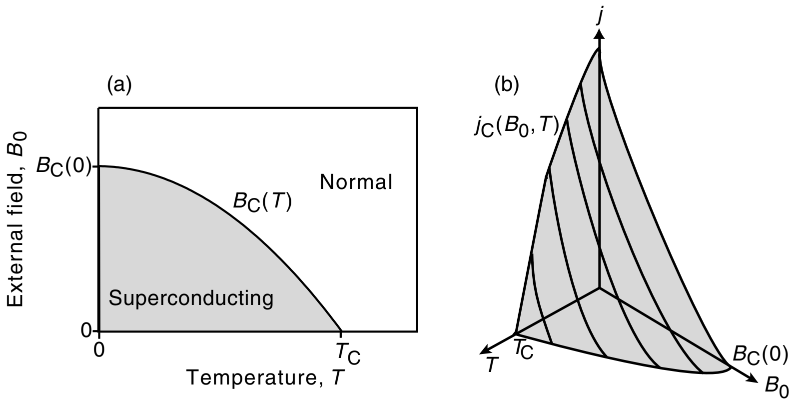

Figure 10.3 (a) Combined effect of a magnetic field and finite temperature on a superconductor. In the region below the curve, that is, for low temperatures and small external fields, the solid is superconducting. Above the curve it is in its normal state. (b) The same but including the effect of a finite current density.

A current through the sample, as expressed by a current density j, has a similar effect as a magnetic field. For too high a current density, the superconductivity breaks down. Again, this critical current density is a function of the temperature. But it is also a function of the applied magnetic field B_0. In a complete description, the critical current density is a function of the temperature and the magnetic field, as shown in Figure 10.3b. The superconducting state is only reached for low temperatures, current densities, and magnetic fields. In a similar way, the critical magnetic field can be viewed as a function of temperature and current density, of course.

Unfortunately, the critical current densities and magnetic fields for most elemental superconductors are so low that these cannot be used in any meaningful technical application. Take for example the electromagnet in a medical magnetic resonance scanner. This magnet has to produce a high field of more than 1 T. In order to do this, very high currents are necessary. It is not possible to build such magnets out of normal conductors because of the large amount of heat that would be produced by ohmic losses. To build them from superconducting materials is not straightforward either because of the high magnetic field and the high current densities. Rather than elemental superconductors, one often uses superconducting alloys that can have critical fields of up to 50 T and a critical current density of up to 10^{11} A m^{-2} at liquid He temperature.

10.1.2
The Meissner Effect

The so-called Meissner effect, discovered in 1933 by W. Meissner and R. Ochsenfeld, is another fundamental property of a superconductor. It simply states that in the superconducting state, the superconductor shows perfect diamagnetism, that is, it has $\chi_m = -1$ and therefore a magnetization $\mathbf{M} = -\mathbf{B}_0/\mu_0$, which totally cancels the external field $\mathbf{B}_0$ inside the superconductor. The origin of the macroscopic magnetization $\mathbf{M}$ is, however, very different from that in normal diamagnetic materials. In the latter, it is caused by microscopic magnetic moments that are induced by the external field throughout the entire solid. In a superconductor, this is not the case. The magnetization $\mathbf{M}$ of a superconductor is rather caused by macroscopic supercurrents that flow close to the surface of the specimen and keep almost the entire inside field free.

The magnetic susceptibility of -1 implies that superconductors are strongly expelled from magnetic fields. You are probably familiar with the Meissner effect from diamagnetic levitation experiments, in which a small high-temperature superconductor is levitated in an inhomogeneous magnetic field (see our discussion of (8.6)).

It is very important to understand that the Meissner effect is a genuinely new effect and not simply a consequence of the vanishing resistance, which can give rise to a supercurrent to keep the inner specimen field free. There is a difference between a (hypothetical) material that becomes merely a perfect conductor with $\rho = 0$ below T_C and a true superconductor that also displays the Meissner effect. For a perfect conductor, the entire physics is given by the Faraday's law which, in its integral form, is

$$\oint \mathbf{E}d\mathbf{l} = -\frac{d\Phi_B}{dt}, \tag{10.2}$$

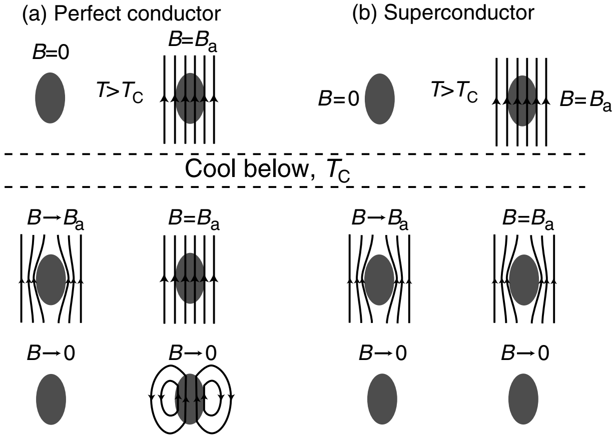

Figure 10.4 The Meissner effect is not merely a consequence of zero resistivity. (a) The behavior of a "perfect conductor," that is, a material that merely has zero resistivity below T_C. For this material, the magnetic field in the specimen below T_C depends on the presence and size of a magnetic field before cooling down below T_C. (b) The situation for a genuine superconductor that displays the Meissner effect. The interior of the specimen is field free below T_C, independent of the sample's history.

where the integral is taken along a closed path in the conductor. For a perfect conductor, no electric field can exist along such a closed path, which means that the entire integral vanishes and the magnetic flux through the loop of integration is constant.

Figure 10.4a illustrates the behavior of a perfect conductor in a magnetic field. We start with two specimens above T_C, one in an applied magnetic field $\mathbf{B}_a$ and one in a field-free region. Both are cooled below T_C, where the resistivity of the conductor drops to zero. Now the magnetic field is also turned on for the conductor that was in the field-free region before. Because of Faraday's law, it expels the field entirely and the inner part is still field free. For the other specimen, nothing changes because the field had been penetrating already before the transition below T_C. In the end, the field is taken to zero for both specimens. Since it cannot change inside the perfect conductor, one remains field free and in the other one a current is induced, which keeps the field in the specimen as it was before. In short, the situation is far from perfect diamagnetism. Below T_C, we can have any field we like in the specimen. It only depends on the history.

Now consider a real superconductor that exhibits the Meissner effect as shown in Figure 10.4b. Again we start with two specimens in the normal state, one without a penetrating magnetic field and one with such a field. Once the specimens are cooled below T_C, the magnetic field is expelled in *both* cases, that is, the interior of the superconductor is field free, independent of the history of the sample. Once the field is turned off, the interior of the specimen remains of course field free. The experimental observation that a superconductor does indeed behave in this way was a very important step for the understanding of superconductivity because it permits the description of the superconducting state as a single thermodynamic phase, which can be described by a few macroscopic variables. A history-dependent magnetization of the sample would not permit such a view.

The bottom line is that the Meissner effect is a genuine effect in itself, which is not implied by zero resistivity. Any theory of superconductivity must be able to explain not only resistance-free current flow but also the Meissner effect.

10.1.3 The Isotope Effect

There are several more experimental observations associated with superconductivity and some will be discussed in a later section. The so-called isotope effect, however, is presented already here because it gives an important clue for the development of amicroscopic theory.

The effect is illustrated in Figure 10.5 that shows the critical temperature of Sn as a function of the ionic mass M, which can be varied by using different isotopes of Sn. The figure is actually a log–log plot on which the data appear as a straight line, equivalent to a power law behavior with $T_C \propto M^{-1/2}$.

The fact that T_C depends on M is remarkable. As a consequence of the Born–Oppenheimer approximation, the electronic structure of a solid should not depend on the mass of the ions, just on their chemical nature. The vibrational properties, in contrast, do depend on the mass of the ions. For a simple harmonic oscillator, we know that $\omega = (\gamma/M)^{1/2}$, that is, the vibrational frequency depends on the mass in the same way as T_C. This can be taken as an important clue for any microscopic theory: It is very likely that the lattice vibrations of the solid play some role in superconductivity.

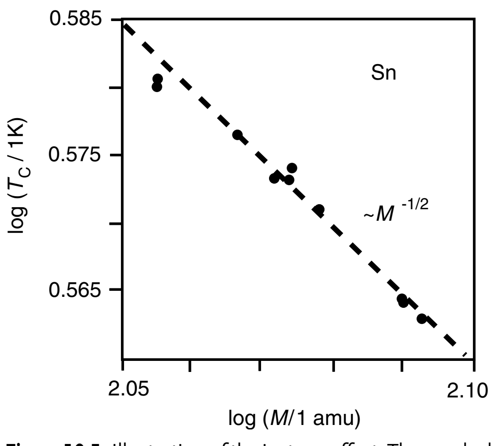

Figure 10.5 Illustration of the isotope effect. The graph shows the critical temperature as a function of isotope mass as a log–log plot. The data points lie on a straight line suggesting a power law behavior with $T_C \propto M^{-1/2}$. Data taken from Refs [8,9].

10.2
Some Theoretical Aspects

10.2.1
Phenomenological Theory

Early theories of superconductivity were not atomistic but merely sought a macroscopic formalism that explains both the resistance-free transport and the Meissner effect. Both were achieved by F. London and H. London in 1935. Their theory can be summarized by the two so-called London equations that we now discuss.

Achieving infinite conductivity for a gas of charged particles is very easy in a macroscopic theory. In fact, in the Drude model the equation of motion for the electrons was taken to be (5.2) and this equation does already lead to an infinite conductivity. In order to prevent this from happening, we had to introduce the concept of the relaxation time. The first London equation is identical to (5.2), only that the equation is now written using a current density **j** instead of the velocity **v** of a single electron:

$$\frac{\partial \mathbf{j}}{\partial t} = \frac{n_s e^2}{m}\mathbf{E}, \tag{10.3}$$

where n_s is the density of superconducting particles, m is their mass and $-e$ their charge.[1] If we take the curl of this equation and combine it with Faraday's law (12.20), we get

$$\frac{\partial}{\partial t}\left(\frac{m}{n_s e^2}\mathrm{curl}\mathbf{j} + \mathbf{B}\right) = 0. \tag{10.4}$$

What this means becomes clear if we integrate it over an area **A** inside the solid and then use Stokes' integral theorem (12.11), giving

$$\frac{\partial}{\partial t}\left(\int \frac{m}{n_s e^2}\mathrm{curl}\,\mathbf{j}\mathrm{d}\mathbf{A} + \int \mathbf{B}\mathrm{d}\mathbf{A}\right) = \frac{\partial}{\partial t}\left(\oint \frac{m}{n_s e^2}\mathbf{j}\mathrm{d}\mathbf{l} + \int \mathbf{B}\mathrm{d}\mathbf{A}\right) = 0. \tag{10.5}$$

The result contains two types of magnetic flux through the area **A**. The first is caused by an electric current density **j** on the perimeter of the area and the second is the actual flux from the external field **B**. We do not know the size of the first contribution but the equation tells us that the sum is constant in time. So if we change the external field **B**, this change is exactly compensated by a change in current density **j** such that the total magnetic flux does not change. The result expresses exactly what we already know to be true for a perfect conductor (see Figure 10.4a).

1) If the particles carrying the superconductivity turn out to have different densities, masses, and charges, the theory remains valid but we have to substitute the correct quantities into the results.

The second London equation is obtained by not only requiring that the partial derivative with respect to time does not change but that (10.4) vanishes altogether, that is,

$$\frac{m}{n_s e^2} \operatorname{curl} \mathbf{j} + \mathbf{B} = 0. \tag{10.6}$$

Using the same arguments as in connection with (10.4), we can see that this equation describes the Meissner effect correctly. Now the current density **j** can be calculated from the external field **B** and the internal field created by this current density exactly compensates the external field.

Expelling the magnetic field from the inside of the superconductor is achieved by a current density in the surface of the superconductor. But how far do these currents penetrate into the material? The London equations permit a more quantitative approach to this. Inside the superconductor where $\mathbf{D} = 0$, we have Ampère's law (12.26) as

$$\operatorname{curl} \mathbf{B} = \mu_0 \mathbf{j}. \tag{10.7}$$

When we combine this with (10.6) and get

$$\operatorname{curl} \operatorname{curl} \mathbf{B} = \mu_0 \operatorname{curl} \mathbf{j} = -\frac{\mu_0 n_s e^2}{m} \mathbf{B}. \tag{10.8}$$

Now we can use the general rule for second derivatives that $\operatorname{curl} \operatorname{curl} \mathbf{B} = \operatorname{grad} \operatorname{div} \mathbf{B}$ $\Delta \mathbf{B}$, so that[2)]

$$\Delta \mathbf{B} = \frac{\mu_0 n_s e^2}{m} \mathbf{B}. \tag{10.9}$$

With this we have a differential equation for the penetrating magnetic field and a very similar equation can be derived for the current density, using the same arguments. It is

$$\Delta \mathbf{j} = \frac{\mu_0 n_s e^2}{m} \mathbf{j}. \tag{10.10}$$

Both equations can be solved by an exponentially decreasing field and current density, respectively. The situation for the field is shown in Figure 10.6. The characteristic length of the exponential decrease is the so-called London penetration depth λ_L, which is given by

$$\lambda_L = \sqrt{m/\mu_0 n_s e^2}. \tag{10.11}$$

Assuming that all the electrons are carriers of supercurrents (which is actually not true as we will see later), one can estimate the London penetration depth to be of the order of 100 nm.

2) You may not be familiar with the notation in which the Laplace operator Δ is applied to a vector field. This notation just means that you again get a vector field that results from applying the Laplace operator individually to each single component of the original field.

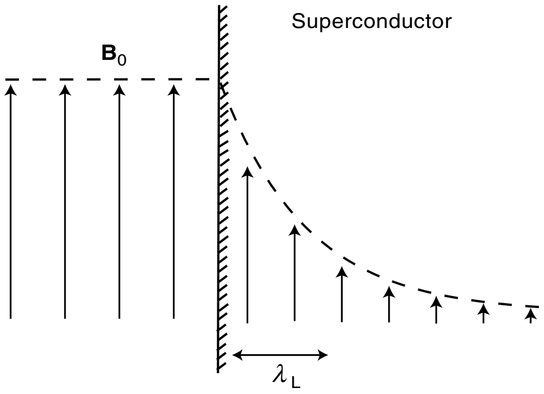

Figure 10.6 Exponential damping of an external magnetic field near the surface of a superconductor.

The agreement between the field penetration calculated by (10.11) and the measured field penetration is by no means satisfactory. There are several reasons for this, which lie mainly in the density of the superconducting carriers n_s. It turns out that this density is not at all constant in a superconducting material. It varies both with position and with temperature.

A considerably more sophisticated theory for n_s was presented in 1950 by V. L. Ginzburg and L. D. Landau. This Ginzburg–Landau theory defines a so-called order parameter $\Psi(\mathbf{r})$, which can be derived from an equation similar to the Schrödinger equation and which has a form that is reminiscent of a free electron wave:

$$\Psi(\mathbf{r}) = \Psi_0 e^{i\phi(\mathbf{r})}, \tag{10.12}$$

in which $\phi(\mathbf{r})$ is a macroscopically changing phase and $|\Psi^*\Psi| = \Psi_0^2$ is equal to the density of superconducting carriers n_s. The spacial character of $\Psi(\mathbf{r})$ means that n_s does no longer need to be constant but it cannot change instantaneously either, for example, from zero to a high value at the surface of a superconductor. Appreciable changes of $\Psi(\mathbf{r})$, and thereby n_s, have to happen over a length scale ξ, the so-called coherence length of the superconductor. The Ginzburg–Landau theory is remarkably successful in describing many phenomena associated with superconductivity even though it is not a microscopic theory. An especially notable feature of the theory is that the order parameter already has the character of a macroscopic quantum mechanical wave function.

10.2.2
Microscopic BCS Theory

The microscopic theory of superconductivity was formulated in 1957, more than 40 years after the discovery of the effect, by J. Bardeen, L. N. Cooper, and J. R. Schrieffer. It is commonly referred to as the BCS theory, after the names of its inventors. Several requirements for a microscopic theory were clear for a long time before the theory was actually developed: It should of course be possible to explain the vanishing resistivity and the Meissner effect. The theory should probably involve lattice vibrations in some way in order to account for the isotope effect. Finally, it should not be specific to any particular material since we have seen that superconductivity is a rather common phenomenon.

Why did it then take such a long time to come up with a microscopic picture of superconductivity? Part of the reason is that two of the most fundamental assumptions for the treatment of solids are not valid in the case of superconductivity and their breakdown is even essential for the theory. The first assumption is the Born–Oppenheimer approximation that allowed us to treat the electronic and the vibrational properties separately. The second is the independent electron approximation in which we considered the properties of one electron in the average potential of all others.[3)]

As already mentioned in the introduction, the superconducting state represents a quantum phenomenon on a macroscopic scale. We will see some direct experimental proof of this in the later sections. Like in a laser or in a Bose–Einstein condensate, the macroscopic quantum state is realized by very many particles occupying the same quantum state. While we have to be careful in comparing the condensing particles in the solid with non-interacting bosons, it is clear that these particles cannot be the free electrons because these are fermions and have to obey the Pauli principle.

It was L. N. Cooper who realized that the formation of electron pairs due to a net attraction, no matter how weak, would resolve this problem and open the possibility for a new ground state of the electron gas with entirely new properties and a slightly lower energy than the original ground state discussed in Chapter 6. Since the energy of the new state is lower, one would expect a phase transition to this state, at least for very low temperatures where the entropy is not important.

A possible mechanism providing a weak attraction between electrons is the interaction between the electrons and lattice vibrations, the so-called electron–phonon interaction. This interaction is weak and usually ignored when we use the Born–Oppenheimer approximation. But for superconductivity it is vital. Figure 10.7a illustrates its origin. When an electron is placed somewhere in the ionic lattice, it slightly deforms the bonds around it due to the electrostatic interaction with the ions. Such a localized lattice distortion can be imagined as a wave package made from phonons, very similar to a localized electron that can be viewed as a wave package from Bloch waves. The deformation is such that the lattice ions are attracted toward the position of the electron. This leads to a local polarization of the lattice, which in turn is attractive for other electrons. In this way, an attractive interaction can exist between two electrons.

The static picture given in Figure 10.7a does, however, not really work because another electron that moves to the polarized part of the lattice would still feel the strong electrostatic repulsion of the first electron. Figure 10.7b shows a dynamic picture that is more appropriate if we consider electrons with kinetic energies near the Fermi energy. An electron moves through the lattice with a high speed, in the order of $10^6\,\mathrm{m\,s^{-1}}$ and attracts the positive lattice ions on its way. The lattice is

3) We have, of course, already abandoned this approximation in the case of magnetic ordering.

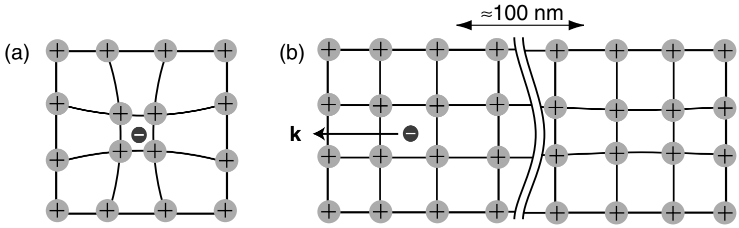

Figure 10.7 Local deformation of the lattice via the electrostatic interaction between the electrons and the ions. Situation for (a) a very slow or static electron and (b) an electron near the Fermi level of a metal.

polarized, but since the ions move much slower than the electrons, the polarization is not established instantaneously. We can estimate that the time for polarizing the lattice has to be on the same timescale as the ion movement, that is, $2\pi/\omega_D \approx 10^{-13}$ s. By the time the maximum polarization is reached, the electron has already traveled around $10^6\,\mathrm{m\,s^{-1}} \times 10^{13}\,\mathrm{s} = 100\,\mathrm{nm}$.

Now it is possible for a second electron to lower its energy by moving in the trail of the first. Electrostatic repulsion between the two electrons will be insignificant because they are so far away from each other. If the second electron is to stay on the same track, its wave vector **k** must either be the same as for the first electron or exactly opposite. It can be shown that the strength of the interaction between two electrons is maximized when they have exactly opposite **k** vectors and opposite spins. The pairs of electrons with **k** and $-\mathbf{k}$ and a total spin of zero are called Cooper pairs.

While these simple arguments give a good visualization of the physical origin of Cooper pair formation, one must be very careful not to take them too far. It is, for example, somewhat stretching the imagination that the wave vector of the second electron should be $-\mathbf{k}$ and not **k**, that is, the second electron moves in the opposite direction of the first. But we have to realize that the question of which electron is where and moves how has already lost its meaning when we study a wave function of just two electrons. It is not obvious either that the total spin of the pair should be zero. In fact, there are some exotic superconductors in which it is one. However, the crucial point still is that it must be integer and not half-integer; that is, the Cooper pairs are bosons, not fermions.

In order to see why we have used the velocity of the electrons at the Fermi level to estimate the separation between the pair of electrons, we describe the electron–phonon interaction in a more quantum mechanical picture. This also explains why the interaction can be so important at low temperatures when essentially all the phonons are "frozen in" and the lattice only performs zero point motions. The interaction between two electrons via phonons can be viewed as the constant emission and absorption of "virtual" phonons of energies up to $\approx \hbar\omega_D$. The emission of a phonon of energy $\hbar\omega$ does not violate energy conservation if it is only short-lived, that is, when it is rapidly absorbed by another electron. The only electrons that can participate in the exchange of such "virtual" phonons and in the formation of Cooper pairs are those close to the Fermi level, within an energy window of approximately $\hbar\omega_D$. All the

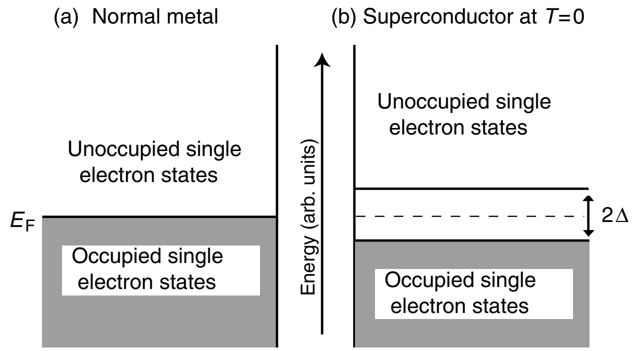

Figure 10.8 Occupation of single-electron levels at zero temperature in (a) a normal metal and (b) a superconductor. For the superconductor, the electrons close to E_F are bound in Cooper pairs and these occupy a single many-body state, the BCS ground state, which cannot be shown in this figure of single-particle levels. In order to excite single electrons out of this ground state, a Cooper pair has to be broken. This costs an energy of Δ per electron and creates two unpaired electrons in the lowest possible unoccupied single-particle states.

other electrons cannot emit or absorb "virtual" phonons because they are trapped by the Fermi–Dirac distribution. All the reachable states around them are already occupied.

The formation of Cooper pairs and their condensation into a common ground state is associated with an energy gain and leads to a new ground state of the system, the so-called BCS ground state. We first discuss this state at zero temperature. The energy levels for a normal metal at zero temperature have the familiar form shown in Figure 10.8a. They are completely filled up to E_F and empty above. In the BCS ground state, the electrons close to the Fermi level are all bound in Cooper pairs and they have gained an average energy Δ per electron by doing so.

The Cooper pairs do not appear in a single-particle energy diagram such as Figure 10.8. Their state is a many-body state with one total energy for all the electrons bound in Cooper pairs. This is similar to the hydrogen molecule energy levels shown in Figure 1.2b. For these molecules there are two energy levels for every possible interatomic distance, one for the singlet and one for the triplet state. It does not make any sense to split these up into single-electron energies. The energy levels are genuine two-electron states.

For the superconductor, we can look at the remaining electrons that are not bound in Cooper pairs. Their energy levels are drastically modified as shown in Figure 10.8b: a gap of size 2Δ is opened around E_F, that is, there are no single particle states close to the Fermi level any more. We can understand this qualitatively by the fact that the electron states just below E_F have been removed to form the Cooper pairs. Above E_F, the lowest possible energy for a single electron is Δ. This is consistent with the energy cost of breaking up a Cooper pair, which is 2Δ and produces two single electrons in the lowest possible energy state, just above the gap.

The gap in the single-electron energies is a very characteristic and central feature of the BCS model. It is frequently referred to as the gap in the single-particle excitation spectrum, because now the minimal excitation energy for an unpaired electron is not zero (like in a normal metal) but 2Δ. In this respect, the situation is similar to a semiconductor that also has a gap, but the gap here is very much smaller and, as we will see in a moment, strongly dependent on the temperature.

We can estimate its size and the total energy gain for the transition to the superconducting state. Only the electrons close to the Fermi level can take part in the exchange of "virtual" phonons, which leads to the formation of Cooper pairs. One can therefore expect electrons with energies of up to $\hbar\omega_D$ below E_F to participate in the pairing, so $\Delta \approx \hbar\omega_D$. This is indeed the case. The BCS theory predicts that the gap at zero temperature is related to the critical temperature T_C as $\Delta = 3.53\, k_B T_C$, which amounts to just a few meV for most superconductors. T_C, in turn, can be predicted from the Debye temperature, the electronic density of states at the Fermi level, and V, a parameter measuring the electron–phonon coupling strength. The BCS result for the transition temperature is

$$T_C = 1.13\Theta_D \exp\frac{-1}{g(E_F)V}. \tag{10.13}$$

This equation tells us a lot about BCS-type superconductivity. First of all, the transition temperature is proportional to the Debye temperature. This readily explains the isotope effect since $\Theta_D \propto \omega_D \propto M^{-1/2}$. Equation 10.13 also shows that the Debye temperature sets the temperature scale of the phenomenon. The exponential cannot become larger than 1, so that one cannot hope to get a transition temperature higher than $1.13\Theta_D$. In reality, the exponential is much smaller than 1. Both the density of states and the interaction strength enter the expression in the same way, meaning that both increase T_C.

We now estimate the absolute number of electrons bound into Cooper pairs and the total energy gain. The number of electrons in Cooper pairs (per volume) must be of the order $g(E_F)\Delta \approx g(E_F)\hbar\omega_D$ and the associated total energy gain must be around $g(E_F)\Delta^2 \approx g(E_F)\hbar^2\omega_D^2$.

At finite temperatures, not all the electrons close to E_F are bound in Cooper pairs and matters are made complicated by a typical phenomenon for the condensation of bosons: The energy gain for the condensation of a particle into the ground state depends on the number of particles already in this state. In other words, the formation energy 2Δ for a Cooper pair depends on the number of pairs already in the ground state and thus Δ becomes a function of temperature. The temperature dependence of Δ is shown in Figure 10.9. When the sample is cooled below T_C, Δ assumes non-zero values and some Cooper pairs start to form. This, in turn, increases the energy gain for the formation of further Cooper pairs. At zero temperature, all the electrons close to E_F are bound in Cooper pairs. Consequently, the size of the gap can also be taken as a measure for how many electrons are condensed in Cooper pairs.

This intricate behavior is reminiscent of the band model for magnetic ordering in the solid where the energy gain for the orientation of a spin depends on the number of spins already present in that orientation. In magnetic ordering as in superconductivity, a self-amplifying process sets in once the sample is cooled under the transition temperature. With this, we also understand the fact that the transition to the superconducting state is so sharp in temperature.

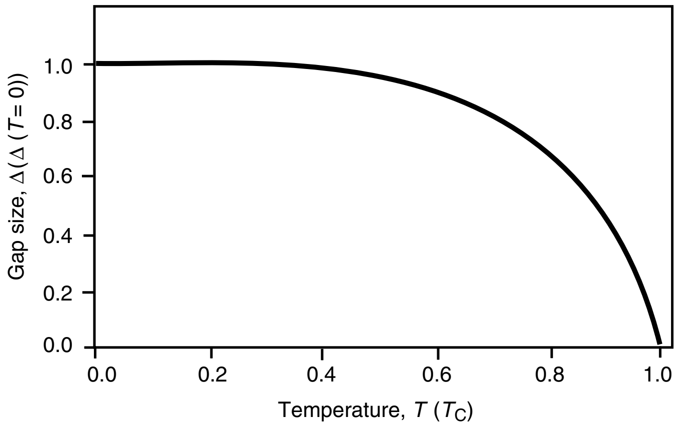

Figure 10.9 Gap size for a superconductor in the BCS model as a function of temperature. At the transition temperature T_C, the gap is closed.

The existence of Cooper pairs and the gap in the singe-particle spectrum can be used to explain all observed phenomena discussed above. Here, we merely discuss the resistance-free transport and the existence of a critical current density and a critical magnetic field. We do not show how the Meissner effect can be explained but the basic idea is that an equation similar to the second London equation (10.6) can be derived in the BCS model.

As already mentioned in connection with the London equations, a resistance-free transport is readily obtained in both the classical and the quantum model of electrons in a solid if we do not introduce some scattering mechanism to keep the resistance finite. Exactly the same applies for Cooper pairs: The whole condensate of pairs is accelerated when an electric field is applied to the sample. Before the field is applied, the Cooper pairs are all in the same quantum state with same total $\mathbf{k} = 0$ per pair. When a current passes through the sample, all the pairs are still in the same state but with another $\mathbf{k}' \neq 0$. The only thing we have to explain is the absence of scattering processes that can lead to a decay of this current. In order tosee the difference between the Cooper pairs and unpaired electrons, it is useful to look back at Figure 6.13. The current decays because the electrons are scattered back to lower lying states by defects or thermal vibrations. This scattering is an individual process, affecting one electron at a time. Such a process does not work for a condensate of Cooper pairs because it is not possible to change $\mathbf{k}$ for one pair without changing it for all the others at the same time, a process that is exceedingly unlikely, unless it is done by applying the same force on all the pairs at the same time, as by an electric field.

Another possibility is a scattering process that splits up a Cooper pair into two individual electrons. This is, in fact, possible but only if the kinetic energy of the pairs is high enough to provide the necessary 2Δ for the destruction of the Cooper pair. The process will thus become important at high current densities for which the number of Cooper pairs in the solid gradually decreases until superconductivity breaks up altogether, when the critical current density $\mathbf{j}_C$ is reached. The existence of a critical magnetic field B_C can be explained in a very similar way. This critical field is reached when the magnetic energy density exceeds the value needed for breaking up the Cooper pairs. Finally, it is clear from Figure 10.9 why the critical magnetic field and the critical current density depend on the temperature. They decrease as T_C is approached because of the shrinking gap size Δ.

10.3
Experimental Detection of the Gap

The concept of a gap in the excitation spectrum is central to the BCS theory. Therefore, experimental tests of the existence and size of the gap have been important for confirming the theoretical predictions. Here, we discuss three approaches to this: single-electron tunneling, optical reflectivity, and the low-temperature heat capacity in the superconducting state.

A tunneling experiment between a superconductor below T_C and a normal metal is illustrated in Figure 10.10. The metal and the superconductor are separated by an insulating barrier, such as an oxide, which is only a few nanometers thick. The wave functions of the metal and the superconductor are not cut off at the interfaces to the

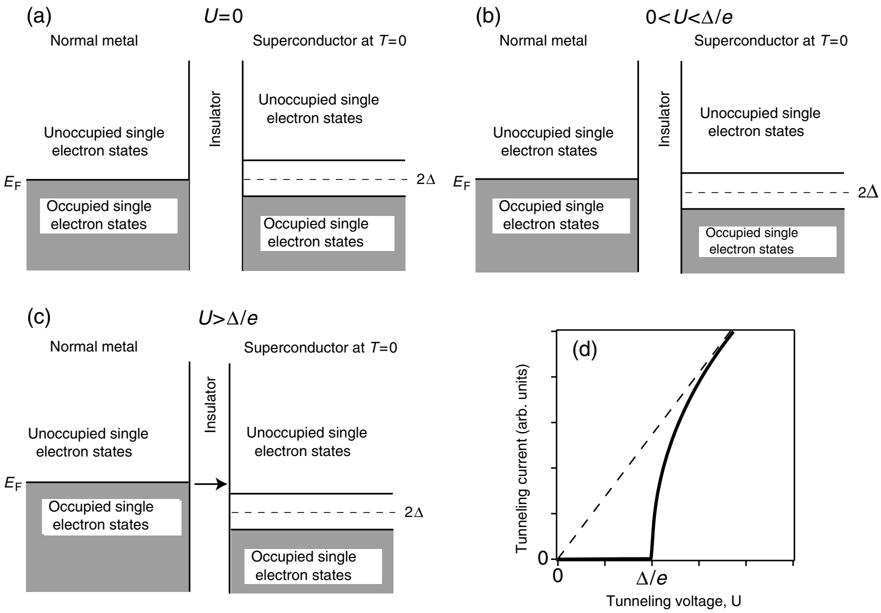

Figure 10.10 Tunneling experiment between a superconductor and a normal metal. The two are separated by a thin insulating oxide. Only elastic single-electron tunneling is considered. (a) Situation without applied voltage. (b) For a small tunneling voltage, no tunneling is possible because of the lack of available states in the gap. (c) As the tunneling voltage exceeds Δ/e, single-electron tunneling becomes possible. (d) Thick line: tunneling current versus voltage for the present junction; dashed line: corresponding curve for tunneling between two metals.

oxide but leak out into the oxide. The small overlap between them permits the tunneling of electrons from one side to the other. Here, we consider only elastic tunneling of single electrons.

If no external voltage is applied, the Fermi level in the metal is aligned with the chemical potential in the superconductor, that is, the middle of the gap (Figure 10.10a). For a small positive tunneling voltage on the superconductor, no current can flow because the electrons from the metal do not find empty states in the superconductor to tunnel into (Figure 10.10b). As the tunneling voltage U reaches a value of $U = \Delta/e$, this situation changes. Now the electrons close to the Fermi level find empty states in the superconductor and a tunneling current is observed (Figure 10.10c). This causes a sudden rise of the tunneling current at Δ/e (Figure 10.10d) that permits the determination of the gap size. Note that the situation would be entirely different for tunneling between two metals: Tunneling would already be possible at very small voltages and the number of electrons able to tunnel would increase continuously with the voltage. This behavior is indicated by the dashed line in Figure 10.10d.

Another possibility for detecting the existence of the gap is to measure the absorption of electromagnetic radiation as it passes through a superconductor. For electromagnetic radiation with an energy of $h\nu < 2\Delta$, electronic transitions across the gap are not possible and hence no absorption is observed. As the photon energy reaches 2Δ, absorption sets in and the transmitted intensity is strongly reduced. Since the gap size is at best a few meV, the electromagnetic radiation required for this experiment lies in the microwave or far infrared region.

Yet another experiment that points toward the existence of a gap is the low-temperature heat capacity of a superconductor. Figure 10.11 shows the heat capacity

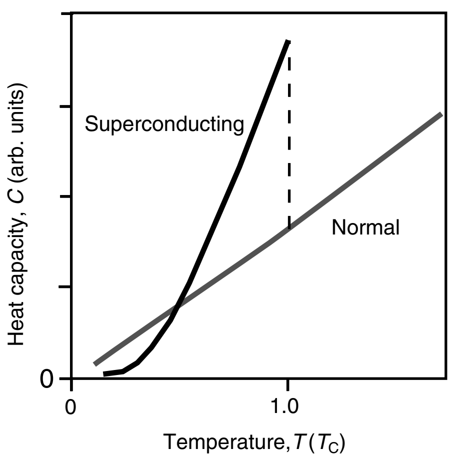

Figure 10.11 Qualitative low-temperature heat capacity of a superconductor in both the superconducting and the normal state. A normal state below T_C can be realized by applying a magnetic field.

for a superconductor at low temperatures both in the superconducting state and in the normal state. Keeping the material in the normal state below T_C can be achieved by applying a weak magnetic field. Upon cooling below T_C, the heat capacity shows a discontinuous change. It also shows a qualitatively different behavior at very low temperatures, where it does not show the linear temperature dependence but an exponential decrease. This is another indication of an excitation gap. We have already encountered the exponential behavior in the case of the Einstein model for the phonon heat capacity where it is also caused by an energy gap, the gap between the ground state and the first excited state of the Einstein oscillators.

10.4 Coherence of the Superconducting State

We have mentioned several times that the superconducting state is associated with a macroscopically coherent wave function, which is built from the Cooper pairs in their common ground state. This macroscopic coherence is, in fact, directly observable in many experiments.

An immediate consequence of the macroscopically coherent wave function is the quantization of magnetic flux through a superconducting ring, as the one shown in Figure 10.12a. If we assume that the wave function of the superconductor is coherent in the entire ring, we can apply the Bohr quantization condition, like for the hydrogen atom but on a macroscopic scale. The Bohr condition states that an integer number of de Broglie wavelengths $\lambda = p/h$ has to fit in the circumference of the ring in Figure 10.12a or that

$$\oint \frac{\mathbf{p}}{h} d\mathbf{r} = n, \tag{10.14}$$

where the integration is carried out around the inner circumference of the ring and n is an integer. If we want to see what happens in the presence of a magnetic field, we

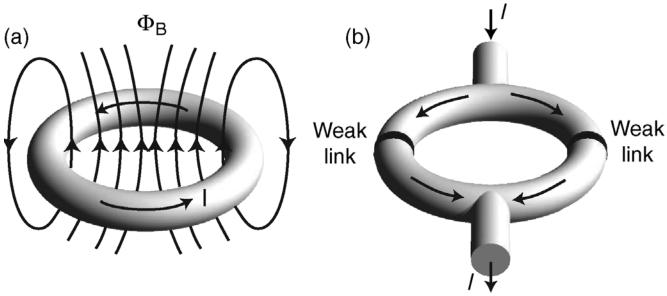

Figure 10.12 (a) A superconducting ring enclosing a magnetic flux. The magnetic flux through such a ring is quantized in multiples of $h/2e$. (b) A superconducting quantum interference device.

can include this field in the equation using the vector potential and the same rules as laid out in Section 8.2 for the quantum mechanical case. Following (8.13), we write

$$\oint \mathbf{p}-q\mathbf{A}d\mathbf{r} = hn. \tag{10.15}$$

For free particles of mass m and charge q have that $\mathbf{p}=m\mathbf{v}$, and the current density associated with these particles is $\mathbf{j}=nq\mathbf{v}$ such that we can write

$$\frac{m}{nq}\oint \mathbf{j}d\mathbf{r}-q\oint \mathbf{A}d\mathbf{r} = nh. \tag{10.16}$$

We now re-write the second integral by using Stokes' integral theorem (12.11), in order to change the line integral into a surface integral over the area in the middle of the ring.

$$\oint \mathbf{A}d\mathbf{r} = \int \text{curl}\,\mathbf{A}d\mathbf{a} = \int \mathbf{B}d\mathbf{a} = \Phi_\mathrm{B}, \tag{10.17}$$

where Φ_B is just the magnetic flux through the hole in the ring. With this, (10.16) becomes

$$\frac{m}{nq^2}\int \mathbf{j}d\mathbf{r}-\Phi_\mathrm{B} = n\frac{h}{q}. \tag{10.18}$$

This implies that the magnetic flux through the ring can only change in units of h/q if the current density in the first integral is constant. We can go one step further and lay the integration path of the first integral a little deeper inside the ring, just outside the screening currents that penetrate up to the London penetration depth λ_L. In many cases, the current density is then vanishingly small and if we furthermore assume that the current is carried by Cooper pairs of charge $-2e$ we get

$$\Phi_\mathrm{B} = n\frac{h}{2e}. \tag{10.19}$$

This means that the magnetic flux through a superconducting ring is quantized in units if $h/2e$ and this was indeed confirmed experimentally in 1961, beautifully confirming this result of the BCS theory. A curious fact is that F. London had already predicted this in 1950, 7 years before the BCS theory.

This so-called flux quantum $h/2e$ is very small, only $2.067 \times 10^{-15}\,\mathrm{T\,m^2}$. To set this into context, consider the Earth's magnetic field that is of the order of $10^{-5}\,\mathrm{T}$. In an area of $1\,\mathrm{mm}^2$, there would still be 10^4 flux quanta or so.

The coherence of the superconducting state can be exploited in so-called superconducting quantum interference devices (SQUIDs), as shown in Figure 10.12b. We cannot describe in detail how these devices work but the key idea is as follows. The device consists of two superconducting "forks" and thin oxide layers or point contacts between them (so-called weak links). The Cooper pairs from one superconductor can tunnel through the weak links into the other superconductor and there is definite relation between the phase of the superconducting state in one fork and that of the other fork. A magnetic field entering perpendicular through the forks gives rise to an additional supercurrent to keep the flux through the ring an integer multiple of the flux quantum. This gives rise to a phase difference for the current going through one weak link and the current going through the other weak link. The maximum current through the entire device is given by the interference of the "left" and "right" currents and one observes oscillations as a function of the applied magnetic field. These correspond to single flux quanta entering the ring. In this way, very small magnetic field (changes) can be measured.

10.5
Type I and Type II Superconductors

In our discussion of superconductors, we have so far assumed that a magnetic field cannot enter the superconductor. It would be completely expelled by supercurrents in the surface of the material. This is actually not always true. There is a class of superconductors for which a magnetic field can enter the bulk specimen while the material still remains in its superconducting state. These superconductors are called type II superconductors, as opposed to the type I superconductors in which the field cannot enter.

Figure 10.13 shows the behavior of a type I and a type II superconductor in an external magnetic field at zero temperature. For the type I material, the superconductivity breaks down above a certain critical field B_C. For higher fields, the external field completely penetrates the sample and the magnetization of the sample vanishes. For lower fields, the material is superconducting and its magnetization exactly compensates the external field such that the inner of the material is field-free. For

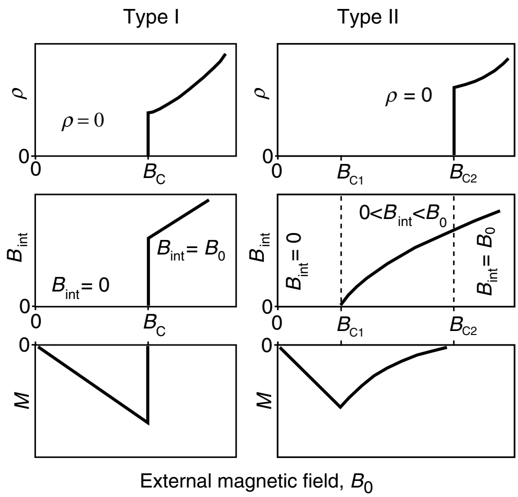

Figure 10.13 Resistivity ρ, internal magnetic field B_{int}, and magnetization M as a function of an external magnetic field for type I and type II superconductors. The temperature is assumed to be 0 K, such that the materials are in their superconducting state when no magnetic field is applied. The type I material has only one critical field B_C whereas the type II material has two different critical fields B_{C1} and B_{C2}.

a type II superconductor, the situation is different. There are two critical fields B_{C1} and B_{C2}. Below B_{C1} the material behaves exactly as a type I superconductor and above B_{C2} the superconductivity is destroyed. Between these two fields, however, the magnetic field partly enters the material. There is a finite magnetization but it is not large enough to compensate the external field. The remarkable fact is that the resistivity is still zero between these two critical fields.

If we compare the critical field B_C for a type I superconductor to the two fields of a type II material, it is typically much closer to B_{C1} than to B_{C2}, meaning that the type II superconductors can tolerate a higher critical field than the type I materials and still remain superconducting. It is this feature that makes type II superconductors very important for technical applications. One example is the coils for superconducting electromagnets. In these, type II materials such as NbTi are used, which have critical fields above 10 T.

How can the superconductivity in the type II superconductor survive a magnetic field entering the material? The answer to this is shown in Figure 10.14. The field penetrates through very thin filaments of normal-state material while the rest of the specimen remains superconducting. The filaments of normal-state material are enclosed by vortices of supercurrent such that the rest of the material remains field free and in its superconducting state. It turns out that the filaments contain only one magnetic flux quantum, so for higher external fields more vortices must enter the sample. Eventually, the density of vortices becomes so large that the superconducting regions of the sample disappear.

The existence of vortices in the superconductor represents a problem to the superconductivity itself. As a current passes through the material, a force similar to the Lorentz force will be exerted on the vortices causing them to move perpendicular to the current and to the magnetic field. This movement leads to an energy dissipation

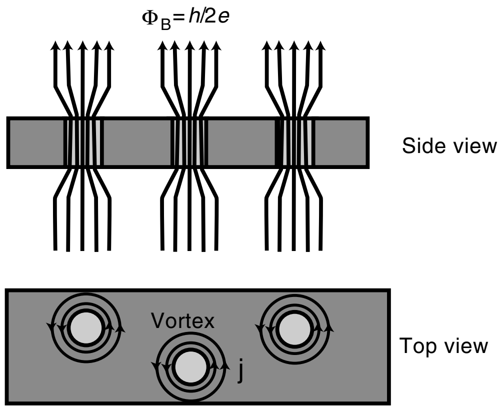

Figure 10.14 Magnetic flux in a type II superconductor. The field penetrates through thin filaments of material in the normal state while the rest of the sample remains superconducting. The filaments are surrounded by vortices of superconducting current that keep the rest of the sample field free.

and hence to a finite resistance even in the superconducting state. However, there is one possibility to avoid this phenomenon and this is to "pin" the vortices by a sufficient number of defects in the crystal. Then a certain energy will be associated with "unpinning" the vortices and as long as the current density is small enough, this energy will not be available. Type II superconductors in technical applications are therefore far from being perfect single crystals. Quite the opposite is desirable. The materials should have a sufficient number of defects such as grain boundaries in order to pin vortices efficiently.

The difference between a material being a type I or a type II superconductor depends on the ratio between the two characteristic length scales in a superconductor, the London penetration depth λ_L and the coherence length ξ in the Ginzburg Landau theory. We have seen that ξ sets the length scale on which the density of Cooper pairs can change appreciably. For our simple treatment here, it is appropriate to think of the coherence length as the distance between the two electrons in a Cooper pair, which we estimated to be in the order of a 100 nm or so. When ξ is smaller than λ_L, the formation of normal-state filaments surrounded by supercurrents becomes favorable and the material tends to be a type II superconductor. If ξ is long with respect to λ_L, the formation of the filaments is not favorable and the material is a type I superconductor. Disorder tends to reduce the coherence length ξ and therefore many intermetallic alloys are type II superconductors.

10.6
High-Temperature Superconductivity

What would we have to do to design a superconductor with a very high T_C, preferably above room temperature? We have already discussed the BCS equation for the transition temperature (10.13) and seen that the scale for T_C is set by the Debye temperature of the solid. Debye temperatures for materials made of light elements with strong bonds can be very high (several hundred kelvin), so this is not the limiting factor. The limitation is in the exponential term in (10.13) where one should try to obtain both a high density of states at the Fermi level and a strong electron–phonon coupling. Estimates of these quantities show that one cannot hope to get a T_C much higher than 30 K or so and, indeed, the progress in designing materials with increasing T_C was very slow initially (Figure 10.15).

A disruptive change of this development came in 1986 when J. G. Bednorz and K. A. Müller discovered superconductivity in the cuprate-perovskite ceramic materials. T_C for the particular material found by Bednorz and Müller was not especially high, only 30 K, but higher than for any compound found before. More importantly, the superconductivity was discovered in an entirely new class of materials. A short time later, superconductors of a similar kind but with a slightly different composition were found, which actually superconduct above the boiling point of liquid nitrogen. This opens (in principle) the possibility for new technological applications because cooling with liquid nitrogen is considerably cheaper than with liquid helium. It is in this sense one has to interpret the term "high-temperature superconductivity."

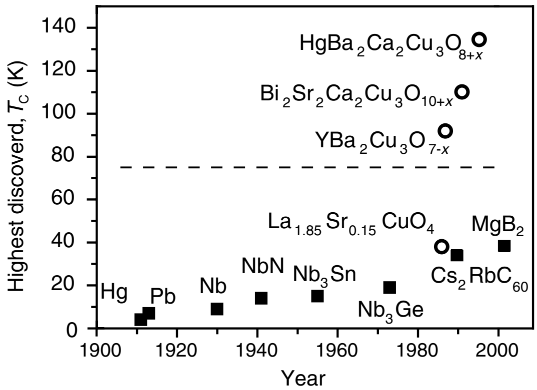

Figure 10.15 Increase of the critical temperature T_C of known superconductors as a function of time. The dashed horizontal line is the boiling temperature of liquid nitrogen. "Conventional" BCS-type superconductors are denoted by squares and novel "high-temperature" superconductors by circles. The *x* denotes a non-stoichiometric composition.

The cuprate perovskite materials are actually insulators that can be doped to become (poor) conductors and, at lower temperatures, superconductors. They have a fairly complicated structure but for us the only important point is that they consist of CuO_2 layers that are separated by other building blocks.

The mechanism of high-temperature superconductivity is not understood yet but there is agreement on a number of points. Most importantly, the supercurrent is also carried by Cooper pairs. It is unclear, however, what mechanism binds them together. Some theories suggest that the electron–phonon interaction can play a role, much like in BCS superconductors, but most favor other mechanisms, such as certain magnetic excitations. The behavior of the high-temperature superconductors in the normal state is not understood either. It is very different from that of a normal metal.

The Cooper pairs in high-temperature superconductors are mainly localized in the CuO_2 planes of the perovskite structure, meaning that the material has very anisotropic properties. In fact, one can view the high-temperature superconductivity as a process that happens practically in two dimensions. Unfortunately, the anisotropy also brings about very low critical current densities for polycrystalline samples, in which the planes in one domain do not match with the planes in the neighboring domain. Nevertheless, many practical hurdles have been overcome and high-temperature superconductors have started to play a role in commercial applications.

10.7 Concluding Remarks

As already mentioned in the introduction of this chapter, we were merely able to introduce some of the very basic ideas about superconductivity and we could not even explain those in any detail. Superconductivity is an extremely rich subject and there are many additional phenomena that were not included here. You can also find exceptions to almost every "rule" we have discussed. There are, for example, many superconductors that are well described by the BCS theory but that fail to show the $M^{-1/2}$ behavior in the isotope effect. Some even show no dependence of T_C on the isotope combination or one that goes in the "wrong" direction. The high-temperature superconductors have a superconducting gap but it is of a very different nature than for the "traditional" BCS superconductors. There is, in fact, no real gap in the sense of Figure 10.8, that is, no energy interval without any single-particle states around the Fermi level. The gap in these materials appears only in certain regions of the three-dimensional Brillouin zone.

This list goes on but we conclude the chapter with a more general question: What is the connection between superconductivity and magnetic ordering? At first it seems that the answer is that there is none – both are mutually exclusive, as we have seen several times in this chapter. But it is really more interesting than that. Magnetic ordering and superconductivity have in common that the electrons with the highest energy, those at the Fermi level, have a possibility to slightly lower their energy, either by establishing magnetic order or by condensing into Cooper pairs. The energy lowering appears small relative to their absolute kinetic energy but it is highly relevant compared to the average thermal energy. In some materials, there is even a close competition between magnetism and superconductivity. Often antiferromagnetic ordering occurs in the same material as superconductivity or the material becomes superconductive at some low temperature and normal, but ferromagnetically ordered, at an even lower temperature.

There are other mechanisms apart from magnetic ordering and superconductivity that give the possibility of an energy lowering for the electrons near the Fermi level. Particularly complex and interesting situations arise when several of these mechanisms compete with each other and this complexity is a very active research area in current solid-state physics.

10.8 Discussion and Problems

10.8.1 Discussion

1. How can you be sure that a superconductor has zero resistance? How can you measure its resistance?
2. What can be the cause for the superconductivity of a material breaking down, even if the temperature is below the critical temperature?
3. Describe the Meissner effect. Why is the Meissner effect "more" than just a consequence of zero resistance?
4. Is it really such that magnetic fields do not penetrate at all into a superconductor?
5. Describe the key ideas behind the microscopic BCS theory for superconductivity.

6. What experimental evidence supports the BCS theory?
7. What is the difference between a type I and a type II superconductor?
8. Why can the critical current in a type II superconductor be much smaller than the current needed to build up a critical magnetic field?

10.8.2 Problems

1. *Zero resistivity*: We have seen that the decay of the superconducting current can be measured by inducing a current in a superconducting ring and measuring how the current decays. (a) How would you induce a current in such a ring? (b) How would you measure if it decays or not? (c) Show that the decay (if any) would be exponential in time. (d) Assume that the ring has a diameter of 0.5 cm and is made of a very thin wire. If you observe that the current decays by less than 1% in a year, estimate the maximum resistance of the wire? *Hint*: For this last part you need to know the strength of the B field through the ring. For simplicity, assume that the field throughout the ring has the same strength as in the center.

2. (*) *Isotope effect*: Calculate the absolute temperature range of T_C for the T_C values in Figure 10.5. Experimentally, it can be difficult to scan the temperature through the superconducting transition for all the samples. Can you suggest an alternative experimental approach that avoids this problem?

3. *London penetration depth*: In connection with Figure 10.6 and Equation 10.9 it was argued that the **B** field decays exponentially inside the superconductor. Show this explicitly for a situation as in Figure 10.6, where the surface is in the $x-y$ plane and the field has only an x component, that is, $\mathbf{B} = (B_x, 0, 0)$.

4. *BCS theory*: The BCS theory predicts an exponential temperature dependence of the heat capacity at very low temperatures, instead of a linear heat capacity usually observed for electrons. (a) Explain qualitatively why this is so. (b)(*) Explain qualitatively why the exponential behavior is only observed at very low temperatures, that is, much lower than T_C.

5. *BCS theory*: We have seen that both the typical dimension of the Cooper pairs and the London penetration depth are of the order of 100 nm and we have assumed that the Ginzburg–Landau coherence length $\boldsymbol{\xi}$ has the same order of magnitude. If $\boldsymbol{\xi}$ is the length scale on which the density of Cooper pairs can change appreciably, how likely is it that the superconducting state breaks down in a volume $\boldsymbol{\xi}^3$? (a) Give a rough estimate of the total energy gain for the condensation of Cooper pairs in a volume of $(100\ \text{nm})^3$. (b) Can you say something about the probability of the superconductivity breaking down spontaneously due to a thermal fluctuation? Assume that the rate R of such breakdowns per second follows a Boltzmann distribution

$$R = f\mathrm{e}^{-E/k_B T}. \tag{10.20}$$

Argue that this rate is extremely small, even though you do not know the actual value of the "attempt frequency" f.

6. *BCS theory*: It is experimentally found that the thermal conductivity of a superconductor well below T_C is smaller than that for the same material in the normal state. Can you explain this?

7. (*) *BCS theory*: Consider the typical temperature dependence of the resistivity $\rho(T)$ for a superconductor and a non-superconducting material. In the case of Figure 10.1, it appears that the resistivity of the superconductor above T_C increases more strongly than for the non-superconducting material. Can you give a plausible explanation?

8. *BCS theory*: How do you expect the London penetration depth λ_L to depend on the temperature (qualitatively)?

11
Finite Solids and Nanostructures

We have so far assumed that we deal with infinite solids or, more precisely, with big but finite solids without any boundaries. This was achieved by using the Born–von Kármán boundary conditions and it has been a very successful concept. The perfect periodicity of the lattice has allowed us to solve many problems that we could not have solved otherwise. One example is the vibrational motion of the atoms: Instead of solving the equations of motion for a practically infinite number of atoms in a linear chain of macroscopic dimensions, we have reduced the problem to a small set of equations. Another example is the electronic structure, where the periodicity of the problem and the introduction of the reciprocal lattice permitted a very simple description of the potential felt by the electrons.

In this chapter, we briefly treat finite solids with a specific emphasis on nanostructured materials such as clusters, ultrathin films, and so on. Currently, there is a strong interest in nanotechnology. In physics and chemistry, much of this is because many materials drastically change their properties on a small scale (electronic, optical, catalytic, etc.). Here, we describe why this is so but we confine our attention to the very basic physical ideas.

We focus on three aspects of the small scale: The first is that the number of possible vibrations or electronic states is drastically reduced. As a consequence, the continuum of allowed energy states in a macroscopic solid is turned into a few discrete levels. This effect is called quantum confinement. The second aspect is that small objects, like clusters, necessarily have a relatively large surface (or interface) area, that is, a much larger ratio of surface area to bulk volume than macroscopic objects. The atoms at the surface (or interface) find themselves in a different environment from that of the bulk atoms and can therefore have other vibrational and electronic properties. If the structure is sufficiently small, these properties will dominate the entire nano-object. Finally, we will see how ferromagnetic ordering in a particle is affected by its size. For very small particles, the spins are still aligned with respect to each other below the Curie temperature, but the magnetization of the entire particle can be rotated by thermal fluctuations at much lower temperatures.

Another interesting aspect that we will not discuss any further is that of reduced dimensionality. In three-dimensional crystals, we succeeded relatively well to describe the electronic states by assuming no interaction between the electrons. As the

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number of dimensions is reduced, to a two-dimensional sheet or a one-dimensional wire, this assumption becomes increasingly problematic and new phenomena arise because of the correlated motion of the electrons. In the case of a strictly one-dimensional system, this is intuitively clear: neglecting the electron–electron interaction will be a bad approximation because the electrons cannot even get past each other.

11.1 Quantum Confinement

In order to see how reducing the size of a system can change its properties, we just have to look back at Figure 4.4 that shows the allowed vibrational frequencies for a finite chain of 10 atoms. The possible vibrational frequencies are denoted by black dots, and they are clearly a very small subset of the possible vibrations for an infinite chain. Formally, this is caused by the requirement of having k-values consistent with the boundary conditions. In the case of Figure 4.4, we had taken these to be (4.20). We see that if the system is sufficiently small, the number of allowed states is reduced; instead of having a quasicontinuum of states we have discrete levels that are separated by a significant energy. This is the essence of quantum confinement.

A particularly good illustration of quantum confinement can be realized by a very thin metal film that is grown on a semiconductor or an insulator (see Figure 11.1a). The wave function $\Psi(\mathbf{r})$ of the electrons in the film can be written as

$$\Psi(\mathbf{r}) = \Psi(z)\Psi(x, y), \tag{11.1}$$

where z is the direction perpendicular to the film, that is, the motion of the electrons perpendicular and parallel to the film can be separated. For the motion parallel to the film, we assume that the electrons behave like free electrons. $\Psi(x, y)$ is then a two-dimensional free electron wave function and the energies associated with the motion

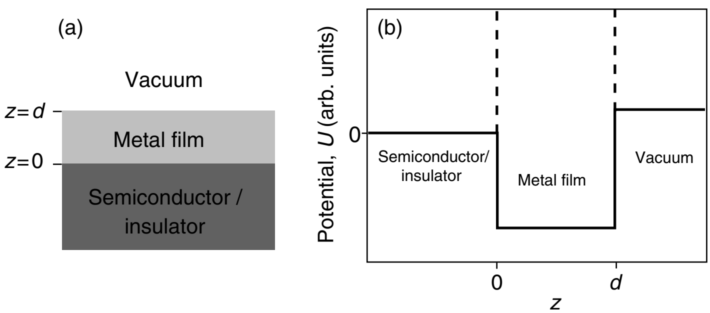

Figure 11.1 (a) A thin metal film on a semiconducting or insulating substrate. (b) Modeling the metal film as a potential well. The dashed lines indicate the simplification of an infinitely deep potential well.

parallel to the film are

$$E_{xy} = \frac{\hbar^2 \mathbf{k}_{xy}^2}{2m_e}, \tag{11.2}$$

where $\mathbf{k}_{xy}$ is the two-dimensional wave vector for the free electron motion parallel to the film. The possible values of $\mathbf{k}_{xy}$ are given by the usual periodic boundary conditions.

More interesting are the wave function and the energy levels for the motion perpendicular to the film, that is, in the z direction. An appropriate model for this is a simple potential well between the metal/semiconductor interface ($z=0$) and the metal/vacuum interface ($z=d$). Such a potential is shown in Figure 11.1b. The potential inside the metal film can be taken to be zero. The potential well has a finite depth and its height on the two sides is, in general, different.

The solutions for an infinitely deep potential well are known from elementary quantum mechanics. The solution inside the well can be written as the superposition of two free electron waves, one moving to the right and the other moving to the left:

$$\Psi(z) = A\mathrm{e}^{ik_z z} + B\mathrm{e}^{-ik_z z}, \tag{11.3}$$

where k_z is the wave number in the z direction and A and B are complex amplitudes. The boundary conditions are that this wave function has to vanish at $z=0$ and $z=d$. This immediately gives the quantization condition for k:

$$k_z d = n\pi, \qquad n = 1, 2, 3, \ldots. \tag{11.4}$$

If we now realize that $k_z = 2\pi/\lambda$, then the quantization condition is $2\mathrm{d} = n\lambda$. This has the familiar meaning that one round trip through the potential well must correspond to an integer number of wavelengths. The possible energies for the states in the z direction are

$$E_z = \frac{\hbar^2 k_z^2}{2m_e}, \tag{11.5}$$

and the total energy for a given k_z and $\mathbf{k}_{xy}$ is the sum of (11.2) and (11.5).

As mentioned above, the infinitely deep potential well is not a very appropriate model and one should really use the potential in Figure 11.1(b). However, the resulting quantization condition is similar to (11.4). It turns out to be

$$2k_z d + \Phi_i + \Phi_v = 2\pi n, \qquad n = 1, 2, 3, \ldots. \tag{11.6}$$

The meaning of this equation is that the total phase change upon one round trip of the electron must be a multiple of 2π. This total phase change must include the phase shifts at the metal/semiconductor interface Φ_i and at the metal/vacuum interface Φ_v. This model describes the resulting k_z values for real thin metal films very well.

In all this, we have only considered the Schrödinger equation for free electrons for which we know the solution already. The only thing we have changed is the boundary conditions for the solution (in the z direction). These new boundary conditions strongly restrict the possible states and give rise to the discrete energy levels in the z direction.

Similar ideas can be used to describe the properties of semiconductor clusters with a radius of less than $\approx$100 nm, so-called quantum dots. In these clusters, the possible electron and hole energies are quantized because of confinement in three dimensions. In a bulk semiconductor, the smallest energy for forming an electron–hole pair is the gap energy E_g. In quantum dots, this energy is larger and it increases as the dot's size decreases. An approximate formula for the minimum energy to separate an electron from a hole is

$$E_{\min} = E_g + \frac{\hbar^2 \pi^2}{2\mu r^2} - \frac{1.8 e^2}{4\pi\varepsilon_0 \varepsilon r}, \tag{11.7}$$

where r is the cluster radius and μ is the reduced mass from the electron and hole effective masses. The second term on the right-hand side is due to the confinement effect: It increases the energy when the electron and hole wave functions are "compressed" in a smaller quantum dot. Remember that in a simple potential well of size r the possible energies are also proportional to r^{-2}. The third term is due to the Coulomb interaction between the electron and the hole. It works in the other direction because the attraction is stronger for a smaller cluster. As the cluster size is decreased, the r^{-2} term eventually wins and becomes the most important term for really small clusters. The change of the smallest excitation energy as a function of cluster size is an extremely useful property, because it allows the production of materials with optical absorption at a desired wavelength.

The small size of the clusters has also another effect on their optical properties. If the cluster is small enough, the electrons are localized and the uncertainty principle dictates that they must have a considerable momentum uncertainty, which (for a free particle) is equivalent to a k uncertainty. In other words, k is not properly defined anymore, consistent with the interpretation of k as the quantum number of infinite translational symmetry. But if k is not well defined, this means that the forbidden transitions across the indirect bandgap in a material such as Si are no longer strictly forbidden and one can start to exploit the optical properties of Si and other materials with an indirect bandgap.

11.2 Surfaces and Interfaces

The surface of a solid can give rise to entirely new, localized electronic (and vibrational) states and this leads to a surface that has different properties than the bulk. We illustrate this for the electronic states in Figure 11.2. Consider first Figure 11.2a that shows the behavior of a bulk state near a surface. The bulk state is a Bloch wave.

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\mathbf{r}} u_{\mathbf{k}}(\mathbf{r}). \tag{11.8}$$

At the surface, this state and its derivative have to be matched continuously to an exponentially decaying wave function in the vacuum.

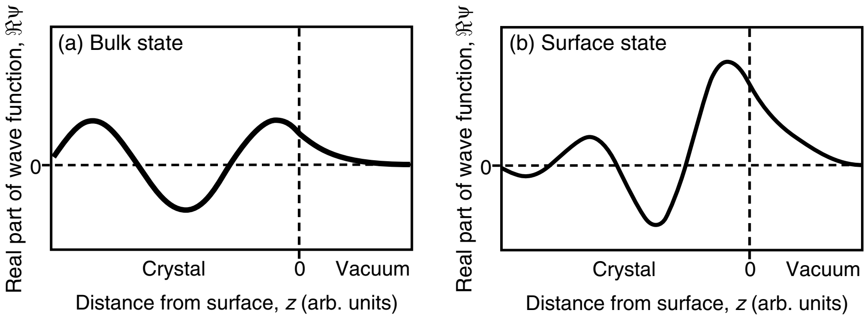

Figure 11.2 (a) Matching of a bulk electronic state (a Bloch wave) to an exponential decay outside the surface. (b) An electronic surface state that is decaying as the distance from the surface is increased, both outside and inside the solid.

The wave vector **k** of the Bloch wave is a purely real number. It is an interesting question to ask what would happen if it was complex. If we assume that the z component of **k** was complex, we could take the imaginary part $\Im(k_z)$ of it out of the plane wave part of the Bloch wave, leading to

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{\Im(k_z)z} e^{i\mathbf{k}' \cdot \mathbf{r}} u_{\mathbf{k}}(\mathbf{r}), \tag{11.9}$$

where $\mathbf{k}'$ is the remaining real part of the wave vector, that is, $(k_x, k_y, \Re(k_z))$. This clearly represents a problem because this solution is exponentially increasing in the $+z$ direction. It can therefore not be normalized and is not physically meaningful.

At the surface, however, such solutions with a complex wave vector are possible and an example is illustrated in Figure 11.2b. The solution has a complex wave vector in the z direction, which means that it is exponentially increasing in that direction. However, as the crystal is terminated by the surface, it has to be matched to an exponentially decreasing wave function outside the surface and it can still be normalized. Parallel to the surface the potential still has translational symmetry, so inside the solid this solution can be written as a two-dimensional Bloch wave times an oscillating damping factor:

$$\psi_{\mathbf{k}_{||}}(\mathbf{r}) = e^{i\mathbf{k}_{||} \cdot \mathbf{r}_{||}} u_{\mathbf{k}_{||}}(\mathbf{r}_{||}) e^{i\kappa z}, \tag{11.10}$$

where $\mathbf{k}_{||} = (k_x, k_y)$ and $\mathbf{r}_{||} = (x, y)$. This solution is a surface-localized electronic state.

Similar arguments can be made for the interface between two solids where localized interface states can exist. Such states (or their avoidance) can have some technical impact on semiconductor devices where they might affect the performance.

For a truly small crystallite, the surfaces confining it are very small and their properties cannot be described by assuming a perfectly two-dimensional situation as in (11.10). Instead, there will be many edges with atoms of low coordination. The chemical reactivity of these atoms can be much higher than that of the atoms in the flat surfaces and this fact is an important contribution to the sometimes improved chemical properties of nano-scale solids.

11.3 Magnetism on the Nanoscale

If a solid is sufficiently small, we have to re-think our approach to its magnetization and this has enormous technological consequences. For a large magnet, the energy required to flip one magnetic moment is of the order of the exchange energy, or the energy corresponding to the Curie temperature $k_B\Theta_C$. This can be much higher than k_BT at room temperature. Hence, large permanent magnets do not loose their magnetization due to some thermal fluctuations. For a small ferromagnetic particle this is still true, but now another possibility arises that can change the magnetization of the particle even if $T \ll \Theta_C$. If the particle is smaller than the typical thickness of a Bloch wall, it will only consist of one domain. Instead of changing the magnetization by flipping all the moments one by one, they can then stay aligned with each other and the entire domain can be rotated *at the same time*. If the particle is sufficiently small, the energy cost for doing this can become small enough to change the magnetization of the entire particle at room temperature, even for ferromagnetic materials with a strong exchange interaction and a high Θ_C. This phenomenon is called superparamagnetism.

We can estimate the energy cost of rotating the entire magnetic moment of a small particle. Assuming that there is no external field, the energy must be comparable to the energy of the particle's magnetization inside the field that is generated by its own magnetization:

$$\Delta E \approx VMB_{\text{inside}}. \tag{11.11}$$

Unfortunately, we do not know B_{inside}, the magnetic field inside the sample. In the chapter about magnetism, we have used (8.3) for the B field in matter. This equation is appropriate for diamagnetic and paramagnetic materials but it does not work for ferromagnetic materials. The reason is essentially the same, which has lead to the introduction of the local field in dielectrics. In diamagnetic and paramagnetic materials, the magnetization is always much smaller than the external field and therefore only the average magnetization and the external field have to be taken into account. In ferromagnetic materials, this is not correct. It turns out that B_{inside} depends on the shape of the magnetic particle but its order of magnitude is $\mu_0 M$, so that we can make the approximation

$$\Delta E \approx V\mu_0 M^2. \tag{11.12}$$

The energy required for rotating the magnetization thus depends linearly on the volume of the particle. For iron particles with a size below a few nanometers, this energy becomes comparable to k_BT at room temperature (see Problem 11.3).

This is obviously a problem for the fabrication of magnetic storage devices. The magnetic particle used for storing one bit must be large enough to prevent thermally induced rotations of the magnetization. On the other hand, superparamagnetic nanoparticles have useful applications because of their large magnetic moments, for example, in medical magnetic resonance imaging.

11.4 Discussion and Problems

11.4.1 Discussion

1. What are the main physical reasons why the properties of nano-scale solids are different from macroscopic solids of the same material?
2. How are the optical properties of semiconductors changed in nano-scale clusters?
3. Explain why it is possible that the magnetization of ferromagnetic nanoparticles can rapidly fluctuate at room temperature, even though the Curie temperature of the material is much higher.

11.4.2 Problems

1. *Boundary conditions*: In this chapter, we have seen that the different properties of solids on the nano-scale are largely due to new boundary conditions, which lead to fewer and discrete solutions of the Schrödinger equation. This is a bit worrying because in the previous chapters, we have just used whatever boundary conditions seemed most convenient. In particular, we have preferred the Born–von Kármán boundary conditions over the fixed boundary conditions where the wave function (or the vibrational amplitude) has to vanish at the boundary of a crystal. Fortunately, the precise choice of boundary conditions is not so important if the solid is only big enough. In order to show this, calculate the density of states for the free electron model in Chapter 6 when assuming fixed boundary conditions. Compare the result to (6.13).

2. *Semiconducting nanoclusters*: (a) Calculate the minimum electron–hole separation energy for CdSe nanoclusters as a function of cluster size. Plot the two contributions (confinement and Coulomb interaction) and their sum separately. (b) Such nanoclusters can be used as fluorescent markers. When exposed to UV light, this leads to the separation of electrons and holes. Eventually, these recombine under the emission of light corresponding to the minimum separation energy. How large would the clusters have to be to emit yellow light?

3. *Superparamagnetism*: (a) How small would an iron particle need to be such that the energy for rotating its entire magnetization becomes comparable to $k_B T$ at room temperature? Assume that each iron atom contributes to the magnetization with a moment of $2.2\boldsymbol{\mu}_B$. (b) Given the energy barrier ΔE, assume that the rate of magnetization rotations R, that is, the number of rotations per second, follows a Boltzmann distribution:

$$R = f e^{-\Delta E / k_B T}. \tag{11.13}$$

The frequency f is of the order $10^9\,\mathrm{s}^{-1}$. How large would the magnetic iron particles in a hard disk have to be if you only want to tolerate one flip every 10 years?

Appendix

This appendix briefly deals with some basics in electromagnetism. It assumes that you are familiar with the Maxwell equations in vacuum in their integral form, as discussed in basic introductory textbooks. What is given here are the explicit forms of vector operations, the so-called differential form of the Maxwell equations and how the Maxwell equations change in the presence of matter. A very good and more in-depth introduction of all three subjects is given in "The Feynman Lectures on Physics."

A.1
Explicit Forms of Vector Operations

In this book, several vector operations are used, which are given here explicitly. They all deal with differentiating a vector field or a scalar field. Let $\mathbf{E}(\mathbf{x})$ be a vector field and $\phi(\mathbf{x})$ a scalar field.

The gradient of the scalar field $\phi(\mathbf{x})$ is a vector with the components:

$$\operatorname{grad}\phi = \nabla\phi = \left(\frac{\partial}{\partial x},\frac{\partial}{\partial y},\frac{\partial}{\partial z}\right)\phi = \left(\frac{\partial\phi}{\partial x},\frac{\partial\phi}{\partial y},\frac{\partial\phi}{\partial z}\right). \tag{A1}$$

The often-used operator ∇ (called "nabla") is defined as

$$\nabla = \left(\frac{\partial}{\partial x},\frac{\partial}{\partial y},\frac{\partial}{\partial z}\right). \tag{A2}$$

The divergence of a vector field $\mathbf{E}(\mathbf{x})$ is a scalar field:

$$\operatorname{div}\mathbf{E} = \nabla\mathbf{E} = \left(\frac{\partial}{\partial x},\frac{\partial}{\partial y},\frac{\partial}{\partial z}\right)\mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}. \tag{A3}$$

The curl of a vector field $\mathbf{E}$ is

$$\operatorname{curl}\mathbf{E} = \nabla\times\mathbf{E} = \left(\frac{\partial}{\partial y}E_3 - \frac{\partial}{\partial z}E_2, \frac{\partial}{\partial z}E_1 - \frac{\partial}{\partial x}E_3, \frac{\partial}{\partial x}E_2 - \frac{\partial}{\partial y}E_1\right). \tag{A4}$$

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It is always such that curl grad $\phi = 0$ and div curl $\mathbf{E} = 0$. Finally, an important second derivative is the divergence of the gradient of a scalar field $\phi(\mathbf{x})$:

$$\text{div grad}\,\phi = \nabla^2\phi = \Delta\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}. \tag{A5}$$

Δ is called the Laplace operator.

A.2
Microscopic Form of the Maxwell Equations

The familiar, integral form of the Maxwell equations is

- Gauss's law

$$\oint \mathbf{E}\mathrm{d}\mathbf{A} = \frac{Q_{\text{encl}}}{\varepsilon_0}; \tag{A6}$$

- Gauss's law for magnetism

$$\oint \mathbf{B}\mathrm{d}\mathbf{A} = 0; \tag{A7}$$

- Ampère's law

$$\oint \mathbf{B}\mathrm{d}\mathbf{l} = \mu_0\left(I_{\text{encl}} + \varepsilon_0\frac{\partial\Phi_{\mathrm{E}}}{\partial t}\right). \tag{A8}$$

- Faraday's law

$$\oint \mathbf{E}\mathrm{d}\mathbf{l} = -\frac{\partial\Phi_{\mathrm{B}}}{\partial t}. \tag{A9}$$

In order to transform these into the differential form frequently used in this book, one makes use of two fundamental theorems. The first is the divergence theorem

$$\oint \mathbf{E}\mathrm{d}\mathbf{A} = \int \text{div}\,\mathbf{E}\,\mathrm{d}V, \tag{A10}$$

which turns a surface integral over vector field into an integral over the enclosed volume and the divergence of the same field.

The second is Stokes' integral theorem, which states

$$\oint \mathbf{E}\mathrm{d}l = \int \text{curl}\,\mathbf{E}\mathrm{d}\mathbf{A}, \tag{A11}$$

which turns an integral along a closed line into an integral over a surface enclosed by the line. As usual, writing $\mathrm{d}\mathbf{A}$ in the integrals means that one integrates over the normal component of the vector field passing through the surfaces.

Using the divergence theorem, we can find a differential version of Gauss' law (A10):

$$\oint \mathbf{E}\mathrm{d}\mathbf{A} = \int \text{div}\,\mathbf{E}\,\mathrm{d}V = \frac{Q_{\text{encl}}}{\varepsilon_0} = \int \frac{\rho}{\varepsilon_0}\mathrm{d}V. \tag{A12}$$

Since this holds for an arbitrary volume, we can also write

$$\operatorname{div}\mathbf{E} = \frac{\rho}{\varepsilon_0}. \tag{A13}$$

Gauss' law of magnetostatics can be treated in the same way to give

$$\operatorname{div}\mathbf{B} = 0. \tag{A14}$$

Equation (A11) can be used in a similar way to rewrite the two other equations. We start with Ampère's law:

$$\oint \mathbf{B}\mathrm{d}\mathbf{l} = \int \operatorname{curl}\mathbf{B}\mathrm{d}\mathbf{A} = \mu_0\left(I_{\mathrm{encl}} + \varepsilon_0\frac{\partial \Phi_{\mathrm{E}}}{\partial t}\right) = \int \mu_0\mathbf{j} + \mu_0\varepsilon_0\frac{\partial \mathbf{E}}{\partial t}\mathrm{d}\mathbf{A}, \tag{A15}$$

which means that we can write

$$\operatorname{curl}\mathbf{B} = \mu_0\mathbf{j} + \mu_0\varepsilon_0\frac{\partial \mathbf{E}}{\partial t}. \tag{A16}$$

Finally, we find Faraday's law in the differential form as

$$\operatorname{curl}\mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \tag{A17}$$

A.3 The Maxwell Equations in Matter

The similarities between the electric and the magnetic polarization are better understood when we inspect the Maxwell equations in matter. Let us start in vacuum and add the effects of matter later on. In vacuum, there are two fields: the **E** field that is created by all charges and the **B** field that is created by all currents. The Maxwell equations are:

- Gauss' law

$$\operatorname{div}\mathbf{E} = \frac{\rho}{\varepsilon_0}; \tag{A18}$$

- Gauss' law for magnetism

$$\operatorname{div}\mathbf{B} = 0; \tag{A19}$$

- Faraday's law

$$\operatorname{curl}\mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0; \tag{A20}$$

- Ampère's law

$$\mathbf{j} = \frac{1}{\mu_0}\operatorname{curl}\mathbf{B} - \varepsilon_0\frac{\partial \mathbf{E}}{\partial t}. \tag{A21}$$

In matter, the most drastic modifications are made to Ampère's law. We can read (A21) such that current densities are created by the curl in magnetic fields and changing electric fields (in time). In a general solid, we have both. The magnetization $\mathbf{M}$ gives rise to currents $\mathbf{j}_m = \text{curl}\,\mathbf{M}$ and a change of the dielectric polarization $\mathbf{P}$ causes currents $\mathbf{j}_j = \partial\mathbf{P}/\partial t$. If we keep calling the "free" current density $\mathbf{j}$, this leads to

$$\mathbf{j} + \mathbf{j}_m + \mathbf{j}_e = \frac{1}{\mu_0}\text{curl}\,\mathbf{B} - \varepsilon_0\frac{\partial\mathbf{E}}{\partial t}, \tag{A22}$$

or

$$\mathbf{j} = \text{curl}\left(\frac{1}{\mu_0}\mathbf{B} - \mathbf{M}\right) - \frac{\partial}{\partial t}(\varepsilon_0\mathbf{E} + \mathbf{P}). \tag{A23}$$

Ampère's law in matter is then written in a more compact form using the usual definitions:

$$\mathbf{H} = \frac{1}{\mu_0}\mathbf{B} - \mathbf{M} \tag{A24}$$

and

$$\mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P}, \tag{A25}$$

where the $\mathbf{D}$ and $\mathbf{H}$ fields correspond to the $\mathbf{E}$ and $\mathbf{B}$ fields with the difference that the latter are caused by all charges and currents whereas $\mathbf{D}$ and $\mathbf{H}$ are only caused by the "free" charges and currents. We get for Ampère's law

$$\mathbf{j} = \text{curl}\,\mathbf{H} - \frac{\partial\mathbf{D}}{\partial t}. \tag{A26}$$

Gauss' law is also modified slightly. One has to take into account that the polarization of the solid can lead to a local increase of charge density $\rho_e = -\text{div}\,\mathbf{P}$.[1] We therefore get

$$\text{div}\,\mathbf{E} = \frac{\rho}{\varepsilon_0} + \frac{\rho_e}{\varepsilon_0} = \frac{\rho}{\varepsilon_0} - \frac{\text{div}\,\mathbf{P}}{\varepsilon_0}, \tag{A27}$$

or, using the usual definition of $\mathbf{D}$(A25),

$$\text{div}\,\mathbf{D} = \rho. \tag{A28}$$

Gauss' law for magnetostatics is unaffected by the presence of materials because there are still no magnetic monopoles. The same is true for Faraday's law as there are no magnetic monopole currents either.

1) The minus sign stems from the sign difference in the definition of $\mathbf{E}$ and $\mathbf{P}$. In the former the vector direction is from the positive to the negative charges and in the latter it is opposite.

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Further Reading

Solid-State Physics

There is no shortage of excellent, more advanced solid-state physics texts. A few are given here:

- Ashcroft, N.W. and Mermin, N.D. (1976) *Solid State Physics*, Holt-Saunders. A classic in the field. Very thorough treatment on an advanced level, both mathematically and conceptually.
- Kittel, C. (2005) *Introduction to Solid State Physics*, 8th edn, Wiley-VCH Verlag GmbH. Another classic. Regularly updated and excellent reference to many phenomena and material properties.
- Ibach, H. and Lüth, H. (2003) *Solid State Physics*, 3rd edn, Springer-Verlag. A modern and advanced text. Excellent in-depth treatment of magnetism and superconductivity.
- Rosenberg, H.M. (1988) *The Solid State*, 3rd. edn, Oxford University Press. A very well written book on a slightly more advanced level as this one. Contains a detailed section about the role of defects in solids.
- Feynman, R.P., Leighton, R.B. and Sand, M. (1966) *The Feynman Lectures on Physics*, Addison-Wesley. This is not a solid-state physics book but contains an excellent fundamental treatment of dielectrics, magnetism, and superconductivity.
- Turton, R.J. (2000) The Physics of Solids, Oxford University Press. A less advanced treatment as this one, not strictly requiring prior knowledge of quantum mechanics. A very good overview on materials' properties.

Statistical Physics

- Mandl, F. (1988) *Statistical Physics*, 2nd edn, Wiley-VCH Verlag GmbH.

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Magnetism

- Buschow, K.H.J. and de Boer, F.R. (2003) *Physics of Magnetism and Magnetic Materials*, Kluwer Academic Publishers.
- Himpsel, F.J., Ortega, J.E., Mankey, G.J. and Willis, R.F. (1998) Magnetic nanotructures. *Advances in Physics*, **47**, 511. Journal article with special emphasis on the nanoscale.

Superconductivity

- Buckel, W. and Kleiner, R. (2004) *Superconductivity*, 2nd edn, Wiley-VCH Verlag GmbH.

Physical Constants and Energy Equivalents

Planck constant	h	$6.6260755 \times 10^{-34}$ J s
		$4.13566743 \times 10^{-15}$ eV s
Boltzmann constant	k_B	1.380658×10^{-23} J K^{-1}
		8.617385×10^{-5} eV K^{-1}
Proton charge	e	$1.60217733 \times 10^{-19}$ C
Bohr radius	a_0	5.29177×10^{-11} m
Bohr magneton	μ_B	$9.2740154 \times 10^{-24}$ J T^{-1}
Avogadro number	N_A	6.0221367×10^{23} particles mol^{-1}
Speed of light	c	2.99792458×10^{8} m s^{-1}
Rest mass of the electron	m_e	$9.1093897 \times 10^{-31}$ kg
Rest mass of the proton	m_p	$1.6726231 \times 10^{-27}$ kg
Rest mass of the neutron	m_n	$1.6749286 \times 10^{-27}$ kg
Atomic mass unit	amu	1.66054×10^{-27} kg
Permeability of vacuum	μ_0	$4\pi \times 10^{-7}$ Vs A^{-1} m^{-1}
Permittivity of vacuum	ε_0	$8.854187817 \times 10^{-12}$ C^2 J^{-1} m^{-1}

$1\,\text{eV} = 1.6021773 \times 10^{-19}\,\text{J}$
$1\,\text{K} = 8.617385 \times 10^{-5}\,\text{eV}$

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Free Solutions Manual available for lectures

Filling a gap in the literature for a brief course in solid state physics, this is a clear and concise introduction that not only describes all the basic phenomena and concepts, but also discusses such advanced issues as magnetism and superconductivity. This textbook assumes only basic mathematical knowledge on the part of the reader and includes more than 100 discussion questions and some 70 problems with solutions as well as further supplementary material available for free to lecturers from the Wiley-VCH website.

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- Chemical Bonding in Solids
- Crystal Structures
- Mechanical Properties
- Thermal Properties of the Lattice
- Electronic Properties of Metals: Classical Approach
- Electronic Properties of Metals: Quantum Mechanical Approach
- Semiconductors
- Magnetism
- Dielectrics
- Superconductivity
- Finite Solids and Nanostructures

Philip Hofmann *studied physics at the Free University, Berlin and did his PhD research at the Fritz-Haber-Institute of the Max Planck Society, also in Berlin. He stayed at the Oak Ridge National Laboratory, USA, as a Feodor Lynen Fellow of the Alexander von Humboldt Foundation. In 1998, he moved to the University of Aarhus, Denmark, where he is associated with the Synchrotron Radiation Source and the Interdisciplinary Nanoscience Center (iNANO). His research is primarily focused on the electronic structure of solids and their surfaces.*

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